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SUPERPOSITION THEOREM

It states that in a linear network containing more than one independent source and dependent source, the resultant current in any element is the algebraic sum of the currents that would be produced by each independent source acting alone, all the other independent sources being represented meanwhile by their respective internal resistances.

The independent voltage sources are represented by their internal resistance if given or short circuits.

The independent current sources are represented by infinite resistance; ie open circuits.

A linear network is one whose parameters are constant, ie they do not change with voltage or current.

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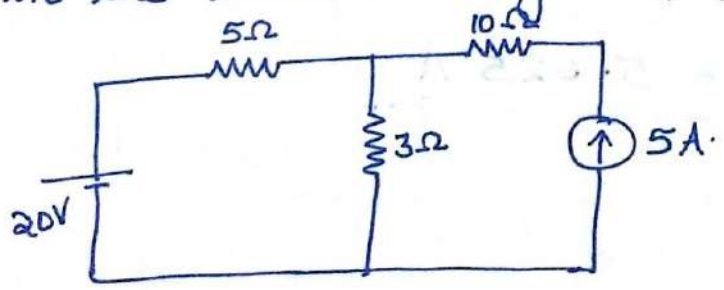
1. Find the current passing through the 3Ω resistor in the circuit using superposition theorem.



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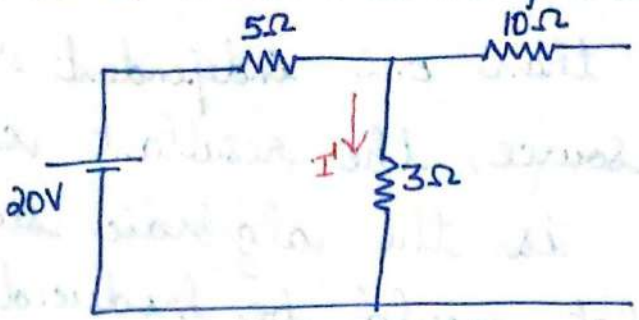
1. Considering 20V voltage source alone

Current source is replaced by an open circuit.

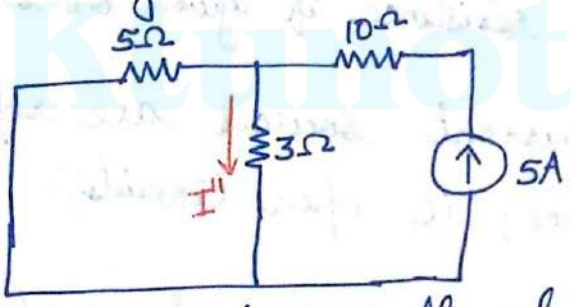


Current passing through 3Ω resistor is I'

$$I' = \frac{20}{5+3} = \frac{20}{8} = \underline{\underline{2.5A}}$$

2. Considering 5A current source alone

Voltage source is replaced by a short circuit.



Current passing through 3Ω resistor is I''

$$I'' = 5 \times \frac{5}{5+3} = \frac{25}{8} = \underline{\underline{3.125A}}$$

3. Total current passing through the 3Ω resistor is

$$I = I' + I''$$

$$= 2.5 + 3.125$$

$$= \underline{\underline{5.625\ A}}$$

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Steps to be followed in Superposition Theorem

1. Find the current through the resistance when only one independent source is acting, replacing all other independent sources by respective internal resistances.
2. Find the current through the resistance for each of the independent sources.
3. Find the resultant current through the resistance by the superposition theorem considering magnitude and direction of each current.

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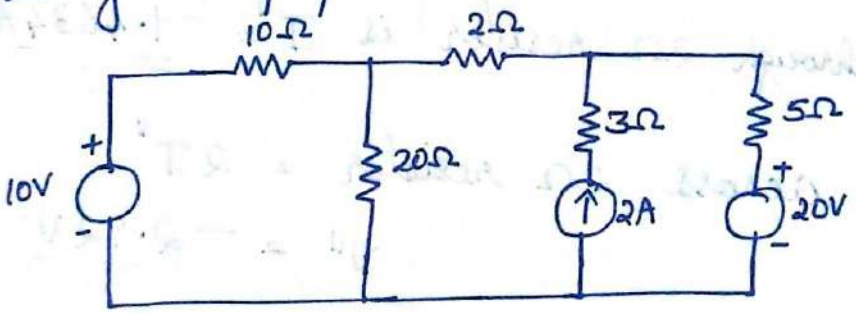
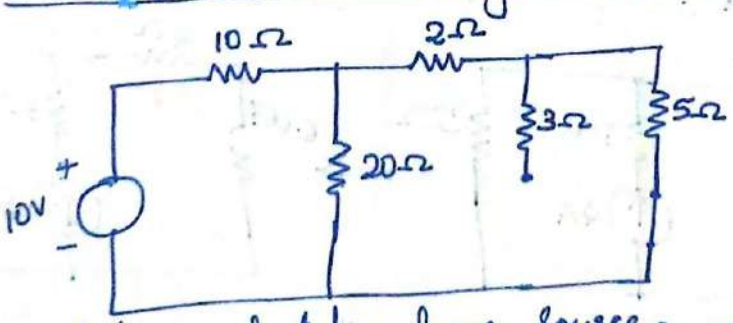
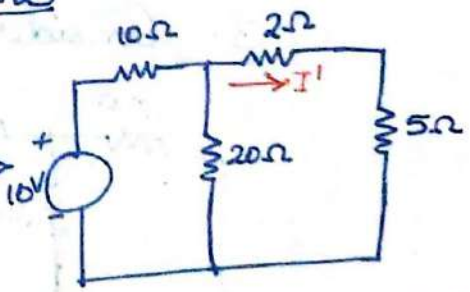
2. Find the voltage across the 2Ω resistor by using superposition theorem.



1. Consider 10V voltage source alone



$\Rightarrow$



Total current taken from source $= \dfrac{10}{(7 \| 20)+10} = \dfrac{10}{5.1851+10} = \underline{0.6585A}$

Current through $2\Omega = 0.6585 \times \dfrac{20}{20+7} = \underline{\underline{0.487\,A}}$

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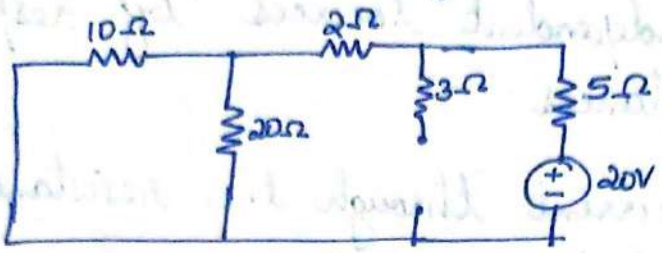
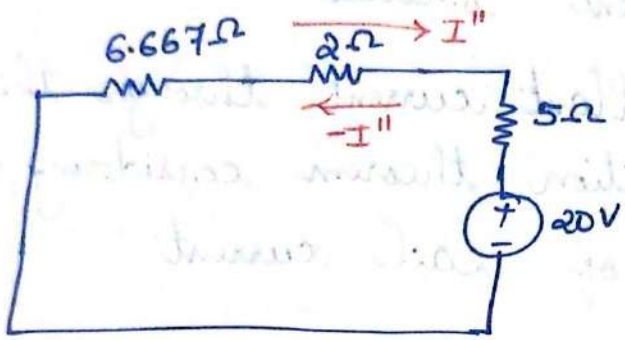
voltage across 2Ω resistor $= 2I'$

$$V' = 2 \times 0.487 = \underline{\underline{0.974V}}$$

2. Consider 20V voltage source alone



$\Rightarrow$



$10 \| 20$

$\frac{10 \times 20}{30} = \underline{\underline{6.667}}$

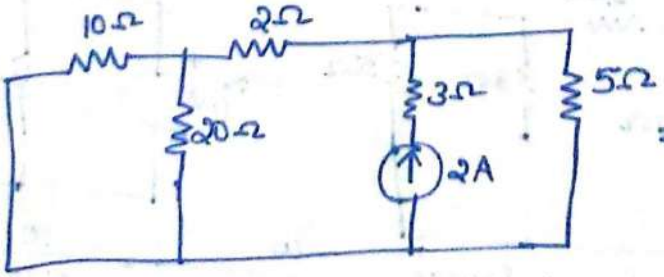
$$-I'' = \frac{20}{5+2+6.667} = 1.4634A$$

Current through 2Ω resistor is $I'' = \underline{\underline{-1.4634A}}$

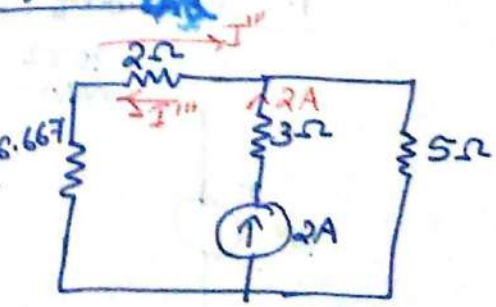
Voltage across 2Ω resistor $= 2I''$

$$V'' = \underline{\underline{-2.92V}}$$

3. Consider 2A Current source alone



$\Longrightarrow$



Current through 2Ω resistor $-I''' = 2 \times \frac{5}{5+8.667} = \underline{\underline{0.732A}}$

$$I''' = \underline{\underline{-0.732A}}$$

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Voltage across 2Ω resistor is $2I'''$

$$V''' = 2 \times -0.732 = \underline{-1.464}$$

4. Total voltage across 2Ω resistor is

$$V = V' + V'' + V'''$$

$$= 0.974 - 2.92 - 1.464$$

$$= \underline{-3.41V}$$

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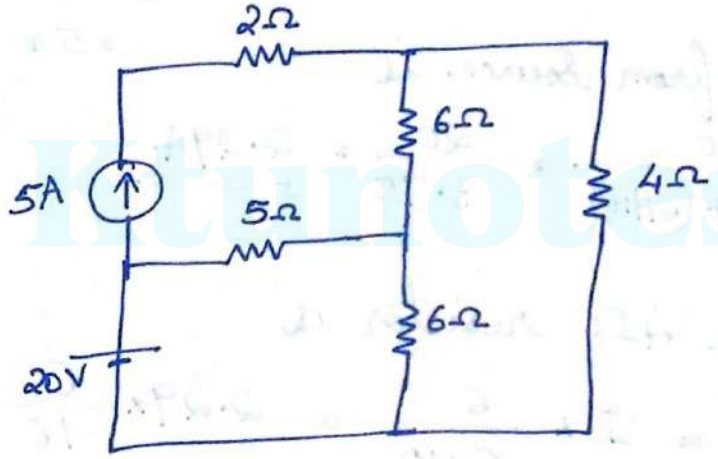
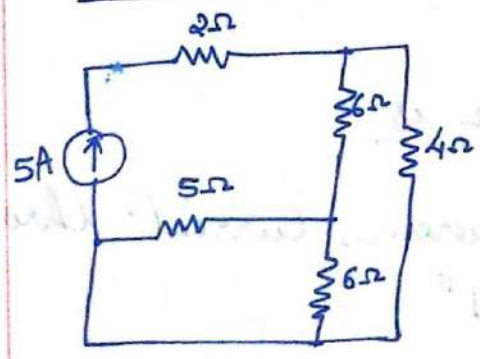
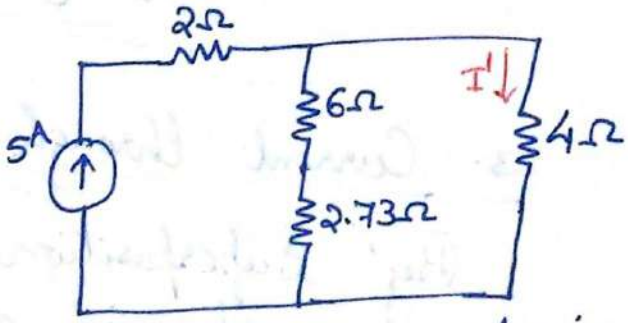
3. Find the current through 4Ω resistor



1. <u>Consider 5A source alone</u>



$\Rightarrow$



$$(5 \| 6) = \frac{5 \times 6}{5+6} = \underline{2.73\Omega}$$

Current through 4Ω resistor is

$$I' = 5 \times \frac{(6 + 2.73)}{6 + 2.73 + 4}$$

$$= \underline{3.43\,A\ (\downarrow)}$$

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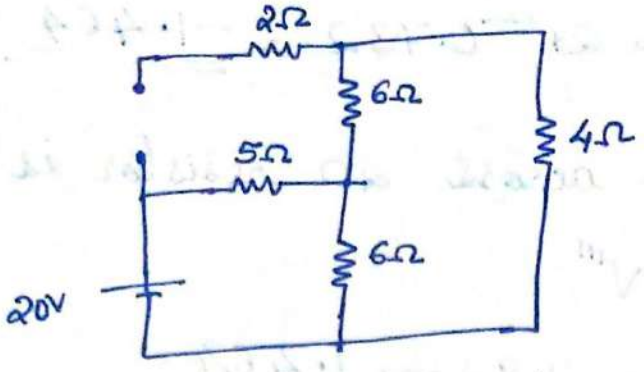
2. Consider 20V source alone



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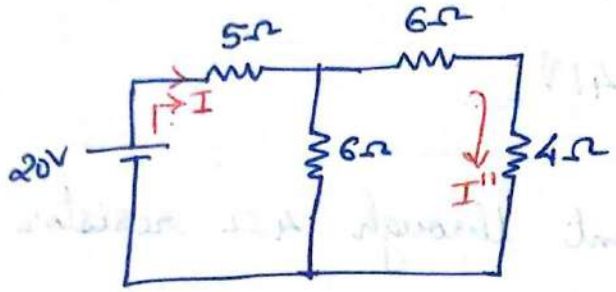
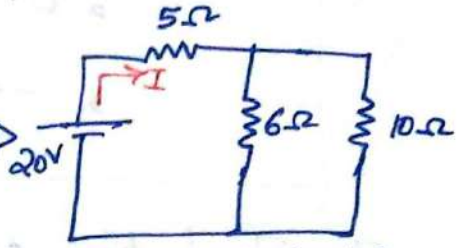


⇒



$R_T = 5 + (6 \| 10)$

$= 5 + \frac{6 \times 10}{16} = \underline{\underline{8.75}}$

Current taken from source is

$$I = \frac{20}{5 + (6 \| 10)} = \frac{20}{8.75} = \underline{\underline{2.29A}}$$

Current through 4Ω resistor is

$$I'' = I \times \frac{6}{6+10} = 2.29 \times \frac{6}{16}$$

$$= \underline{\underline{0.86A \ (\downarrow)}}$$

3. Current through 4Ω is

By Superposition theorem, Current through 4Ω resistor is $I' + I''$

$$= 3.43 + 0.86 = \underline{\underline{4.29A \ (\downarrow)}}$$

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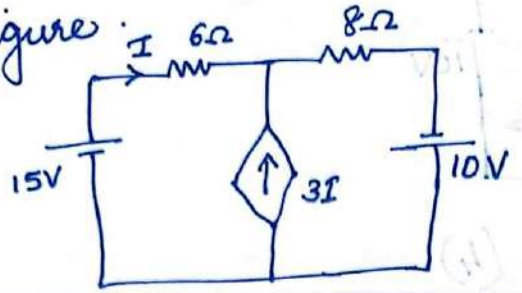
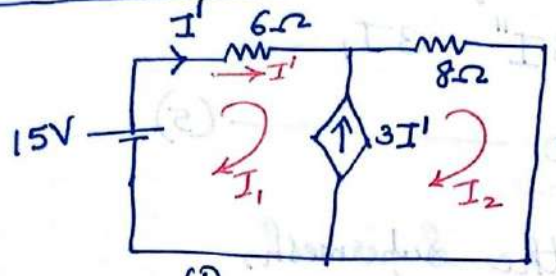
Example with Dependent Sources

4. Find the current through the 6Ω resistor in figure.



1. When the 15V source is acting alone



$I' = I_1$ —— (1)

Meshes 1 & 2 will form a supermesh,

$I_2 - I_1 = 3I' = 3I_1$

$I_2 = 4I_1$

$4I_1 - I_2 = 0$ —— (2)

Applying KVL to the outerpath of supermesh

$15 - 6I_1 - 8I_2 = 0$

$6I_1 + 8I_2 = 15$ —— (3)

Solving eqns (2) & (3)

$I_1 = 0.39A$, $I_2 = 1.57A$.

So $I' = I_1 = 0.39A$ (→)

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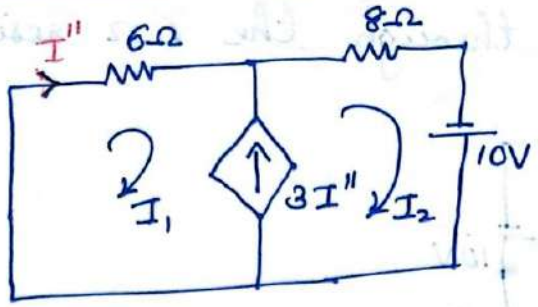
2. When the 10V source is acting alone



$I'' = I_1$ ——— (4)

Meshes 1 & 2 will form a supermesh.

$I_2 - I_1 = 3I'' = 3I_1$

$4I_1 - I_2 = 0$ ——— (5)

Applying KVL to the supermesh,

$-6I_1 - 8I_2 + 10 = 0$

$6I_1 + 8I_2 = 10$ ——— (6)

Solving (5) & (6)

$\Rightarrow I_1 = 0.26A, \quad I_2 = 1.05A$

$I'' = I_1 = 0.26A$

3. By superposition theorem

$I = I' + I'' = \cancel{0.39 + 1.05}$

$= 0.39 + 0.26 = 0.65A (\rightarrow)$

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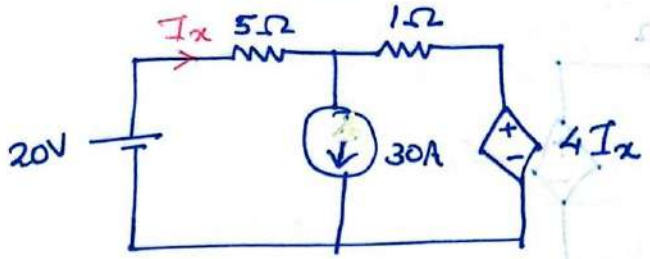
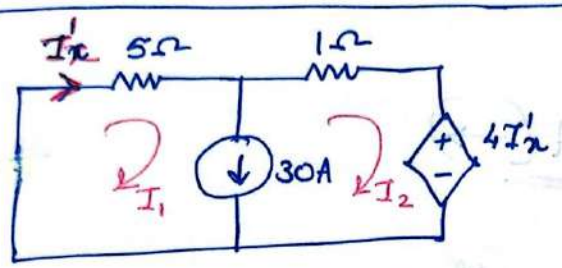
5 Find the current I_x



1. When 30A source is acting alone



$I_x' = I_1$ ——— (1)

Meshes 1 & 2 will form a supermesh

$I_1 - I_2 = 30$ ——— (2)

Applying KVL to path of supermesh

$-5I_1 - I_2 - 4I_x' = 0$

$-5I_1 - I_2 - 4I_1 = 0$

$9I_1 + I_2 = 0$ ——— (3)

Solving (2) & (3)

$\Rightarrow I_1 = 3A, \quad I_2 = -27A$

So $I_x' = I_1 = 3A \ (\rightarrow)$

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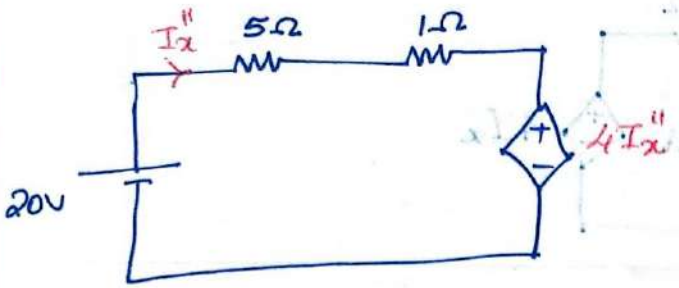
2. When the 20V source is acting alone.



$$20 - 5I_x'' - 1I_x'' - 4I_x'' = 0$$

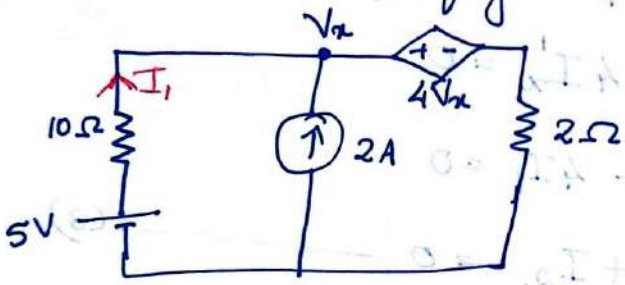
$$10I_x'' = 20$$

$$I_x'' = 2\,A\,(\rightarrow)$$

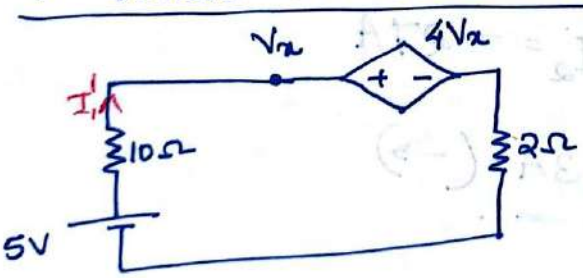
3. By Superposition theorem

$$I_x = I_x' + I_x'' = 3+2 = 5A\,(\rightarrow)$$

6. Find I_1 in the figure.



1. When 5V source is acting alone



$$5 - 10I_1' = V_x \quad \text{—— (1)}$$

Applying KVL to mesh

$$5 - 10I_1' - 4V_x - 2I_1' = 0$$

$$5 - 10I_1' - 4(5 - 10I_1') - 2I_1' = 0$$

$$5 - 10I_1' - 20 + 40I_1' - 2I_1' = 0$$

$$28I_1' = 15; \quad I_1' = \frac{15}{28}\,A\,(\uparrow) = 0.54A\,(\uparrow)$$

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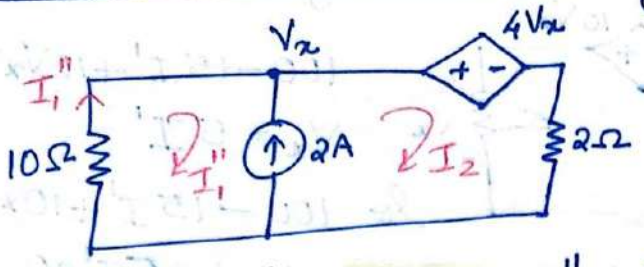
2. When 2A source is acting alone



$V_x = -10\,I_1'' \Rightarrow$ (1) $\quad -I_1'' + I_2 = 2$ —— (2)

$-10I_1'' - 4V_x - 2I_2 = 0$

$-10I_1'' - 4(-10I_1'') - 2I_2 = 0$

$-10I_1'' + 40I_1'' - 2I_2 = 0$

$30I_1'' - 2I_2 = 0$ —— (3)

Solving (2) & (3)

$I_1'' = 0.14A(\uparrow)$

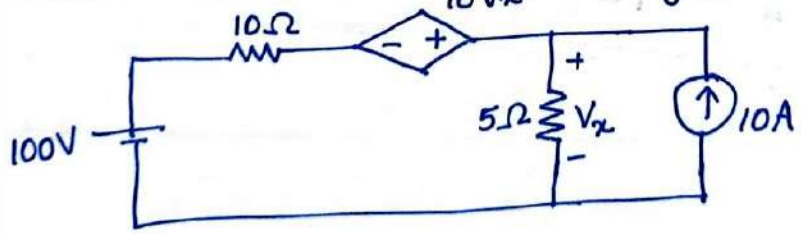
$I_2 = 2.14\,A\ (\circlearrowleft)$

3. By Superposition Theorem

$I_1 = I_1' + I_1''$

$= 0.54 + 0.14 = 0.68A\ (\uparrow)$

7. Determine the current through the 10Ω resistor



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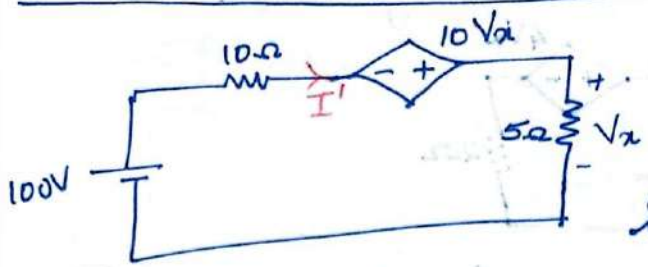
1. When 100V source is acting alone



$100 - 15I' + 10V_x = 0$

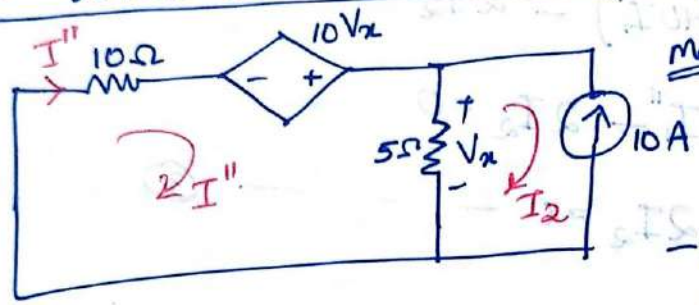
$V_x = 5I'$

So $100 - 15I' + 10 \times 5I' = 0$

$100 + 35I' = 0$

$I' = \frac{-100}{35} = -2.86A (\rightarrow)$

2. When 10A source is acting alone



$V_x = 5(I'' - I_2)$

Mesh 1 $-10I'' + 10V_x - V_x = 0$

$-10I'' + 9 \times 5(I'' - I_2) = 0$

$-10I'' + 45I'' - 45I_2 = 0$

$35I'' - 45I_2 = 0$ —— (1)

Mesh 2

$I_2 = -10A$ —— (2)

Solving (1) & (2) $35I'' + 45 \times 10 = 0$

$I'' = -12.86A (\rightarrow)$

3. By Superposition Theorem

$I = I' + I''$

$= -2.86 - 12.86$

$= -15.72A (\rightarrow)$

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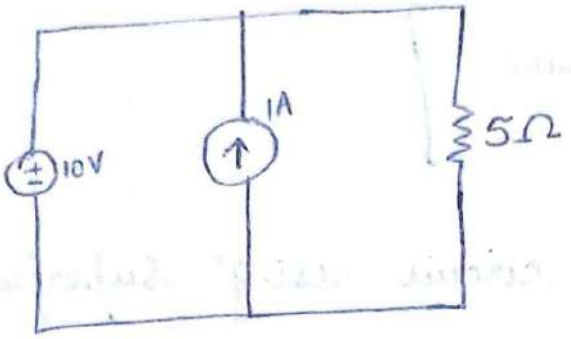
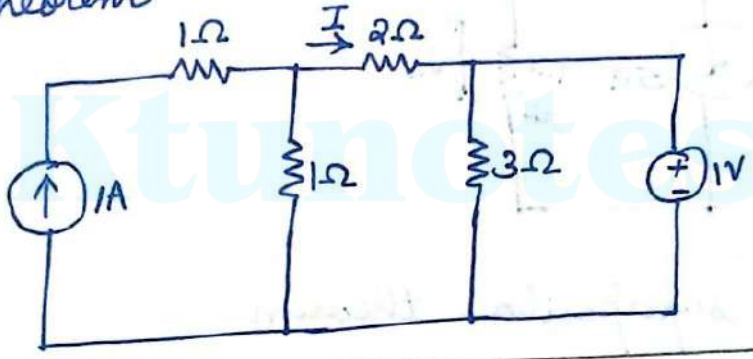
Tutorial Questions

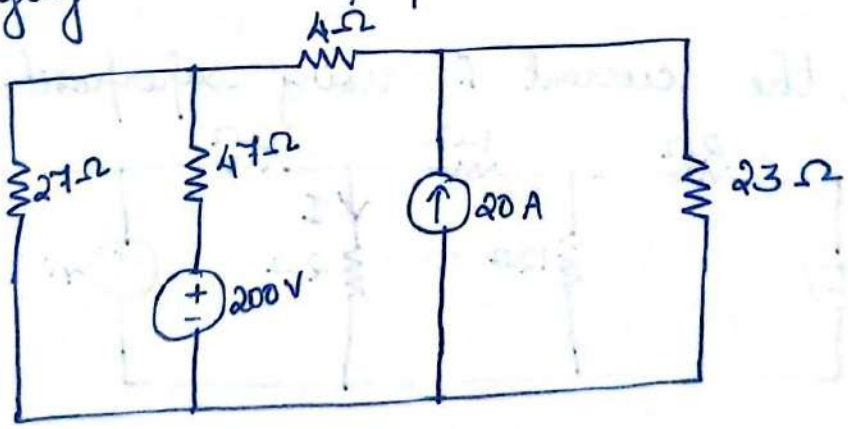
1. In the circuit, find the current through the 5Ω resistor using the principle of superposition.



2. Find I in the circuit shown using superposition theorem



3. Compute the current in the 23Ω resistor by applying the superposition principle.



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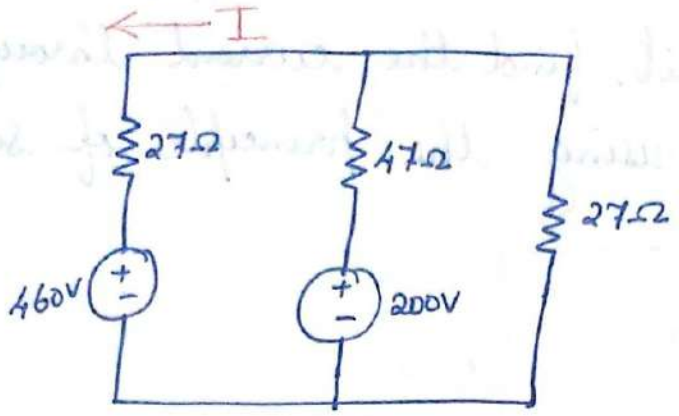
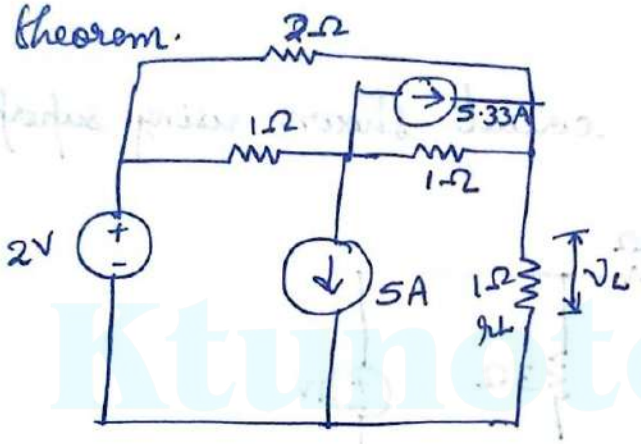
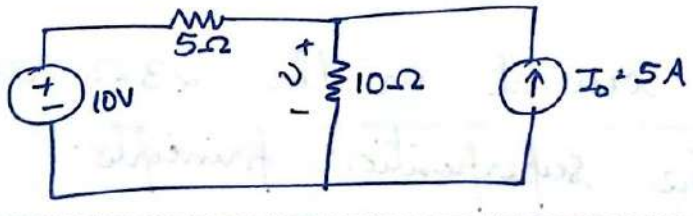
4. Use superposition theorem to find I.



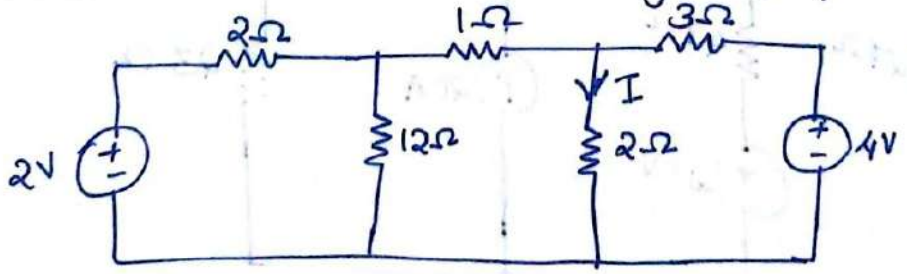
5. Find v_L in the circuit using superposition theorem.



6. Find v by superposition theorem.



7. Find the current I using superposition theorem



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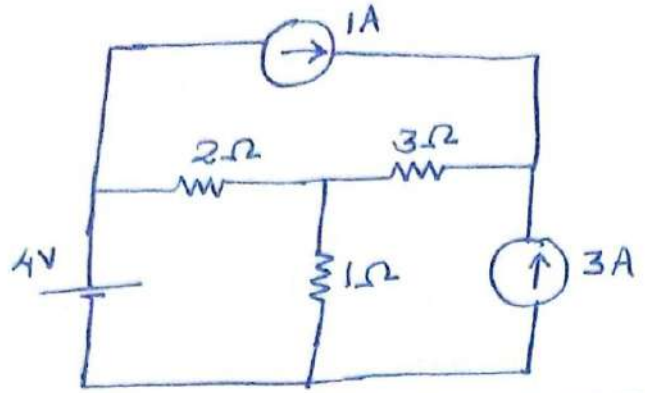
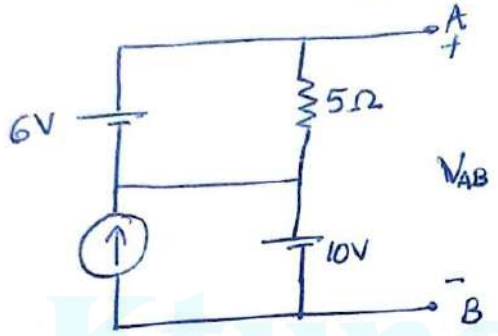
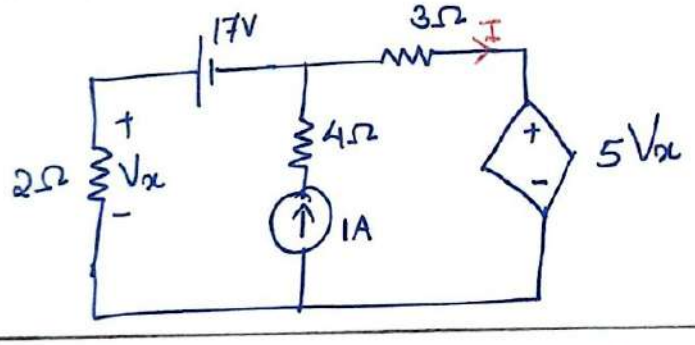
8. Find the current in the 1Ω resistor



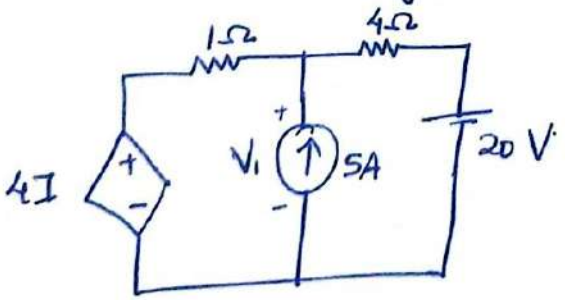
9. Find the voltage V_{AB} in figure.



10. Find the current I in the network of figure



11. Find the voltage V_1 in figure



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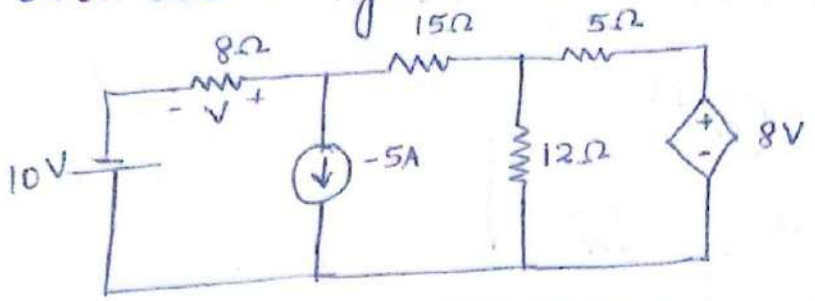
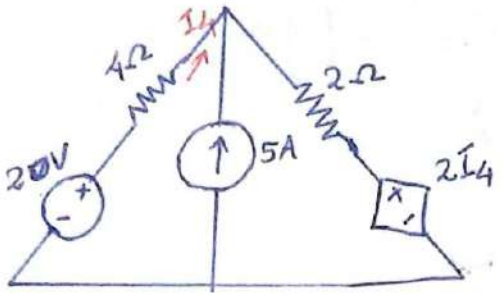
12 Find the Voltage V in the network



13. For the circuit in figure, Find Current I_4 using superposition principle.



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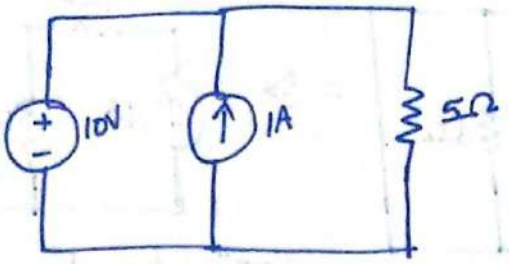
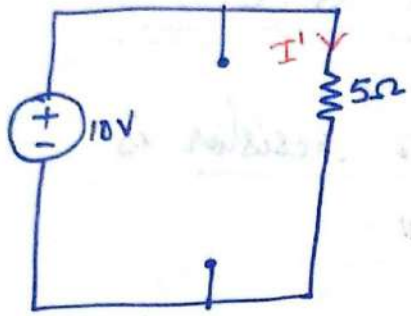
Solutions — Tutorial on Superposition Theorem

1.

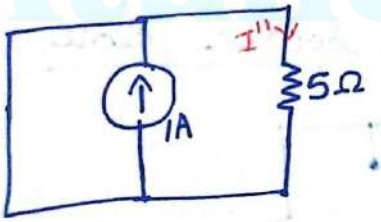


1. Considering 10V source alone



$I' = \frac{10}{5} = \underline{2A}$.

2. Considering 1A source alone



The entire current will pass through the short circuited path so $\underline{I'' = 0}$.

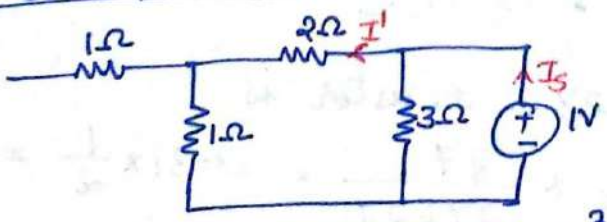
3. Current through 5Ω resistor is

$I = I' + I''$ (Superposition theorem)

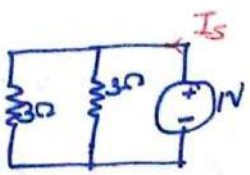
$= 2 + 0 = \underline{2A}$.

---

2. 1. Consider 1V source alone



$\Rightarrow$



$I_s = \frac{1}{(3//3)} = \frac{1}{3/2} = \underline{\frac{2}{3}A}$

Current through 2Ω, $I' = I_s \times \frac{3}{3+3} = \frac{2}{3} \times \frac{3}{6} = \underline{\frac{1}{3}A}$. (←)

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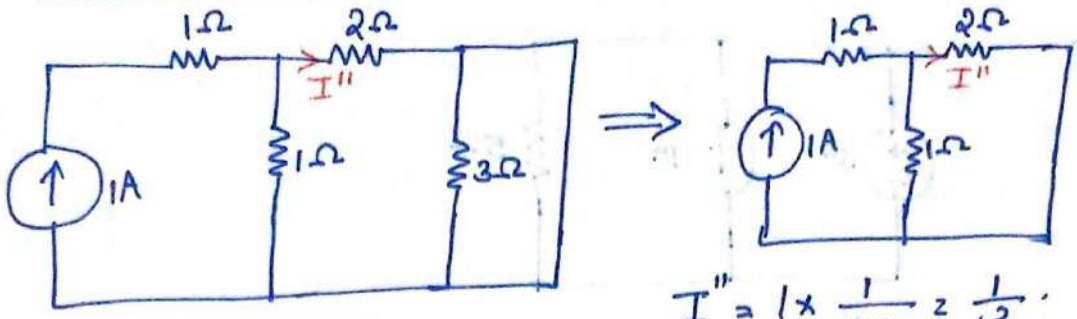
2. Consider 1A source alone



$$I'' = 1 \times \frac{1}{2+1} = \frac{1}{3}$$

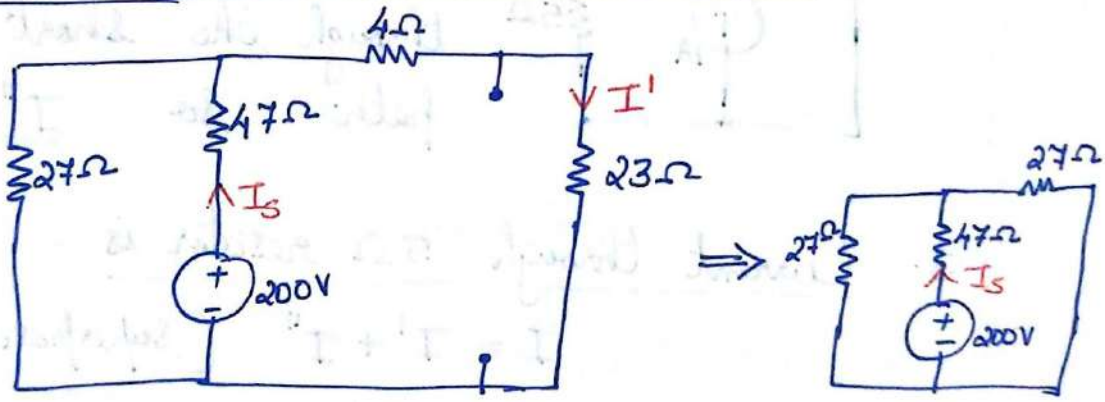
Current through 2Ω, $I'' = \frac{1}{3}$ (→)

3. Current through 2Ω resistor is

$$I = I' + I''$$

$$= -\frac{1}{3} + \frac{1}{3} = 0\,A$$

3. 1. Consider the 200V source alone



$$I_s = \frac{200}{47+(27\|27)} = \frac{200}{47+\frac{27\times27}{27+27}} = \frac{200}{47+\frac{27}{2}}$$

$$= \frac{200\times2}{121} = 3.31A$$

Current through 23Ω resistor is

$$I' = I_s \times \frac{27}{27+27} = 3.31\times\frac{1}{2} = 1.65A\ (\downarrow)$$

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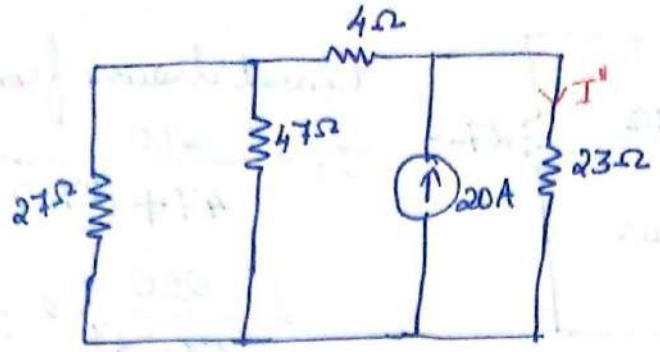
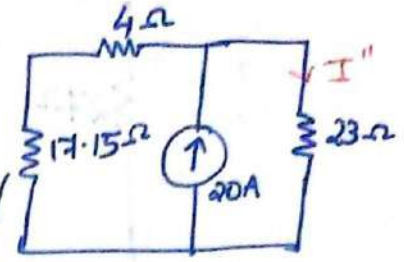
2. Consider 20A source alone



⇒



$$(27\|47) = \frac{27\times47}{27+47} = \underline{17.15\Omega}$$

$$I'' = 20\times\frac{(17.15+4)}{(17.15+4+23)}$$

$$= 20\times\frac{21.15}{44.15} = \underline{\underline{9.58\ A\ (\downarrow)}}$$

3. Total Current through 23 Ω resistor is

$$I = I' + I'' = 1.65 + 9.58$$

$$= \underline{\underline{11.23\ A}}$$

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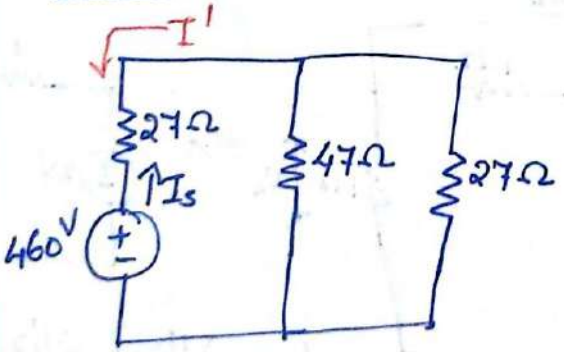
4. 1. Consider 460V source alone



Current drawn from the source $= \dfrac{460}{27+(47\|27)}$

$$= \frac{460}{27+\frac{47\times27}{47+27}} = \frac{460}{44.149}$$

$$= \underline{\underline{10.42A}}$$

Current drawn from source

$I_s = \underline{\underline{10.42A}}$

$\therefore I' = \underline{\underline{-10.42\ A}}$

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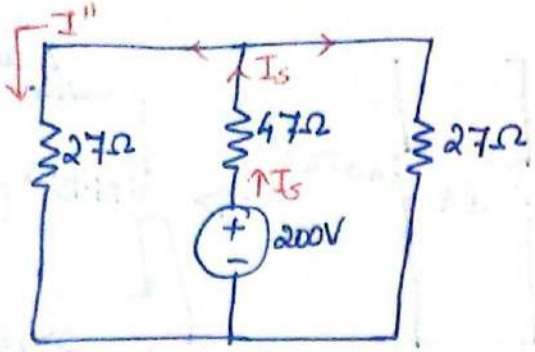
2. Consider 200V source alone



Current drawn from source

$$I_s = \frac{200}{47 + (27 \| 27)}$$

$$= \frac{200}{47 + \frac{27}{2}} = \underline{\underline{3.3A}}$$

Current through 27Ω resistor, $I'' = I_s \times \frac{27}{27+27}$

$$I'' = 3.3 \times \frac{1}{2} = \underline{\underline{1.65A}}$$

3. Net Current in 27Ω resistor is

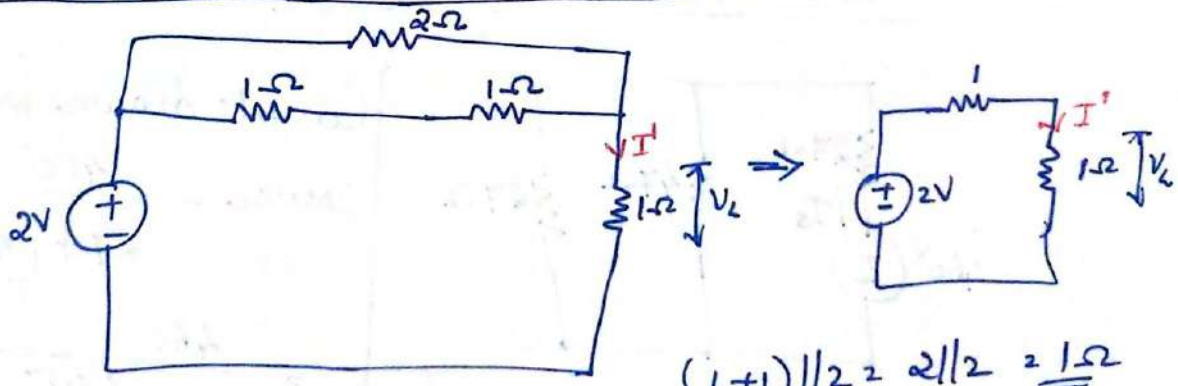
$$I = I' + I''$$

$$= -10.42 + 1.65$$

$$= \underline{\underline{-8.77A}}$$

---

5. 1. Consider 2V source alone



$(1+1) \| 2 = 2 \| 2 = \underline{\underline{1\Omega}}$

$$I' = \frac{2}{1+1} = \underline{\underline{1A}}$$

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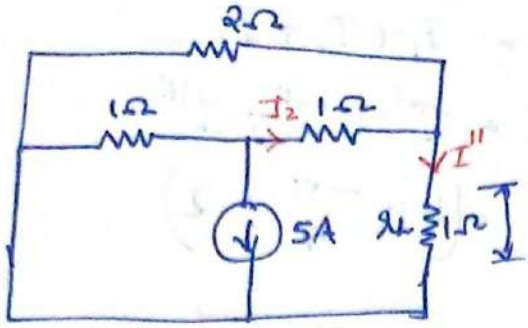
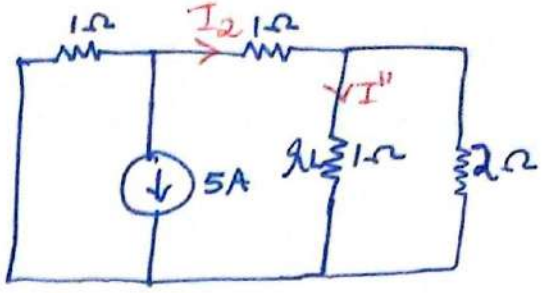
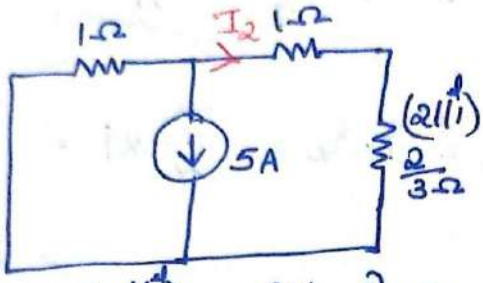
2. Consider 5A source alone



⇒

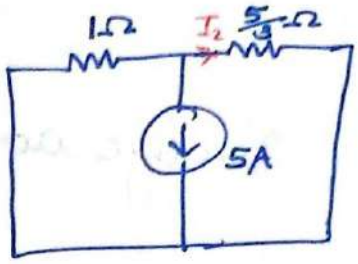


⇒



$(2||1) = \frac{2\times1}{3} = \frac{2}{3}\Omega$

⇒



$I_2 = -5 \times \frac{1}{1+\frac{5}{3}} = -5 \times \frac{1}{8/3}$

$= \frac{-15}{8} A.$

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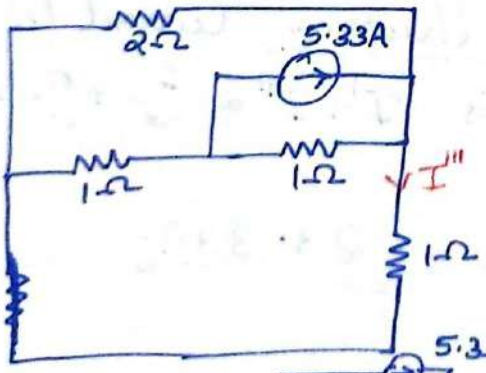
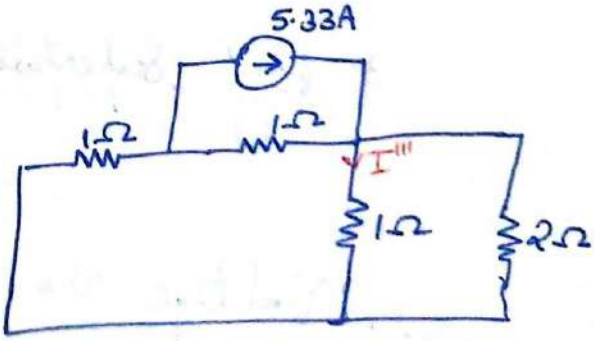
Current $I'' = I_2 \times \frac{2}{1+2} = I_2 \times \frac{2}{3}$

$= \frac{-15}{8} \times \frac{2}{3} = \frac{-5}{4} A.$

3. Consider 5.33A source alone



⇒



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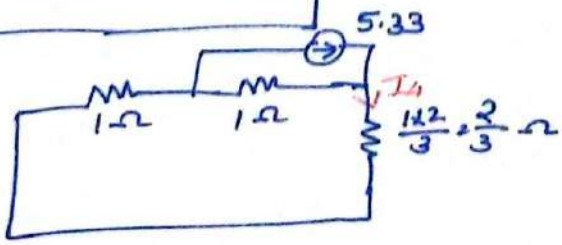

$I_4 = \frac{5.33\times1}{5/3} = 3.198A.$

$I''' = 3.198 \times \frac{2}{3} = 2A.$

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4. Net Current through r_L

$$\cancel{I = I_1 + I_2 + I_3}$$

$$I = I' + I'' + I'''$$

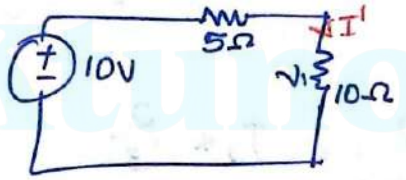
$$= \left(1 + \frac{-5}{4} + 2\right)$$

$$= 3 - \frac{5}{4} = \frac{7}{4} A.$$

Voltage across $r_L = I \cdot r_L = \frac{7}{4} \times 1 = \frac{7}{4} V$

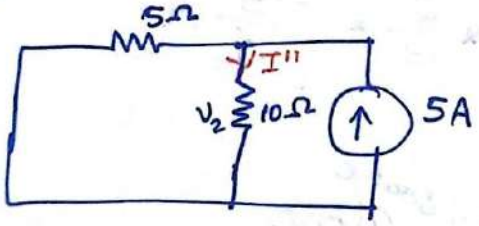
$$= 1.75 V$$

6. 1. Consider 10V source only



$$I' = \frac{10}{15} = \frac{2}{3} A.$$

2. Consider 5A source alone



$$I'' = 5 \times \frac{5}{5+10} = 5 \times \frac{5}{15}$$

$$= \frac{5}{3} A.$$

3. By Superposition theorem, Current through 10Ω is $I' + I'' = \frac{2}{3} + \frac{5}{3} = \frac{7}{3} A.$

$\therefore$ Voltage $v = \frac{7}{3} \times 10 = 23.33 A.$

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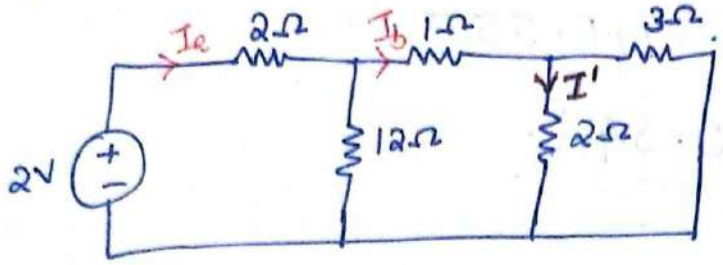
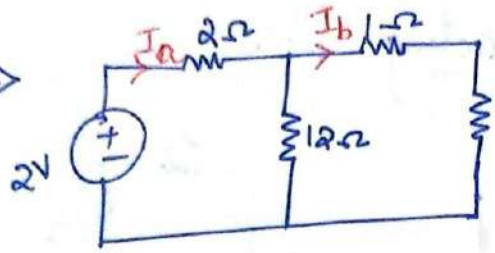
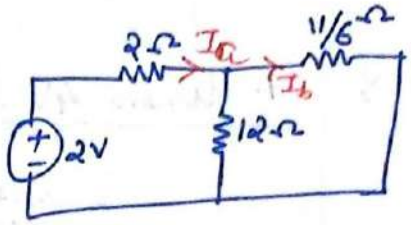
7. 1. Consider 2V source alone



$\Rightarrow$



$\Rightarrow$

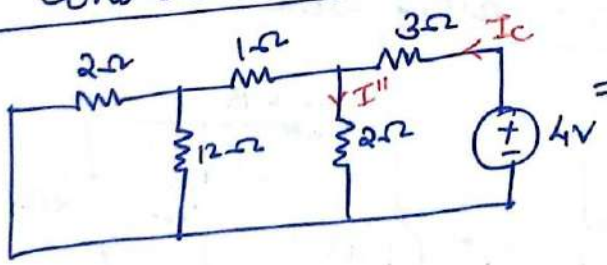


$$I_a = \frac{2}{2+\left(12 \| \frac{11}{6}\right)} = \frac{2}{2+\frac{12\times\frac{11}{6}}{12+\frac{11}{6}}} = \frac{2}{2+1.6} = \underline{\underline{0.56A}}$$

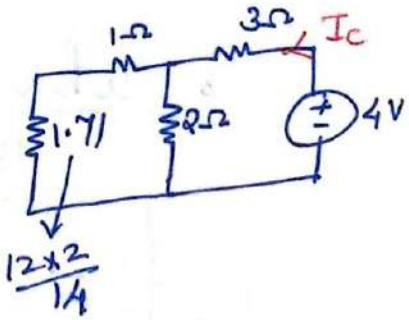
$$I_b = I_a \times \frac{12}{12+\frac{11}{6}} = 0.56 \times \frac{12}{12+\frac{11}{6}} = \underline{\underline{0.486A}}$$

$$I' = I_b \times \frac{3}{5} = 0.486 \times \frac{3}{5} = \underline{\underline{0.29A}}$$

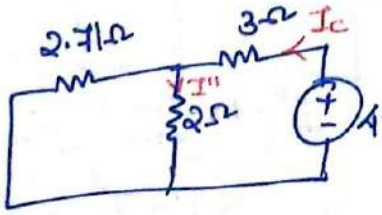
2. Consider 4V source alone



$\Rightarrow$



$\Rightarrow$



$$I_c = \frac{4}{3+(2\|2.71)} = \frac{4}{3+1.15}$$

$$= \underline{\underline{0.96A}}$$

$$I'' = 0.96 \times \frac{2.71}{4.71} = \underline{\underline{0.555A}}.$$

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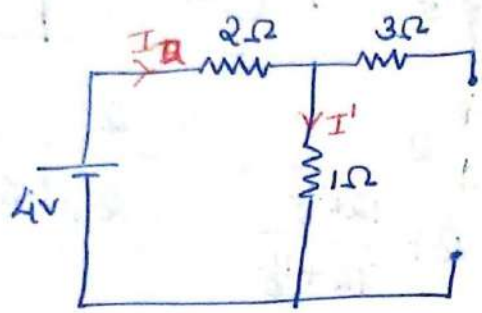
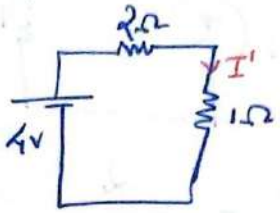
3. Net Current I is

$$I = 0.29 + 0.555$$
$$= 0.84\,A.$$

8. 1. When 4V Source acting alone

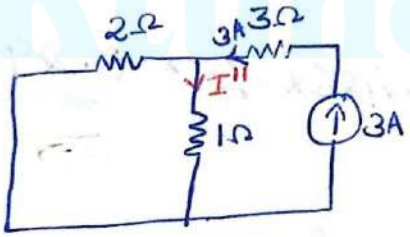


$\Rightarrow$



$$I' = \frac{4}{2+1} = \frac{4}{3}\,A.(\downarrow)$$

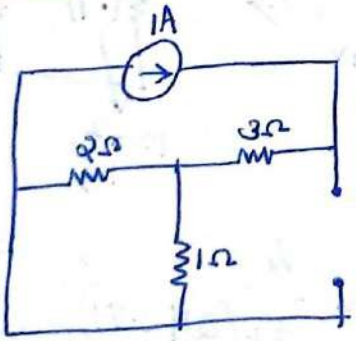
2. When 3A Source is acting alone



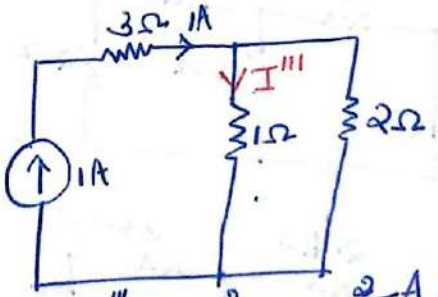
$$\Rightarrow I'' = 3 \times \frac{2}{(2+1)} = 3 \times \frac{2}{3}$$

$$I'' = 2A\ (\downarrow)$$

3. When 1A Source acting alone



$\Rightarrow$



$$I''' = 1 \times \frac{2}{(2+1)} = \frac{2}{3}\,A\ (\downarrow)$$

4. By Superposition Theorem $I = I' + I'' + I'''$

$$= \frac{4}{3} + 2 + \frac{2}{3} = 4A\ (\downarrow)$$

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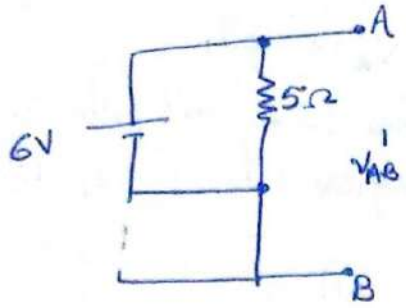
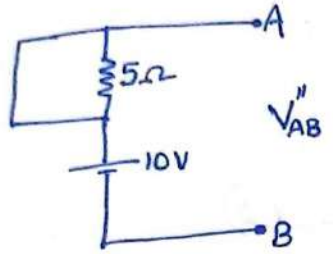
9. 1. When 6V source is acting alone



$V'_{AB} = \underline{\underline{6V}}$

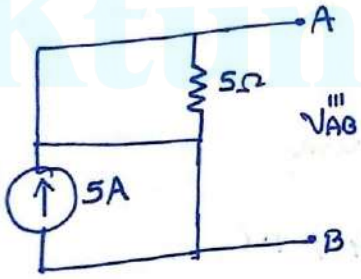
2. When 10V source is acting alone



Since the resistor 5Ω is shorted, the voltage across it is zero.

$V''_{AB} = \underline{\underline{10V}}$.

3. When 5A source is acting alone

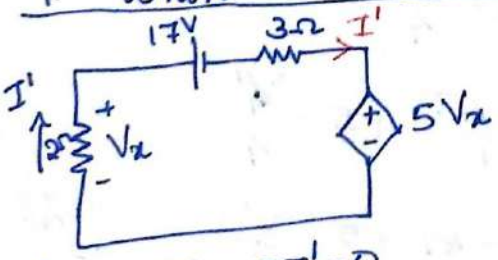


Due to short circuit in both parts, $\underline{\underline{V'''_{AB} = 0\ V}}$

4. By Superposition theorem

$V_{AB} = V'_{AB} + V''_{AB} + V'''_{AB} = 6+10+0 = \underline{\underline{16V}}$.

10. 1. When 17V source is acting alone



$-2I' = V_x$.

Consider the Mesh

$-V_x - 17 - 3I' - 5V_x = 0$

$-4V_x - 17 - 3I' = 0$

$-4 \times -2I' - 17 - 3I' = 0$

$8I' - 17 - 3I' = 0$

$5I' = 17$.

$I' = \frac{17}{5} = \underline{3.4A(\rightarrow)}$

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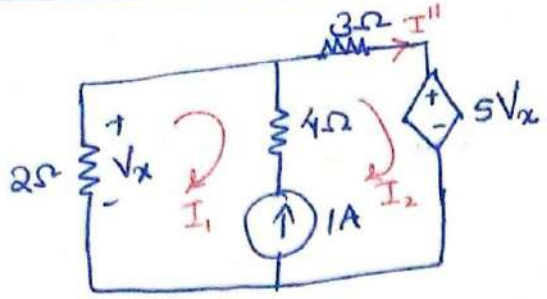
2. When 1A source is acting alone



$V_x = -2I_1$

Meshes 1 & 2 will form a Supermesh $I_2 - I_1 = 1A$

$I_2 - I_1 = 1A$ ——— (1)

KVL $\Rightarrow -2I_1 - 3I_2 - 5V_x = 0$

$-2I_1 - 3I_2 - 5(-2I_1) = 0$

$-2I_1 - 3I_2 + 10I_1 = 0$

$8I_1 - 3I_2 = 0$ ——— (2)

Solving (1) & (2)

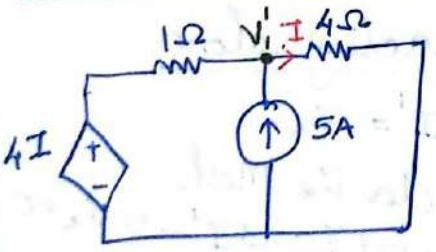
$\Rightarrow I_1 = 0.6A, \quad I_2 = 1.6A.$ $\quad I'' = 1.6A\ (\rightarrow)$

3. By Superposition Theorem

$I = I' + I'' = 3.4 + 1.6$

$= 5A\ (\rightarrow).$

II. 1. When 5A source is acting alone



$I = \frac{V_1'}{4}.$

$\frac{V_1' - 4I}{1} + 5 = \frac{V_1'}{4}$

$V_1' - 4I + 5 = \frac{V_1'}{4}.$ $\quad (4I = V_1')$

$V_1' - V_1' + 5 = \frac{V_1'}{4}.$

$V_1' = 20\ V$

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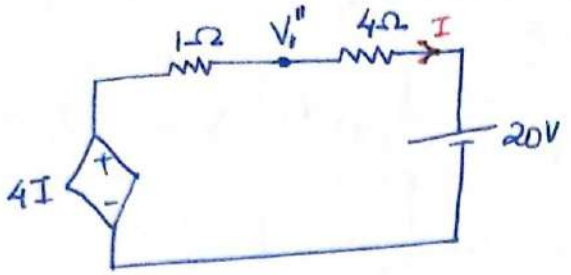
2. When 20V source is acting alone



$4I - I - 4I - 20 = 0$

$I = -20A$

$V_1'' = 4I - I = 3I$

$= -60V$

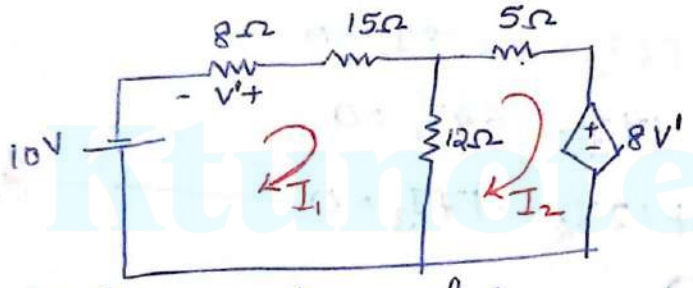
3. By Superposition theorem $V_1 = V_1' + V_1''$

$= 20 - 60$

$= -40V$

---

12 1. When 10V source is acting alone



From the figure

$V' = -8I_1$ ——— (1)

Apply KVL to mesh 1

$-10 - 8I_1 - 15I_1 - 12(I_1 - I_2) = 0$

$-35I_1 + 12I_2 = 10$ ——— (2)

Apply KVL to mesh 2

$-12(I_2 - I_1) - 5I_2 - 8V' = 0$

$12I_1 - 17I_2 - 8(-8I_1) = 0$

$76I_1 - 17I_2 = 0$ ——— (3)

Solving (2) & (3) $\Rightarrow$ $I_1 = 0.54A$, $I_2 = 2.4A$

$V' = -8I_1 = -4.32V$

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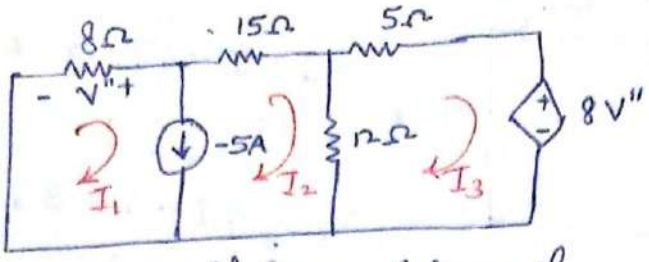
2. When 5A source is acting alone



$V'' = -8I_1$ —— (1)

Mesh 1 & 2 will form supermesh.

$$I_1 - I_2 = -5 \quad \text{—— (2)}$$

$$-8I_1 - 15I_2 - 12(I_2 - I_3) = 0$$

$$-8I_1 - 27I_2 + 12I_3 = 0 \quad \text{—— (3)}$$

KVL to mesh 3

$$-12(I_3 - I_2) - 5I_3 - 8V'' = 0$$

$$12I_2 - 17I_3 - 8 \times -8I_1 = 0$$

$$12I_2 - 17I_3 + 64I_1 = 0$$

$$64I_1 + 12I_2 - 17I_3 = 0 \quad \text{—— (4)}$$

Solve (2), (3) & (4)

$\Rightarrow I_1 = 4.97A$, $I_2 = 9.97A$, $I_3 = 25.74A$

$V'' = -8I_1 = -39.76V$

3. By Superposition Theorem

$$V = V' + V'' = -4.32 - 39.76$$

$$= -44.08\,V$$

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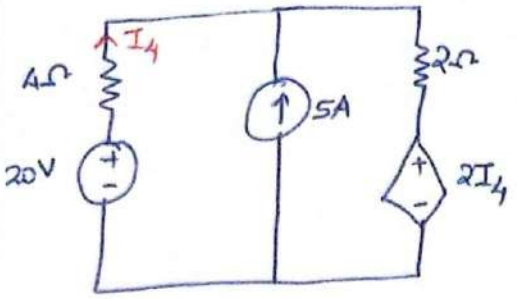
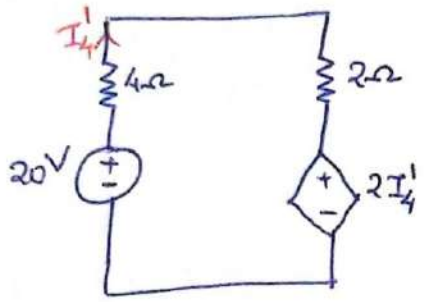
13. Redraw



1. Considering 20V Source alone

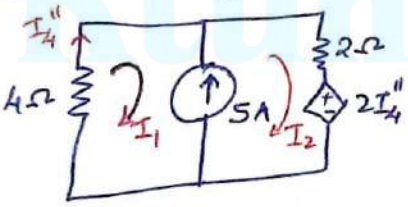


$$20 - 4I_4' - 2I_4' - 2I_4' = 0$$

$$20 = 8I_4'$$

$$I_4' = \frac{20}{8} = \underline{2.5A}$$

2. Considering 5A alone



$$I_2 - I_1 = 5A$$

$$I_2 - I_4'' = \underline{5A} \quad \text{——— (1)}$$

$$-4I_4'' - 2I_2 - 2I_4'' = 0$$

$$-6I_4'' - 2I_2 = 0$$

$$+2I_2 + 6I_4'' = 0 \quad \text{——— (2)}$$

Solving (1) & (II)

$I_2 = 3.75A$, ~~$I_4'' = -0.714$~~

$I_4'' = \underline{-1.25A}$

3. By Superposition Theorem

$$I_4 = I_4' + I_4''$$

$$= 2.5 - 1.25 = \underline{1.25A}$$

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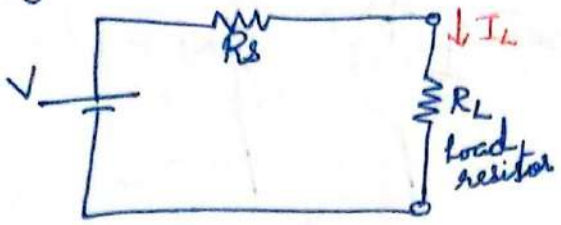
Practical Voltage Source

Practical Voltage Source is represented by Voltage Source (V) in series with source resistance (R_s)

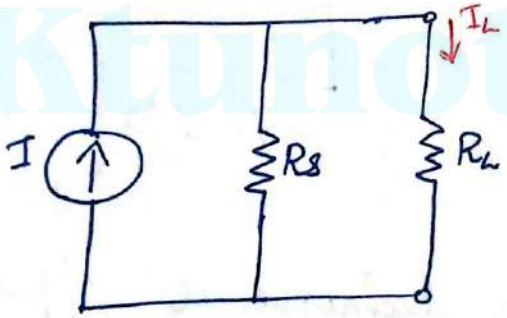


load current,

$$I_L = \frac{V}{R_s + R_L}$$

Practical Current Source

Practical current source is represented by current source (I) parallel with source resistance R_s.



$R_L \rightarrow$ load resistor

load current, I_L is

$$I_L = I \times \frac{R_s}{R_s + R_L}$$

Source Transformation

If a voltage source has a resistance connected in series, it can be transformed into an equivalent current source with the resistance connected in parallel.

The practical voltage and current sources are equivalent sources only when

$$\frac{V}{R_s + R_L} = I \cdot \frac{R_s}{R_s + R_L}$$

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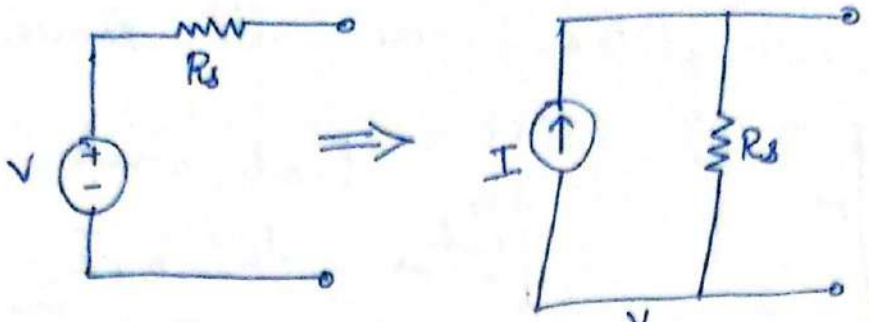
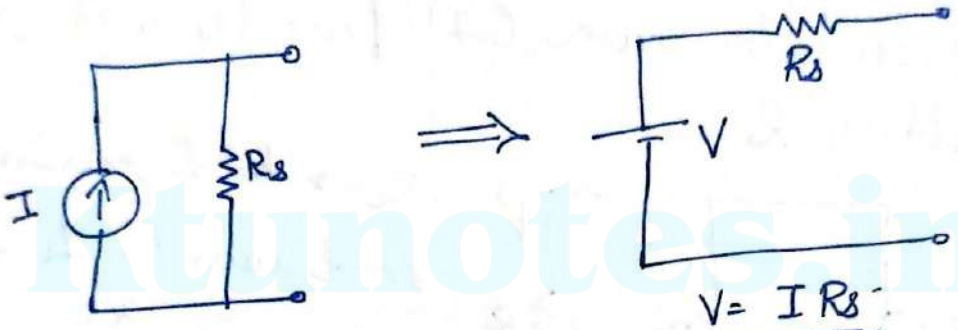
Voltage Source to Current Source



$$I = \frac{V}{R_s}$$

Current Source to Voltage Source



$$V = I R_s$$

This transformation technique is applicable to linear networks containing voltage or current sources of constant magnitude.

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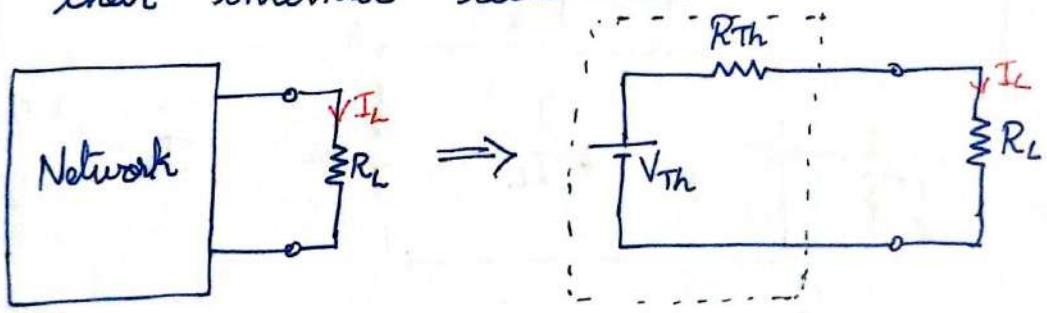
THEVENIN'S THEOREM

Thevenin's Theorem states that any two terminal linear network having a number of voltage and current sources and resistances can be replaced by a simple equivalent circuit consisting of a single voltage source in series with a resistance.

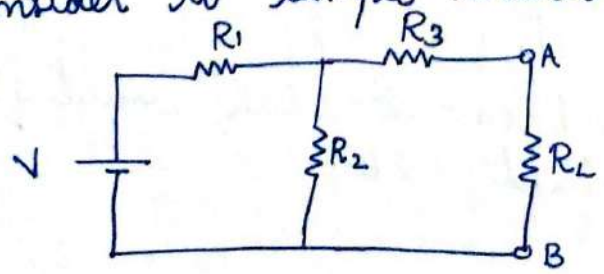
The value of the voltage source is equal to the open circuit voltage across the two terminals of the network.

The resistance is equal to the equivalent resistance measured between the terminals with all the energy sources are replaced by their internal resistances.



Explanation

Consider a simple network as shown



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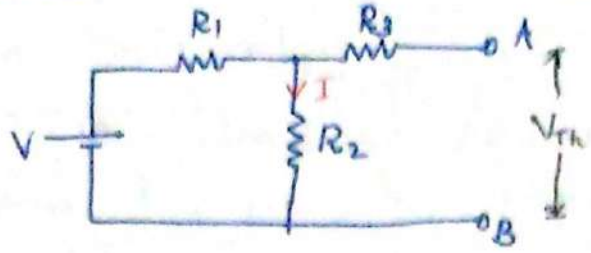
First remove R_L and calculate V_{Th}

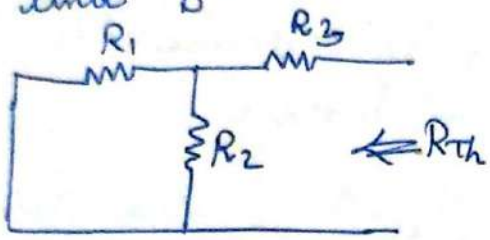


$$V_{Th} = I R_2$$

$$= \frac{V}{(R_1+R_2)} \times R_2$$

Find R_{Th}

Replace voltage source by a short circuit and calculate the resistance between points A and B.



$$R_{Th} = R_3 + (R_1 \| R_2)$$

$$= R_3 + \frac{R_1 R_2}{R_1 + R_2}$$

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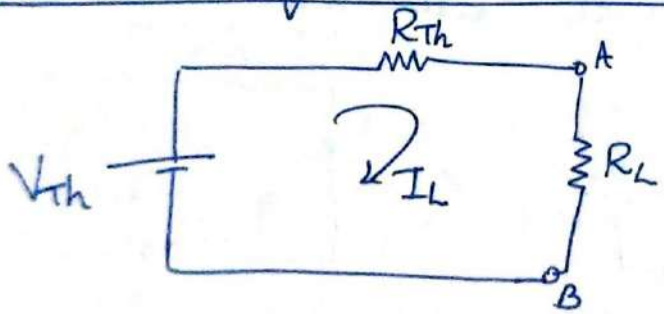
Thevenin's equivalent network



$$I_L = \frac{V_{Th}}{R_{Th} + R_L}$$

If the network contains both independent and dependent sources, Thevenin's resistance R_{Th} is calculated as $R_{Th} = \frac{V_{Th}}{I_N}$

$I_N \rightarrow$ short circuit current which would flow in a short circuit placed A & B.

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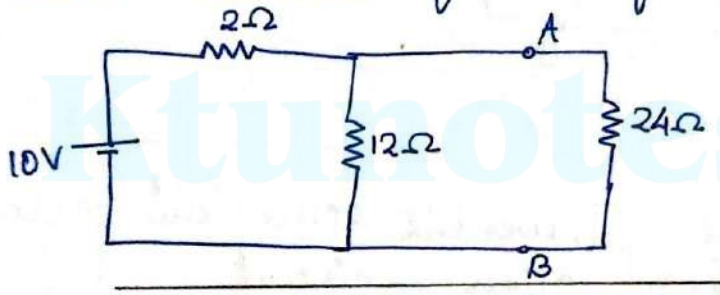
Steps to be followed in Thevenin's Theorem

1. Remove the load resistance R_L
2. Find the open circuit voltage V_{Th} across A & B
3. Find the resistance R_{Th} as seen from A & B
4. Replace the network by a voltage source V_{Th} in series with resistance R_{Th}.
5. Find the current through R_L using Ohm's law.

$$I_L = \frac{V_{Th}}{R_{Th} + R_L}$$

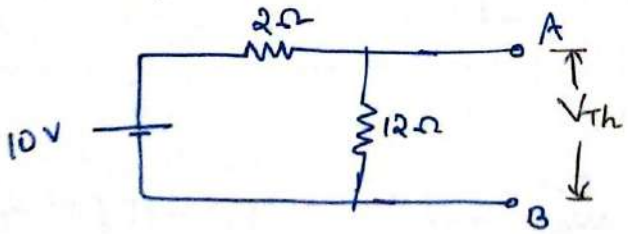
---

1. Find Thevenin's equivent for the given circuit.



$R_L = 24\Omega$

To find V_{Th}



V_{Th} = Voltage across 12Ω resistor

= Current through 12Ω resistor × R (V=IR)

$$= \frac{10}{(2+12)} \times 12 = \frac{10}{14} \times 12 = 8.57V$$

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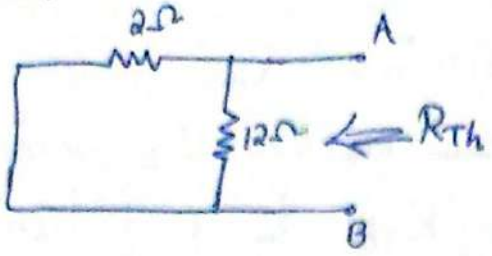
To find R_{Th}

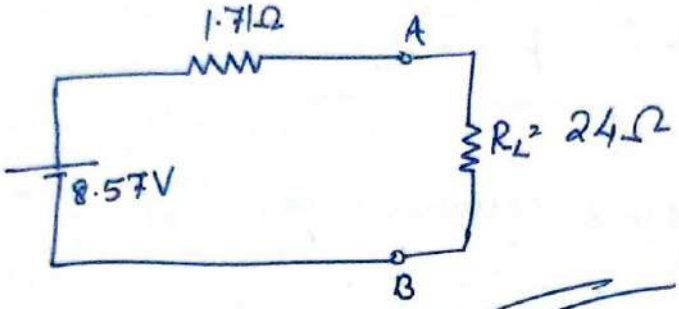


$R_{Th} = (2 \| 12)$

$= \frac{2 \times 12}{2+12} = \frac{24}{14}$

$= 1.71\,\Omega$

Thevenin's Equivalent Circuit



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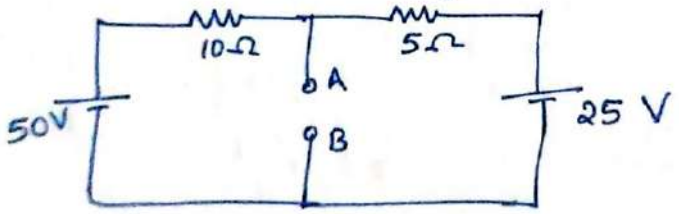
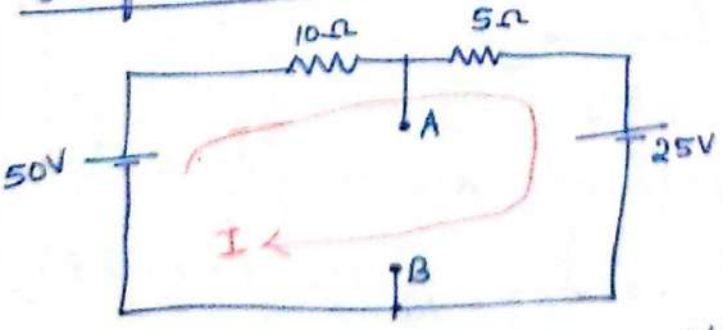
2. Determine the Thevenin's equivalent circuit across AB for given circuit



To find V_{Th}



$50 - 10I - 5I - 25 = 0$

$25 = 15I$

$I = \frac{25}{15}$ A

Voltage across A & B $= 50 -$ Voltage across 10Ω

$= 50 - I \times 10 = 50 - \frac{25}{15} \times 10$

$V_{Th} = 33.33V$

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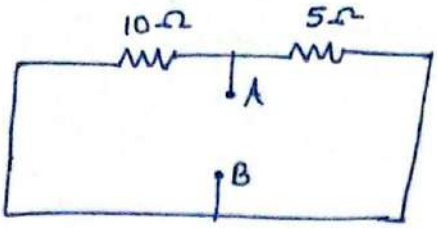
To find R_{Th}

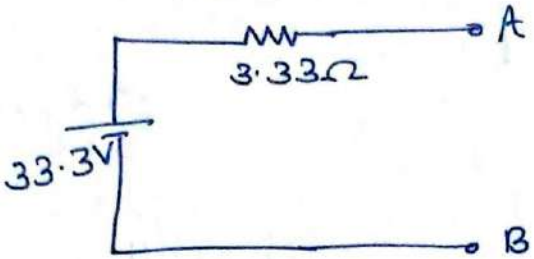


while looking from A & B

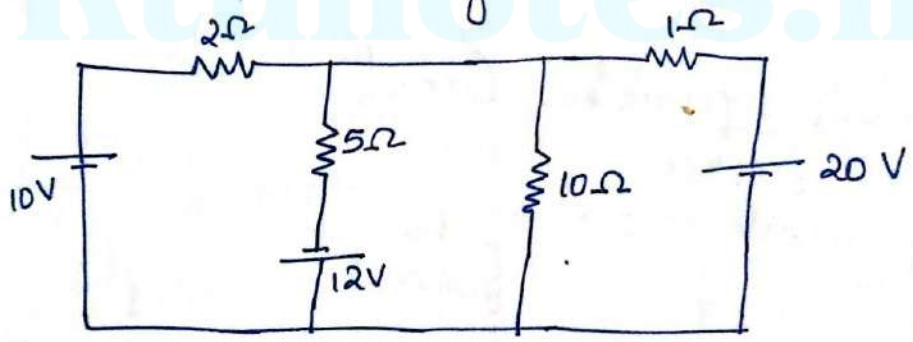
$R_{Th} = 10 // 5$

$= \frac{10 \times 5}{10+5} = \frac{50}{15} = 3.33\Omega$

The Thevenin's equivalent Circuit

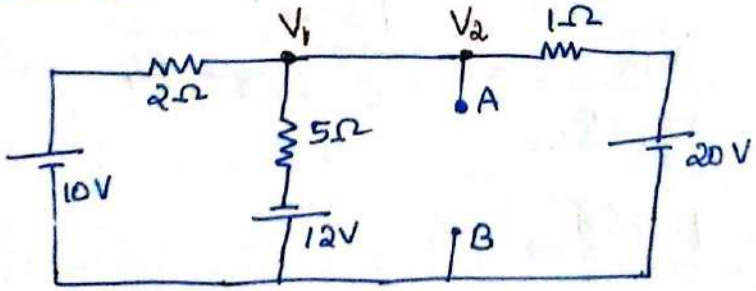


3. In the given circuit, find the current through 10Ω resistor utilising Thevenin's Theorem.



To find V_{Th}

$R_L = 10\Omega$



$V_1 = V_2$

$$\frac{V_1 - 10}{2} + \frac{V_1 + 12}{5} + \frac{V_1 - 20}{1} = 0$$

$$5V_1 - 50 + 2V_1 + 24 + 10V_1 - 200 = 0$$

$$V_1 = \frac{226}{17} = 13.29V$$

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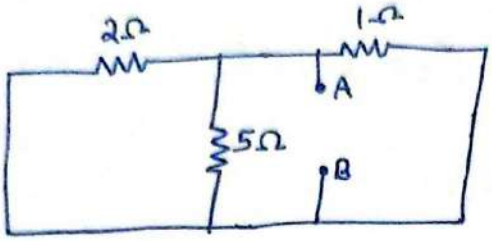
V_{Th} = Voltage across A & B $= V_2$
$= V_1$

$= \underline{\underline{13.29V}}$

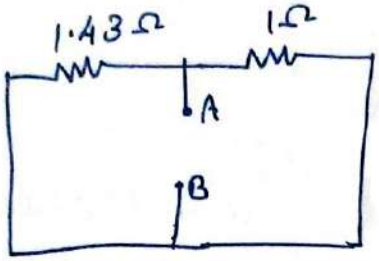
To find R_{Th}



Looking from AB.
5 & 2 are in ||el.

$\frac{5\times2}{5+2} = \underline{1.43\Omega}$

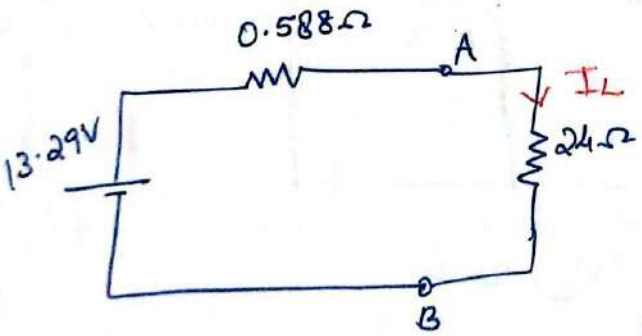
So



$R_{AB} = (1.43)||1$

$= \frac{1.43\times1}{1.43+1} = \underline{\underline{0.588\Omega}}$

Thevenin's Equivalent Circuit



$I_L = \frac{V_{Th}}{R_{Th}+R_L} = \frac{13.29}{0.588+10}$

$= \underline{\underline{1.26A}}$

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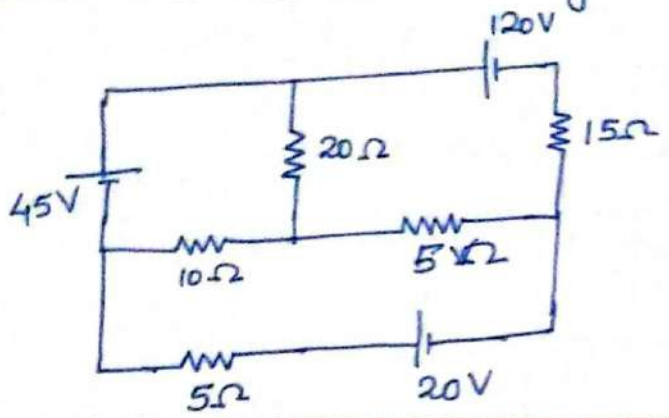
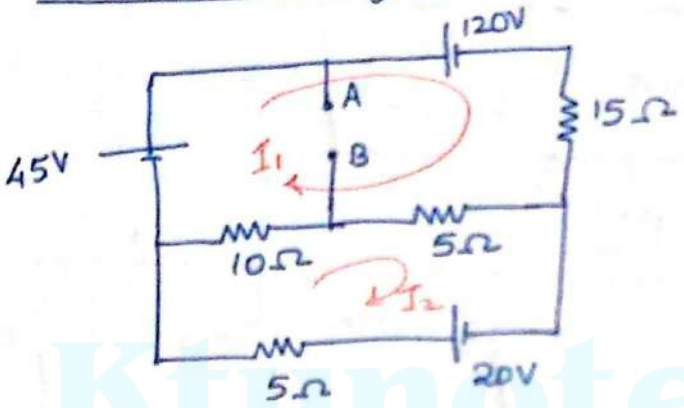
4. Find the current through the 20 Ω resistor



1. Calculation of V_{Th}



KVL to Mesh 1

$$45 - 120 - 15I_1 - 5(I_1 - I_2) - 10(I_1 - I_2) = 0$$

$$-30I_1 + 15I_2 = 75 \quad \text{——— (1)}$$

KVL to Mesh 2

$$-10(I_2 - I_1) - 5(I_2 - I_1) + 20 - 5I_2 = 0$$

$$15I_1 - 20I_2 = -20 \quad \text{——— (2)}$$

Solve ① & ② for I_1 & I_2

$$I_1 = -3.2A \quad \& \quad I_2 = -1.4\,A$$

Writing equation for V_{Th}

$$45 - V_{Th} - 10(I_1 - I_2) = 0$$

$$V_{Th} = 45 - 10(-3.2 + 1.4) = 45 + 18$$

$$= 63V$$

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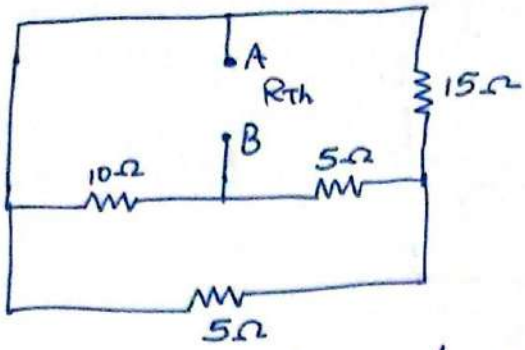
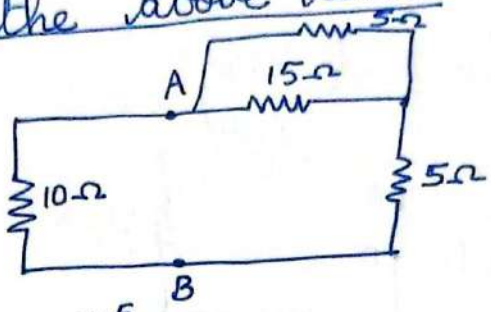
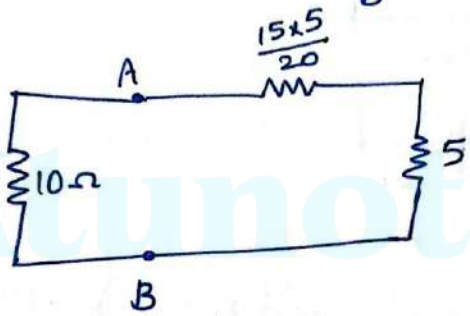
2. Calculation of R_{Th}



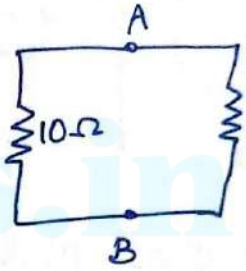
Redrawing the above circuit



$\Rightarrow$



$\Rightarrow$



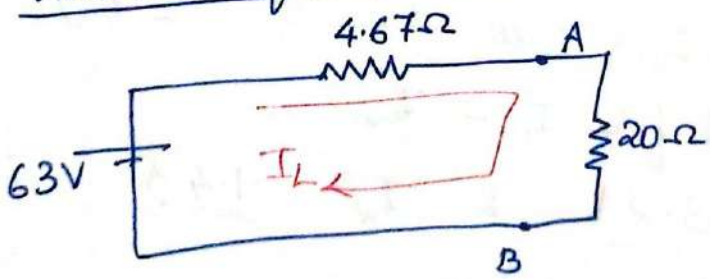
$5 + \frac{75}{20}$

$= \underline{\underline{8.75\Omega}}$

R_{Th} = Resistance b/w A & B

$= (10 \,//\, 8.75) = \dfrac{10 \times 8.75}{10 + 8.75} = \underline{\underline{4.67\Omega}}$

Thevenin Equivalent Circuit



$$I_L = \frac{V_{Th}}{R_{Th} + R_L}$$

$$= \frac{63}{4.67 + 20}$$

$$= \underline{\underline{2.55\,A}}$$

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Examples with Dependent Sources

If the network contains both independent and dependent sources, Thevenin's resistance is calculated as, $R_{Th} = \frac{V_{Th}}{I_N}$

where I_N is short circuit current which would flow in a short circuit placed across A & B.

Dependent sources are active at all times.

If the network contains only dependent sources

$R_{Th} = \frac{V}{I}$; where $V \rightarrow$ a known voltage applied across A & B

$I \rightarrow$ Current through the path AB

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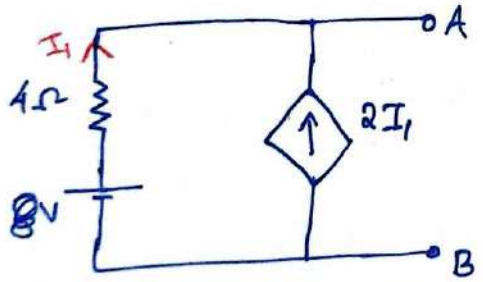
5. Obtain the Thevenin equivalent network



1. Calculation of V_{Th}

From Fig. $I_1 = -2I_1$

$3I_1 = 0$

$I_1 = 0$

V_{Th} eqn

$8 - 4I_1 - V_{Th} = 0$

$V_{Th} = 8 - 4I_1$

$= 8\,V$

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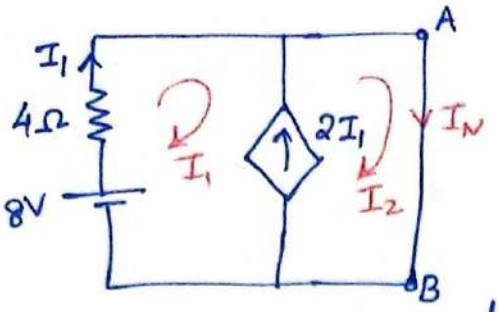
2. Calculation of I_N



$$-I_1 + I_2 = 2I_1$$

$$3I_1 - I_2 = 0 \quad \text{——— (1)}$$

Applying KVL to the outer mesh

$$8 - 4I_1 = 0 \quad \text{——— (2)}$$

$$I_1 = 2$$

Solving (1) & (2) $\quad 3 \times 2 - I_2 = 0; \quad I_2 = 6$

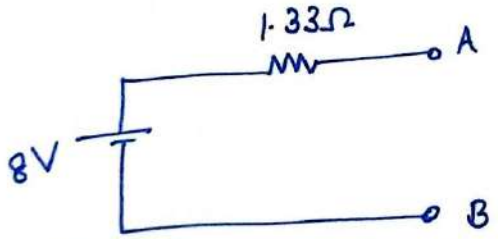
$$I_1 = 2A, \quad I_2 = 6A$$

$$I_N = I_2 = 6A.$$

3. Calculation of R_{Th}

$$R_{Th} = \frac{V_{Th}}{I_N} = \frac{8}{6} = 1.33\Omega$$

4. Thevenin's equivalent Circuit



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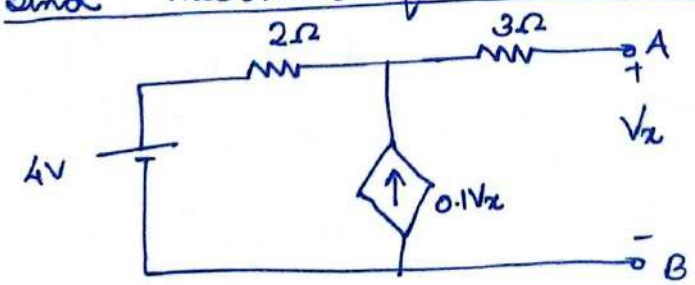
6. Find Thevenin's equivalent network



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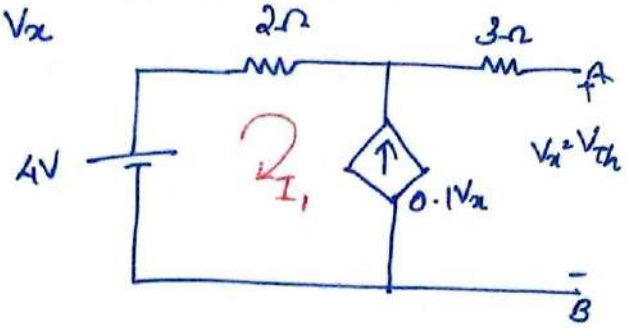
1. Calculation of V_{Th}

$V_x = V_{Th}$; $I_1 = -0.1 V_x$



Writing V_{Th} equation

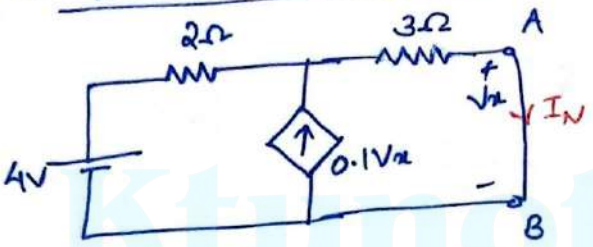
$$4 - 2I_1 - V_x = 0$$

$$4 - 2(-0.1V_x) - V_x = 0$$

$$4 + 0.2V_x - V_x = 0$$

$$V_x = V_{Th} = 5V$$

2. Calculation of I_N



$V_x = 0$

As $V_x = 0$, dependent source voltage $0.1V_x \rightarrow 0$.

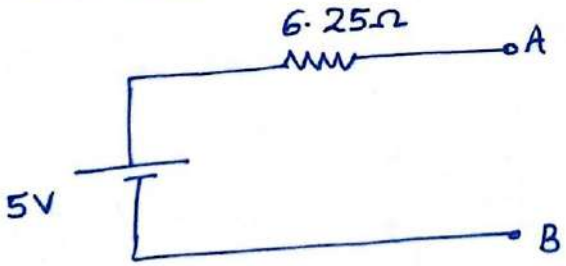
The dependent source vanishes.

$$I_N = \frac{4}{2+3} = 0.8A$$

3. Calculation of R_{Th}

$$R_{Th} = \frac{V_{Th}}{I_N} = \frac{5}{0.8} = 6.25\,\Omega$$

4. Thevenin equivalent circuit



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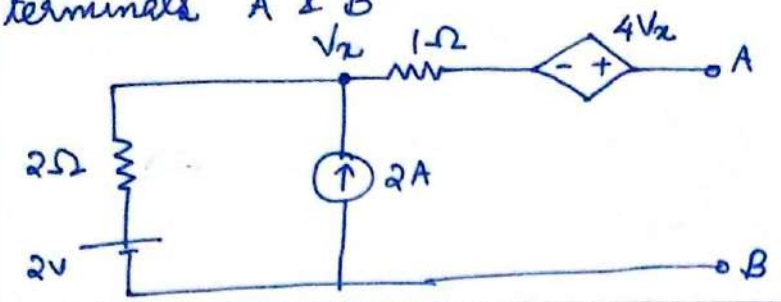
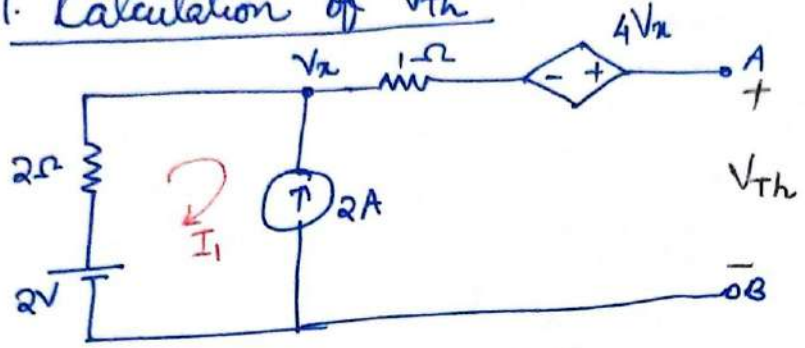
7. Obtain the Thevenin equivalent network for the terminals A & B



1. Calculation of V_{Th}



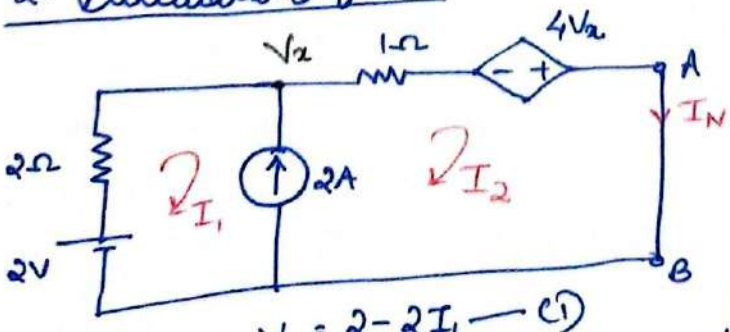
From Fig. $I_1 = -2A$.

$$V_x = 2 - 2I_1 = 2 - 2\times -2$$
$$= \underline{\underline{6V}}$$

From outer mesh

$$2 - 2I_1 - 0 + 4V_x - V_{Th} = 0$$
$$4V_x = 2I_1 - 2 + V_{Th}$$
$$V_{Th} = 2 - 2I_1 + 4V_x$$
$$= 2 - 2\times(-2) + 4\times 6$$
$$= 2 + 4 + 24 = \underline{\underline{30V}}$$

2. Calculation of I_N



$V_x = 2 - 2I_1$ — (1)

$-I_1 + I_2 = 2$ — (2)

Applying KVL

$$2 - 2I_1 - I_2 + 4V_x = 0$$
$$2 - 2I_1 - I_2 + 4(2 - 2I_1) = 0$$
$$+10I_1 + I_2 = 10 \quad \text{— (3)}$$

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Solving ② & ③

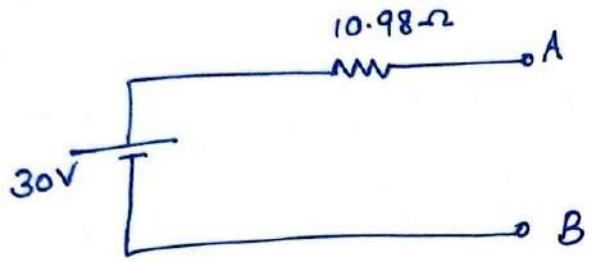
$I_1 = 0.73A$, $I_2 = 2.73\ A$

$I_N = I_2 = \underline{\underline{2.73\ A}}$

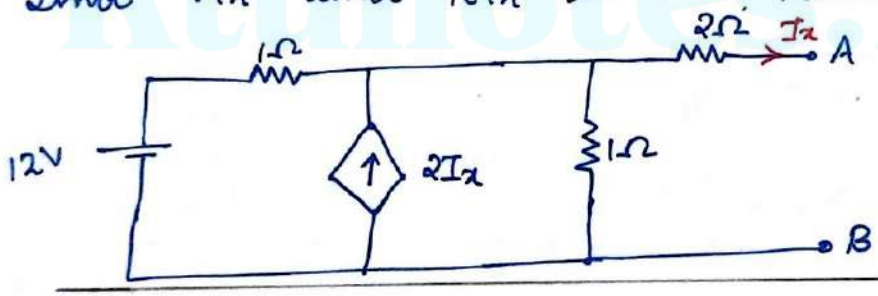
3. Calculation of R_{Th}

$$R_{Th} = \frac{V_{Th}}{I_N} = \frac{30}{2.73} = \underline{\underline{10.98\ \Omega}}$$

4. Thevenin equivalent circuit

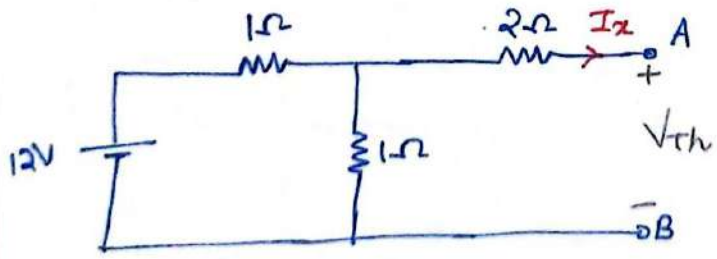


8. Find V_{Th} and R_{Th} between terminals A & B



1. Calculation of V_{Th}

Dependent source vanishes as $I_x = 0$ ($\because 2I_x = 0$)



$I_x = 0$

V_{Th} equation

$$V_{Th} = \frac{12}{1+1} \times 1 = \underline{\underline{6\ V}}$$

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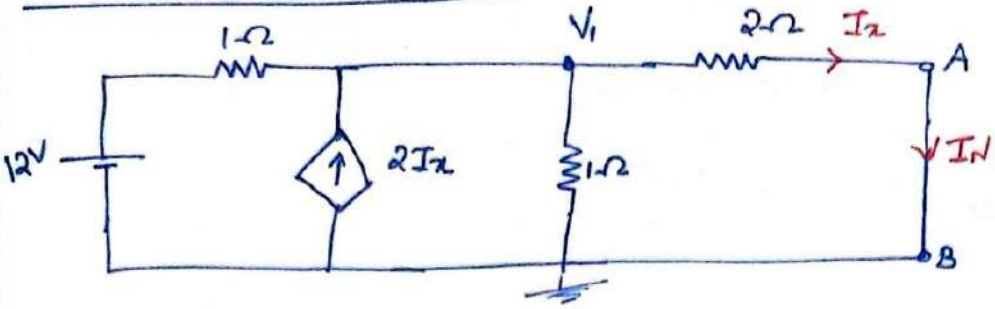
2. Calculation of I_N



$$I_x = \frac{V_1}{2}$$

$$\frac{V_1 - 12}{1} - 2I_x + \frac{V_1}{2} + \frac{V_1}{1} = 0$$

$$V_1 - 12 + \frac{V_1}{2} + V_1 = 2I_x$$

$$2V_1 - 24 + V_1 + 2V_1 = 4I_x$$

$$5V_1 - 24 = 4I_x$$

$$5V_1 - 24 = 4 \times \frac{V_1}{2}$$

$$V_1 = \frac{24}{3} = 8V$$

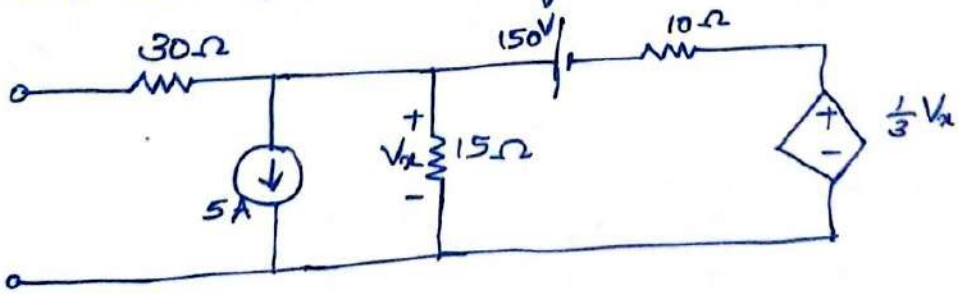
$$I_N = \frac{V_1}{2} = 4A.$$

3. Calculation of R_{Th}

$$R_{Th} = \frac{V_{Th}}{I_N} = \frac{6}{4} = 1.5\Omega$$

---

9. Obtain the Thevenin equivalent network.



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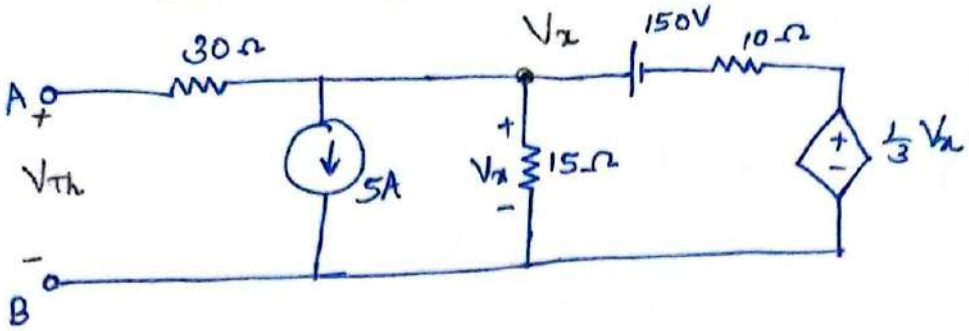
1. Calculation of V_{Th}



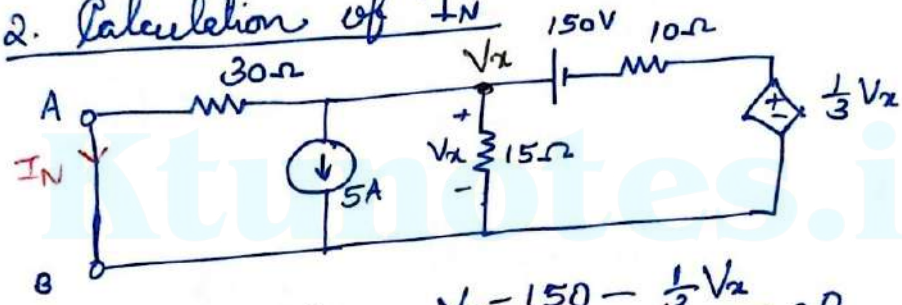
$$\frac{V_x - 150 - \frac{1}{3}V_x}{10} + \frac{V_x}{15} + 5 = 0$$

$$3V_x - 450 - V_x + 2V_x + 150 = 0$$

$$4V_x = 300; \qquad V_x = 75V$$

$$V_{Th} = V_x = 75V$$

2. Calculation of I_N



$$\frac{V_x}{30} + 5 + \frac{V_x}{15} + \frac{V_x - 150 - \frac{1}{3}V_x}{10} = 0$$

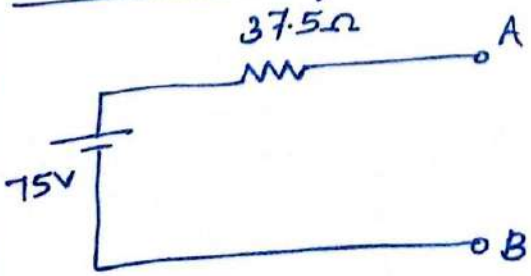
$$V_x + 150 + 2V_x + 3V_x - 450 - V_x = 0$$

$$5V_x = 300; \qquad V_x = 60V$$

$$I_N = \frac{V_x}{30} = \cancel{\frac{75}{30}}\ \frac{60}{30} = 2A$$

$$R_{Th} = \frac{V_{Th}}{I_N} = \frac{75}{2} = 37.5\Omega$$

3. Thevenin equivalent Circuit



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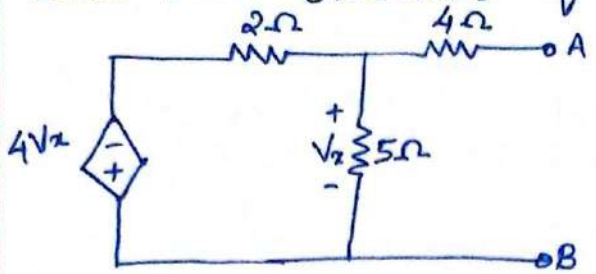
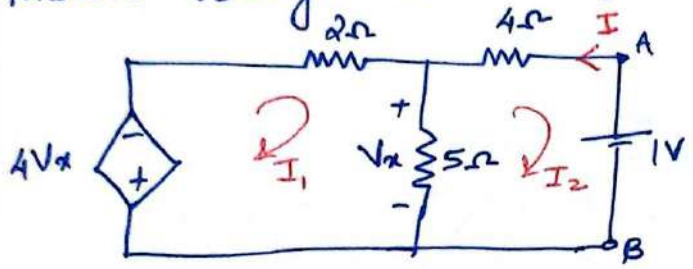
10. Find the Thevenin equivalent network at A & B



Since the network does not contain independent source, $V_{Th} = 0$

$I_N = 0$

But the R_{Th} can be calculated by applying a known voltage source of 1V at A & B



$R_{Th} = \frac{V}{I} = \frac{1}{I}$

$V_x = 5(I_1 - I_2)$

KVL to Mesh 1

$-4V_x - 2I_1 - 5(I_1 - I_2) = 0$

$-4 \times 5(I_1 - I_2) - 2I_1 - 5(I_1 - I_2) = 0$

$-27I_1 + 25I_2 = 0$ ——— (1)

KVL to Mesh 2

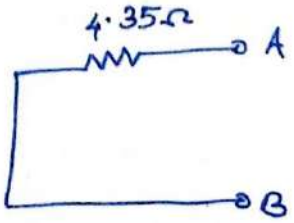
$-5(I_2 - I_1) - 4I_2 - 1 = 0$

$-5(I_2 - I_1) - 4I_2 = 1$; $5I_1 - 9I_2 = 1$ ——— (2)

Solving (1) & (2) $\Rightarrow$ $I_1 = -0.21A$; $I_2 = -0.23A$

$R_{Th} = \frac{1}{I} = \frac{1}{0.23} = 4.35\Omega$

Thevenin equivalent network



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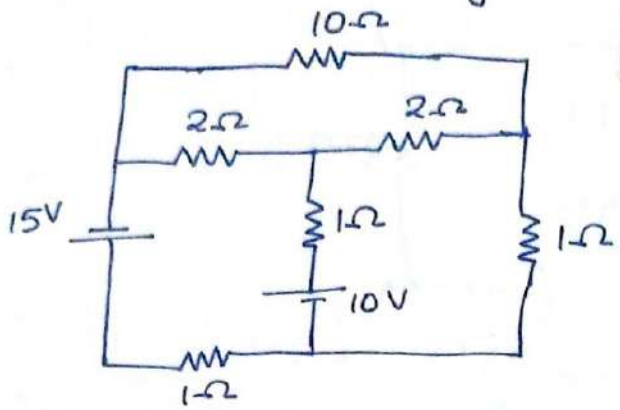
Tutorial Questions

1. Find the current through 10Ω resistor

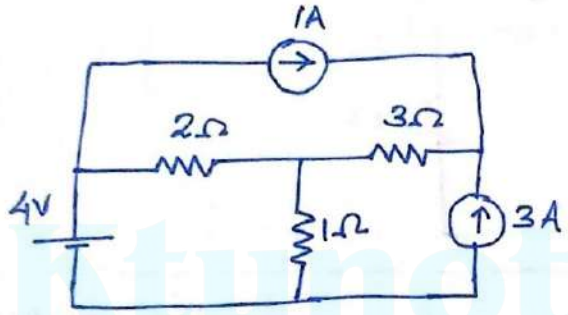


[Ans: $V_{Th} = -10V$, $R_{Th} = 1.33\Omega$]

$I_L = -0.88A$

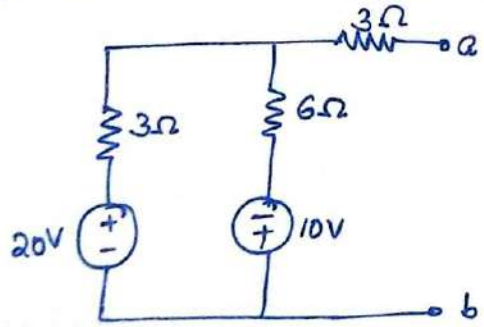
$I_L = 0.88A (\uparrow)$

2. Find current through the 1Ω resistor in figure.



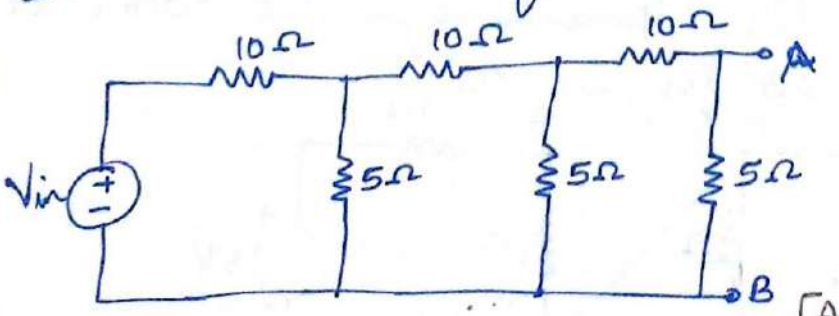
[Ans: $V_{Th} = 12V$, $R_{Th} = 2\Omega$, $I_L = 4A$]

3. Obtain the Thevenin equivalent circuit.



[Ans: $V_{Th} = 10V$, $R_{Th} = 5\Omega$]

4. Obtain Thevenin equivalent Circuit



[Ans: $V_{Th} = \frac{V_{in}}{41}$, $R_{Th} = \frac{150}{41}\Omega$]

A

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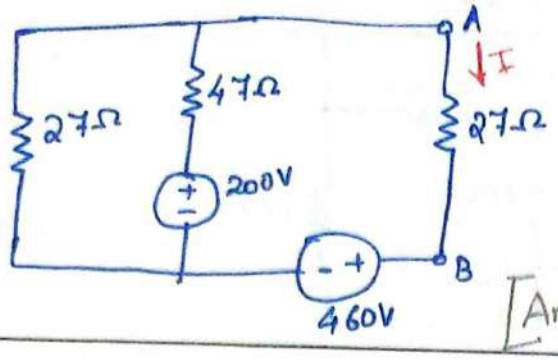
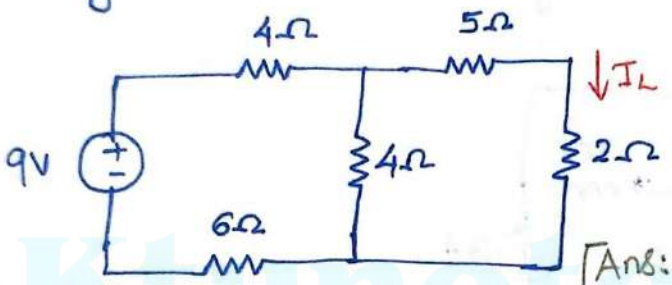
5. Draw the Thevenin equivalent circuit and solve for the current I



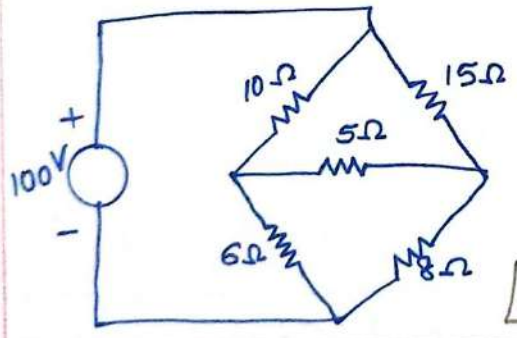
[Ans: −386.9V, R_{Th} = −8.76 A]

6. Use Thevenin's theorem to find the current through the 2Ω resistor in the circuit



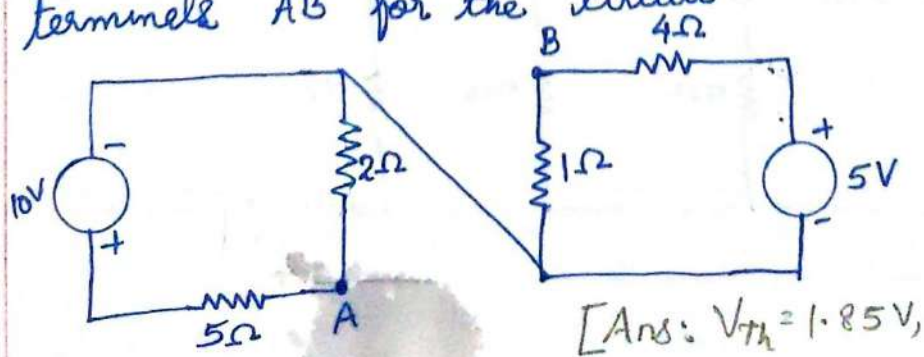
[Ans: V_{Th} = 2.571, R_{Th} = 7.857, I_L = 0.2608]

7. Use Thevenin's Theorem to find the current through the 5Ω resistor.



[Ans: V_{Th} = 2.72V, R_{Th} = 8.97Ω, I_L = 0.193]

8. Determine the Thevenin's equivalent circuit across terminals AB for the circuit.



[Ans: V_{Th} = 1.85V, R_{Th} = 2.23Ω]

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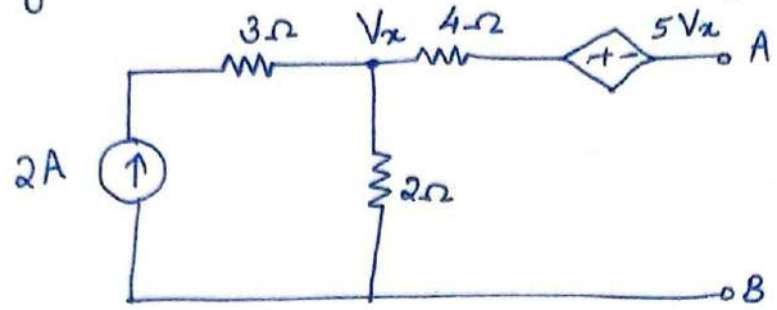
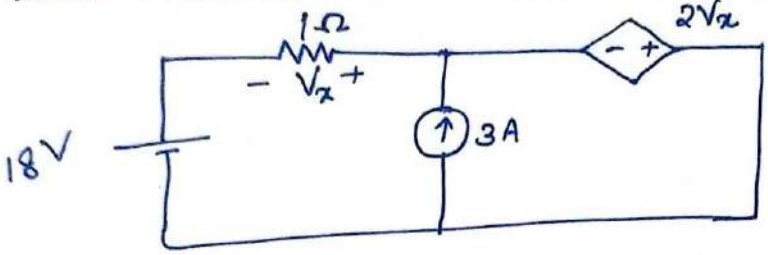
13. Obtain the Thevenin equivalent network for the given network at terminals A & B



[Ans: $V_{Th} = -16V$, $R_{Th} = -4\Omega$]

14. Find current in the 6Ω resistor.



[Ans: $V_{Th} = 27V$, $R_{Th} = 3\Omega$, $I_L = 3A$]

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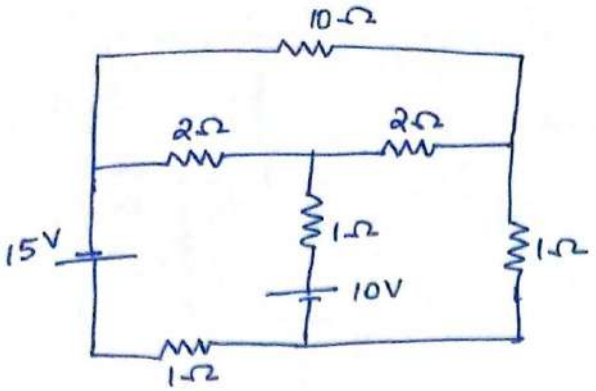
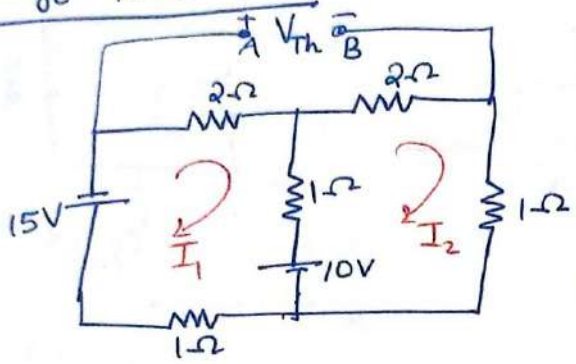
Answers to Tutorial Questions on Thevenin

1.



1. To Find V_{Th}



Mesh 1

$$-15 - 2I_1 - 1(I_1 - I_2) - 10 - 1I_1 = 0$$

$$-4I_1 + I_2 = 25 \quad \text{——— (1)}$$

Mesh 2

$$10 - 1(I_2 - I_1) - 2I_2 - I_2 = 0$$

$$I_1 - 4I_2 = -10 \quad \text{——— (2)}$$

Solving ① & ②

$\Rightarrow$ $I_1 = -6A, \quad I_2 = 1A.$

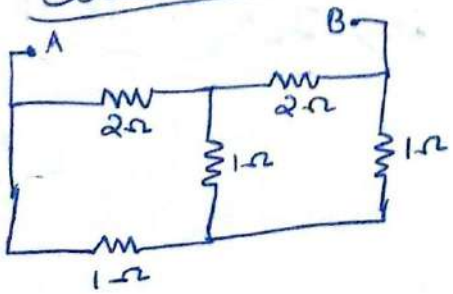
Writing eqn for V_{Th}

$$\Rightarrow -V_{Th} + 2I_2 + 2I_1 = 0$$

$$V_{Th} = 2I_1 + 2I_2$$

$$= 2 \times -6 + 2 \times 1 = -10V$$

2. Calculation of R_{Th}



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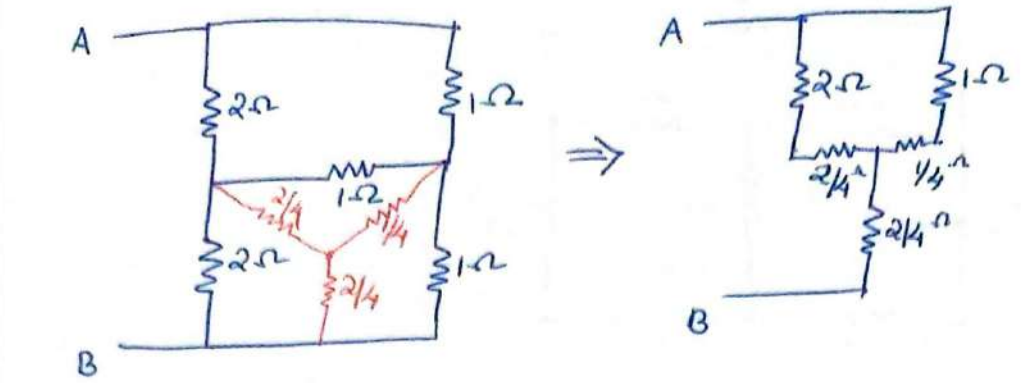
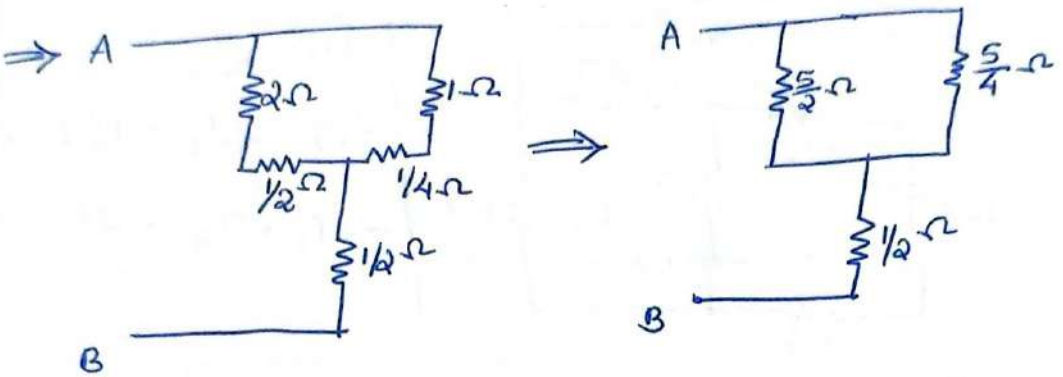
Redraw





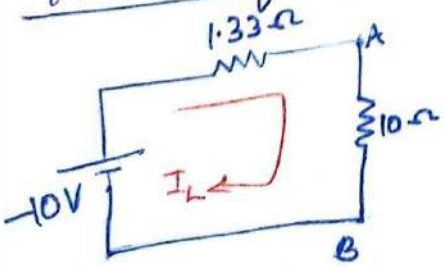
$$R_{Th} = R_{AB} = \left(\frac{5}{2} \parallel \frac{5}{4}\right) + \frac{1}{2}$$

$$= \frac{\frac{5}{2} \times \frac{5}{4}}{\frac{5}{2} + \frac{5}{4}} + \frac{1}{2}$$

$$= \frac{25}{30} + \frac{1}{2} = \frac{25+15}{30} = \frac{40}{30}$$

$$= 1.33\,\Omega$$

Thevenin equivalent Circuit



$$I_L = \frac{-10}{1.33+10} = -0.88\,A \ (\downarrow)$$

ie; $I_L = 0.88\,A\ (\uparrow)$

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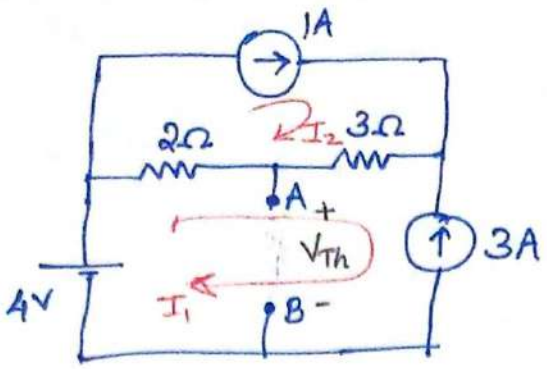
2. 1. Calculate V_{Th}



From Meshes 1 & 2

$I_1 = -3A$

$I_2 = 1A$

KVL to Mesh 1

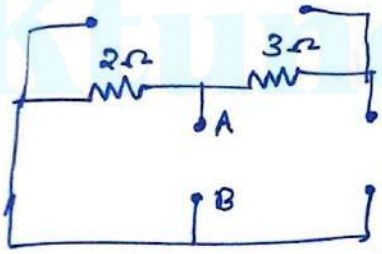
$$4 - 2(I_1 - I_2) - V_{Th} = 0$$

$$V_{Th} = 4 - 2(I_1 - I_2)$$

$$= 4 - 2(-3-1) = 4 - 2\times -4$$

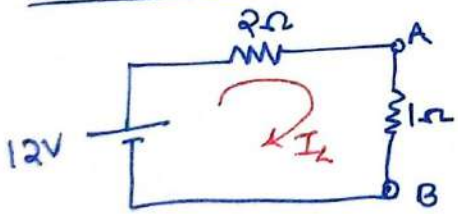
$$= \underline{\underline{12V}}$$

2. Calculation of R_{Th}



$R_{Th} = \underline{\underline{2\Omega}}$

3. Calculation of I_L



$$I_L = \frac{V_{Th}}{R_{Th}+R_L}$$

$$= \frac{12}{2+1} = \underline{\underline{4A}}$$

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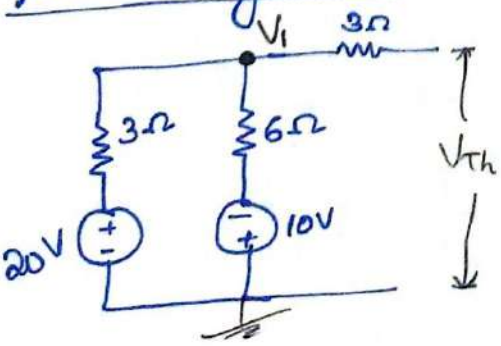
3. Calculating V_{Th}



From the circuit, $V_{Th} = V_1$

$$\frac{V_1 - 20}{3} + \frac{V_1 - -10}{6} = 0$$

$$\frac{V_1 - 20}{3} + \frac{V_1 + 10}{6} = 0$$

$$2V_1 - 40 + V_1 + 10 = 0$$

$$V_1 = \underline{\underline{10V}}$$

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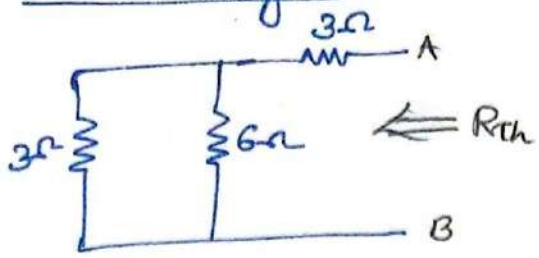
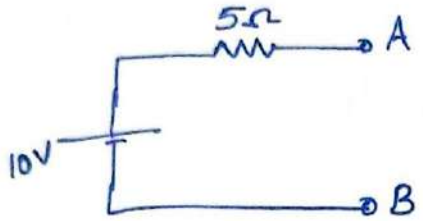
2. Calculating R_{Th}



$$R_{Th} = 3 + (3 \| 6)$$

$$= 3 + \frac{3 \times 6}{9} = \underline{\underline{5\Omega}}$$

3. Thevenin equivalent circuit



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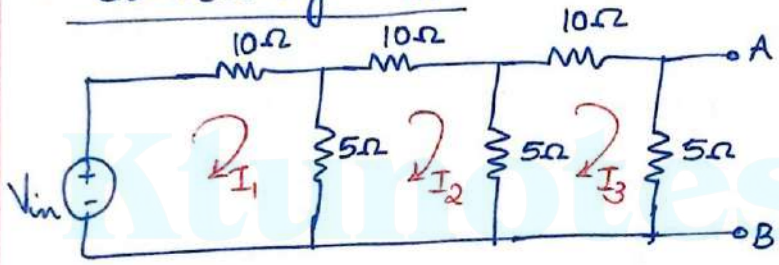
4. 1. Calculating V_{Th}



$$15 I_1 - 5 I_2 = V_{in} \quad \text{——— (1)}$$

$$-5 I_1 + 20 I_2 - 5 I_3 = 0 \quad \text{——— (2)}$$

$$-5 I_2 + 20 I_3 = 0 \quad \text{——— (3)}$$

Solving ①, ② & ③

$$I_3 = \frac{\begin{vmatrix} 15 & -5 & V_{in} \\ -5 & 20 & 0 \\ 0 & -5 & 0 \end{vmatrix}}{\begin{vmatrix} 15 & -5 & 0 \\ -5 & 20 & -5 \\ 0 & -5 & 20 \end{vmatrix}} = \frac{V_{in}(25-0)}{15(400-25)+5(-100)}$$

$$= \frac{25 V_{in}}{5125} = \underline{\underline{\frac{1}{205} V_{in}}}$$

$$V_{Th} = I_3 \times 5 = \frac{V_{in}}{205} \times 5 = \underline{\underline{\frac{V_{in}}{41}}}$$

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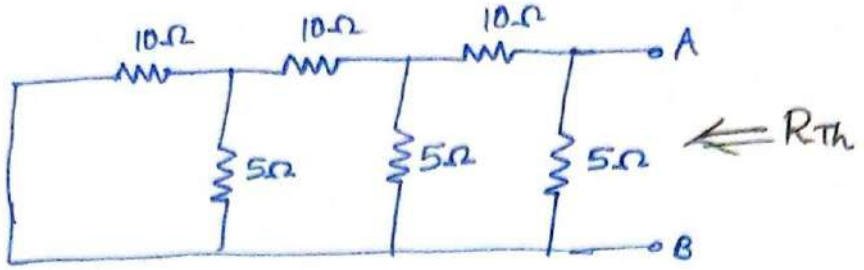
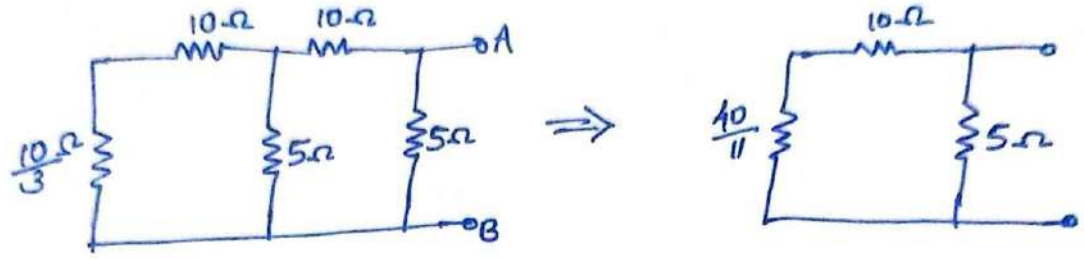
2. Calculating R_{Th}





$$R_{Th} = \left(\frac{150}{11} \parallel 5\right)$$

$$= \frac{\frac{150}{11}\times 5}{\frac{150}{11}+5} = \frac{150\times 5}{\frac{205}{41}} = \frac{150}{41}\,\Omega$$

$$\frac{\frac{40}{3}\times 5}{\frac{40}{3}+5} = \frac{40\times 5}{55/11}$$

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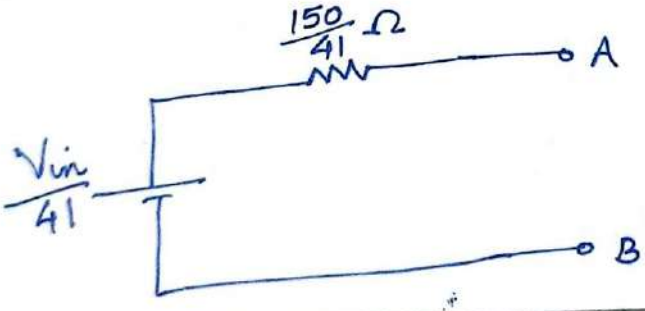
3. Thevenin's equivalent Circuit



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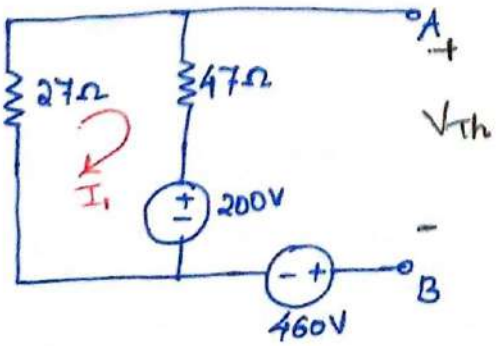
5. 1. Calculating V_{Th}



KVL to Mesh 1

$$-27I_1 - 47I_1 - 200 = 0$$

$$I_1 = -2.70$$

KVL for V_{Th}

$$-V_{Th}\ 460 + 200 + 47I_1 = 0$$

$$V_{Th} = -460 + 200 + 47(-2.7)$$

$$= -386.9\,V$$

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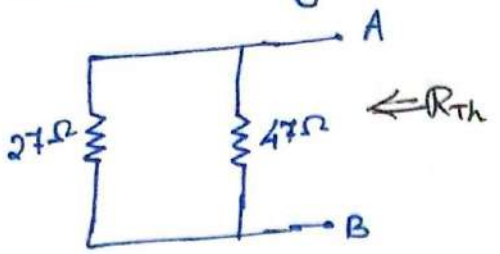
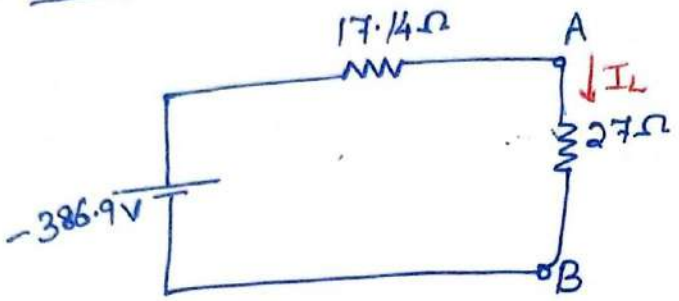
2. Calculating R_{Th}



$$R_{Th} = (27 \| 47)$$

$$= \frac{27 \times 47}{27+47} = \underline{\underline{17.14\Omega}}$$

3. Thevenin equivalent Circuit



$$I_L = \frac{V_{Th}}{R_{Th}+R_L}$$

$$= \frac{-386.9}{17.14+27}$$

$$= \underline{\underline{-8.76A}}$$

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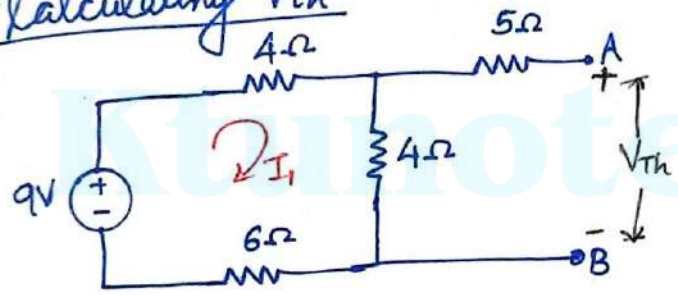
6. 1. Calculating V_{Th}



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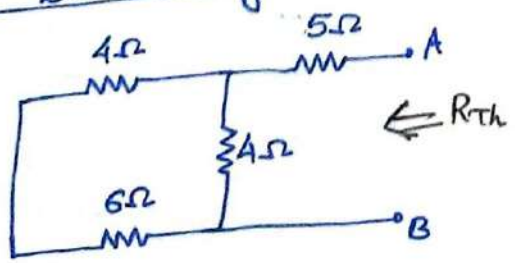
Applying KVL to Mesh 1

$$9 - 4I_1 - 4I_1 - 6I_1 = 0$$

$$I_1 = \underline{\underline{\frac{9}{14} A}}$$

$$V_{Th} = 4I_1 = 4 \times \frac{9}{14} = \underline{\underline{2.571V}}$$

2. Calculating R_{Th}



$$R_{Th} = 5 + (4 \| (4+6))$$

$$= 5 + (4 \| 10) = 5 + \frac{4 \times 10}{4+10}$$

$$= 5 + \frac{40}{14} = \frac{110}{14}$$

$$= \underline{\underline{7.857\Omega}}$$

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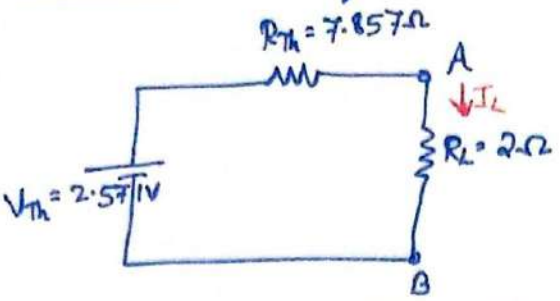
3. Thevenin equivalent Circuit



$$I_L = \frac{V_{Th}}{R_{Th}+R_L}$$

$$= \frac{2.571}{7.857+2} = \underline{\underline{0.2608\,A}}$$

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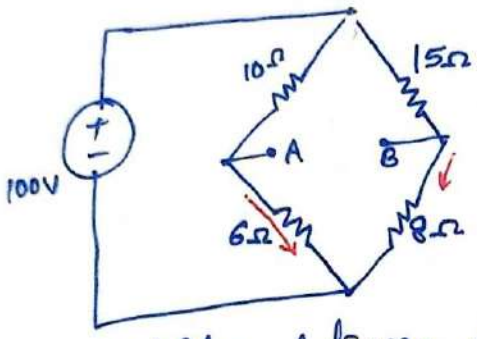
7. 1. Calculating V_{Th}



Current through $6\Omega = \frac{100}{16}$ A

Current through $8\Omega = \frac{100}{23}$ A

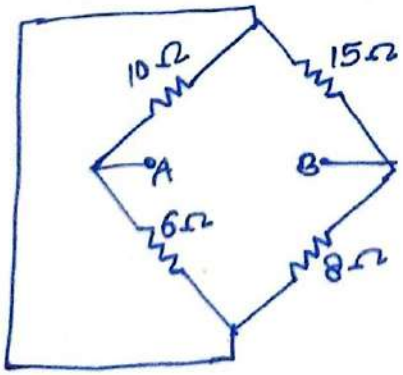
V_{Th} = Voltage between A & B

= Voltage at A − Voltage at B

= Voltage across 6Ω − Voltage across 8Ω

$$= 6 \times \frac{100}{16} - 8 \times \frac{100}{23} = \underline{\underline{2.72V}}$$

2. Calculating R_{Th}



$$R_{Th} = (10 \| 6) + (15 \| 8)$$

$$= 10 \| 6 + 15 \| 8$$

$$= \frac{10 \times 6}{10+6} + \frac{15 \times 8}{15+8}$$

$$= \frac{60}{16} + \frac{120}{23} = \underline{\underline{8.97\Omega}}$$

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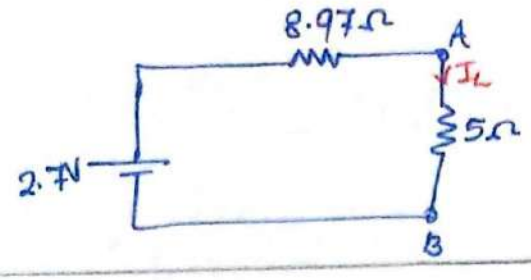
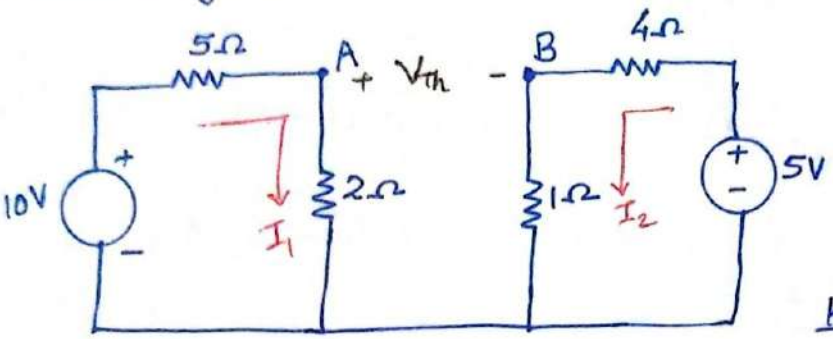
3. Thevenin Equivalent Circuit



$$I_L = \frac{2.7}{(8.97+5)}$$

$$= \underline{0.193\,A}$$

8. Redrawing the circuit



KVL → $V_{Th} + V_1 + V_2 = 0$

$V_{Th} - 2I_1 + 1I_2 = 0$

$V_{Th} = 2I_1 - I_2$

$$V_{Th} = V_{AB} = V_A - V_B$$

$$= \frac{10}{7} \times 2 - \frac{5}{5} \times 1$$

$$= \frac{20}{7} - 1 = \underline{\underline{1.85\,V}}$$

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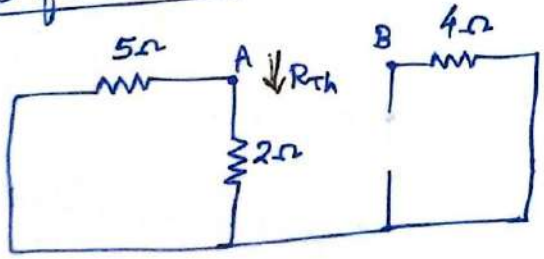
To find R_{Th}

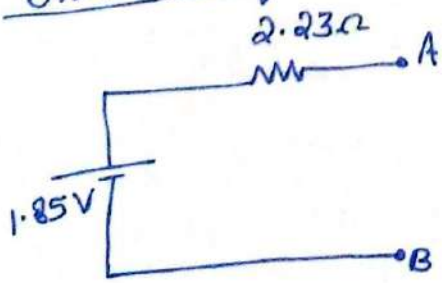


$$R_{Th} = (5 \| 2) + (4 \| 1)$$

$$= \frac{5 \times 2}{5+2} + \frac{4 \times 1}{4+1}$$

$$= \frac{10}{7} + \frac{4}{5} = \underline{\underline{2.23\,\Omega}}$$

Thevenins equivalent circuit



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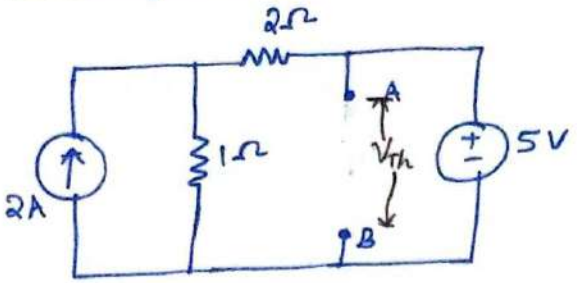
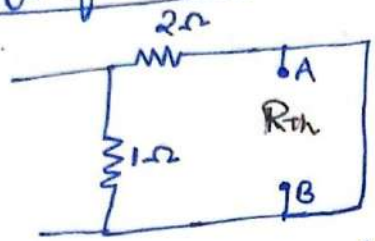
9. 1. To find V_{Th}



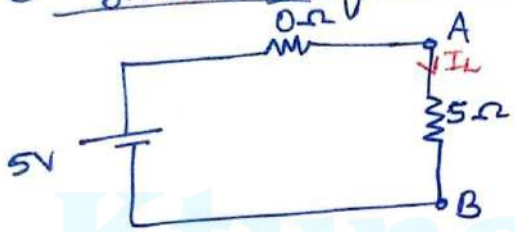
From the circuit, it is clear that $V_{Th} = \underline{5V}$

2. To find R_{Th}



It is evident from figure that $R_{Th} = 0$

3. Thevenin's equivalent circuit



$$I_L = \frac{V_{Th}}{R_{Th}+R_L}$$

$$= \frac{5}{0+5} = \underline{1A.}$$

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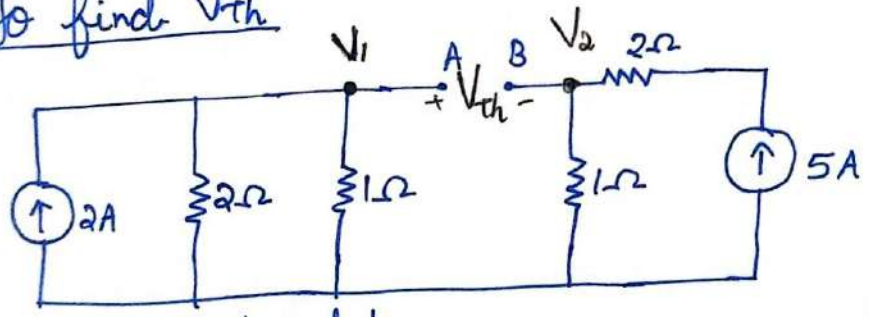
10. To find V_{Th}



$V_A = V_1$

$V_B = V_2$

Applying KCL to node 1

$$2 = \frac{V_1}{2} + \frac{V_1}{1} \; ; \quad 3V_1 = 4$$

$$V_1 = \underline{4/3 \; V.}$$

Applying KCL to node 2

$$5 = \frac{V_2}{1} \; ; \quad V_2 = \underline{5V.}$$

$$V_{Th} = V_{AB} = V_A - V_B = V_1 - V_2 = \frac{4}{3} - 5 = \underline{\frac{-11}{3} V}$$

$$V_{Th} = \underline{-3.67 \, V}$$

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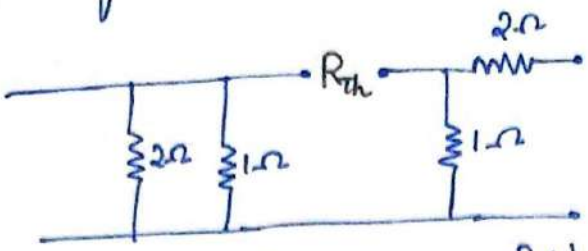
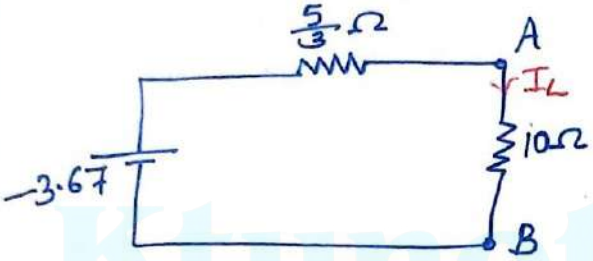
2. To find R_{Th}



$$R_{Th} = (2 \| 1) + 1 = \frac{2 \times 1}{2+1} + 1$$

$$= \frac{2}{3} + 1 = \frac{5}{3} = \frac{5}{3}\,\Omega$$

3. Thevenin's Equivalent Circuit



$$I_L = \frac{-3.67}{\frac{5}{3} + 10}$$

$$= \frac{-3.67 \times 3}{35} = -0.315$$

$I_L = -0.315A\ (\downarrow)$

ie; $I_L = 0.315A\ (\uparrow)$

Power loss in 10Ω resistor

$= I_L^2 R = (0.315)^2 \times 10 = 0.993$

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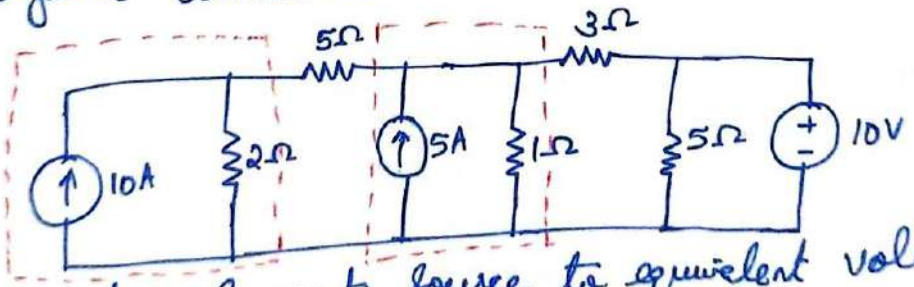
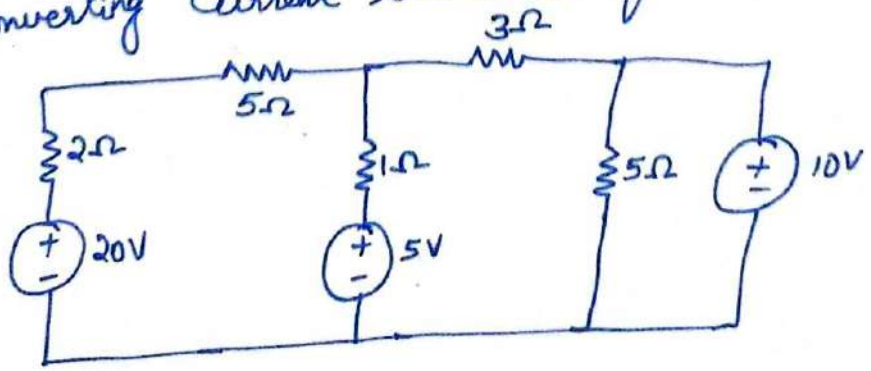
11. Given Circuit is



Converting Current Source to equivalent voltage source



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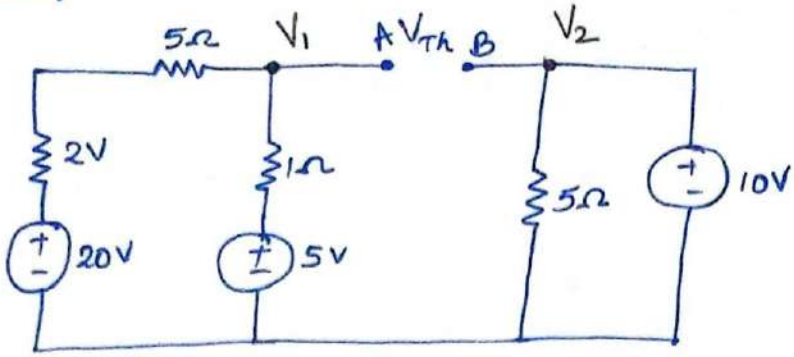
1. To find V_{Th}



Node 1 : $\frac{V_1 - 20}{7} + \frac{V_1 - 5}{1} = 0$

$V_1 - 20 + 7V_1 - 35 = 0$

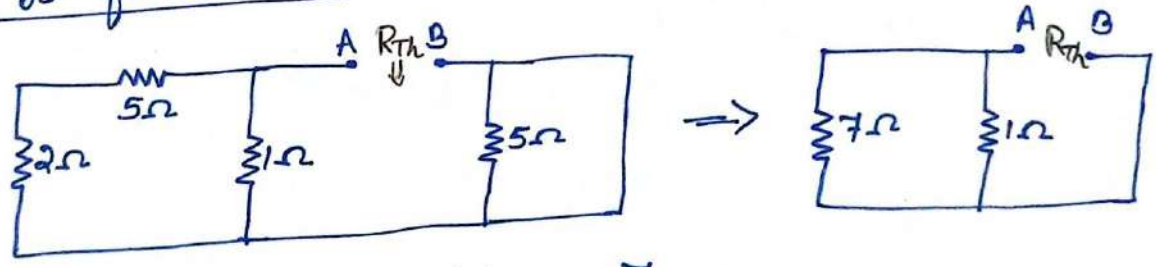
$V_1 = 6.875V$

Node 2 : $V_2 = 10\ V$

$V_{Th} = V_1 - V_2 = 6.875 - 10$

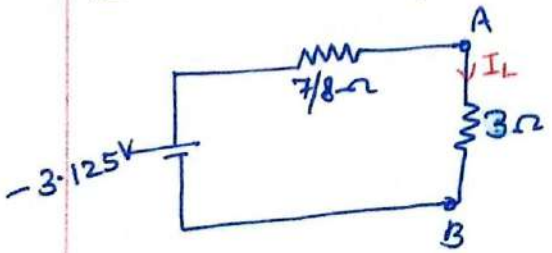
$= -3.125\ V$

2. To find R_{Th}



$R_{Th} = 7 \| 1 + 0 = \frac{7 \times 1}{7+1} = \frac{7}{8}\ \Omega$

3. Thevenin equivalent circuit



$I_L = \frac{-3.125}{\frac{7}{8}+3} = -0.806A\ (\downarrow)$

$I_L = 0.806A\ (\uparrow)$

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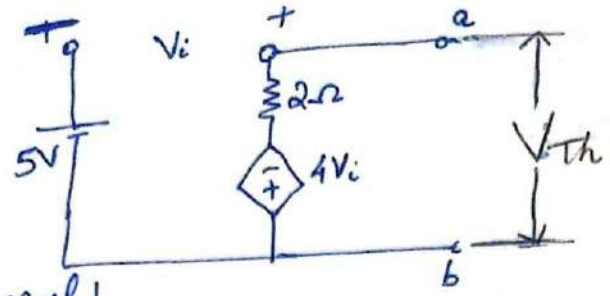
12. 1. To Find V_{Th}



$-4V_i - 2\times 0 - V_{Th} = 0$

$V_{Th} = -4V_i$

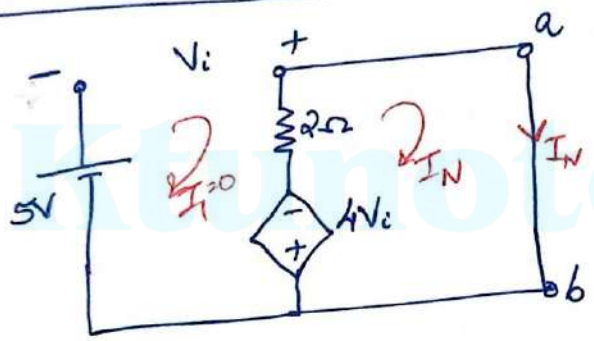
Mesh 1

$5 + V_i - 0 + 4V_i = 0$

$5V_i = -5; \quad V_i = -1V$

So $V_{Th} = -4V_i = -4\times -1 = +4V$

2. Find I_N



Loop 1

$5 + V_i + 2I_N + 4V_i = 0$

$5V_i + 2I_N = -5$ —— 1

$-4V_i - 2I_N = 0$

$4V_i + 2I_N = 0$ —— 2

$I_N = \frac{-4V_i}{2} = -2V_i$

$I_N = -2\times -1 = 2A$

3. Find R_{Th}

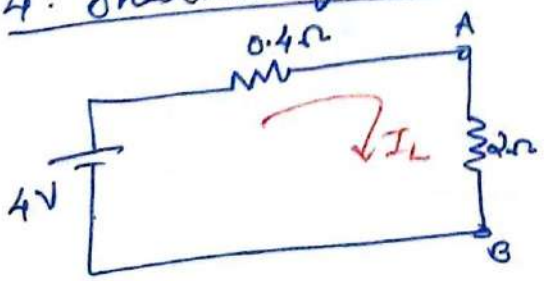
$R_{Th} = \frac{V_{Th}}{I_N} = \frac{4}{2} = 2\Omega$

Solve ① & ② $\Rightarrow$ $V_i = -5$, $I_N = 10$

3. To find R_{Th}

$R_{Th} = \frac{V_{Th}}{I_N} = \frac{4}{10} = 0.4\Omega$

4. Thevenin equivalent Circuit



$I_L = \frac{4}{0.4+2} = 1.67A$

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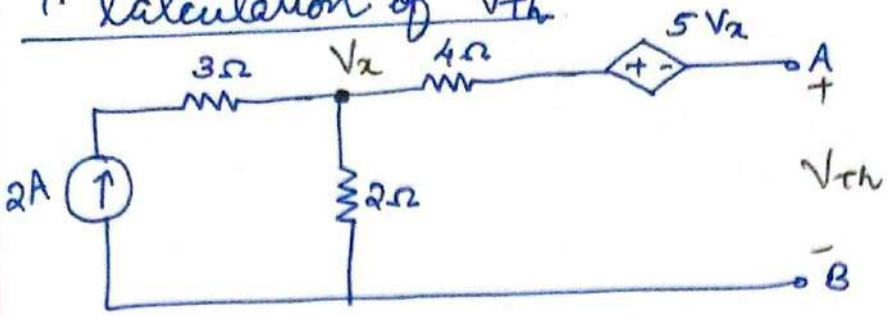
13. 1. Calculation of V_{Th}



KCL at Node x

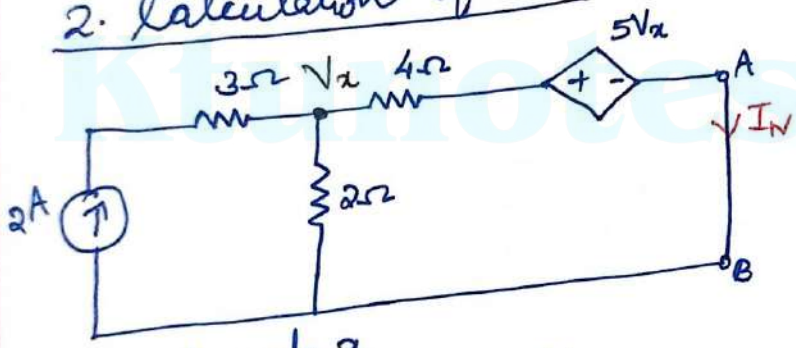
$2 = \frac{V_x}{2} + 0; \quad V_x = 4V$

Writing V_{Th} equation

$V_x - 4 \times 0 - 5V_x - V_{Th} = 0$

$V_{Th} = -4V_x = -16V$

2. Calculation of I_N



KCL at node x

$2 = \frac{V_x}{2} + \frac{V_x - 5V_x}{4}$

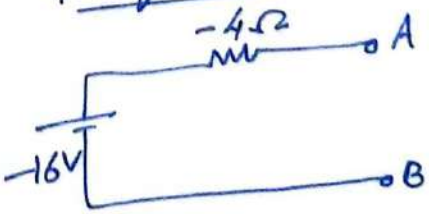
$8 = 4V_x + V_x - 5V_x; \quad 8 = -2V_x$

So $V_x = -4V; \quad I_N = \frac{V_x - 5V_x}{4} = \frac{-4 - 5 \times -4}{4} = 4A.$

3. Calculation of R_{Th}

$R_{Th} = \frac{V_{Th}}{I_N} = \frac{-16}{4} = -4\Omega$

4. Equivalent Thevenin Circuit



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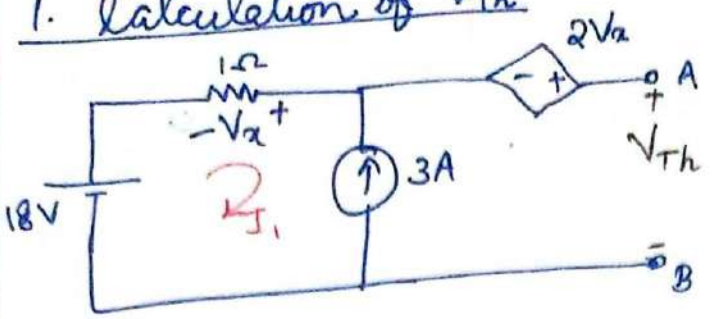
14. 1. Calculation of V_{Th}



KVL to Mesh (outer)

$18 + V_x + 2V_x - V_{Th} = 0$

$V_{Th} = 18 - 3V_x$ —— 1

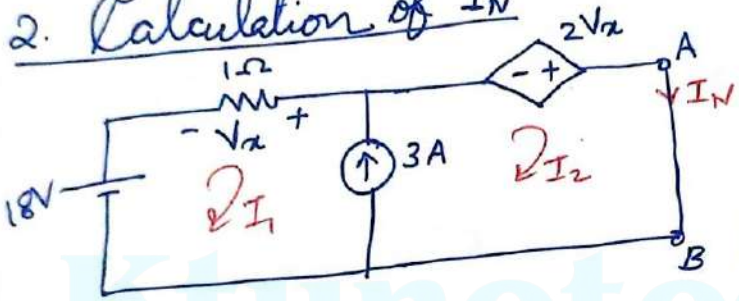
$I_1 = -3A$ —— 2

$V_x = -1 \times I_1 = 3V$

Substitute ② in ①

$V_{Th} = 18 + 3V_x = 18 + 3\times3 = 27V$

2. Calculation of I_N



KVL to mesh (outer)

$18 + V_x + 2V_x = 0$

$18 + 3V_x = 0$

As $V_x = -I_1$

$18 = 3I_1$; $I_1 = 6A$

$-I_1 + I_2 = 3$

$V_x = -I_1$

$-I_1 + I_2 = 3$

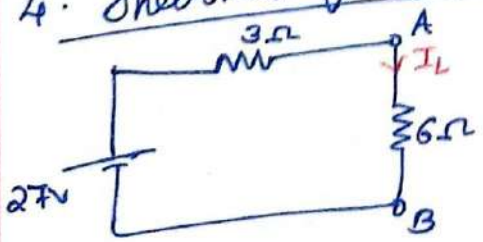
so $I_2 = 9A$

$I_N = I_2 = 9A$

3. Calculation of R_{Th}

$R_{Th} = \frac{V_{Th}}{I_N} = \frac{27}{9} = 3\Omega$

4. Thevenin equ. circuit



$I_L = \frac{V_{Th}}{R_{Th}+R_L}$

$= \frac{27}{3+6} = 3A$

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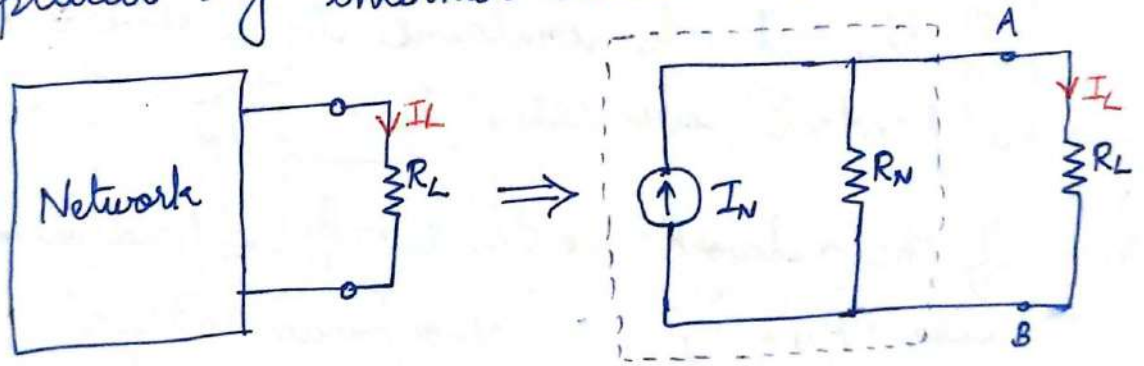
NORTON'S THEOREM

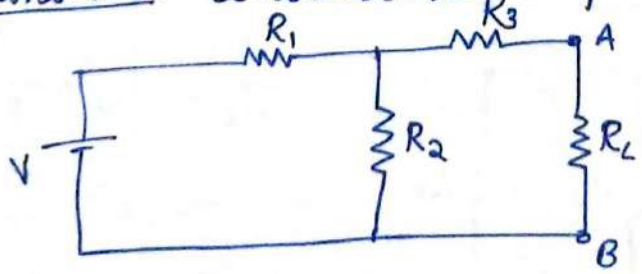
It states that any two terminals of a network can be replaced by an equivalent current source and an equivalent parallel resistance.

The constant current is equal to the current which would flow in a short circuit placed across the terminals.

The parallel resistance is the resistance of the network when veived from these open circuited terminals after all voltage and current sources have been removed and replaced by internal resistances.



Explanation: Consider a simple network in Fig.



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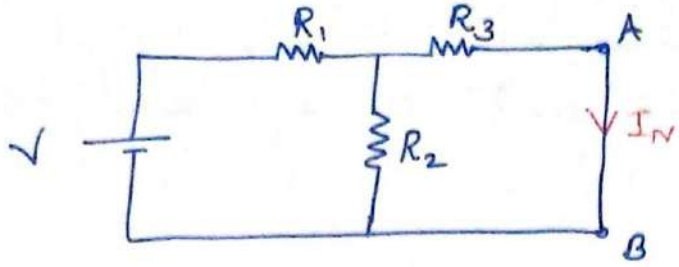
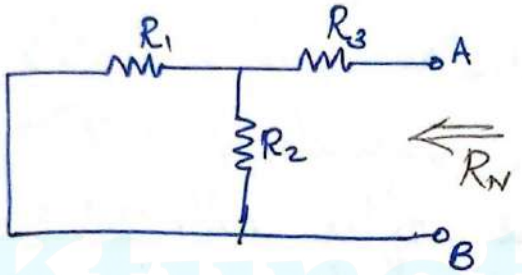
First remove the load resistor R_L, and calculate the short circuit current I_N



For finding R_N, replace voltage source by a short circuit and calculate the resistance between points A & B



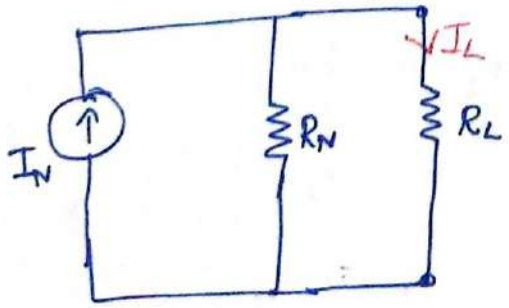
$$R_N = (R_1 \| R_2) + R_3$$

$$= \frac{R_1 R_2}{R_1 + R_2} + R_3$$

If the network contains both independent and dependent sources, $R_N = \frac{V_{Th}}{I_N}$

If the network contains only dependent sources, then $R_N = \frac{V}{I}$ $\quad V \rightarrow$ known voltage

Nortons equivalent network



$$I_L = I_N \times \frac{R_N}{R_N + R_L}$$

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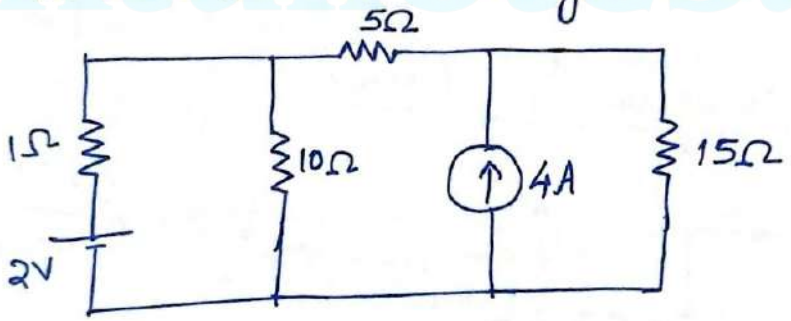
Steps to be followed in Norton's Theorem

1. Remove the load resistance R_L and put a short circuit across the terminals.
2. Find the short circuit current I_N
3. Find the resistance R_N as seen from A&B
4. Replace the network by a current source I_N in parallel with resistance R_N.
5. Find current through R_L by current-division rule.

$$I_L = \frac{I_N R_N}{R_N + R_L}$$

---

1. Find the current through the 10Ω resistor



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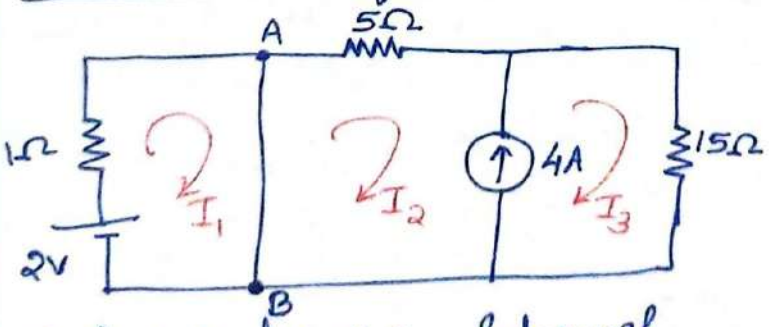
1. Calculation of I_N



KVL to Mesh 1

$2 - I_1 = 0$

$I_1 = 2A$ —— (1)

Mesh 2 & 3 forms a Supermesh

$-I_2 + I_3 = 4$ —— (2)

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$-5I_2 - 15I_3 = 0$ ——— (3)

Solving eqn ①, ② & ③

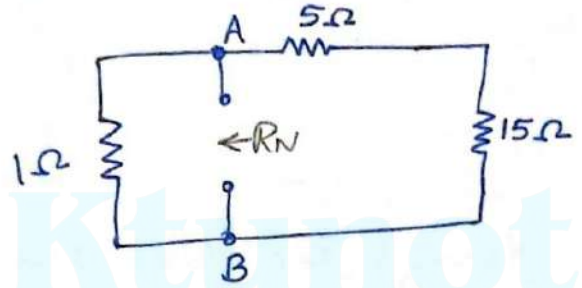
$\Rightarrow I_1 = 2A$

$I_2 = -3A$

$I_3 = 1A$

$I_N = I_1 - I_2 = 2 - -3 = 5A$

2. Calculation of R_N

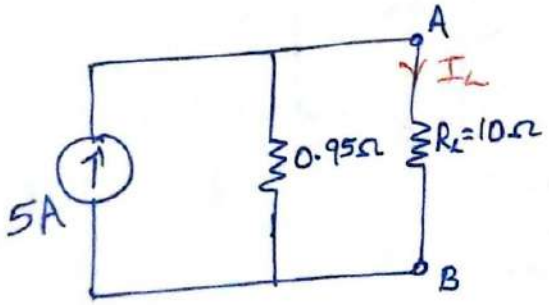


$R_N = 1 \,//\, (5+15)$

$= 1 \,//\, 20$

$= \frac{1 \times 20}{1+20} = 0.95\Omega$

3. Norton's equivalent Circuit



$I_L = 5 \times \frac{0.95}{0.95+10}$

$= 0.43A$

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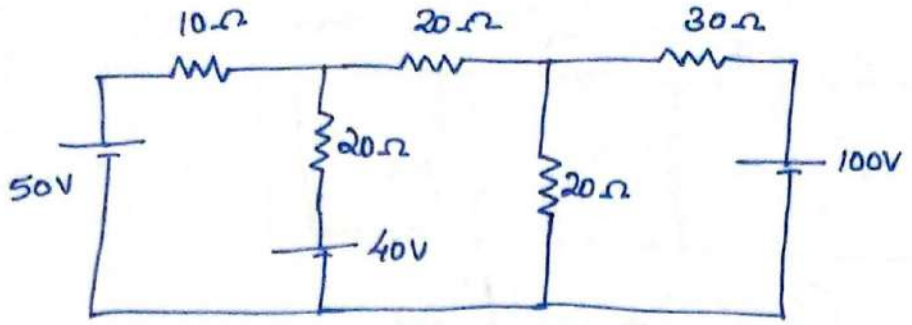
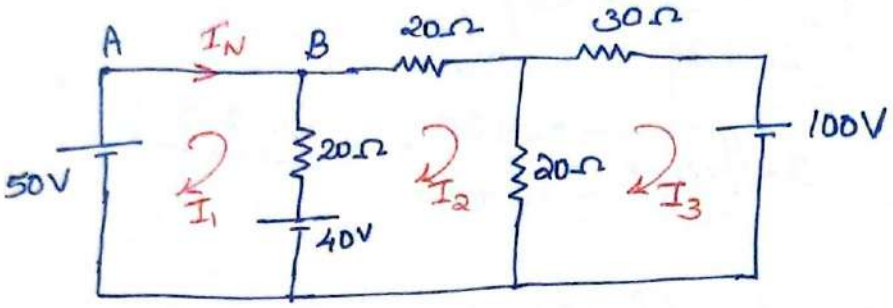
2. Find the current through 10Ω resistor



1. Calculation of I_N



Applying KVL to Mesh 1

$$50 - 20(I_1 - I_2) - 40 = 0$$

$$20I_1 - 20I_2 = 10 \quad \text{——— (1)}$$

KVL to Mesh 2

$$40 - 20(I_2 - I_1) - 20I_2 - 20(I_2 - I_3) = 0$$

$$-20I_1 + 60I_2 - 20I_3 = 40 \quad \text{——— (2)}$$

KVL to Mesh 3

$$-20(I_3 - I_2) - 30I_3 - 100 = 0$$

$$-20I_2 + 50I_3 = -100 \quad \text{——— (3)}$$

Solving (1), (2) and (3)

$I_1 = 0.8125\,A$

$I_2 = 0.3125\,A$

$I_3 = -1.875\,A$

$I_N = I_1 = 0.8125\,A$

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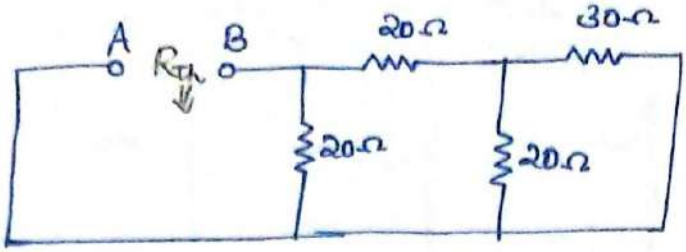
2. Calculation of R_N

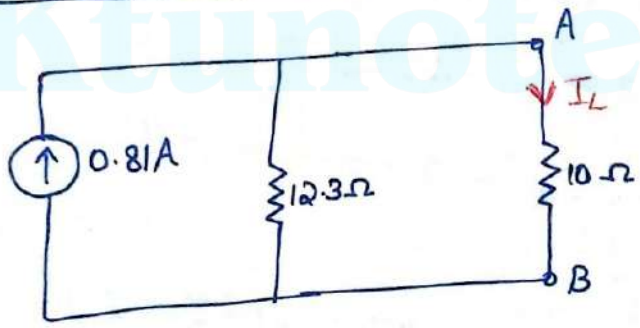


$$R_{Th} = \left[(20 \| 30) + 20\right] \| 20$$

$$= \left(\frac{20 \times 30}{20+30} + 20\right) \| 20$$

$$= 32 \| 20 = \frac{32 \times 20}{32+20} = \underline{12.3\Omega}$$

3. Calculation of I_L

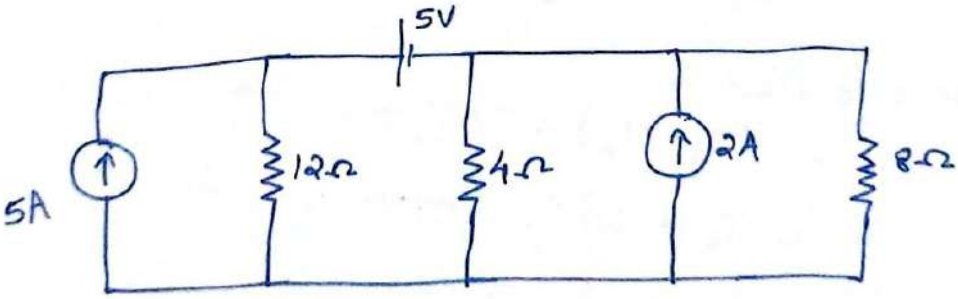


$$I_L = \frac{0.81 \times 12.3}{12.3+10}$$

$$= \underline{\underline{0.45\,A}}$$

---

3. Find the current through the 8Ω resistor



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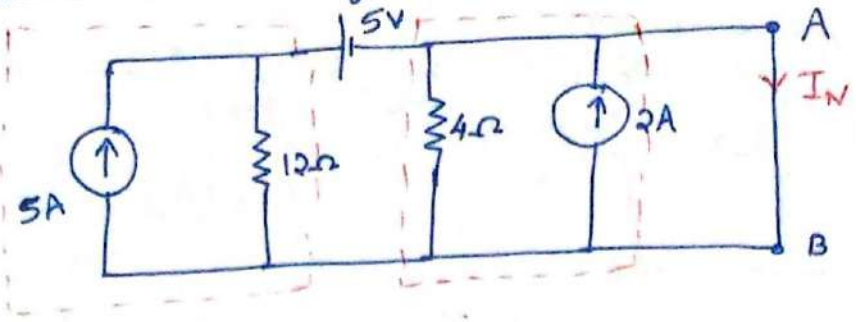
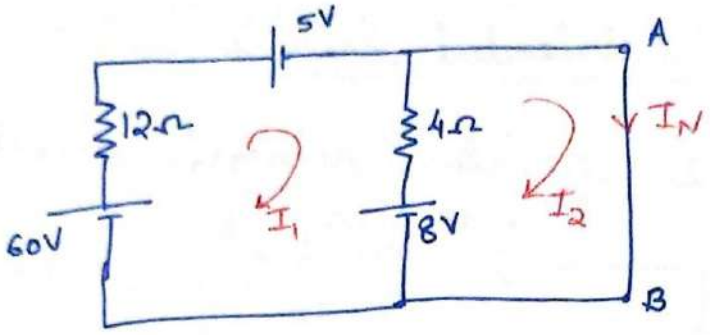
1. Calculation of I_N





Mesh 1: $60 - 12I_1 - 5 - 4(I_1 - I_2) - 8 = 0$

$$16I_1 - 4I_2 = 47 \quad \text{——— (1)}$$

Mesh 2: $8 - 4(I_2 - I_1) = 0$

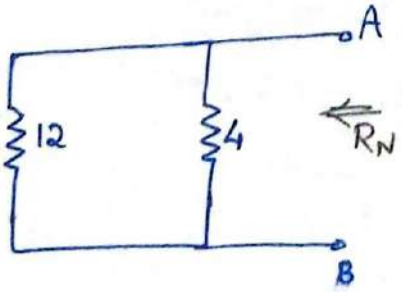
$$-4I_1 + 4I_2 = 8 \quad \text{——— (2)}$$

Solve (1) & (2) $\Rightarrow$ $I_1 = 4.58\ A$

$$I_2 = 6.58\ A$$

$$I_N = I_2 = 6.58A$$

2. Calculation of R_N



$R_N = 12 \| 4$

$$= \frac{12 \times 4}{12 + 4} = 3\Omega$$

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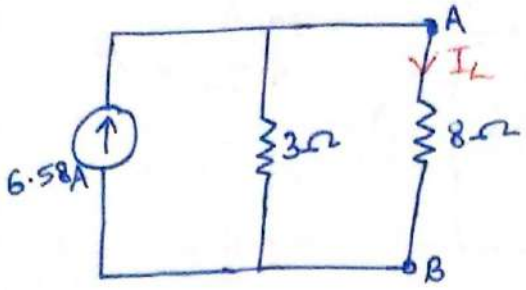
3. Calculation of I_L

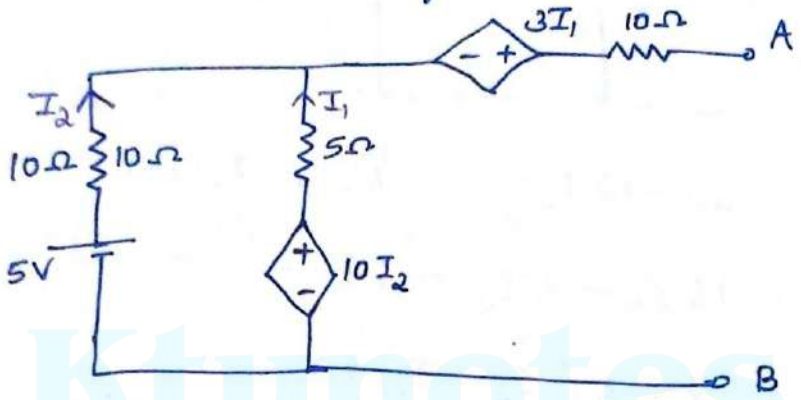


$$I_L = \frac{6.58 \times 3}{3+8}$$

$$= \underline{\underline{1.79A}}$$

Examples With Dependent Sources

4. Find Nortons equivalent network across A&B



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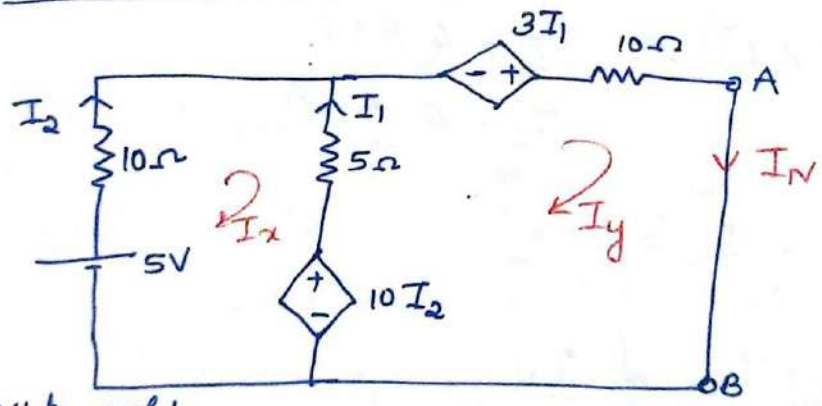
1. Calculation of I_N



$I_2 = I_x$ —(1)

$I_1 = I_y - I_x$ —(2)

KVL to mesh 1

$$5 - 10I_x - 5(I_x - I_y) - 10I_2 = 0$$

$$5 - 10I_x - 5I_x + 5I_y - 10I_x = 0$$

$$25I_x - 5I_y = 5 \quad \text{—— (3)}$$

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KVL to mesh 2

$$10 I_2 - 5(I_y - I_x) + 3I_1 - 10 I_y = 0$$

$$10 I_x - 5 I_y + 5 I_x + 3(I_y - I_x) - 10 I_y = 0$$

$$12 I_x - 12 I_y = 0 \quad \text{———} \quad (4)$$

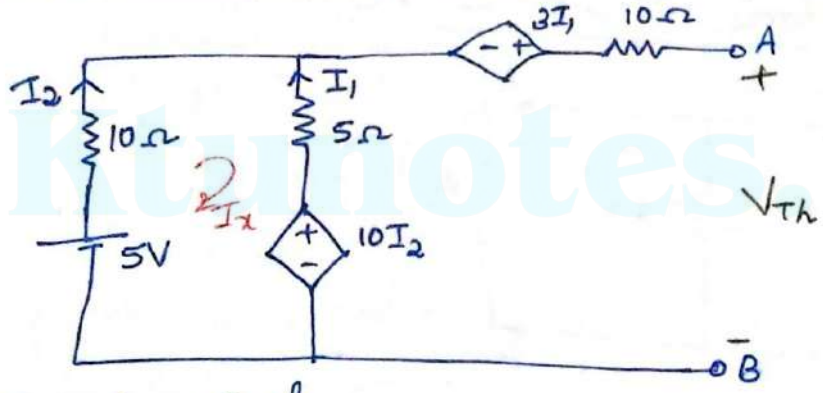
Solving (3) & (4)

$I_x = 0.25\,A$

$I_y = \underline{0.25A}$

$I_N = I_y = \underline{0.25A}$

2. Calculation of V_{Th}



$I_x = I_2$

$I_1 = -I_x$

KVL to Mesh

$$5 - 10 I_x - 5(I_x) - 10 I_2 = 0$$

$$5 - 15 I_x - 10 I_x = 0$$

$$25 I_x = 5; \qquad I_x = \underline{0.2A}$$

~~Writing V_{Th} eqn~~

$$\cancel{10 I_2 - 5 I_1 + 3 I_1 - 0 - V_{Th} = 0}$$

$$\cancel{V_{Th} = 10 I_2 - 5 I_1 + 3 I_1}$$

$$\cancel{= 10 I_x + 5 I_x + 3 \times -I_x}$$

$$\cancel{= 8 I_x = 12 \times 0.2 = 2.4V}$$

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Writing V_{Th} eqn

$$5 - 10 I_x + 3 I_1 - V_{Th} = 0$$

$$5 - 10 \times 0.2 - 3 I_x = V_{Th}$$

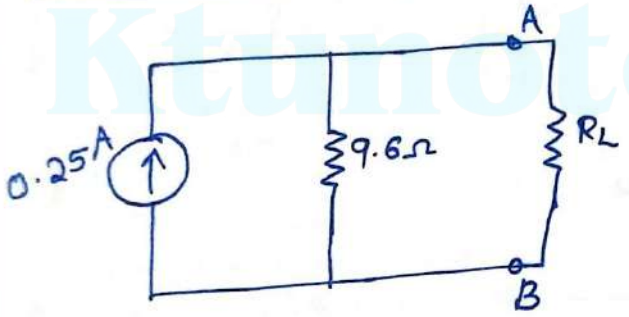
$$5 - 10 \times 0.2 - 3 \times 0.2 = V_{Th}$$

$$V_{Th} = \underline{\underline{2.4 V}}$$

3. Calculation of R_N

$$R_N = \frac{V_{Th}}{I_N} = \frac{2.4}{0.25} = \underline{\underline{9.6 \Omega}}$$

4. Norton's equivalent network



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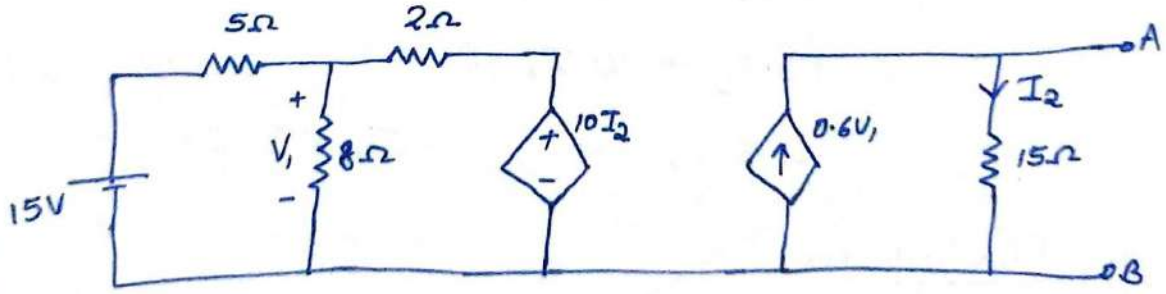
5. Obtain Norton's equivalent network across AB.



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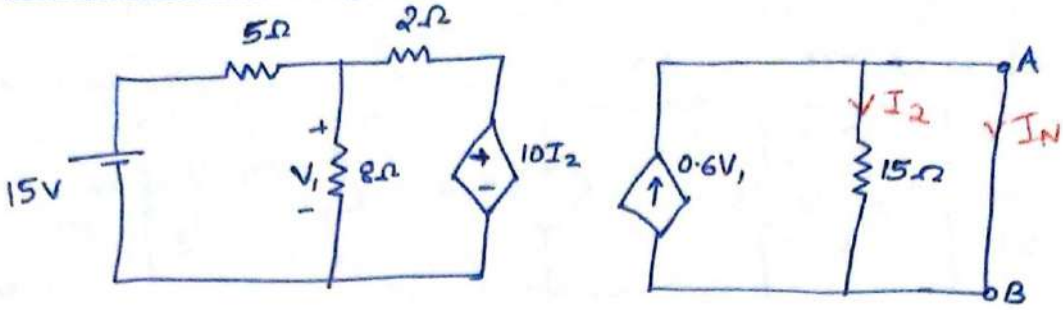
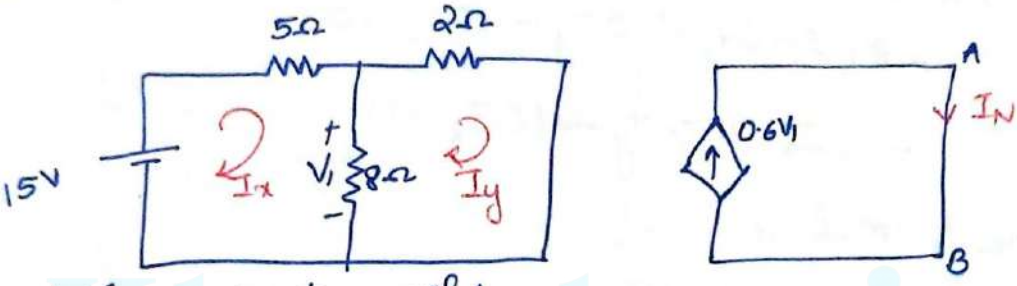
1. Calculation of I_N



From Fig. $I_2 = 0$.

The dependent source of $10I_2$ depends on the controlling variable I_2. When $I_2 = 0$, the dependent source vanishes. i.e; $10I_2 = 0$.



Applying KVL to mesh 1

$$15 - 5I_x - 8(I_x - I_y) = 0$$

$$13I_x - 8I_y = 15 \quad \text{——— (1)}$$

Applying KVL to mesh 2

$$-8(I_y - I_x) - 2I_y = 0$$

$$-8I_x + 10I_y = 0 \quad \text{——— (2)}$$

Solving (1) & (2)

$$I_x = 2.27A, \quad I_y = 1.82A$$

$$V_1 = 8(I_x - I_y) = 8(2.27 - 1.82) = 3.6V$$

Mesh 3

$$I_N = 0.6V_1 = 0.6 \times 3.6$$

$$= 2.16A$$

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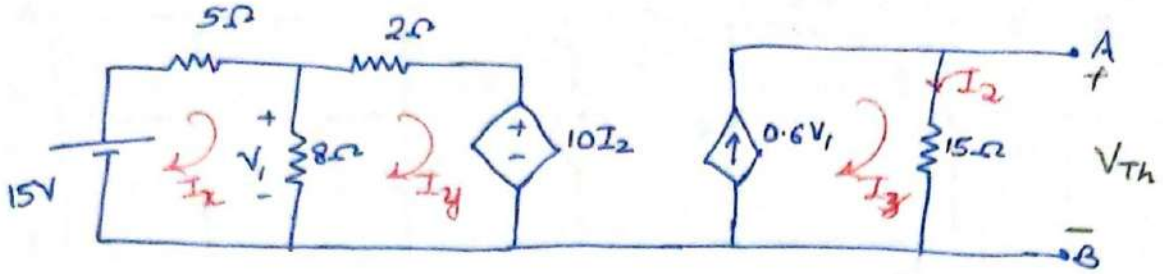
2. Calculation of V_{Th}



KVL to Mesh 1 $\qquad V_1 = (I_x - I_y)8$ —— (1)

$$15 - 5I_x - 8(I_x - I_y) = 0$$

$$13I_x - 8I_y = 15 \quad \text{——— (2)}$$

KVL to Mesh 2

$$-8(I_y - I_x) - 2I_y - 10I_2 = 0$$

$$-8I_x + 10I_y - 10I_2 = 0 \quad \text{——— (3)}$$

KVL to Mesh 3

$$I_2 = 0.6V_1$$

$$= 0.6(I_x - I_y) \times 8 = 4.8(I_x - I_y)$$

$$4.8I_x - 4.8I_y - I_2 = 0 \quad \text{——— (4)}$$

Solving (2), (3) & (4)

$$I_x = 3.28A$$

$$I_y = 3.45A$$

$$I_2 = -0.83A$$

$$V_{Th} = 15I_2 = 15 \times -0.83$$

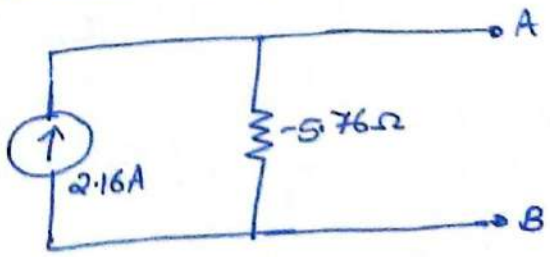
$$= -12.45V$$

3. Calculation of R_N

$$R_N = \frac{V_{Th}}{I_N} = \frac{-12.45}{2.16}$$

$$= -5.76\Omega$$

4. Norton's Equ Circuit



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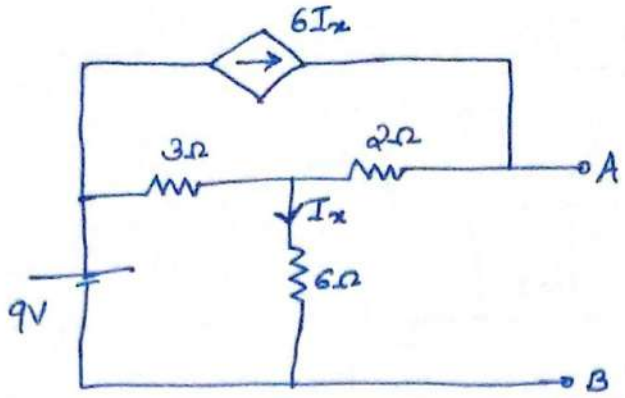
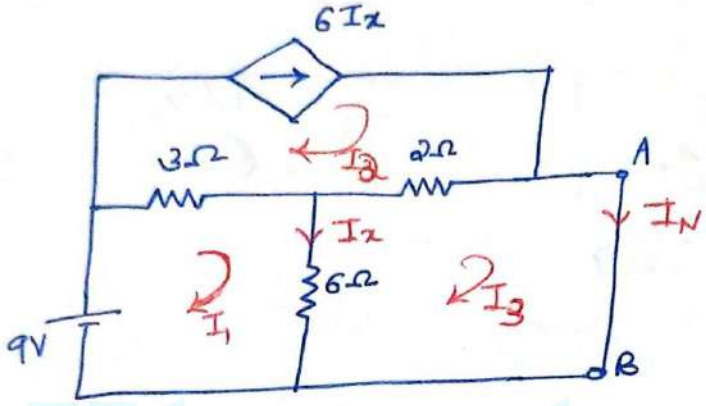
6.



Find Norton's equivalent network at terminals A & B

1. Calculation of I_N



$I_x = I_1 - I_3$ —— (1)

mesh 1 $9I_1 - 6I_3 - 3I_2 = 9$

$9I_1 - 3I_2 - 6I_3 = 9$ —— 2

~~$-3I_1$~~

Mesh 2 $I_2 = 6I_x = 6(I_1 - I_3)$

$I_2 = 6I_1 - 6I_3$

$6I_1 - I_2 - 6I_3 = 0$ —— 3

mesh 3

$-6I_1 - 2I_2 + 8I_3 = 0$ —— 4

Solving 2, 3 & 4

$I_1 = 5A$

$I_2 = 3A$

$I_3 = 4.5A$

$I_N = I_3$

$= 4.5A.$

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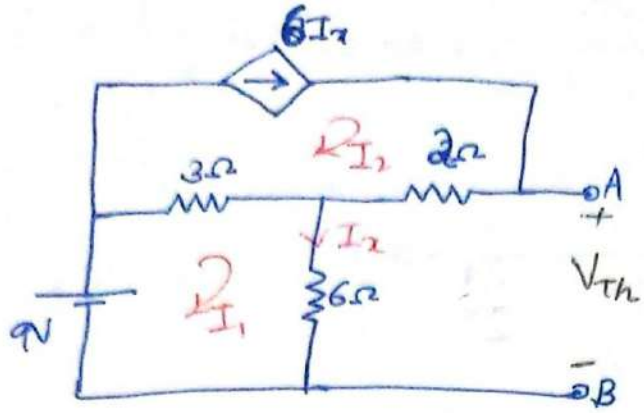
2. Calculation of V_{Th}



$I_x = I_1$

$9I_1 - 3I_2 = 9$ $\qquad I_2 = 6I_x$

$9I_1 - 3\times 6I_1 = 9$ $\qquad = 6I_1$

$\cancel{-15I_1}\ -9I_1 = 9$

$I_1 = \underline{-1A}$

$I_2 = 6I_1 = \underline{-6A}$

Writing V_{Th} eqn

$9 - 3(I_1 - I_2) + 2I_2 - V_{Th} = 0$

$V_{Th} = -3I_1 - I_2 + 9$

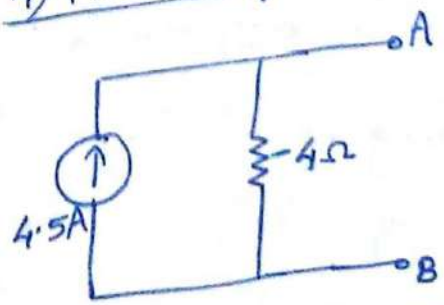
$= -3\times -1 - -6 + 9 = \underline{-18V}$

3. Calculation of R_N

$R_N = \frac{V_{Th}}{I_N} = \frac{-18}{4.5}$

$= \underline{-4\Omega}$

4) Norton's equ ckt



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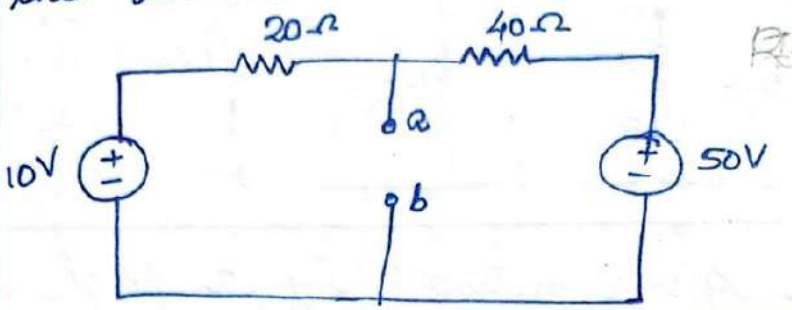
Tutorial Questions from Norton's Theorem

1. Find the Thevenin's and Norton's equivalents for the circuit.

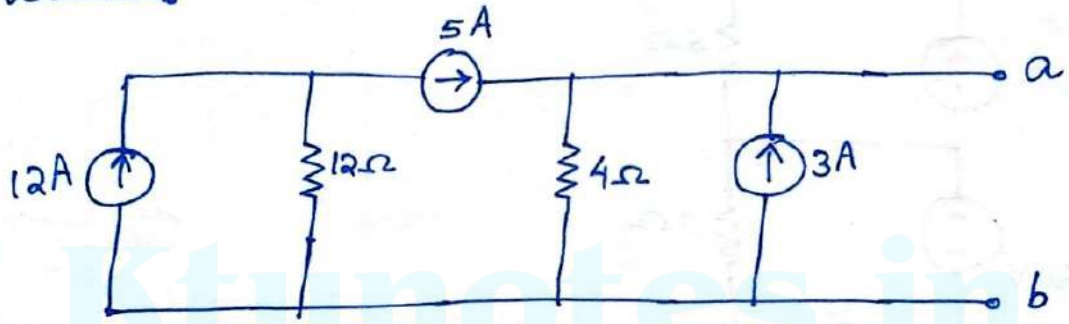
$V_{Th} = 23.33V$ $I_N = 1.75A$

$R_{Th} = 13.33\Omega$ $R_N = 13.33\Omega$



2. Determine the Thevenin's and Nortons equivalent circuits

$V_{Th} = 32$
$R_{Th} = 4$
$I_N = 8$
$R_N = 4$



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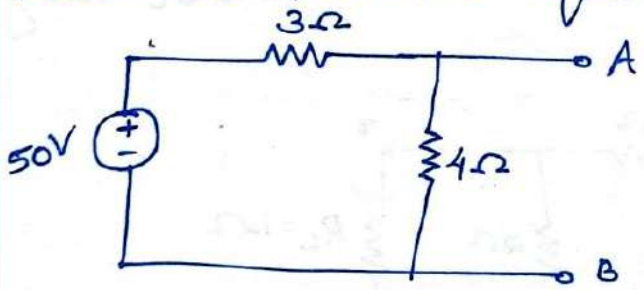
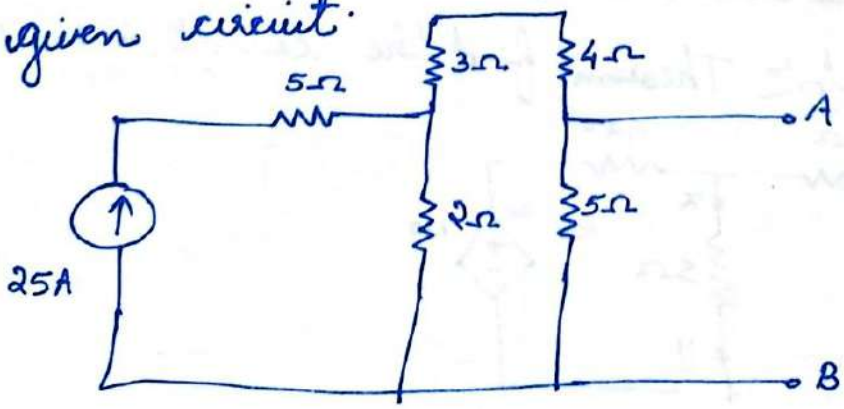
3. Determine Norton's equivalent circuit



$I_N = 16.67A$
$R_N = 1.71\Omega$

4. Determine Norton's equivalent circuit for the given circuit.



$I_N = 5.55A$
$R_N = 3.21\Omega$

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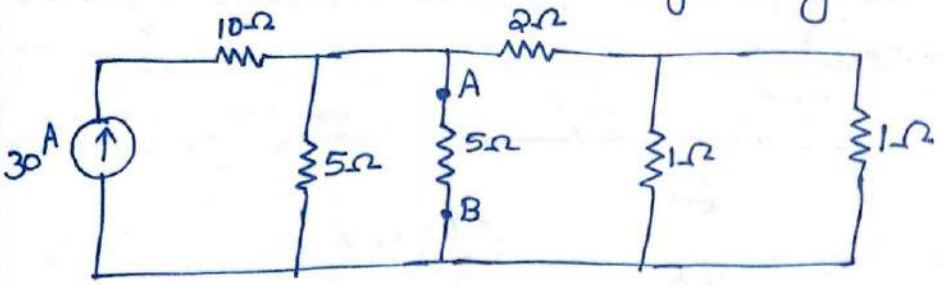
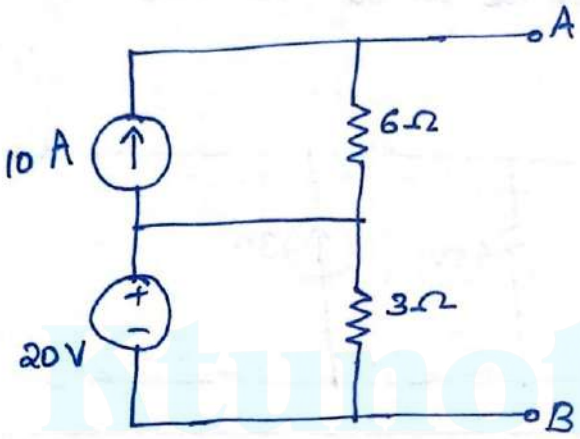
5. Determine the current flowing through the 5Ω resistor in the circuit by using Norton's Theorem.



$I_N = 30A$
$R_N = 1.67\Omega$
$I_L = 7.51\Omega$

6. Replace the given network by a single current source in parallel with a resistance:



$I_N = 13.33\ A$
$R_N = 6\Omega$

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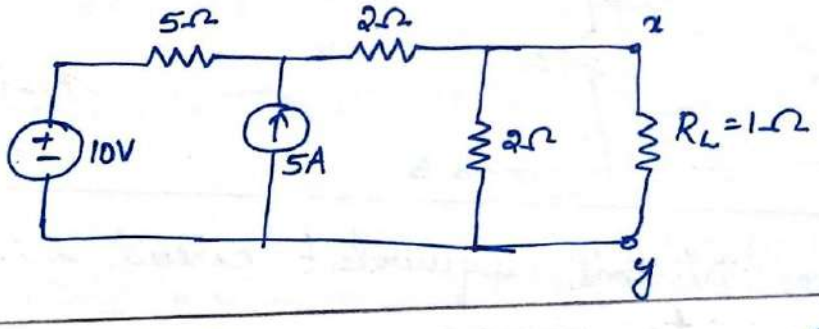
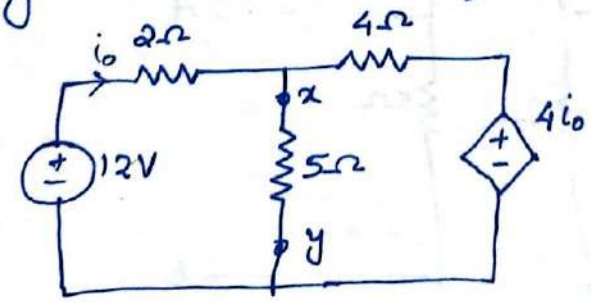
7. Find the powerloss in 1Ω resistor using Norton's Theorem



$I_N = 5A$
$R_N = 1.56\Omega$
$I_L = 3.04A$
Power Loss = 9.24W

8. Using Norton's Theorem find the current in the 5Ω



$I_N = 12\ A$
$R_N = 0.8\Omega$
$I_L = 1.66A$

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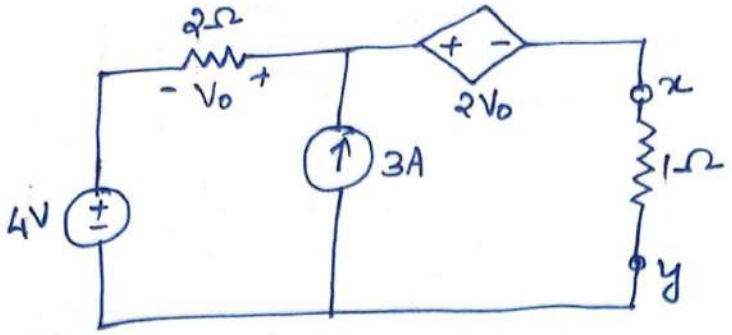
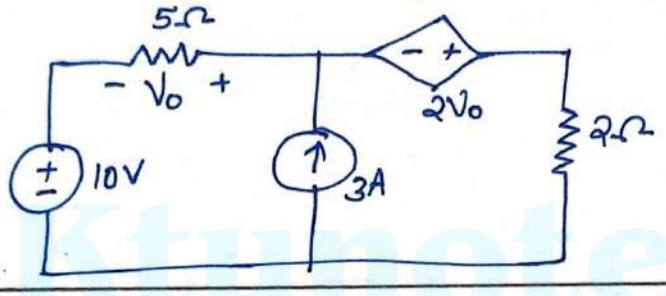
9. Determine the power loss in 1Ω resistor using Norton's Theorem.



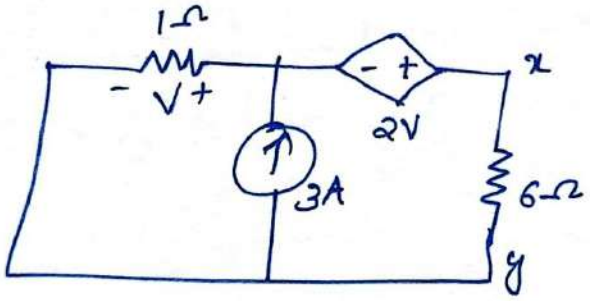
$I_N = 1A$
$R_N = \infty A$
$I_L = 0.67 A$
$P_{loss} = 0.45 W$

10. Find the current in the 2Ω resistance in the circuit of Fig. using Norton's Theorem



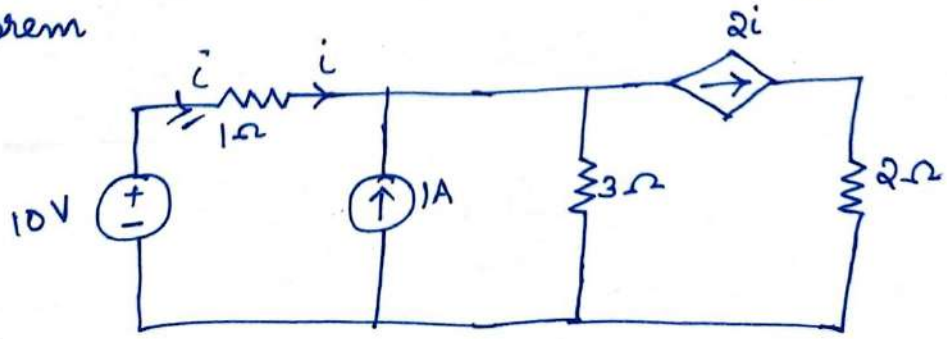
$I_N = \frac{11}{3} A$
$R_N = 15\Omega$
$I_L = 3.235 A$

11. Find the current in the 6Ω resistor



$I_N = 3A$
$R_N = 3\Omega$
$I_L = 1A$

12. Find the current in 2Ω resistor using Thortons Theorem



$I_N = -7A$
$R_N = -1.178$
$I_L = 10.03$

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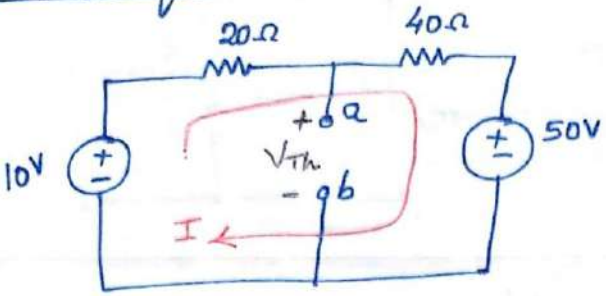
Answers – Tutorial from Norton's Theorem

1. To find V_{Th}



$$10 - 20I - 40I - 50 = 0$$

$$60I = -40$$

$$I = -2/3\ A$$

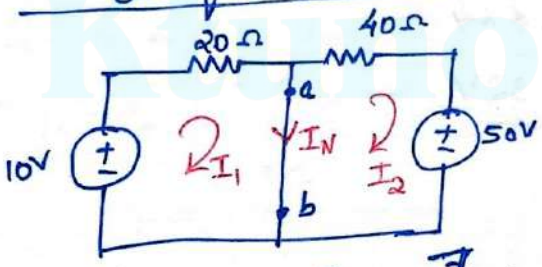
Writing V_{Th} equation

$$10 - 20I - V_{Th} = 0$$

$$V_{Th} = 10 - 20I = 10 - 20 \times \frac{-2}{3} = 10 + \frac{40}{3}$$

$$= 23.33V.$$

2. To find I_N



$$I_N = I_1 - I_2$$

$$20I_1 = 10; \quad I_1 = 1/2.$$

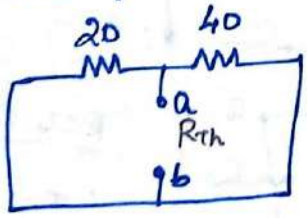
$$-50 - 40I_2 = 0$$

$$I_2 = -5/4.$$

$$I_N = \frac{1}{2} + \frac{5}{4} = \frac{7}{4}A.$$

$$= 1.75A.$$

3. To find $R_{Th} = R_N$



$$R_{Th} = R_N = 20 // 40$$

$$= \frac{20 \times 40}{20 + 40}$$

$$= 13.33\Omega$$

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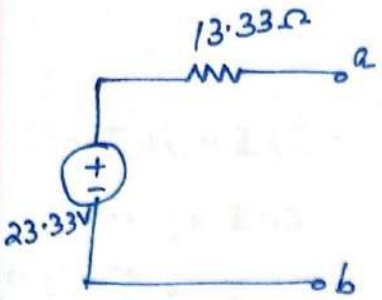
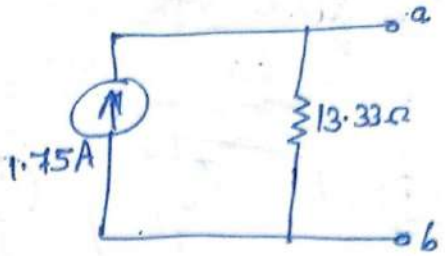
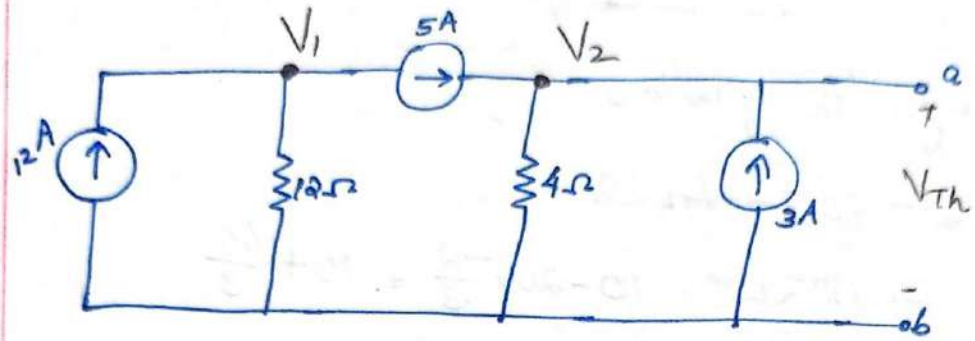
Thevenin equivalent



Norton equivalent



2.



Finding V_{Th}

$V_{Th} = V_{ab} = V_2$

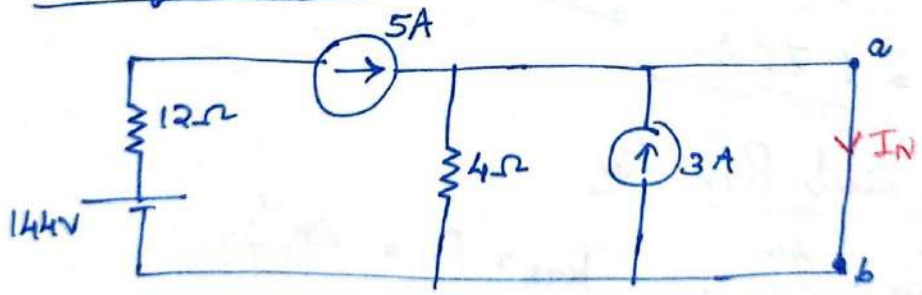
KCL to node 2;

$3+5 = \frac{V_2}{4}$; $V_2 = 8 \times 4$

$= 32V$

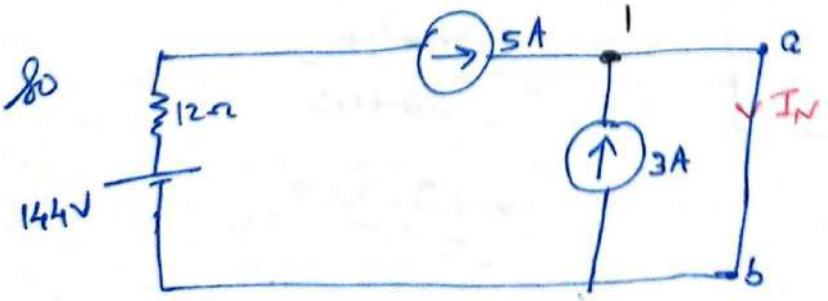
$V_{Th} = 32V$

To find I_N



There is a short ckt between terminals of 4Ω

So



KCL to Node 1

$5+3 = I_N$

$I_N = 8A$

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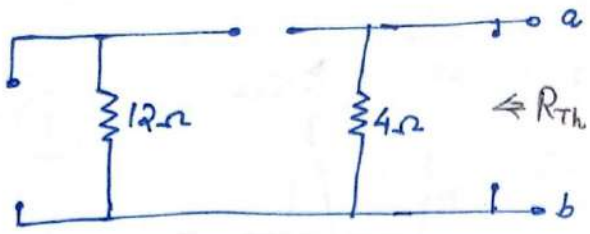
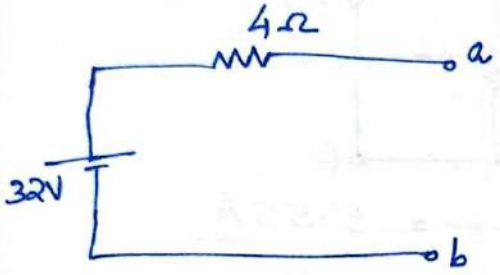
Calculating R_{Th} or R_N

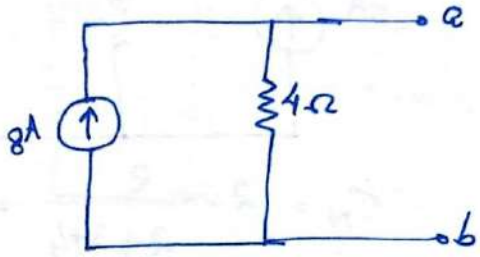


$R_{Th} = R_N = 4\Omega$

Thevenin Equivalent



Norton Equivalent



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3. Determine Norton's equivalent Circuit.

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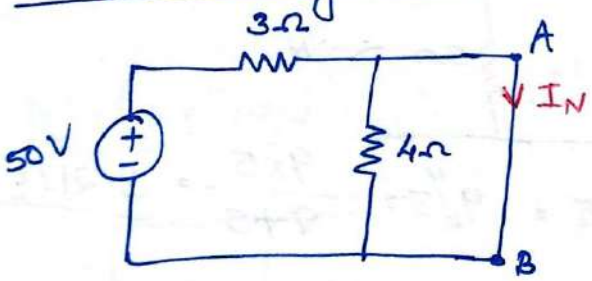
1. Calculating I_N



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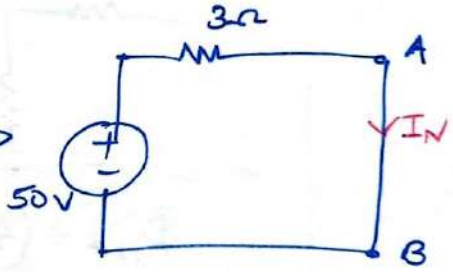
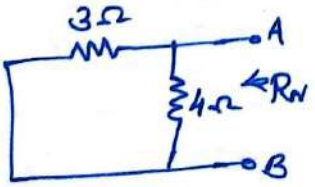


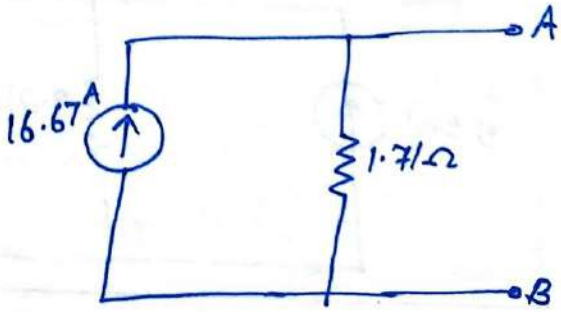
$$I_N = \frac{50}{3} A = 16.67 A$$

2. Calculating R_N



$$R_N = \frac{4 \times 3}{4+3} = 1.71\Omega$$

3. Norton's equivalent Circuit



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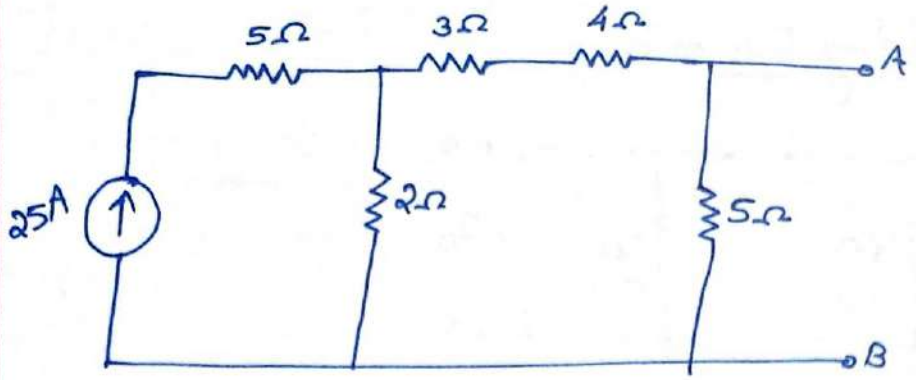
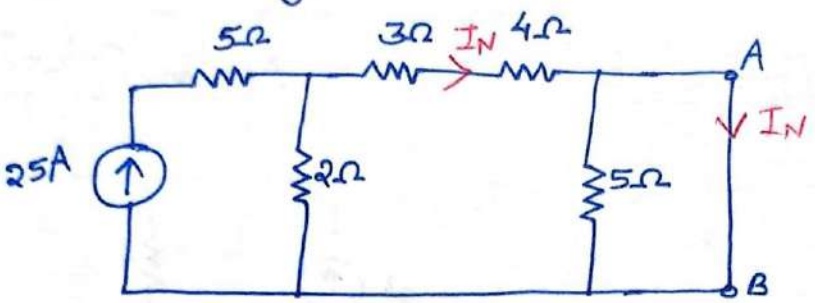
4.

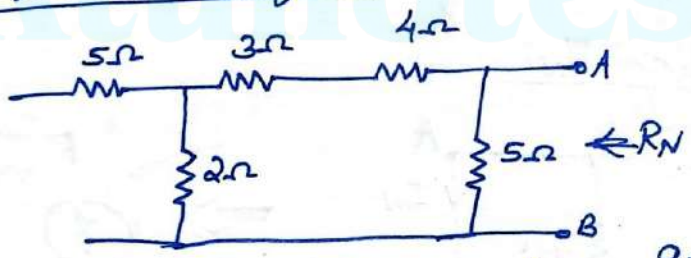


1. Calculating I_N



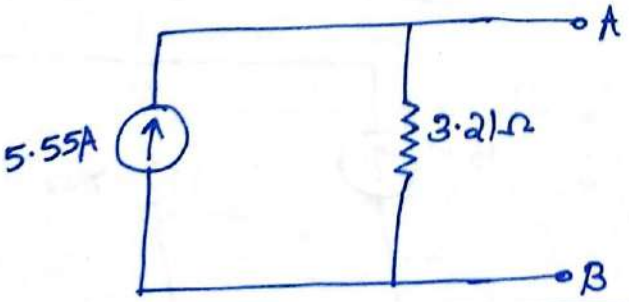
$$I_N = 25 \times \frac{2}{2+3+4} = \frac{25 \times 2}{9} = \underline{\underline{5.55A}}$$

2. Calculation of R_N



$$R_N = (4+3+2) // 5 = 9 // 5 = \frac{9 \times 5}{9+5} = \underline{\underline{3.21\Omega}}$$

3. Norton's equivalent circuit



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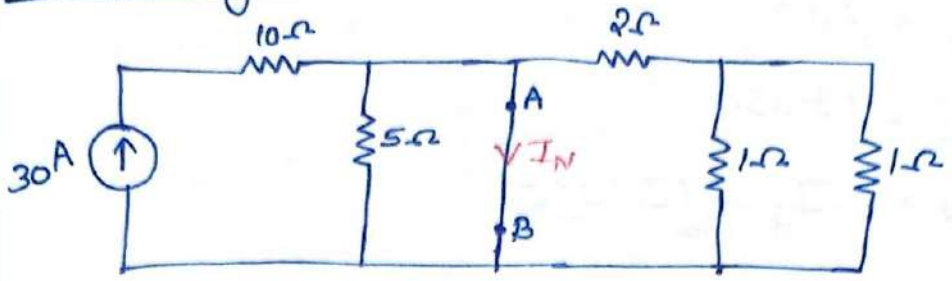
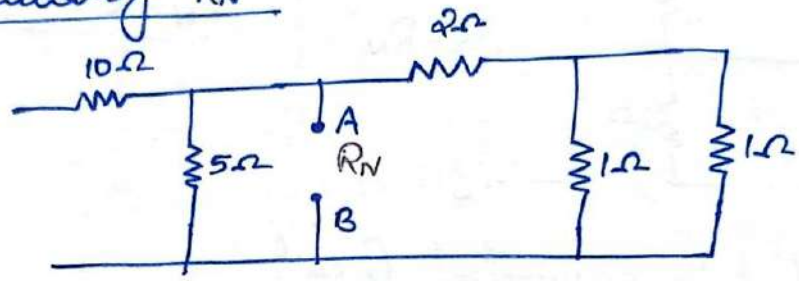
5. Calculating I_N



$I_N = 30A.$

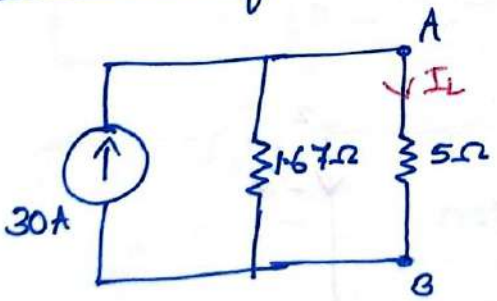
Calculating R_N



$$R_N = \left[(1 \| 1) + 2\right] \| 5$$

$$= \frac{2.5 \times 5}{7.5} = 1.67\Omega$$

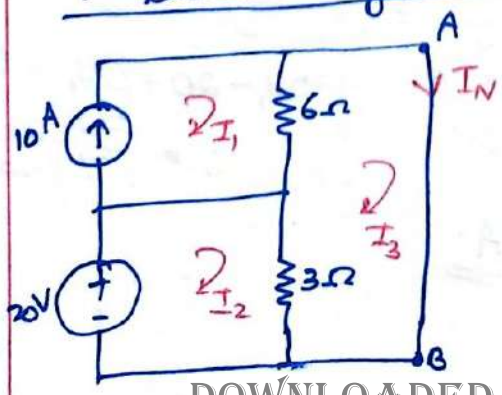
Norton's equivalent Circuit



$$I_L = \frac{30 \times 1.67}{1.67 + 5}$$

$$= 7.51A.$$

6. 1. Calculating I_N



$I_1 = 10A.$ ——— (1)

$20 - 3(I_2 - I_3) = 0$

$20 = 3I_2 - 3I_3$ ——— (2)

$-3(I_3 - I_2) - 6(I_3 - I_1) = 0$

$-3I_3 + 3I_2 - 6I_3 + 6 \times 10 = 0$

$-60 = 3I_2 - 9I_3$ ——— (3)

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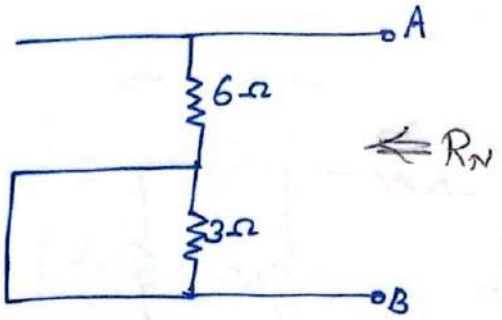
Solving (2) & (3)

$I_2 = 20A$

$I_3 = 13.33A$.

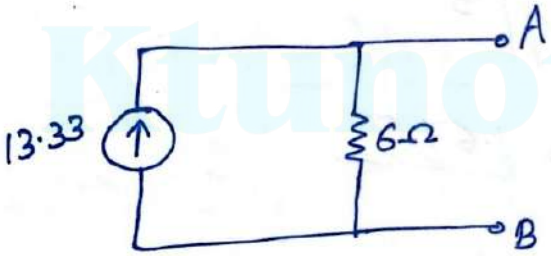
$I_N = I_3 = \underline{13.33A}$.

2. Calculation of R_N



$R_N = 6\Omega$

3. Norton's equivalent Circuit



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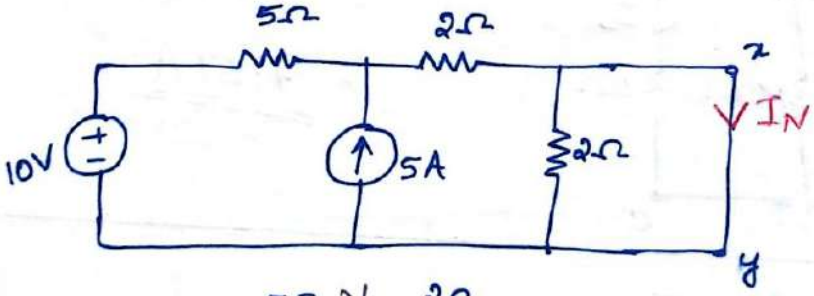
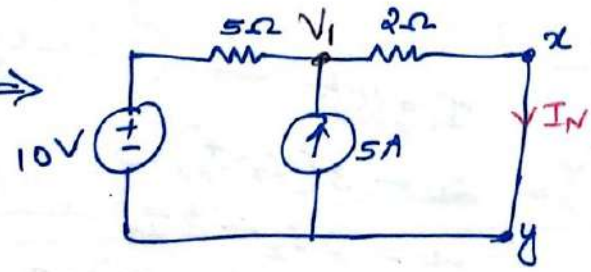
7. 1. Calculation of I_N



$\Rightarrow$



$\Rightarrow \frac{V_1 - 10}{5} + \frac{V_1}{2} = 5$

$2V_1 - 20 + 5V_1 = 50$

$V_1 = \frac{70}{7} = \underline{10V}$

$I_N = \frac{V_1}{2} = \frac{10}{2} = \underline{5A}$.

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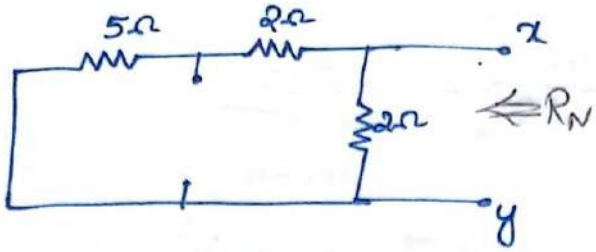
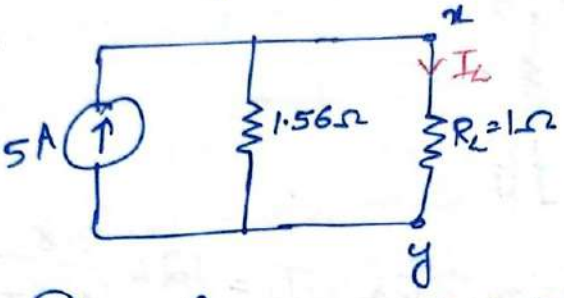
2. Calculation of R_N



$R_N = 7 // 2$

$= \frac{14}{9} = \underline{\underline{1.56\Omega}}$

3. Norton equivalent Circuit



$I_L = \frac{5 \times 1.56}{1.56 + 1} = \underline{\underline{3.04\,A}}$

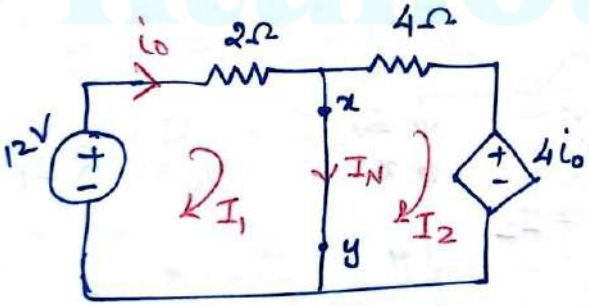
$\boxed{P = I^2 R}$

Power loss in 1Ω resistor $= (3.04)^2 \times 1$

$= \underline{\underline{9.24\,W}}$

---

8. 1. Calculating I_N



Mesh 1

$12 - 2I_1 = 0$; $\underline{\underline{I_1 = 6A}}$

Mesh 2

$-4I_2 - 4i_o = 0$

$-4I_2 - 4I_1 = 0$

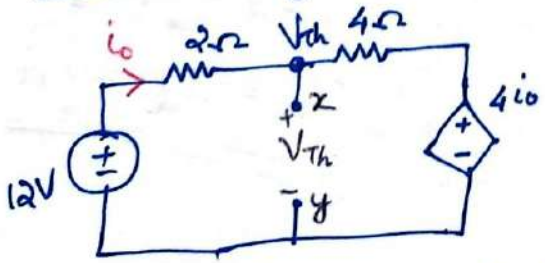
$4I_2 = -4I_1$

$I_2 = -I_1 = \underline{\underline{-6A}}$

$I_N = I_1 - I_2$

$= 6 - -6 = \underline{\underline{12A}}$

2. Calculating V_{Th}



$\cancel{\frac{V_{Th} - 12}{2} + \frac{V_{Th} - 4i_o}{4} = 0}$

$12 - 2i_o - 4i_o - 4i_o = 0$

$12 - 10i_o = 0$

$i_o = \underline{\underline{1.2A}}$

$12 - 2i_o = V_{Th} = 12 - 2 \times 1.2$

$= \underline{\underline{9.6V}}$

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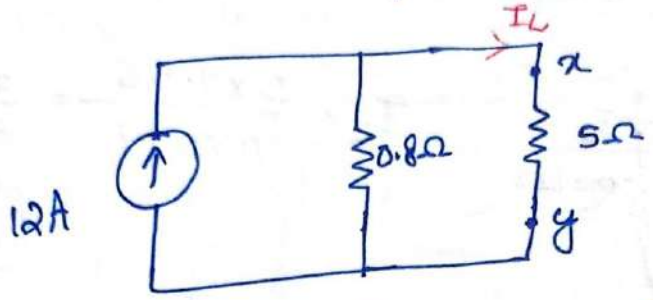
3. Calculating R_N

$$R_N = \frac{V_{Th}}{I_N} = \frac{9.6}{12} = 0.8\Omega$$

4. Norton's equivalent circuit

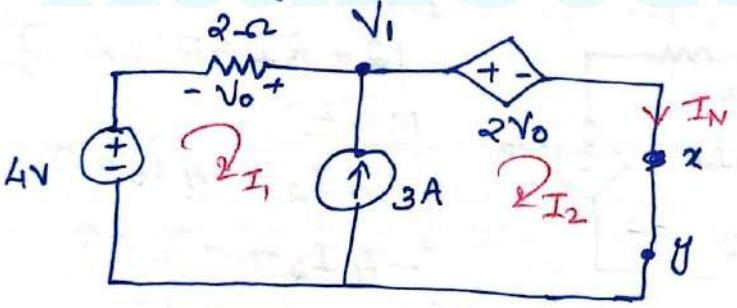


Current through 5Ω resistor; $I_L = 12 \times \frac{0.8}{(0.8+5)}$

$= 1.66A$

---

9. 1. Calculating I_N



$V_o = -2I_1$.

KVL to supermesh

$4 + V_o - 2V_o = 0$

$V_o = 4V = -2I_1$

So $I_1 = -2A$.

KCL to Node 1

$I_1 + 3 = I_2$

$-2 + 3 = I_2$

$I_2 = +1A$.

$I_N = I_2 = 1A$

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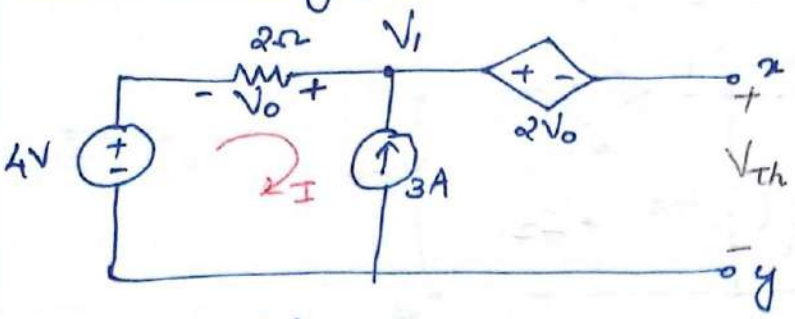
2. Calculating V_{Th}



$I = -3A$

$V_o = -2I$

$V_1 = 4 - 2I = 4 - 2 \times -3$

$V_1 = \underline{10V}$

$V_{Th} = V_1 - 2V_o$

$= V_1 - 2 \times -2I$

$= V_1 + 4I$

$= V_1 + 4 \times -3$

$= 10 - 12 = \underline{-2V}$

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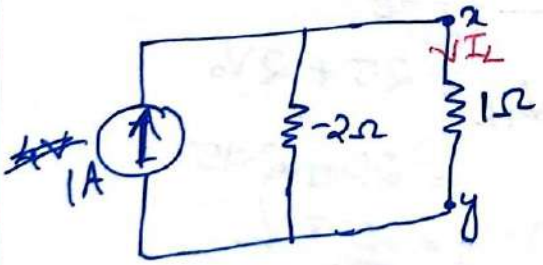
3. Calculating R_N

$R_N = \frac{V_{Th}}{I_N} = \frac{-2}{1} = \underline{-2\Omega}$

4. Norton equivalent Circuit



$I_L = 1 \times \frac{+2}{+2+1}$

$= \underline{0.67A}$

Powerloss in 1Ω resistor $= (0.67)^2 \times 1$

$= \underline{0.45W}$

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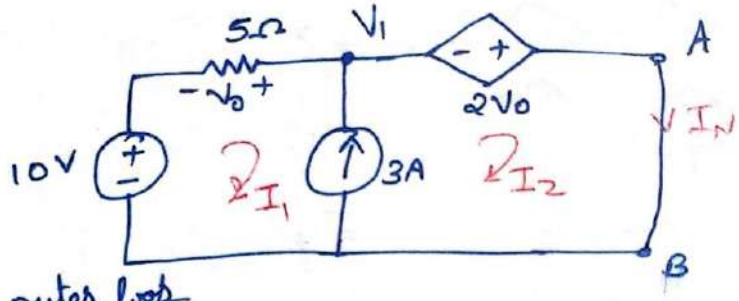
10. 1. Calculation of I_N



$V_0 = -5I_1$

outer loop

$10 + V_0 + 2V_0 = 0$

$V_0 = -10/3$

KCL to Node 1

$I_1 + 3 = I_N$

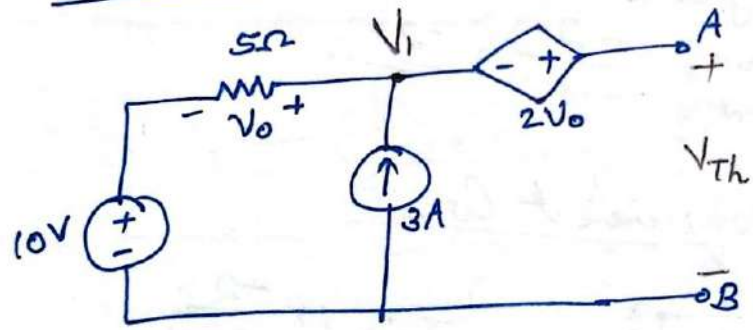
$-\frac{V_0}{5} + 3 = I_N$

$\frac{10/3}{5} + 3 = I_N$

$I_N = \frac{10}{15} + 3 = \frac{2}{3} + 3$

$= \frac{11}{3} A.$

2. Calculation of V_{Th}



$\frac{V_1 - 10}{5} = 3$

$V_1 = 15 + 10$

$= 25V$

$V_0 = 5 \times \frac{V_1 - 10}{5}$

$= 25 - 10$

$= 15V$

$V_{Th} = 25 + 2V_0$

$= 25 + 2 \times 15$

$= 55V$

3. Calculation of R_N

$R_N = \frac{V_{Th}}{I_N} = \frac{55}{11/3}$

$= \frac{55 \times 3}{11} = 15\Omega$

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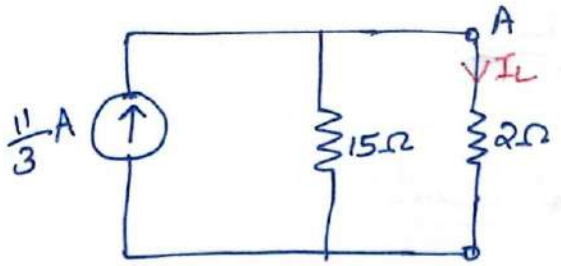
4. Norton equivalent Circuit

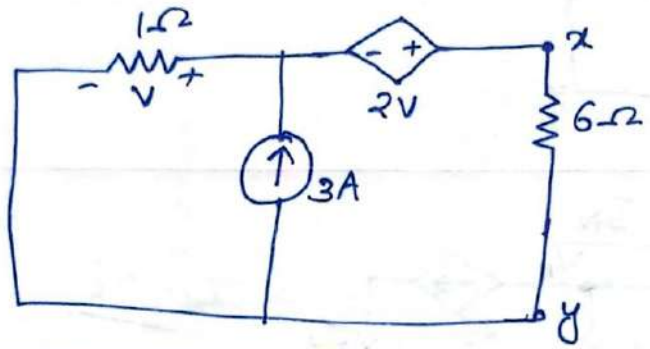


$I_L = \frac{11}{3} \times \frac{15}{17}$

$= 3.235A$

---

11. Find the current in the 6Ω resistor.



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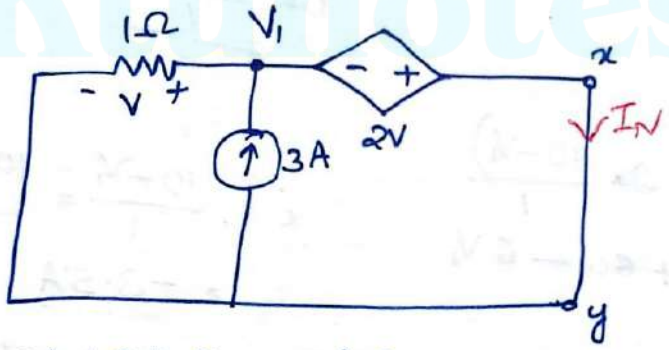
1. Calculation of I_N



$\frac{V_1}{1} + I_N = 3$

$V + 2V = 0$; $3V = 0$

$V = 0$

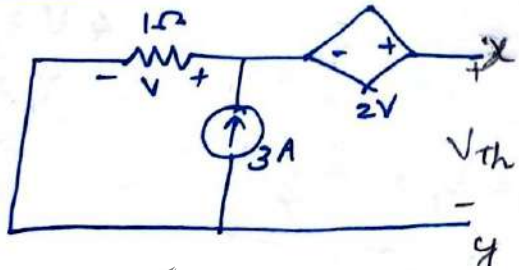
$0 + I_N = 3$

$V = 0 \Rightarrow I_{1\Omega} = 0.$

$I_N = 3A.$

$\frac{V_1}{1} = 0$; $V_1 = 0$

2. Calculation of V_{Th}



~~3×1~~

~~$V - V_{Th} = 0$~~

~~3×1 = 2×3~~

~~3×1 = V_{Th}~~

$V + 2V - V_{Th} = 0$

$3V = V_{Th}$

$V_{Th} = 3 \times 3 \times 1$

$= 9V$

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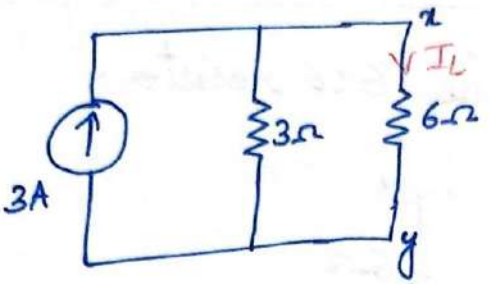
3. Calculation of R_N

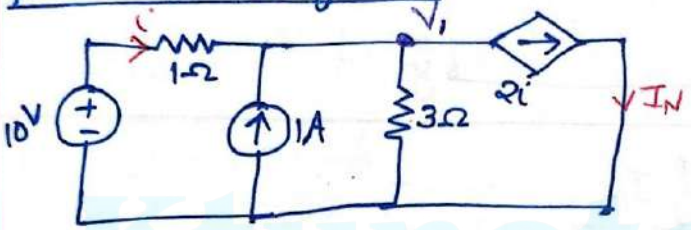
$$R_N = \frac{V_{Th}}{I_N} = \frac{9}{3} = 3\Omega$$

4. Norton's equivalent Circuit



$$I_L = 3 \times \frac{3}{9} = 1A.$$

12. Calculation of I_N



$$I_N = 2i.$$

KCL to Node1

$$i = \frac{10 - V_1}{1}$$

$$i + 1 = \frac{V_1}{3} + 2i$$

$$\frac{10 - V_1}{1} + 1 = \frac{V_1}{3} + 2 \times \frac{(10 - V_1)}{1}$$

$$30 - 3V_1 + 3 = V_1 + 60 - 6V_1$$

$$2V_1 = 27.$$

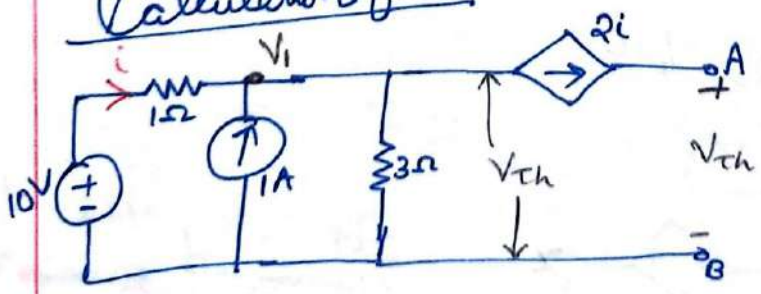
$$V_1 = \frac{27}{2} V.$$

$$i = \frac{10 - V_1}{1} = \frac{10 - \frac{27}{2}}{1}$$

$$= -3.5A.$$

$$I_N = 2i = -7A$$

Calculation of V_{Th}



$$\frac{10 - V_1}{1} + 1 = \frac{V_1}{3}$$

$$30 - 3V_1 + 3 = V_1$$

$$4V_1 = 33$$

$$V_1 = 33/4$$

From the Circuit; $V_{Th} = V_1 = 33/4$ V

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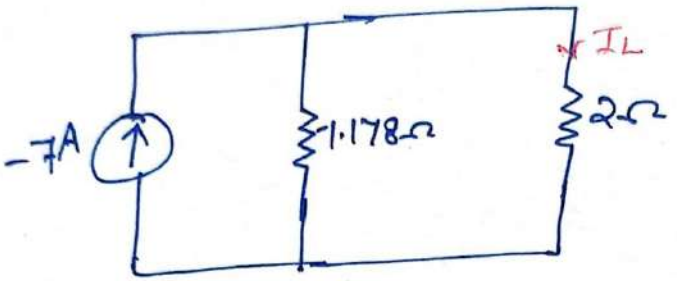
Calculation of R_N

$$R_N = \frac{V_{Th}}{I_N}$$

$$= \frac{33/4}{-7} = -1.178\,\Omega$$

Norton's equivalent Circuit



$$I_L = -7 \times \frac{-1.178}{-1.178+2} = \frac{7 \times 1.178}{0.822}$$

$$= 10.03\,A.$$

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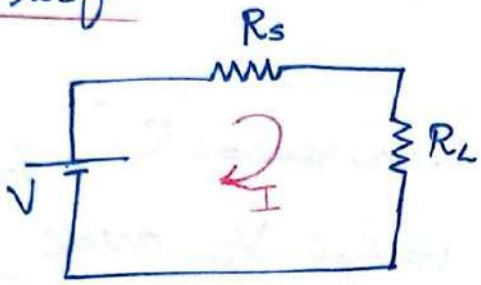
MAXIMUM POWER TRANSFER THEOREM

It states that the maximum power is delivered from a source to a load when the load resistance is equal to the source resistance.

Proof



$$I = \frac{V}{R_s + R_L}$$

Power delivered to the Load R_L; $P = I^2 R_L$

$$= \frac{V^2}{(R_s+R_L)^2} R_L$$

To determine the value of R_L for maximum power to be transferred to the load

$$\frac{dP}{dR_L} = 0$$

$$\frac{d}{dR_L}\left\{\frac{V^2}{(R_s+R_L)^2} \cdot R_L\right\} = 0$$

$$V^2 \left\{\frac{(R_s+R_L)^2 \times 1 - R_L \times 2(R_s+R_L)}{(R_s+R_L)^4}\right\} = 0$$

ie; $R_s^2 + R_L^2 + 2R_sR_L - 2R_sR_L - 2R_L^2 = 0$

$\Rightarrow R_s^2 - R_L^2 = 0$; $\boxed{R_s = R_L}$

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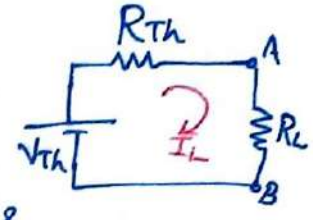
$R_S = R_L$.

Hence, the maximum power will be transferred to the load when load resistance is equal to the source resistance.

Steps to be followed in Maximum Power Transfer Theorem



1. Remove the variable load resistor R_L
2. Find the open Circuit voltage V_{Th} across points A & B
3. Find the resistance R_{Th} as seen from points A & B
4. Find the resistance R_L for maximum power transfer.

$$R_L = R_{Th}$$

5. Find the maximum power

$$I_L = \frac{V_{Th}}{R_{Th}+R_L} = \frac{V_{Th}}{2R_{Th}}$$

$$P_{max} = I_L^2 R_L = \frac{V_{Th}^2}{4R_{Th}^2} \times R_{Th}$$

$$\boxed{P_{max} = \frac{V_{Th}^2}{4R_{Th}}}$$

Efficiency of ckt $= \frac{o/p \text{ Power}}{i/p \text{ Power}} = \frac{I^2(R_L)}{I^2(R_L+R_{Th})} = \frac{I^2 R_{Th}}{I^2 2R_{Th}}$

$= \frac{1}{2} = 50\%$

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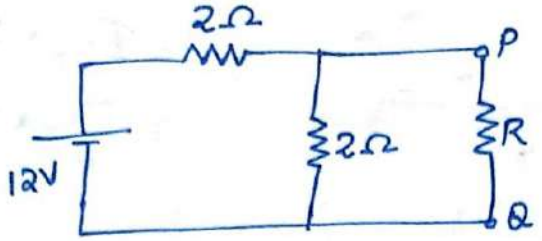
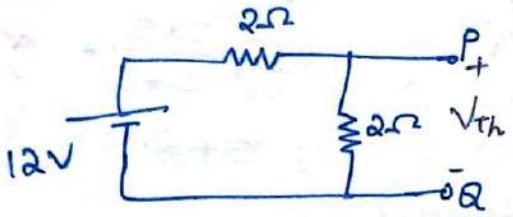
1. For what value of R in the given circuit will it receive the maximum power? Find the maximum power.



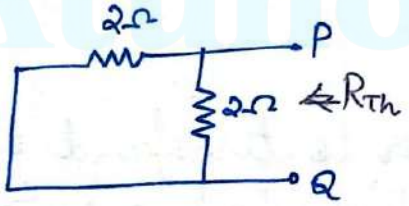
1. Calculation of V_{Th}



$$V_{Th} = \frac{12}{(2+2)} \times 2$$

$$= 3 \times 2 = \underline{\underline{6V}}$$

2. Calculation of R_{Th}



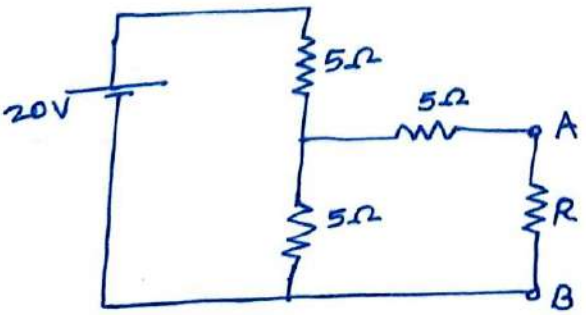
$$R_{Th} = 2 \| 2$$

$$= \frac{2 \times 2}{2+2} = \underline{\underline{1\Omega}}$$

3. Maximum Power ; P_{max}

$$P_{max} = \frac{V_{Th}^2}{4 R_{Th}} = \frac{6^2}{4 \times 1} = \underline{\underline{9W}}$$

2. Find the value of R for maximum power to R. Determine the maximum power to R for given n/w.



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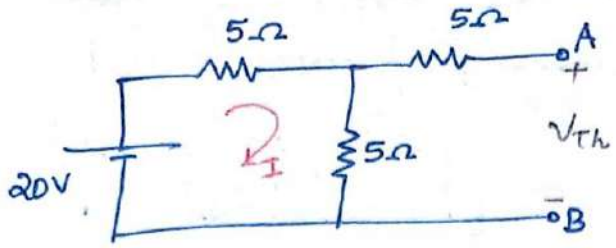
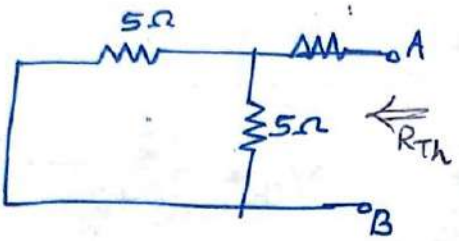
1. Calculation of V_{Th}



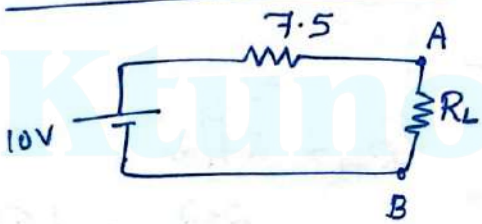
$$V_{Th} = V_{5\Omega}$$
$$= I_{5\Omega} \times 5$$
$$= \frac{20}{10} \times 5 = \underline{\underline{10V}}$$

2. Calculation of R_{Th}



$$R_{Th} = 5 \| 5 + 5$$
$$= 5 + \frac{5 \times 5}{5+5}$$
$$= 5 + 2.5 = \underline{\underline{7.5\Omega}}$$

3. Reduced Circuit (Thevenin)



The maximum power is transferred through the load resistor only when load resistance equals Thevenin's resistance. $\underline{\underline{R_L = R_{Th} = 7.5\Omega}}$

4. Calculation of P_{max}

$$P_{max} = \frac{V_{Th}^2}{4R_{Th}} = \frac{(10)^2}{4 \times 7.5} = \underline{\underline{3.33\,W}}$$

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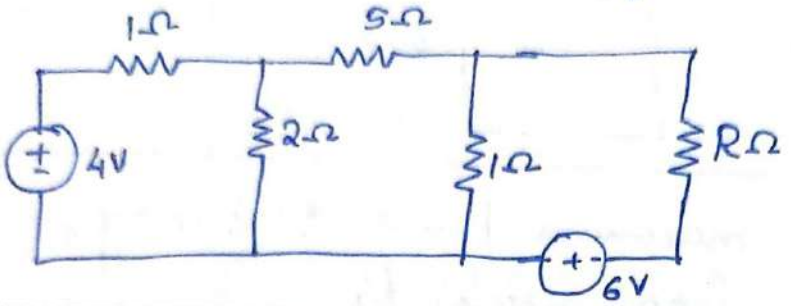
Tutorial Questions from Maximum Power Transfer Theorem

1. Find the value of R in the circuit of figure, such that Maximum Power Transfer takes place. What is the amount of this power.

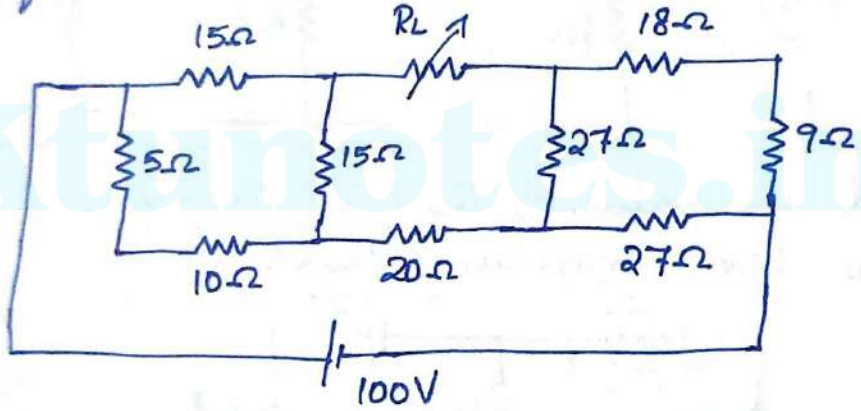


$V_{Th} = 6.4$ V

$R_{Th} = R_L = 0.85\Omega$

$P_{max} = 12$ W

2. Find the value of R_L for maximum power Transfer and calculate the maximum Power.

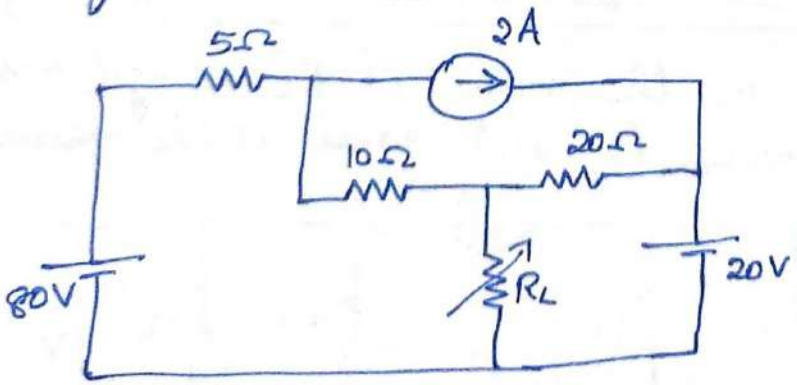


$V_{Th} = 70.88$ V

$R_L = R_{Th} = 23.92\Omega$

$P_{max} = 52.51$ W

3. Find the value of R_L for maximum Power Transfer. Calculate the maximum Power



$V_{Th} = 48.6$ V

$R_L = R_{Th} = 8.57\Omega$

$P_{max} = 68.9$ W

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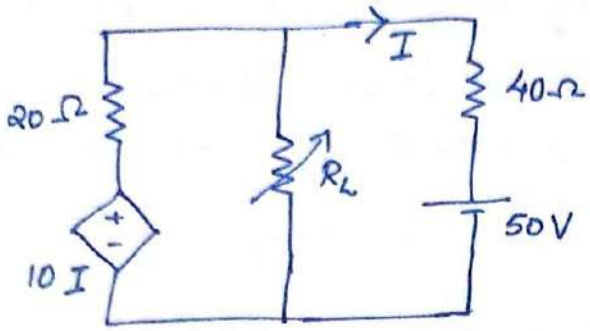
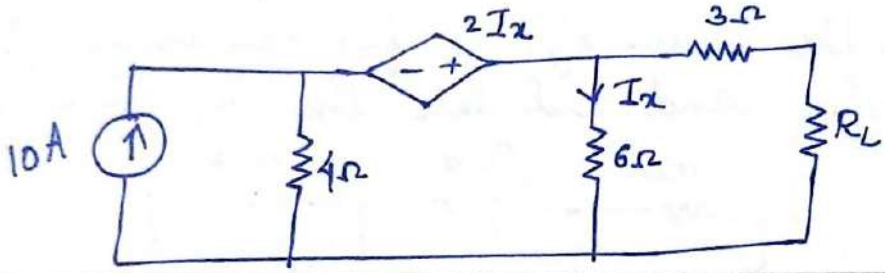
4. Find the value of R_L for maximum power transfer. Also calculate maximum power.



$V_{Th} = 10V$
$R_L = R_{Th} = 16\Omega$
$P_{max} = 1.56\ W$

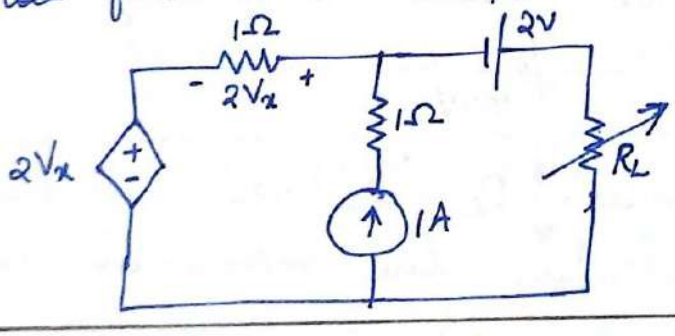
5. Calculate maximum Power that may be dissipated in the load resistor R_L

$V_{Th} = 30V$
$R_L = R_{Th} = 6\Omega$
$P_{max} = 37.5W$



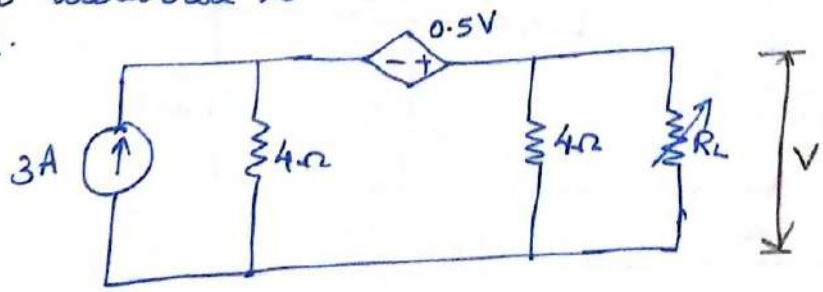
6. Find the value of R_L for maximum Power Transfer. Also find maximum Power

$V_{Th} = 5V$
$R_L = R_{Th} = 3\Omega$
$P_{max} = 2.08W$



7. What will be the value of R_L to get maximum power delivered to it? what is the value of this power.

$V_{Th} = 8V$
$R_{Th} = R_L = 2.67\Omega$
$P_{max} = 6W$



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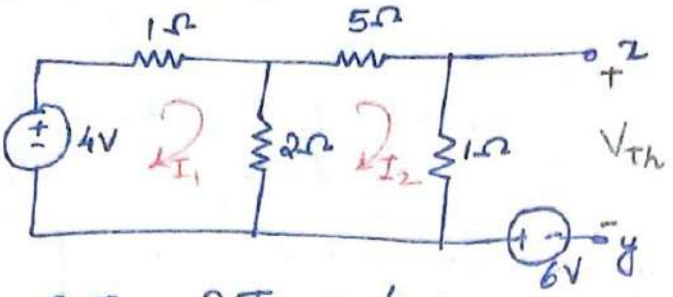
Answers to Tutorial Questions from Maximum Power Transfer theorem

1. **Calculation of V_{Th}**



$$3I_1 - 2I_2 = 4 \quad \text{——— (1)}$$

$$-2I_1 + 8I_2 = 0 \quad \text{——— (2)}$$

Solving (1) & (2)

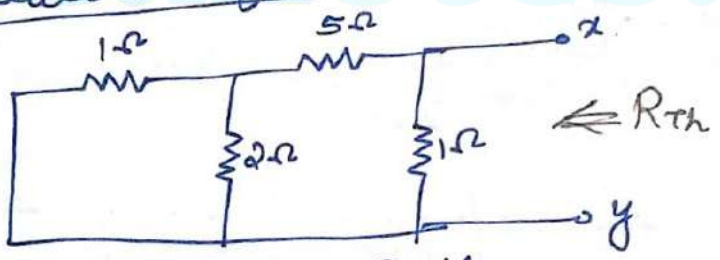
$$I_1 = 1.6A, \qquad I_2 = 0.4A$$

V_{Th} equation

$$6 + 1I_2 - V_{Th} = 0$$

$$V_{Th} = 6 + 1 \times 0.4 = 6.4V$$

Calculation of R_{Th}



$$R_{Th} = (2 \| 1 + 5) \| 1$$

$$= \left[\frac{2 \times 1}{2+1} + 5\right] \| 1 = \left(\frac{2}{3} + 5\right) \| 1$$

$$= \frac{17}{3} \| 1 = \frac{\frac{17}{3} \times 1}{\frac{17}{3} + 1} = \frac{17}{20} = 0.85\Omega$$

As per Maximum Power Transfer Theorem

$$R = R_{Th} = 0.85\Omega$$

$$P_{max} = \frac{V_{Th}^2}{4R_{Th}} = \frac{(6.4)^2}{4 \times 0.85} = 12W$$

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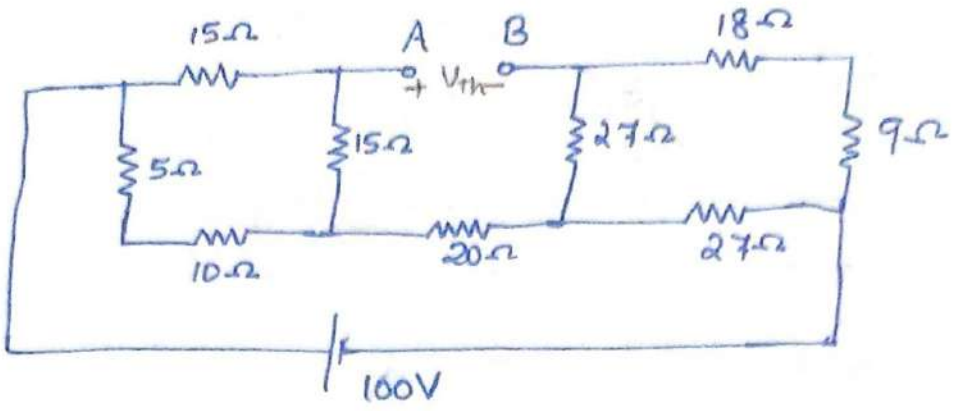
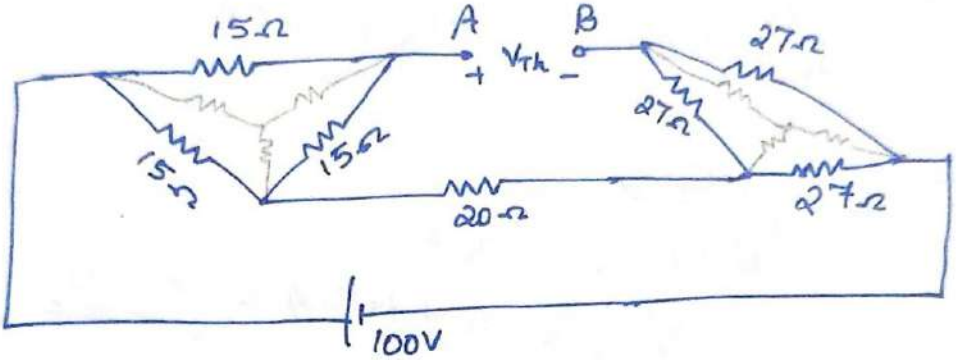
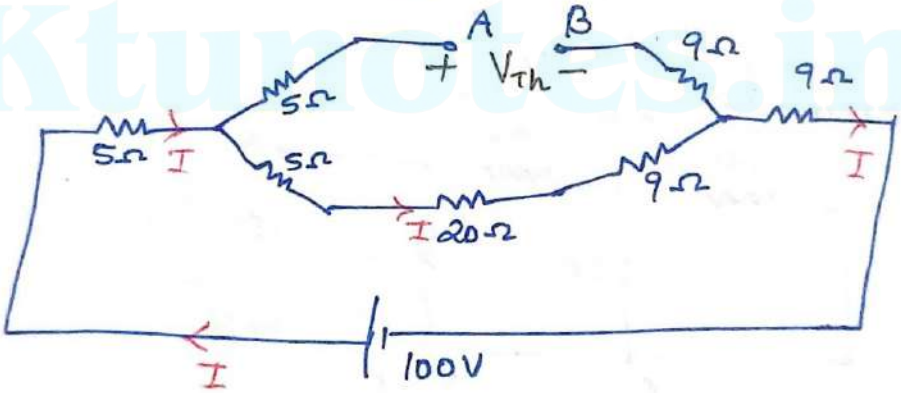
2. Calculation of V_{Th}







$$I = \frac{100}{5+5+20+9+9} = \underline{\underline{2.08\,A}}$$

V_{Th} equation

$$100 - 5I - V_{Th} - V_{Th} - 0 - 9I = 0$$

$$V_{Th} = 100 - 14I = 100 - 14 \times 2.08$$

$$= \underline{\underline{70.88\,V}}$$

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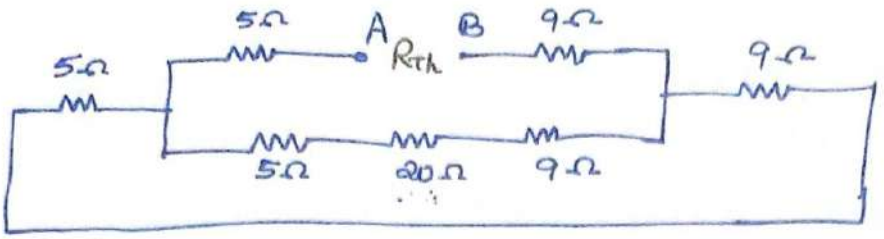
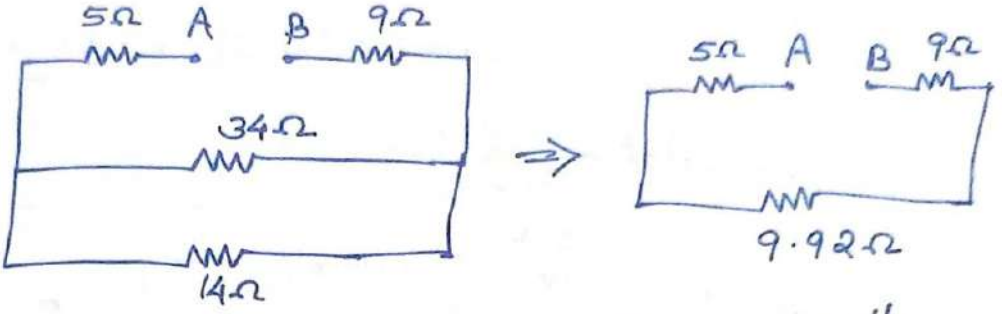
Calculation of R_{Th}

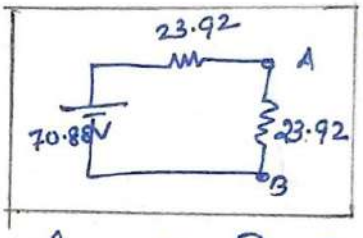




$34 \| 14 = \underline{9.92}$

$\Rightarrow R_{Th} = 5 + 9.92 + 9$

$= \underline{\underline{23.92\Omega}}$

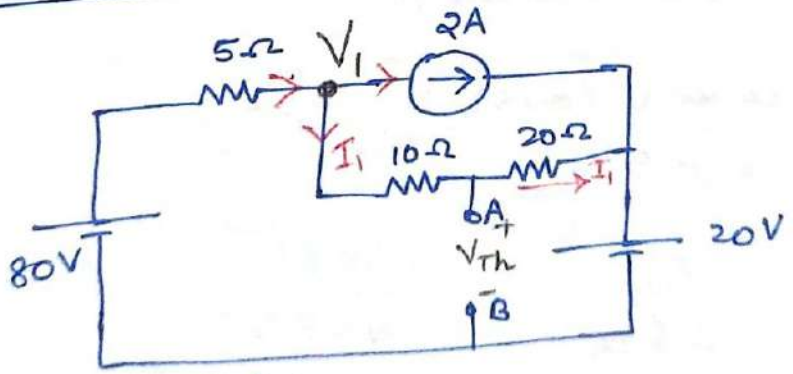


For Maximum Power Transfer, $R_L = R_{Th} = \underline{\underline{23.92\Omega}}$

$$P_{max} = \frac{V_{Th}^2}{4R_{Th}} = \frac{(70.88)^2}{4 \times 23.92} = \underline{\underline{52.51W}}$$

---

3. ## Calculation of V_{Th}.



KCL to Node 1

$$\frac{80 - V_1}{5} = 2 + \frac{V_1 - 20}{30}$$

$480 - 6V_1 = 60 + V_1 - 20$

$7V_1 = 440;$

~~$V_1 = 74.2857V$~~

$V_1 = \underline{\underline{62.8571V}}$

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$$I_1 = \frac{V_1 - 20}{30} = \frac{62.8571 - 20}{30} = \underline{\underline{1.43A}}$$

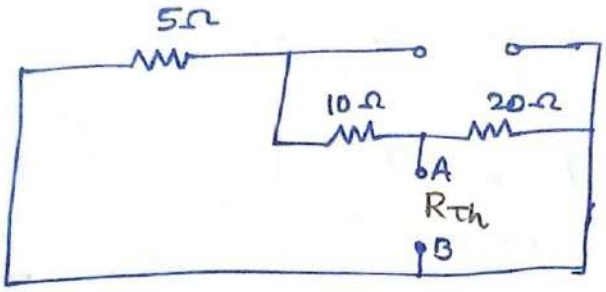
V_{Th} equation

$$V_{Th} - 20I_1 - 20 = 0$$

$$V_{Th} = 20I_1 + 20$$

$$= 20 \times 1.43 + 20 = \underline{\underline{48.6V}}$$

Calculation of R_{Th}



$$R_{Th} = (10+5)//20$$

$$= 15//20$$

$$= \frac{15 \times 20}{15+20}$$

$$= \underline{\underline{8.57\Omega}}$$

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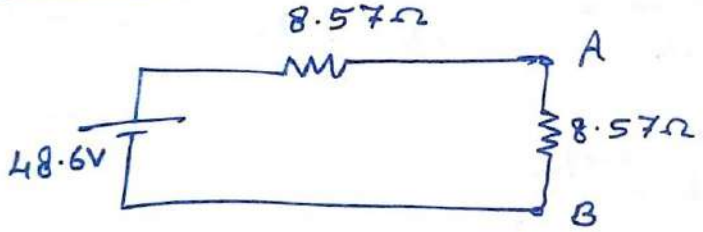
Reduced Circuit (Thevenin's)



For Maximum Power Transfer

$$R_L = \underline{\underline{8.57\Omega}}$$

$$P_{max} = \frac{V_{Th}^2}{4R_{Th}} = \frac{(48.6)^2}{4 \times 8.57} = \underline{\underline{68.9W}}$$

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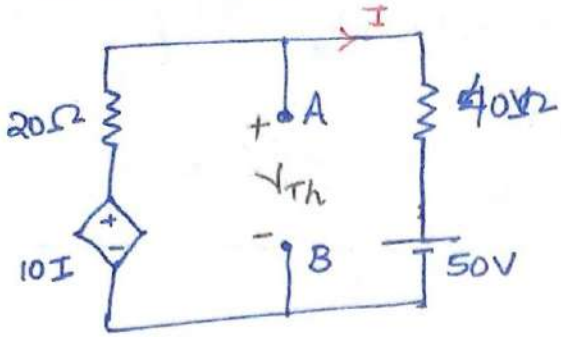
4. Calculation of V_{Th}



Applying KVL to the mesh

$$10I - 20I - 40I - 50 = 0$$

$$-50I = 50$$

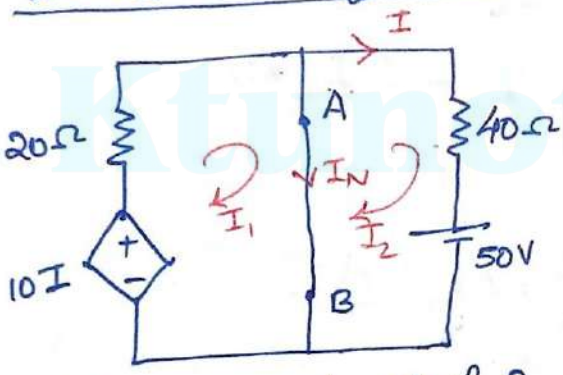
$$I = -1$$

Writing V_{Th} equation

$$10I - 20I - V_{Th} = 0$$

$$V_{Th} = -10I = 10\ V$$

Calculation of I_N



Applying KVL to mesh 1

$$10I - 20I_1 = 0 \quad [I = I_2]$$

$$10I_2 - 20I_1 = 0 \quad \text{—(1)}$$

Applying KVL to Mesh 2

$$-40I_2 - 50 = 0$$

$$I_2 = -1.25A$$

From eqn (1); $20I_1 = 10I_2$; $I_1 = \frac{10 \times 1.25}{20} = -0.625A$

$$I_N = I_1 - I_2$$

$$= -0.625 - -1.25 = 0.625A$$

Calculation of R_{Th}

$$R_{Th} = \frac{V_{Th}}{I_N} = \frac{10}{0.625} = 16\Omega$$

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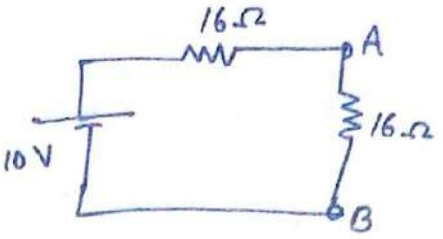
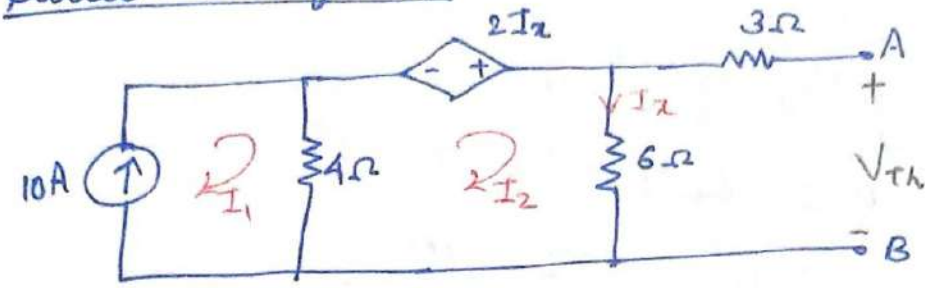
For maximum Power,

$R_L = R_{Th} = \underline{16\Omega}$



$$P_{max} = \frac{V_{Th}^2}{4R_{Th}} = \frac{(10)^2}{4 \times 16} = \underline{1.56\,W}$$

5. <u>Calculation of V_{Th}</u>



$\underline{I_x = I_2}$ $\quad I_1 = \underline{10\,A}$

$$-4(I_2 - I_1) + 2I_x - 6I_2 = 0$$

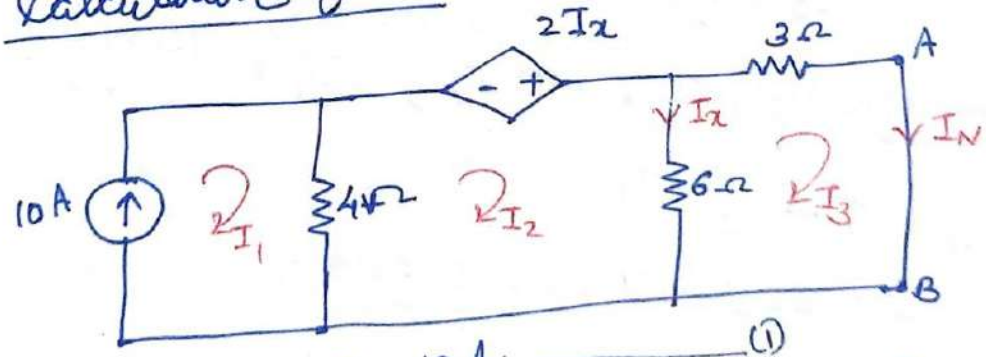
$$-4I_2 + 4I_1 + 2I_2 - 6I_2 = 0$$

$$4I_1 - 8I_2 = 0$$

$$I_2 = \frac{4I_1}{8} = \frac{4}{8} \times 10 = \underline{5\,A}$$

$$V_{Th} = 6I_x = 6 \times 5 = \underline{30V}$$

<u>Calculation of I_N</u>



$I_x = I_2 - I_3$

Mesh 1 $\longrightarrow$ $I_1 = 10\,A$ ——— (1)

Mesh 2 $\longrightarrow$ $-4(I_2 - I_1) + 2I_x - 6(I_2 - I_3) = 0$

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$$-4I_2 + 4I_1 + 2(I_2 - I_3) - 6(I_2 - I_3) = 0$$

$$4I_1 - 8I_2 + 4I_3 = 0 \quad \text{———} \quad (2)$$

Mesh 3 $\rightarrow$ $-6(I_3 - I_2) - 3I_3 = 0$

$$6I_2 - 9I_3 = 0 \quad \text{———} \quad (3)$$

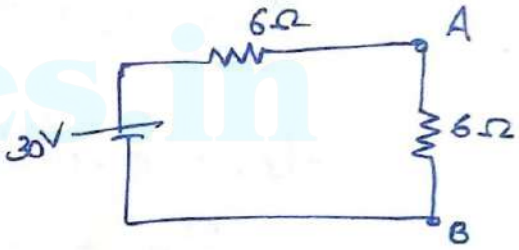
Solving (1), (2) and (3)

$\Longrightarrow$ $I_1 = 10A$, $I_2 = 7.5A$ $I_3 = 5A$.

$I_N = I_3 = 5A$

Calculation of R_{Th}

$$R_{Th} = \frac{V_{Th}}{I_N} = \frac{30}{5} = 6\Omega$$



For maximum Power Transfer

$R_L = 6\Omega$

$$P_{max} = \frac{V_{Th}^2}{4R_{Th}} = \frac{30^2}{4 \times 6} = 37.5\,W$$

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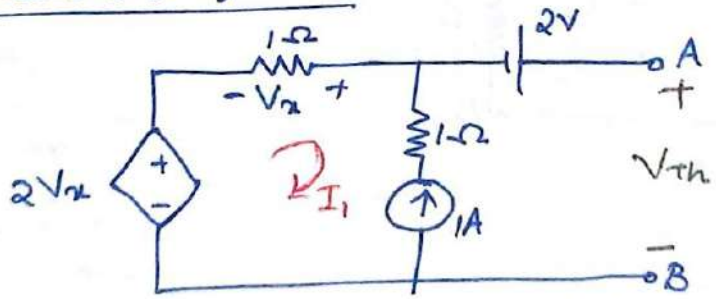
6. Calculation of V_{Th}



$I_1 = -1$ A

$V_x = -I_1 \times 1 = 1V$

V_{Th} eqn

$2V_x + V_x + 2 - V_{Th} = 0$

$V_{Th} = 3V_x + 2 = 3 \times 1 + 2$

$= 5V$

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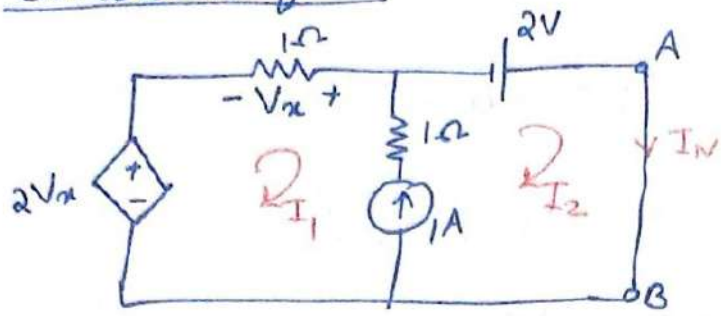
Calculation of I_N



$V_x = -1 I_1$

$V_x = -I_1$ —— (1)

$I_2 - I_1 = 1$ —— (2)

Mesh 1 ⟶ $2V_x - 1I_1 + 2 = 0$

$2 \times -I_1 - I_1 + 2 = 0$

$-3I_1 = -2$

$I_1 = 2/3$

(2) ⟹

$I_2 = 1 + I_1 = 1 + \frac{2}{3} = \frac{5}{3}$

$I_N = I_2 = \frac{5}{3} = 1.67\,A$

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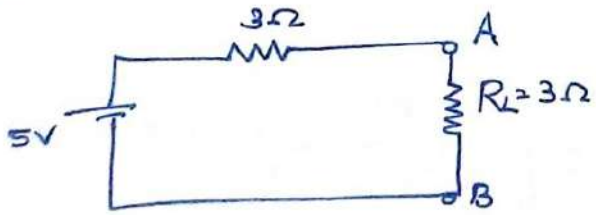
Calculation of R_{Th}

$$R_{Th} = \frac{V_{Th}}{I_N} = \frac{5}{1.67} = 3\Omega$$

Calculation of P_{max}



$$P_{max} = \frac{V_{Th}^2}{4R_{Th}}$$

$$= \frac{5^2}{4 \times 3}$$

$$= 2.08\,W$$

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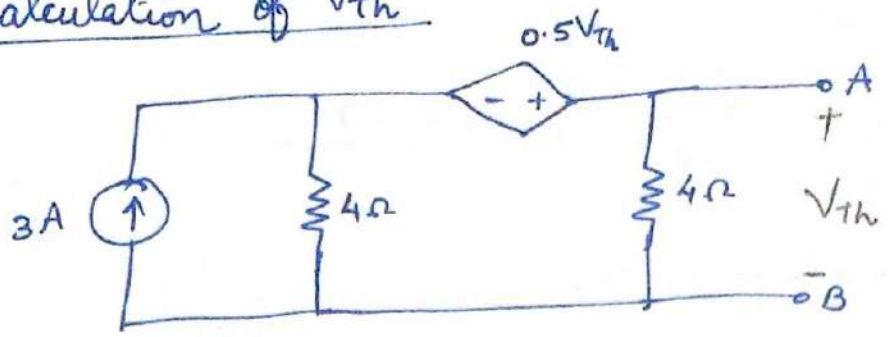
7. Calculation of V_{Th}



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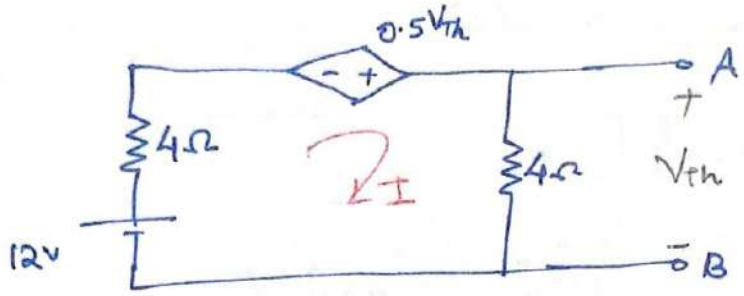


$V_{Th} = 4I$

$12 - 4I + 0.5V_{Th} - 4I = 0$

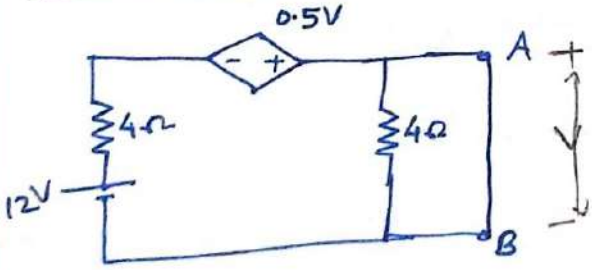
$12 - 4I + 0.5 \times 4I - 4I = 0$

$12 - 6I = 0$

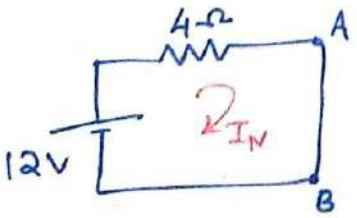
$I = 2A$

$V_{Th} = 4I = 4 \times 2 = 8V$

Calculation of I_N



A & B are shorted.
So 4Ω resistor gets shorted.
$V = 0$
When $V=0$, dependent source (or $V=0$) vanishes.



$I_N = \frac{12}{4} = 3A$

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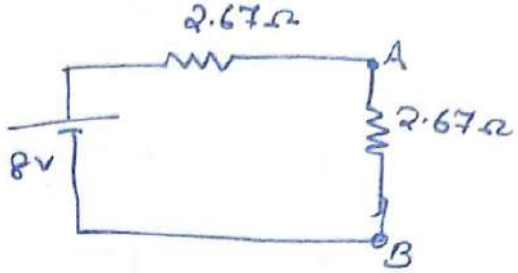
Calculation of R_{Th}

$$R_{Th} = \frac{V_{Th}}{I_N} = \frac{8}{3} = 2.67\,\Omega$$



Calculation of R_L

$$R_L = R_{Th} = 2.67\,\Omega$$

Calculation of P_{max}

$$P_{max} = \frac{V_{Th}^2}{4R_{Th}} = \frac{8^2}{4 \times 2.67}$$

$$= 6\ W$$

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RECIPROCITY THEOREM

It states that, "in a linear, bilateral, active, single source network, the ratio of excitation to response remains same when the positions of excitation and response are interchanged".

If a single voltage source V_a in the branch 'a' produces a current I_b in the branch 'b' then if the voltage source removed and inserted in the branch 'b', it will produce a current I_b in branch 'a'.

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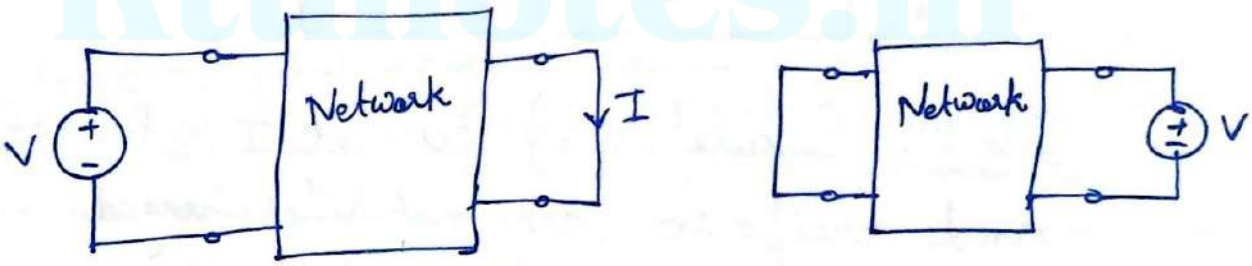

Limitation of the Theorem

→ Applicable only to a single-source network.

→ Not applicable in the network which has a dependent source.

→ Applicable only in linear and bilateral networks.

Donot change the positions of any passive elements (R, L, C). Only excitation and response are interchanged.

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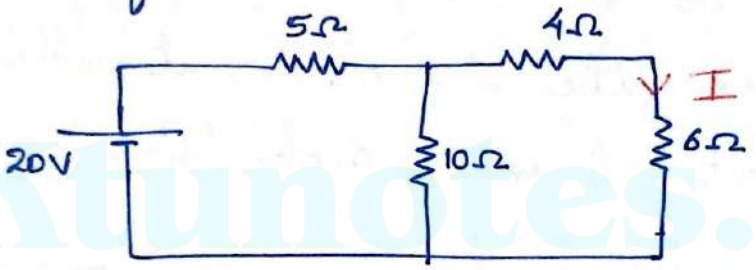
Steps to be followed in Reciprocity Theorem

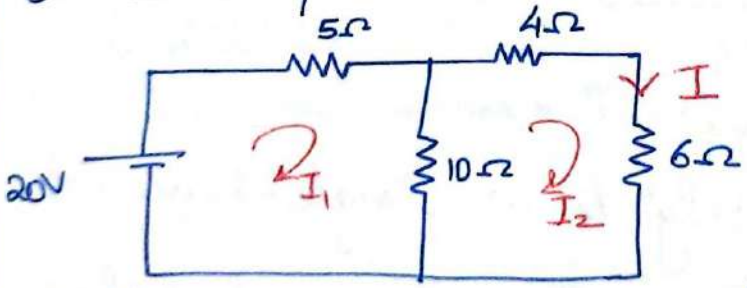
1. Identify the branches between which reciprocity is to be established.
2. Find the current in the branch when excitation and response are not interchanged.
3. Find the current in the branch when excitation and response are interchanged.

---

1. Calculate current I and verify the Reciprocity Theorem for the network.



Case 1: Calculation of current I when excitation and response are not interchanged



Mesh 1 →

$$20 - 5I_1 - 10(I_1 - I_2) = 0$$

$$15I_1 - 10I_2 = 20 \rightarrow (1)$$

Mesh 2 →

$$-10(I_2 - I_1) - 4I_2 - 6I_2 = 0$$

$$-10I_1 + 20I_2 = 0 \rightarrow (2)$$

Solving (1) & (2)

$\Rightarrow I_1 = 2A$

$I_2 = 1A$

$I = I_2 = 1A$

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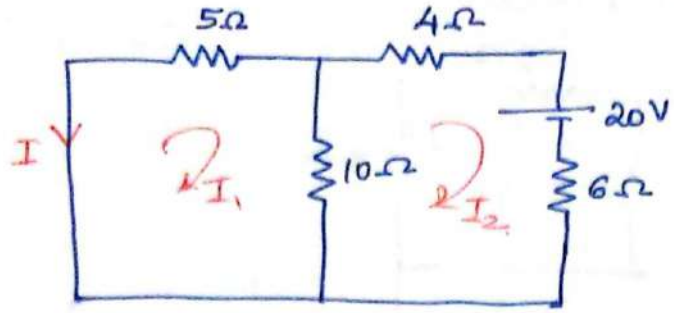
Case 2 : Calculation of current I when excitation and response are interchanged



Mesh 1→

$-5I_1 - 10(I_1 - I_2) = 0$

$15I_1 - 10I_2 = 0 \longrightarrow (1)$

Mesh 2→

$-10(I_2 - I_1) - 4I_2 - 20 - 6I_2 = 0$

$-10I_1 + 20I_2 = -20 \rightarrow (2)$

Solving (1) & (2)

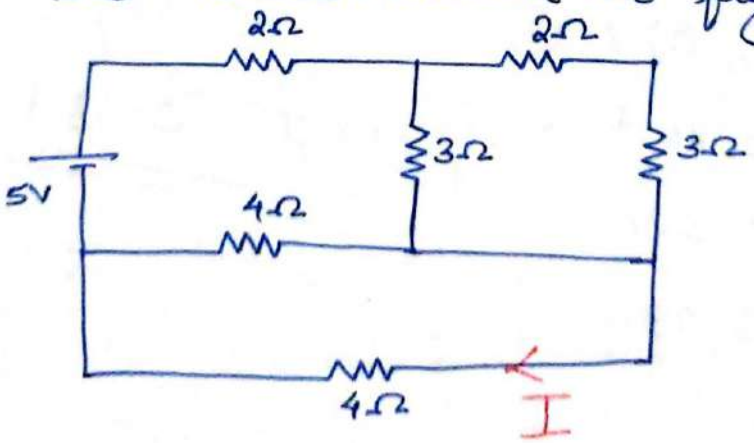
$I_1 = -1A$

$I_2 = -1.5A$

$I = -I_1 = 1A$

Since the current I remains the same in both the cases, Reciprocity theorem is verified

2. Find the current I and verify reciprocity theorem for the network shown in figure.



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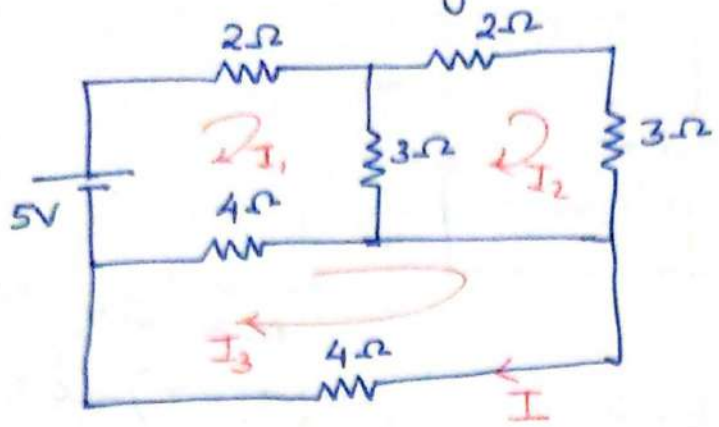
Case 1:

Calculation of I when excitation and response are not interchanged.



Mesh 1

$$5 - 2I_1 - 3(I_1 - I_2) - 4(I_1 - I_3) = 0$$

$$9I_1 - 3I_2 - 4I_3 = 5 \quad \text{——— (1)}$$

Mesh 2

$$-3(I_2 - I_1) - 2I_2 - 3I_2 = 0$$

$$-3I_1 + 8I_2 = 0 \quad \text{——— (2)}$$

Mesh 3

$$-4I_3 - 4(I_3 - I_1) = 0$$

$$-4I_1 + 8I_3 = 0 \quad \text{——— (3)}$$

Solving (1), (2) & (3)

$I_1 = 0.85A$

$I_2 = 0.32A$

$I_3 = 0.43A$

$I = I_3 = 0.43A$

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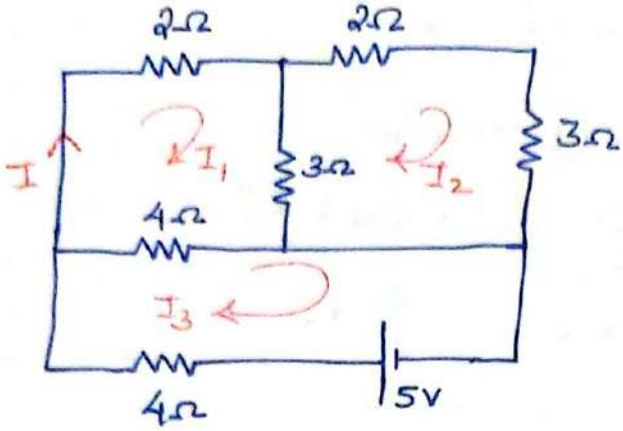
Case 2:

Calculation of I when excitation and response are interchanged.



Mesh 1 → $-2I_1 - 3(I_1 - I_2) - 4(I_1 - I_3) = 0$

$$9I_1 - 3I_2 - 4I_3 = 0 \quad \text{——— (1)}$$

Mesh 2 → $-3(I_2 - I_1) - 2I_2 - 3I_2 = 0$

$$-3I_1 + 8I_2 = 0 \quad \text{——— (2)}$$

Mesh 3 → $-4I_3 - 4(I_3 - I_1) + 5 = 0$

$$-4I_1 + 8I_3 = 5 \quad \text{——— (3)}$$

Solving (1), (2) & (3)

$I_1 = 0.43A$

$I_2 = 0.16A$

$I_3 = 0.84A$

$I = I_1 = 0.43A$

Since the current I remains the same in both the cases, Reciprocity Theorem is verified.

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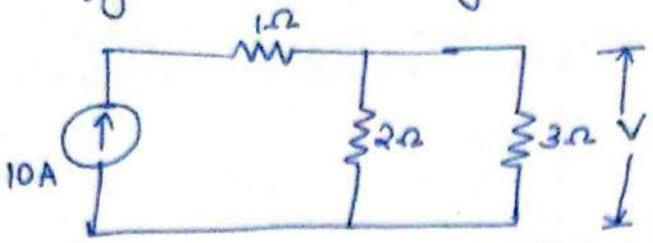
3. Verify the reciprocity theorem in the circuit.



Case 1

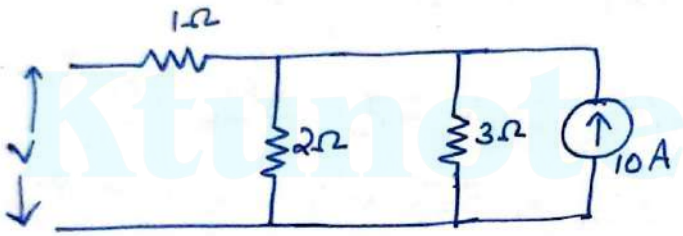
V = Voltage across 3Ω resistor

= Current through 3Ω resistor × 3

$= 10 \times \frac{2}{2+3} \times 3 = \frac{20}{5} \times 3$

$= 12\,V$

Case 2 Source & Response are interchanged.



$V = I_{2\Omega} \times 2$

$= 10 \times \frac{3}{3+2} \times 2 = 6 \times 2$

$= 12V$

Since the voltage remains same in both the cases, Reciprocity Theorem is verified.

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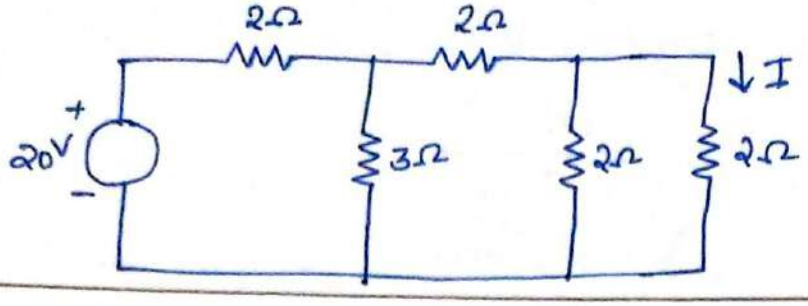
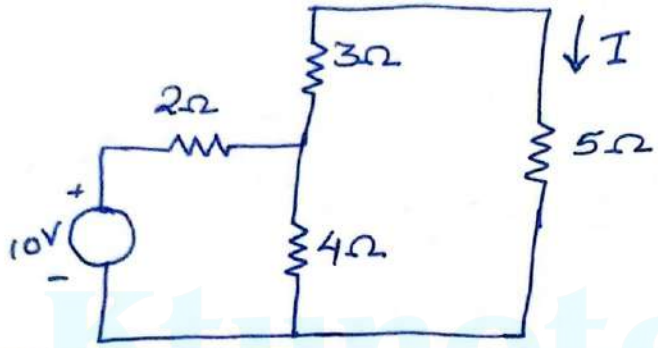
Tutorial Questions on Reciprocity Theorem

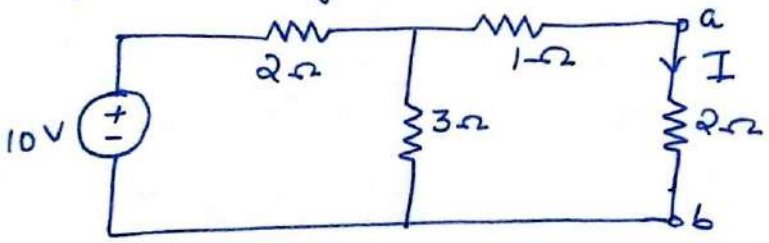
1. Verify the reciprocity Theorem for the network shown.



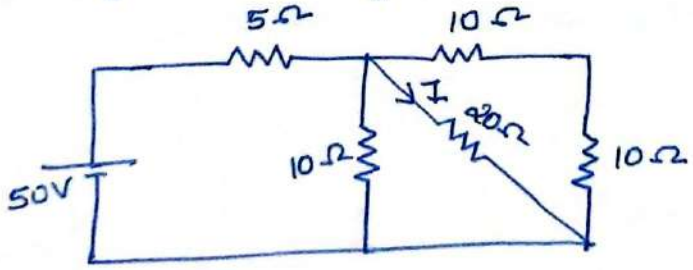
2. Verify the reciprocity theorem for the given circuit.



3. Verify Reciprocity Theorem



4. Verify reciprocity theorem



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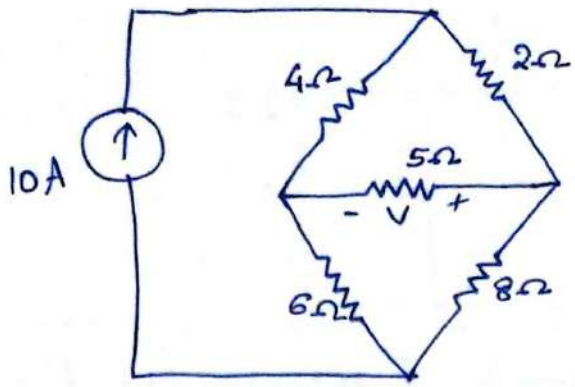
5. Find the voltage V and verify reciprocity theorem.



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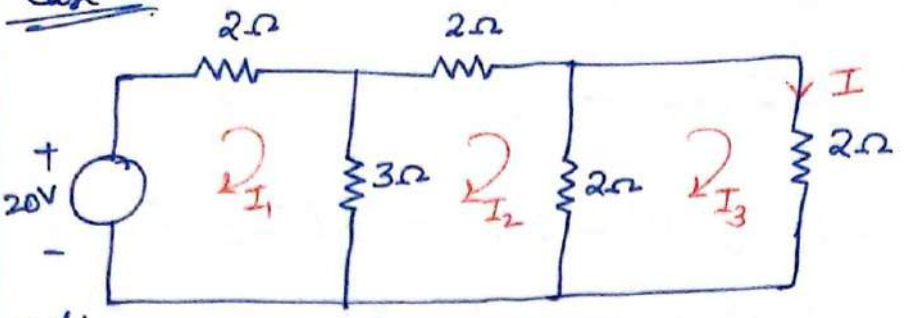
Answers for Tutorial Questions from Reciprocity Theorem

Case 1

1.



Mesh 1

$$20 - 2I_1 - 3(I_1 - I_2) = 0$$

$$5I_1 - 3I_2 = 20 \quad \text{——— (1)}$$

Mesh 2

$$-3(I_2 - I_1) - 2I_2 - 2(I_2 - I_3) = 0$$

$$-3I_1 + 7I_2 - 2I_3 = 0 \quad \text{——— (2)}$$

Mesh 3

$$-2(I_3 - I_2) - 2I_3 = 0$$

$$-2I_2 + 4I_3 = 0 \quad \text{——— (3)}$$

Solving (1), (2) & (3)

$I_1 = 5.71A$

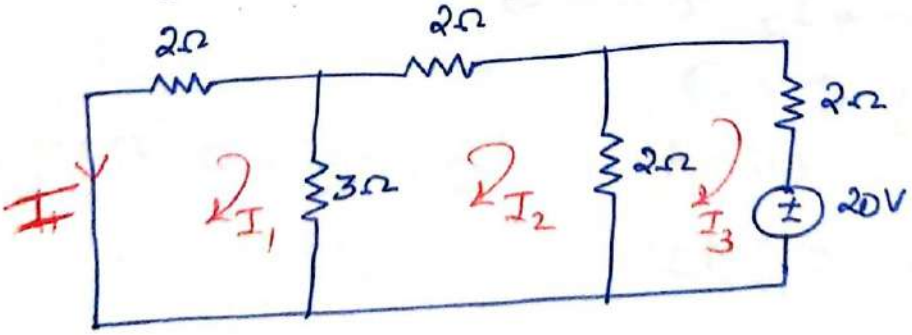
$I_2 = 2.86A$

$I_3 = \underline{1.43A}$

$I = I_3 = \underline{1.43A}$

Case 2

Source & Response are interchanged.



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Mesh 1

$-2I_1 - 3(I_1 - I_2) = 0$

$5I_1 - 3I_2 = 0$ ——— (1)

Mesh 2

$-3(I_2 - I_1) - 2I_2 - 2(I_2 - I_3) = 0$

$-3I_1 + 7I_2 - 2I_3 = 0$ ——— (2)

Mesh 3

$-2(I_3 - I_2) - 2I_3 - 20 = 0$

$-2I_2 + 4I_3 = -20$ ——— (3)

Solving ①, ② & ③

$I_1 = -1.43$

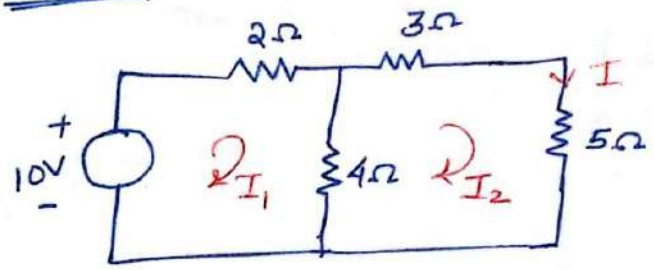
$I_2 = -2.38$

$I_3 = -6.19$

$I = -I_1 = \underline{1.43A}$

I remains same. Hence Proved.

2. Case 1



$10 - 2I_1 - 4(I_1 - I_2) = 0$

$6I_1 - 4I_2 = 10$ ——— (1)

$-4(I_2 - I_1) - 3I_2 - 5I_2 = 0$

$-4I_1 + 12I_2 = 0$ ——— (2)

Solving ① & ②

$I_1 = 2.143A$

$I_2 = 0.714A$

$I = I_2 = \underline{0.71A}$

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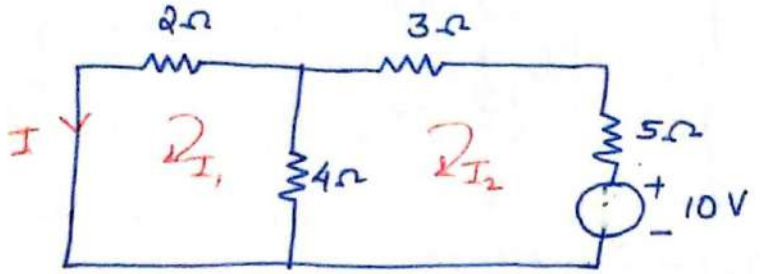
Case 2

Source and Response are interchanged.



$-2I_1 - 4(I_1 - I_2) = 0$

$6I_1 - 4I_2 = 0$ —(1)

$-4(I_2 - I_1) - 3I_2 - 5I_2 - 10 = 0$

$-4I_1 + 12I_2 = -10$ —(2)

Solving (1) & (2)

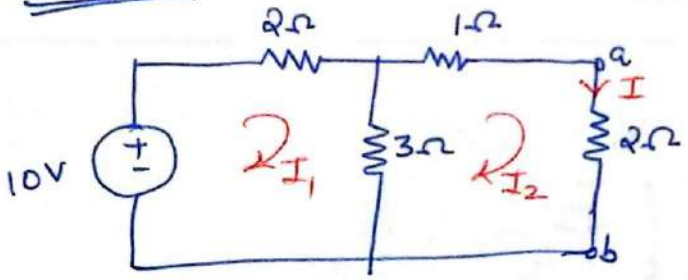
$I_1 = -0.714\,A$

$I_2 = -1.071\,A$

$I = -I_1 = 0.714\,A$

Since the current remains same in both the cases, Reciprocity Theorem is verified.

3. Case 1



$10 - 2I_1 - 3(I_1 - I_2) = 0$

$5I_1 - 3I_2 = 10$

$-3(I_2 - I_1) - 1 \cdot I_2 - 2I_2 = 0$

$-3I_1 + 6I_2 = 0$

$5I_1 - 3I_2 = 10$

$-3I_1 + 6I_2 = 0$

Solving

$I_1 = 2.86\,A$

$I_2 = 1.43\,A$

$I = I_2 = 1.43\,A$

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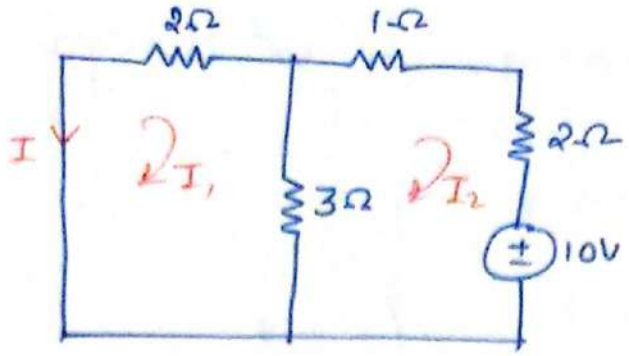
Case 2



$$-3(I_1 - I_2) - 2I_1 = 0$$

$$5I_1 - 3I_2 = 0 \quad \text{——— (1)}$$

$$-3(I_2 - I_1) - I_2 - 2I_2 - 10 = 0$$

$$-3I_1 + 6I_2 = -10 \quad \text{——— (2)}$$

Solving (1) & (2)

$I_1 = -1.43A$ $\qquad$ $I = -I_1 = \underline{1.43A}$

$I_2 = \underline{-2.38A}$

Since the current remains same in both the cases, Reciprocity Theorem is verified.

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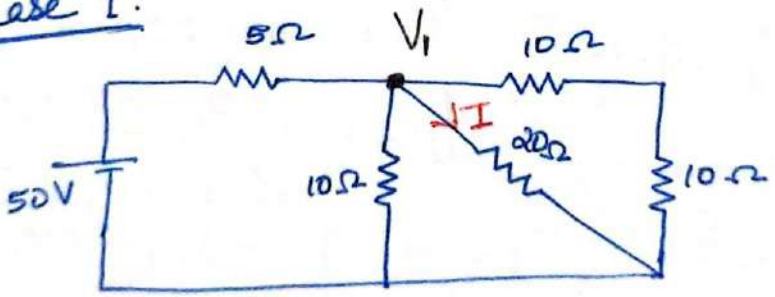
4. Case 1:



KCL to Node 1

$$\frac{V_1 - 50}{5} + \frac{V_1}{10} + \frac{V_1}{20} + \frac{V_1}{(10+10)} = 0$$

$$4V_1 - 200 + 2V_1 + V_1 + V_1 = 0$$

$$8V_1 = 200; \qquad V_1 = \underline{25V}$$

$$I = \frac{V_1}{20} = \frac{25}{20} = \underline{1.25A}$$

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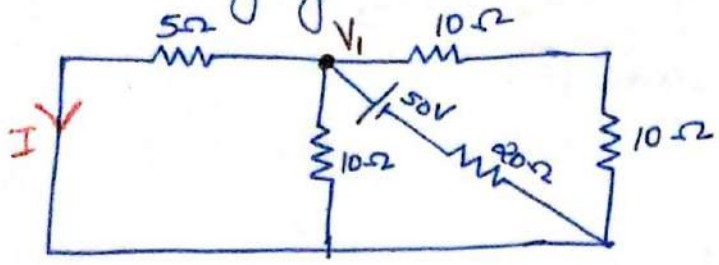
Case 2

Interchanging the source & response



$$\frac{V_1}{5} + \frac{V_1}{10} + \frac{V_1 - 50}{20} + \frac{V_1}{(10+10)} = 0$$

$$4V_1 + 2V_1 + V_1 - 50 + V_1 = 0$$

$$8V_1 = 50$$

$$V_1 = \frac{50}{8} V$$

$$I = \frac{V_1}{5} = \frac{50/8}{5} = \frac{10}{8} = 1.25 A$$

Since Current remains same in both the cases, Reciprocity Theorem is verified.

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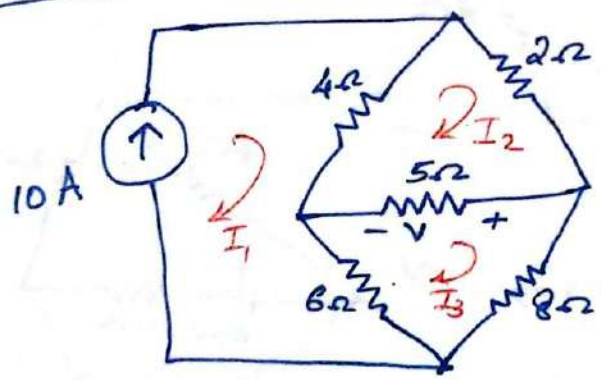
5. Case 1:



Mesh 1

$I_1 = 10A$ —— (1)

Mesh 2 → $-4(I_2 - I_1) - 2I_2 - 5(I_2 - I_3) = 0$

$-4I_1 + 11I_2 - 5I_3 = 0$ —— (2)

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Mesh 3 → $-6(I_3-I_1)-5(I_3-I_2)-8I_3=0$

$$-6I_1-5I_2+19I_3=0 \quad \text{——— (3)}$$

Solving (1), (2) and (3)

$I_1 = 10A$

$I_2 = 5.76A$

$I_3 = \underline{\underline{4.67A}}$

$V = 5(I_2-I_3) = 5(5.76-4.67)$

$= \underline{\underline{5.45\,V}}$

<u>Case 2:</u>

Excitation and Response are interchanged.

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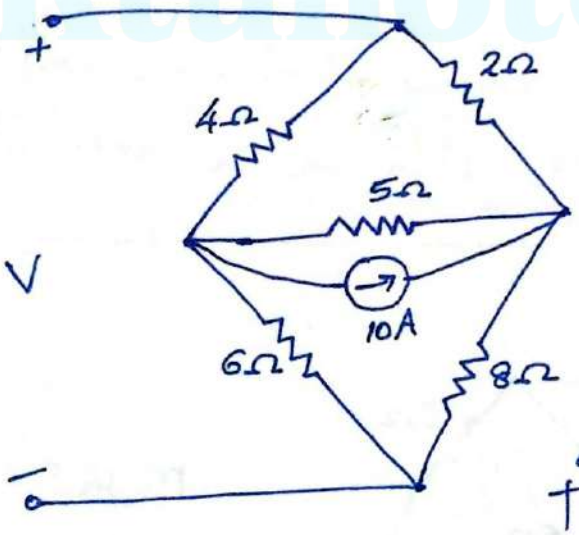
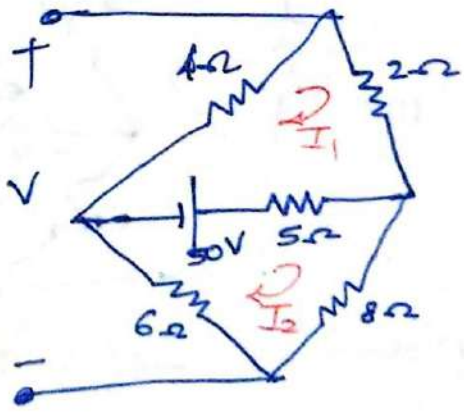


Source Transformation ⇒



<u>Mesh 1</u>

$-4I_1-2I_1-5(I_1-I_2)-50=0$

$11I_1-5I_2=-50$ ——— (1)

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Mesh 2 →

$$-5(I_2 - I_1) - 8I_2 - 6I_2 + 50 = 0$$

$$-5I_1 + 19I_2 = 50 \quad \text{——} \quad (2)$$

Solving (1) & (2)

$I_1 = -3.8$

$I_2 = 1.63$

From the Circuit

$$V + 4I_1 + 6I_2 = 0$$

$$V + 4 \times -3.8 + 6 \times 1.63 = 0$$

$$V = 5.42V$$

Since the voltage V is same in both the cases, the reciprocity theorem is verified.

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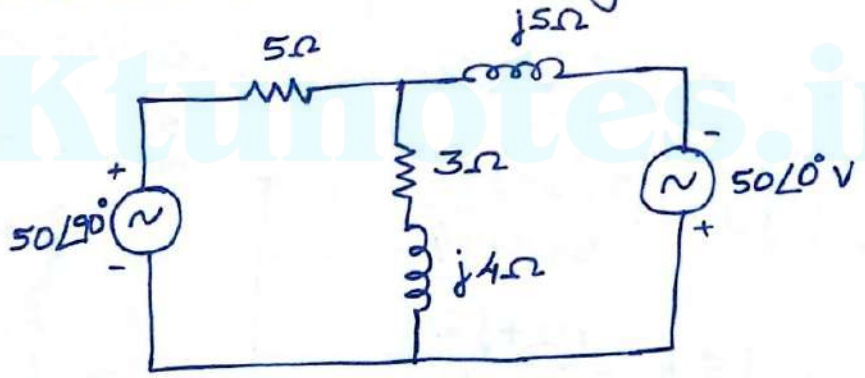
Analysis of AC Circuits

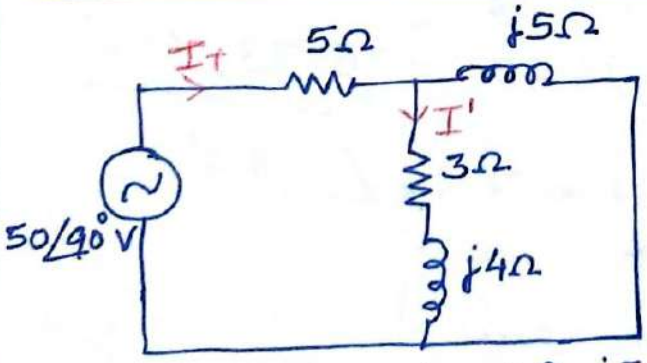
SUPERPOSITION THEOREM

The superposition theorem states that in a network containing more than one voltage source or current source, the total current or voltage in any branch of the network is the phasor sum of currents or voltages produced in that branch by each source acting separately. This theorem is valid only for linear systems.

1. Find the current through the $(3+j4)\,\Omega$ impedence.



1. When $50\angle 90^\circ$ V source is acting alone



$$Z_T = 5 + \frac{(3+j4)\cdot j5}{3+j4+j5} = 5 + \frac{-20+j15}{3+j9}$$

$$= \frac{15+j45-20+j15}{3+j9} = \frac{-5+j60}{3+j9} = 5.833+j2.5$$

$$= 6.35\angle 23.2^\circ\ \Omega$$

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$$I_T = \frac{50\angle 90^\circ}{6.35\angle 23.2^\circ} = \underline{\underline{7.87\angle 66.8^\circ A}}$$

By Current division rule

$$I' = (7.87\angle 66.8^\circ) \times \frac{j5}{3+j4+j5}$$

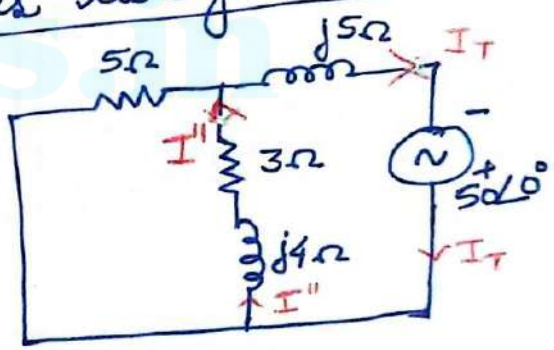
$$= 7.87\angle 66.8^\circ \times \frac{j5}{3+j9}$$

$$= 7.87\angle 66.8^\circ \times (0.5+j0.167)$$

$$= 7.87\angle 66.8^\circ \times 0.527\angle 18.435^\circ$$

$$= \underline{\underline{4.15\angle 85.24^\circ}}\ A\ (\downarrow)$$

2. When $50\angle 0^\circ$ V source is acting alone



$$Z_T = j5 + \frac{5(3+j4)}{5+3+j4}$$

$$= j5 + \frac{15+j20}{8+j4}$$

$$= j5 + 2.5 + j1.25$$

$$= 2.5 + j6.25 = \underline{\underline{6.73\angle 68.2^\circ}}$$

$$I_T = \frac{50\angle 0}{6.73\angle 68.2} = \underline{\underline{7.42\angle -68.2^\circ}}$$

By Current division rule;

$$I'' = 7.42\angle -68.2^\circ \times \frac{5}{8+j4}$$

$$= \underline{\underline{4.15\angle -94.77^\circ}}\ A\ (\uparrow)$$

$$= \underline{\underline{-4.15\angle -94.77^\circ}}\ A\ (\downarrow)$$

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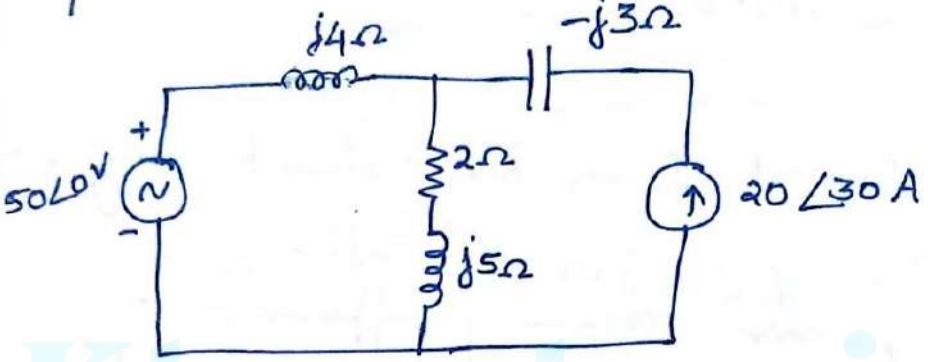
3. By Superposition Theorem

$$I = I' + I''$$

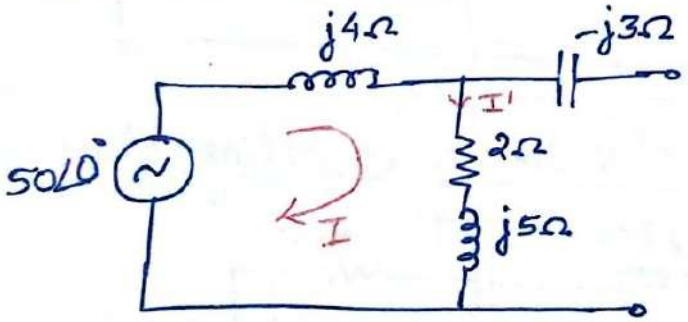
$$= 4.15\angle 85.3^\circ + {}^{-}4.15\angle{}^{-}94.77^\circ$$

$$= 8.3\angle 85.3^\circ \text{ A } (\downarrow)$$

2. Determine the voltage across the $(2+j5)\Omega$ impedance.



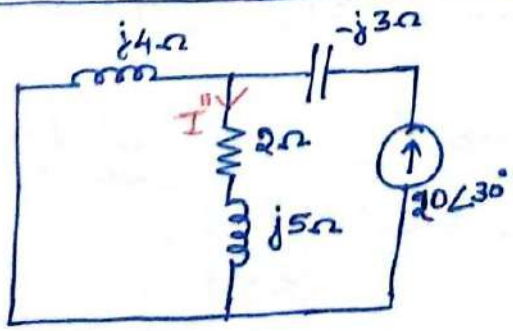
1. When $50\angle 0^\circ$ V source is acting alone



$$I' = \frac{50\angle 0}{j4+2+j5} = \frac{50\angle 0}{2+j9} = 5.42\angle{}^{-}77.47 \text{ A}$$

2. When $20\angle 30^\circ$ A source is acting alone

Current division rule



$$I'' = 20\angle 30^\circ \times \frac{j4}{j4+2+j5}$$

$$= 20\angle 30 \times \frac{j4}{2+j9}$$

$$= 8.68\angle 42.53^\circ \text{ A}$$

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By Superposition Theorem

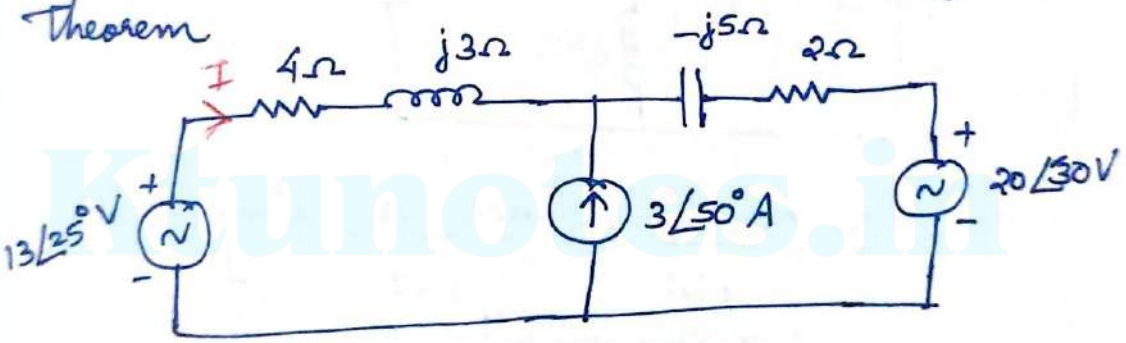
Current through $(2+j5)\Omega$ is; $I = I' + I''$

Voltage across $(2+j5)\Omega$ is; $V = (2+j5)(I'+I'')$

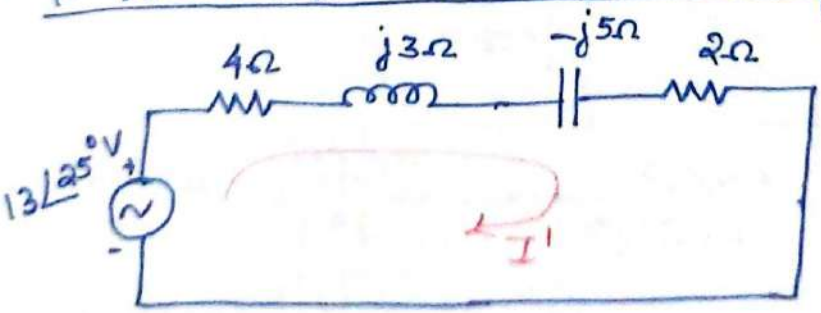
$$V = (2+j5)\left[5.42\angle -77.47^\circ + 8.68\angle 42.53^\circ\right]$$

$$= 40.89\angle 72.55^\circ \text{ V}$$

3. Find Current I in the network using Superposition Theorem



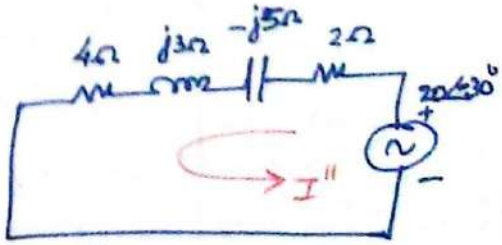
1. When $13\angle 25^\circ$ V source is acting alone



$$I' = \frac{13\angle 25}{(6-j2)} = 2.056\angle 43.43^\circ \text{ A } (\rightarrow)$$

2. When $20\angle -30^\circ$ V source alone

$$I'' = \frac{20\angle -30^\circ}{6-j2} = 3.16\angle -11.57 \text{ A } (\leftarrow)$$



$$= -3.16\angle 11.57 \text{ A } (\rightarrow)$$

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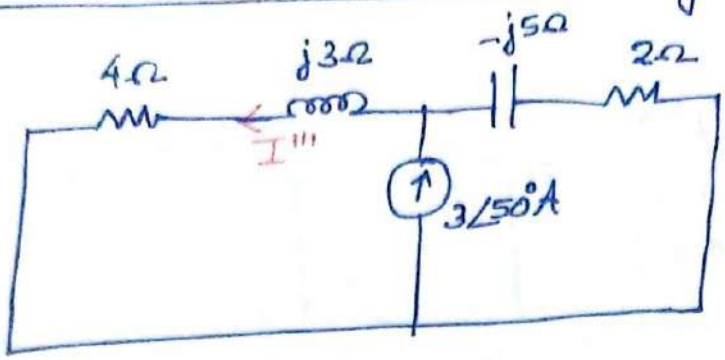
3. When $3\angle 50^\circ$ A Source is acting alone



$$I''' = 3\angle 50 \times \frac{(2-j5)}{6-j2} = 2.56\angle 0.23^\circ A \ (\leftarrow)$$

$$= -2.56\angle 0.23^\circ A \ (\rightarrow)$$

4. By Superposition Theorem

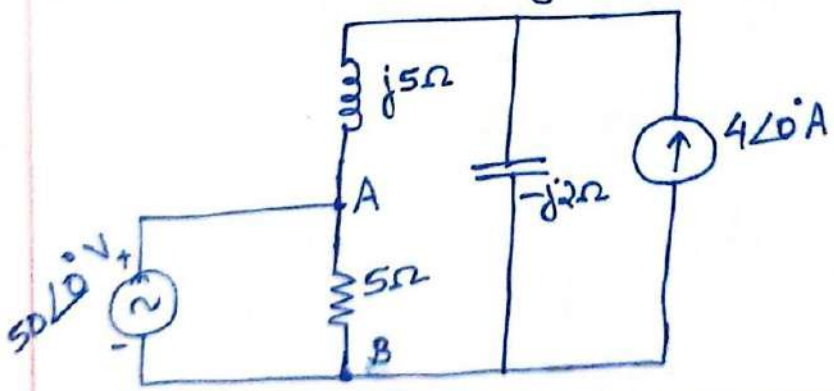
$$I = I' + I'' + I'''$$

$$= 2.056\angle 43.43 - 3.16\angle -11.57 - 2.56\angle 0.23)$$

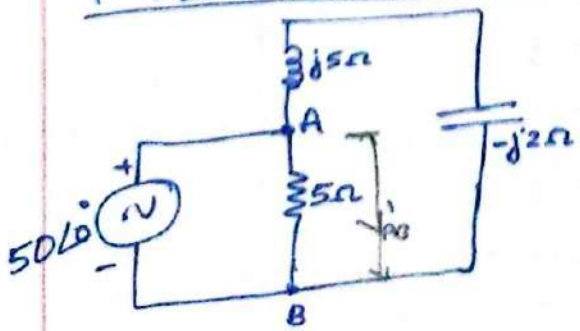
$$= 4.63\angle 153.93^\circ A \ (\rightarrow)$$

---

4. Determine the voltage V_{AB} for the network shown



1. $50\angle 0^\circ$ V Source alone



$$V'_{AB} = 50\angle 0^\circ V$$

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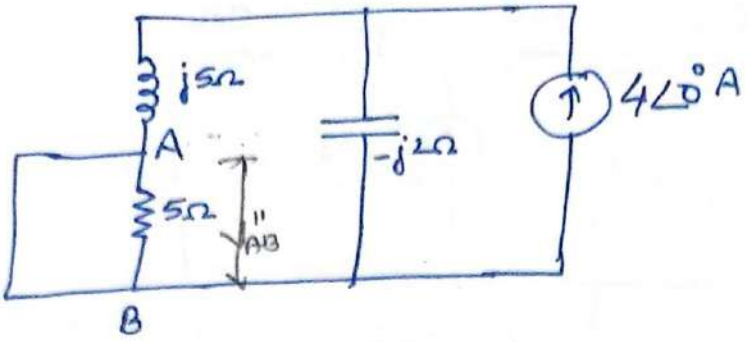
2. $4\angle 0^\circ A$ source alone



$V''_{AB} = 0$

3. By Superposition Theorem

$V_{AB} = V'_{AB} + V''_{AB}$

$= 50\angle 0^\circ + 0$

$= 50\angle 0^\circ \ V$

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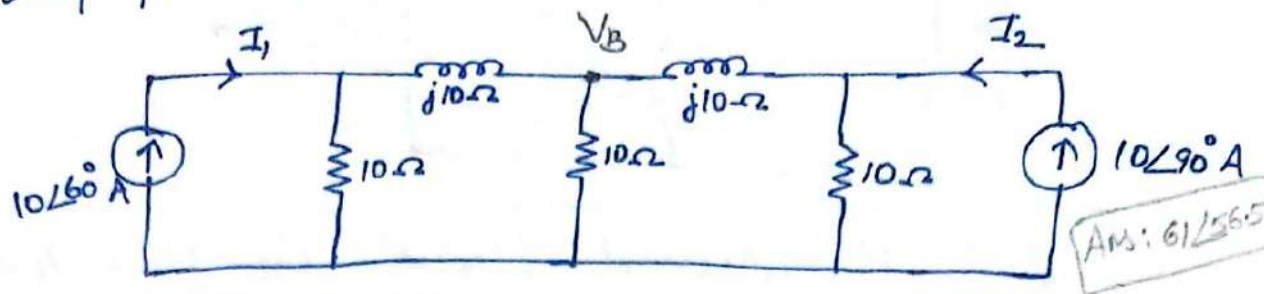
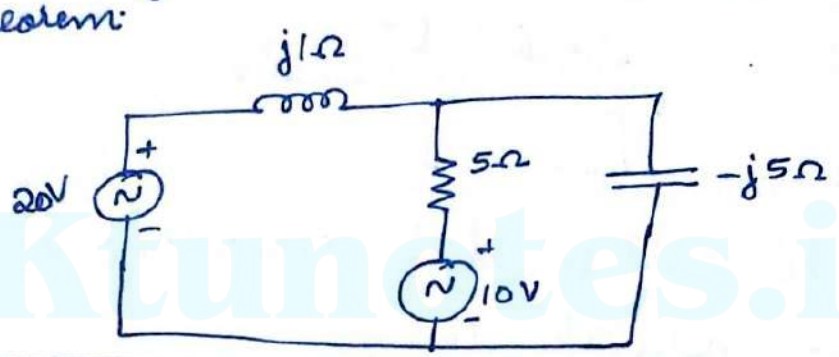
Tutorial Questions → Superposition Theorem (AC)

1. Determine V_B by means of Principle of Superposition



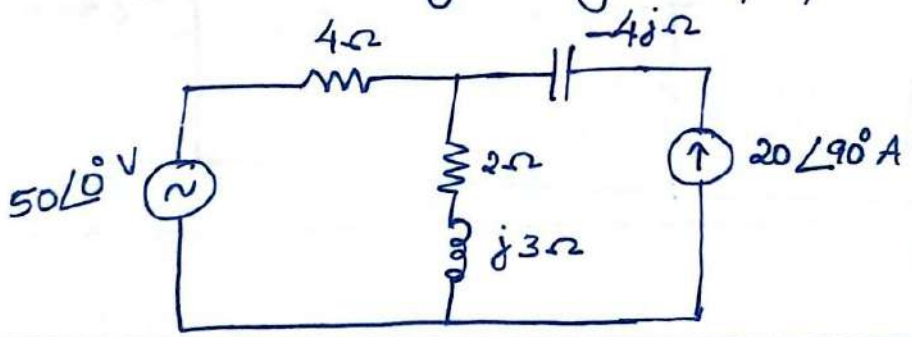
Ans: $61\angle 56.58$

2. Calculate current flowing through the branch containing the capacitor using Superposition Theorem.



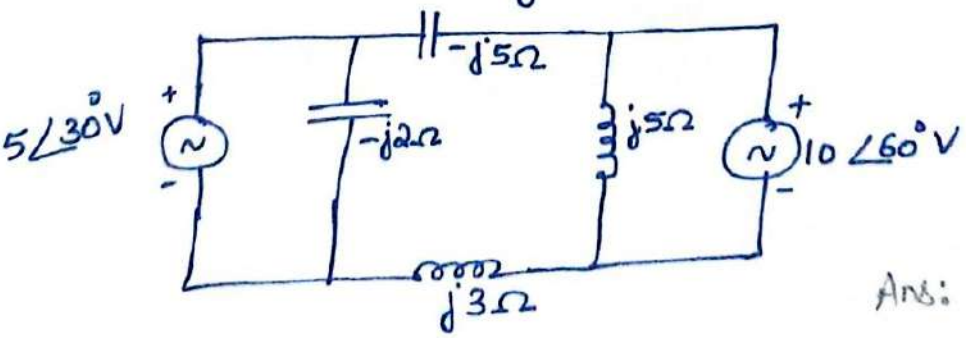
Ans: $4.87\angle 81.68^\circ$ A

3. For the circuit shown determine the current in $(2+j3)\,\Omega$ by using Superposition Theorem



Ans: $14.07\angle 31.4^\circ$ A

4. Find the current through $j3\Omega$ reactance.



Ans: $3.1\angle -6.2^\circ$ A

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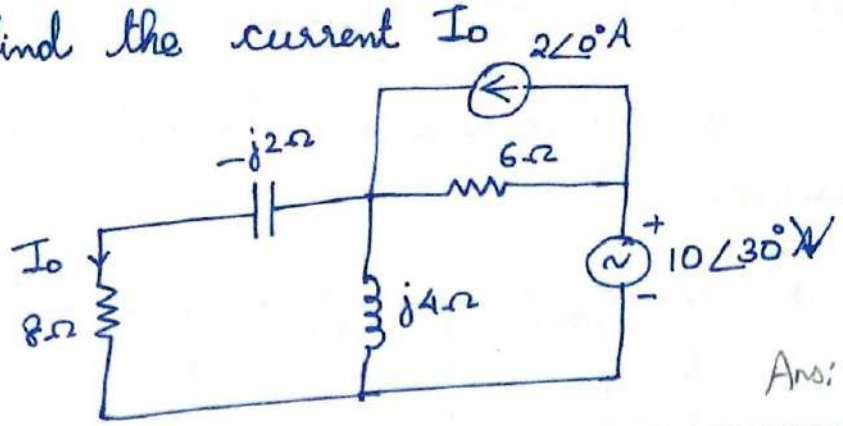
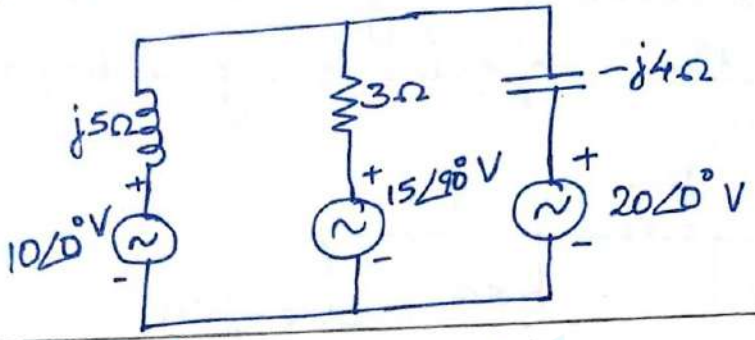
5. Find the current I_o



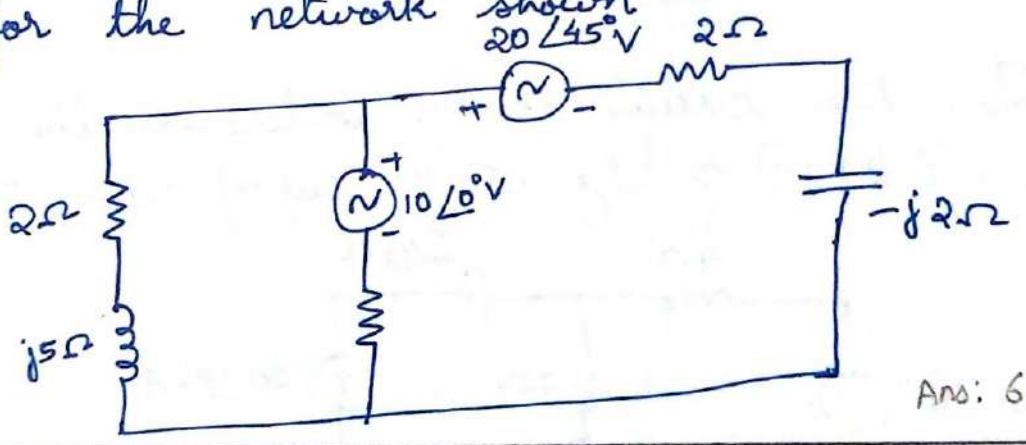
Ans: $I_o = 1.19\angle 65.48°$ A (↓)

6. Find the current through the j5Ω branch for the network.



Ans: $I = 4.86\angle -164.41°$ A

7. Find the voltage drop across the capacitor for the network shown



Ans: $6.02\angle 152.14°$ V

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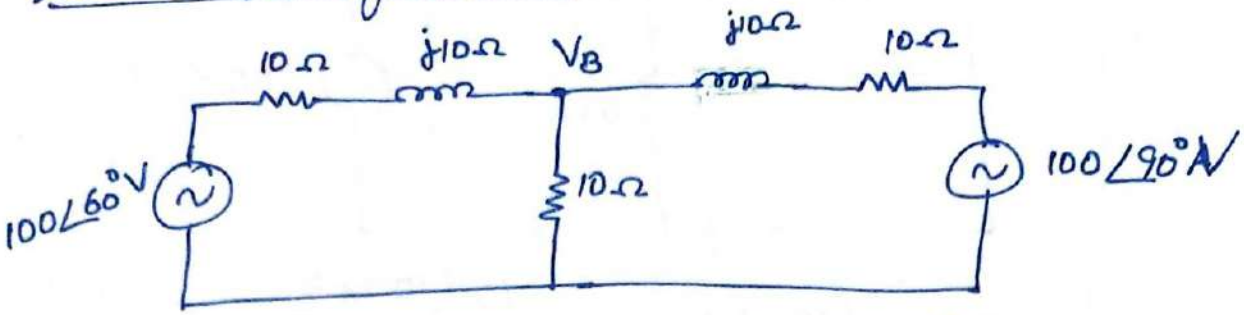
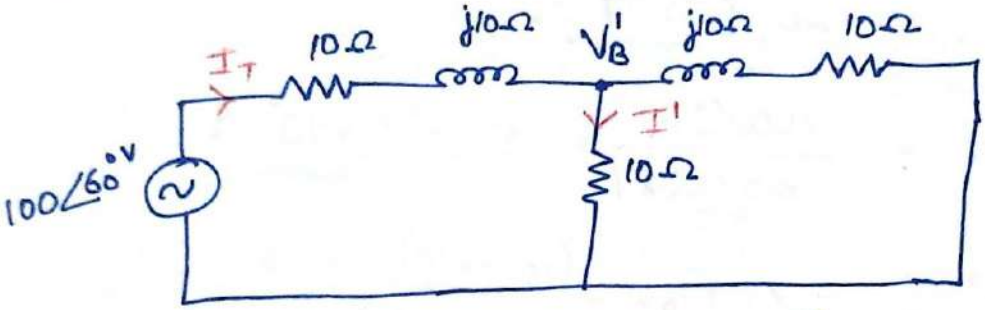
Answers for Tutorial → Superposition (AC)

1. Source Transformation



1. when 100∠60° V source acting alone



$$Z_T = (10+j10) + \frac{10\,(10+j10)}{10+(10+j10)}$$

$$= (10+j10) + \frac{100+j100}{20+j10}$$

$$= \underline{\underline{20\angle 36.87^\circ\ \Omega}}$$

$$I_T = \frac{100\angle 60}{20\angle 36.87} = \underline{\underline{5\angle 23.13^\circ\ A}}$$

$$I' = 5\angle 23.13^\circ \times \frac{(10+j10)}{10+10+j10}$$

$$= 5\angle 23.13^\circ \times \frac{(10+j10)}{20+j10} = \underline{\underline{3.16\angle 41.56^\circ}}\ (\downarrow)$$

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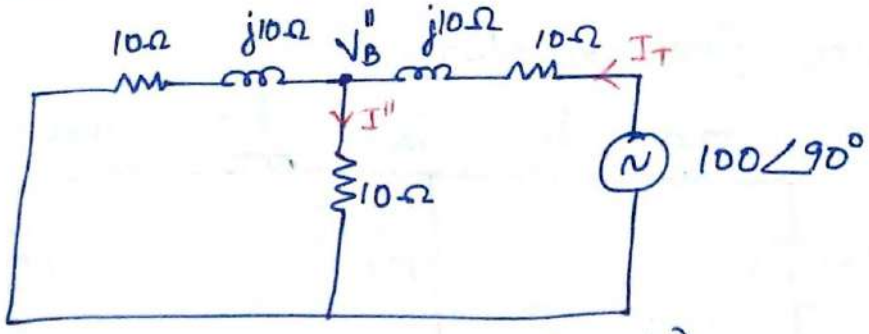
2. When $100\angle 90^\circ$ Source alone



$$Z_T = (10+j10) + \frac{10(10+j10)}{10+(10+j10)}$$

$$= \underline{\underline{20\angle 36.87^\circ\ \Omega}}$$

$$I_T = \frac{100\angle 90}{20\angle 36.87} = \underline{\underline{5\angle 53.13^\circ\ A}}$$

$$I'' = 5\angle 53.16 \times \frac{(10+j10)}{10+(10+j10)}$$

$$= 5\angle 53.16 \times \frac{(10+j10)}{20+j10}$$

$$= \underline{\underline{3.16\angle 71.6^\circ}}\ (\downarrow)$$

3. By Superposition Theorem

Current through 10Ω resistor; $I = I' + I''$

$$I = 3.16\angle 41.56 + 3.16\angle 71.6$$

$$= \underline{\underline{6.1\angle 56.58^\circ\ A}}\ (\downarrow)$$

$$V_B = 6.1\angle 56.58 \times 10$$

$$= \underline{\underline{61\angle 56.58^\circ\ V}}.$$

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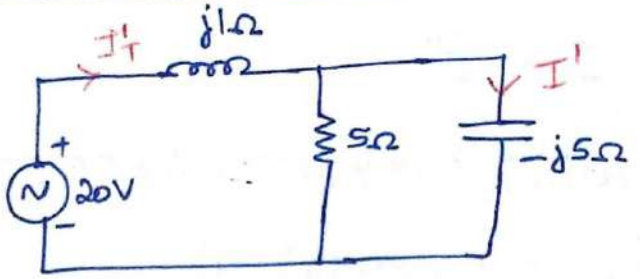
2. 1. 20V Source alone



$$Z_T = j1\Omega + \frac{5 \times -j5}{5 + -j5} = j1 - \frac{j25}{(5-j5)}$$

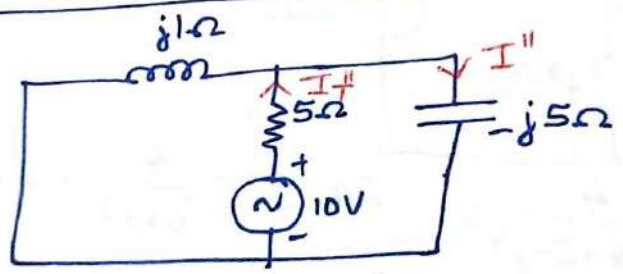
$$= \underline{\underline{2.92\angle -30.96}}\ \Omega$$

$$I_T' = \frac{20\angle 0}{2.92\angle -30.96} = \underline{\underline{6.85\angle 30.96^\circ}}\ A$$

$$I' = 6.85\angle 30.96 \times \frac{5}{(5-j5)}$$

$$= \underline{\underline{4.844\angle 75.96^\circ}}\ A \ (\downarrow)$$

2. 10V Source alone



$$Z_T = 5 + \frac{j1 \times -j5}{j1 - j5} = 5 + \frac{-5j^2}{-j4}$$

$$= 5 + \frac{5}{-j4} = \underline{\underline{5.15\angle 14^\circ}}\ \Omega$$

$$I_T'' = \frac{10}{5.15\angle 14^\circ} = \underline{\underline{1.942\angle -14^\circ}}\ A$$

$$I'' = 1.942\angle -14 \times \frac{j1}{j1 - j5} = 1.942\angle -14 \times \frac{1}{-4} = \underline{\underline{-0.4855\angle -14}}$$

(↓)

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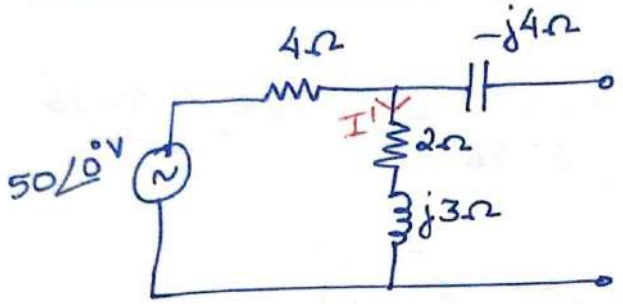
3. By Superposition Theorem

$$I_{j5\Omega} = I' + I''$$

$$= 4.844\angle 75.96 + 0.4855\angle -14$$

$$= 4.87\angle 81.68^\circ \text{ A}$$

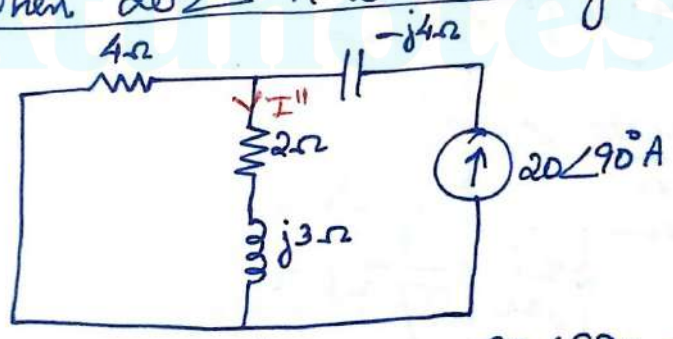
3. 1. When $50\angle 0^\circ$V source acting alone



$$I' = \frac{50\angle 0}{4+2+j3}$$

$$= \frac{50}{6+j3} = 7.46\angle -26.56\text{A}$$

2. When $20\angle 90^\circ$A source acting alone



$$I'' = 20\angle 90 \times \frac{4}{4+2+j3} \quad 20\angle 90 \times \frac{4}{6+j3} = 11.93\angle 63.43\text{A}$$

3. Superposition Theorem

$$I = I' + I''$$

$$= 7.46\angle -26.56 + 11.93\angle 63.43$$

$$= 14.07\angle 31.4^\circ \text{A}$$

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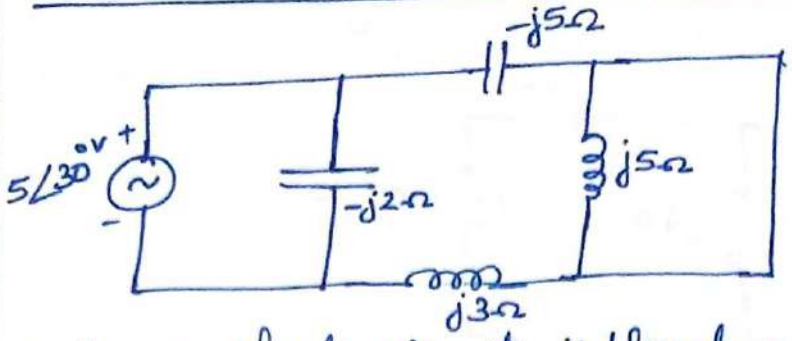
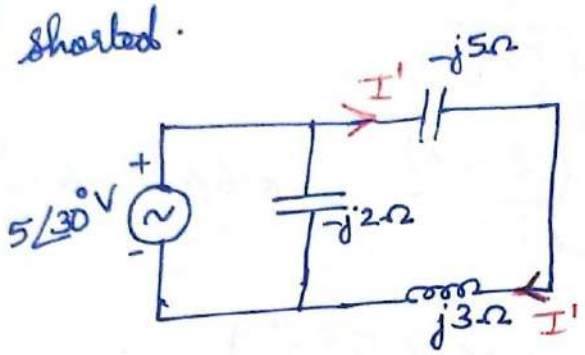
4. 1. When $5\angle 30^\circ$ V source is acting alone



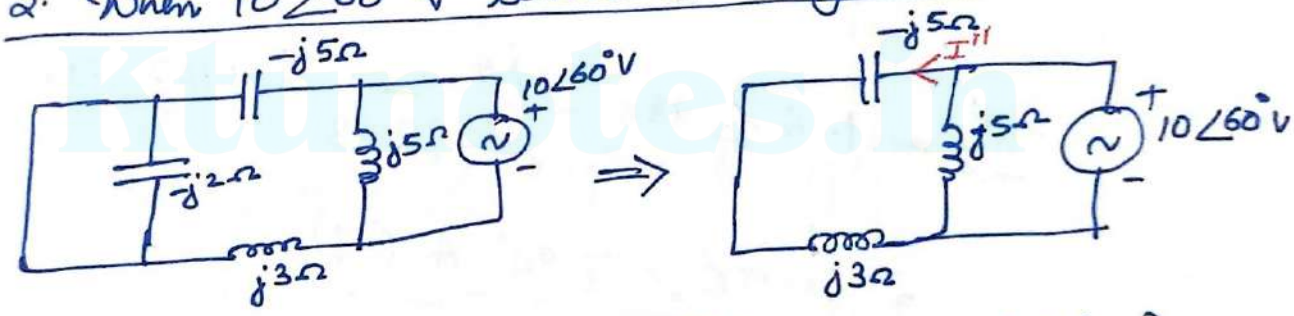
When a short circuit is placed across $j5\Omega$, it gets shorted.



$$I' = \frac{5\angle 30^\circ}{-j5+j3}$$

$$= \frac{5\angle 30^\circ}{-j2} = 2.5\angle 120\ A\ (\leftarrow)$$

2. When $10\angle 60^\circ$ V source is acting alone



$$I'' = \frac{10\angle 60}{-j5+j3} = \frac{10\angle 60}{-j2} = 5\angle 150^\circ A\ (\rightarrow)$$

$$= -5\angle 150^\circ\ (\leftarrow)$$

3. Superposition Theorem

$$I = 2.5\angle 120 - 5\angle 150$$

$$= 3.1\angle -6.21^\circ\ A$$

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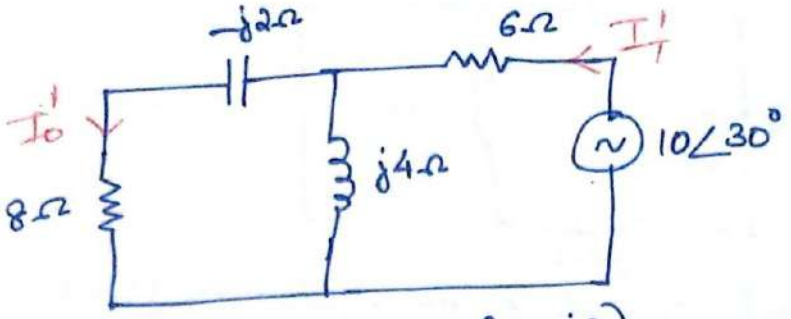
5. 1. When $10\angle 30^\circ$ V source acting alone



$$Z_T = 6 + \frac{j4 \times (8-j2)}{j4+(8-j2)}$$

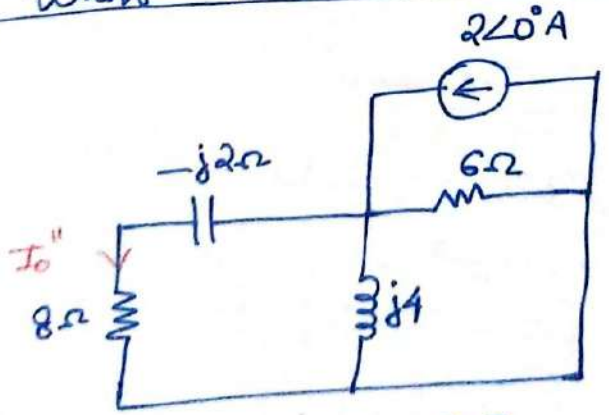
$$= 6 + \frac{j4(8-j2)}{8+j2} = 8.64\angle 24.12^\circ\ \Omega$$

$$I'_T = \frac{10\angle 30}{8.64\angle 24.12} = 1.16\angle 5.88^\circ\ A$$

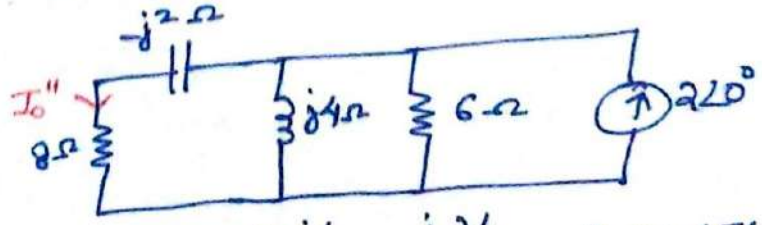
$$I'_o = 1.16\angle 5.88 \times \frac{j4}{j4-j2+8} = 1.16\angle 5.88 \times \frac{j4}{(8+j2)}$$

$$= 0.56\angle 81.84^\circ\ A\ (\downarrow)$$

2. When $2\angle 0^\circ$ A source is acting alone



Redrawn as ⟹



$$j4 \parallel 6 = \frac{6 \times j4}{\ldots} = \frac{j24}{\ldots} = 3.33\angle 56.31$$

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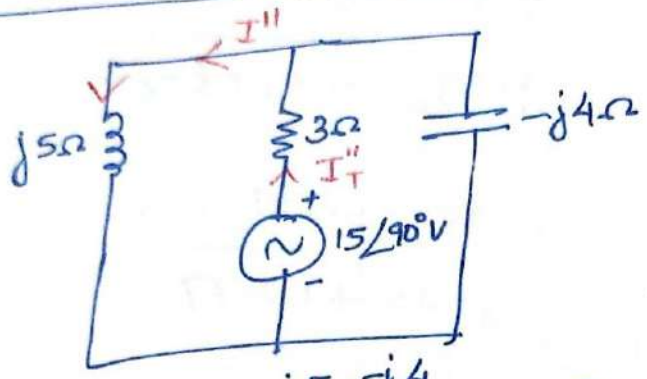
$$I' = \frac{10\angle 0^\circ}{4.04\angle 61.66^\circ} = \underline{\underline{2.48\angle -61.66^\circ A}} \ (\uparrow)$$

2. When $15\angle 90^\circ$ V source alone



$$Z_T = 3 + \frac{j5 \times j4}{j5 - j4} = 3 + \frac{20}{j}$$

$$= 3 - j20 = \underline{\underline{20.22\angle -81.47^\circ \Omega}}$$

$$I''_T = \frac{15\angle 90^\circ}{20.22\angle -81.47^\circ} = \underline{\underline{0.74\angle 171.47}} \ A$$

Current division rule

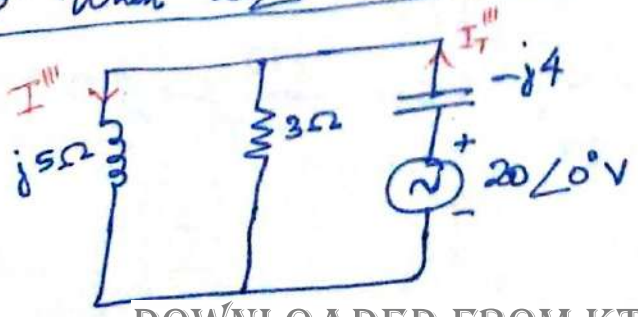
$$I'' = 0.74\angle 171.47 \times \frac{-j4}{j5 - j4}$$

$$= 0.74\angle 171.47 \times \frac{-4}{1}$$

$$= -2.96\angle 171.47^\circ \ (\downarrow)$$

$$= \underline{2.96\angle 171.47^\circ} \ (\uparrow)$$

3. When $20\angle 0^\circ$ V source alone



$$Z_T = -j4 + \frac{3 \times j5}{3 + j5}$$

$$= -j4 + \frac{j15}{3 + j5}$$

$$= \underline{\underline{3.47\angle -50.51^\circ \Omega}}$$

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$$I_T''' = \frac{20\angle 0}{3.47\angle -50.51^\circ} = 5.76\angle 50.51^\circ \text{ A}$$

$$I''' = 5.76\angle 50.51 \times \frac{3}{(3+j5)}$$

$$= 2.963\angle -8.53^\circ \text{ A } (\downarrow)$$

$$= -2.963\angle -8.53^\circ (\uparrow)$$

4. By Superposition Theorem

$$I = I' + I'' + I'''$$

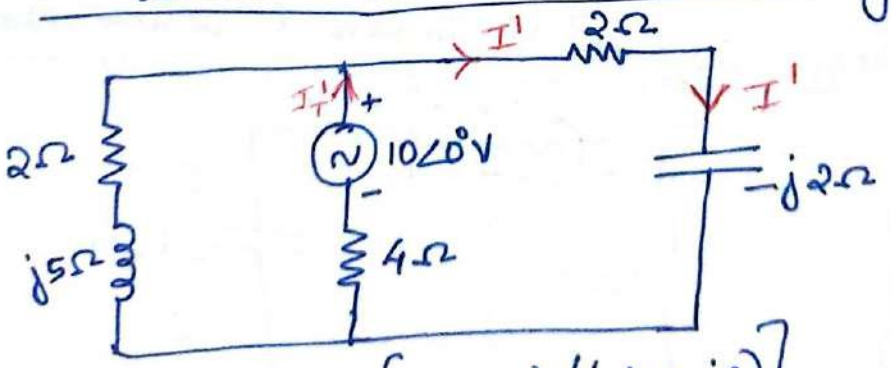
$$= 2.48\angle -61.66 (\uparrow) + 2.96\angle 71.47 (\uparrow)$$

$$- 2.963\angle -8.53 (\uparrow)$$

$$= 4.86\angle -164.41^\circ \text{ A}.$$

---

7. 1. When $10\angle 0^\circ$ V source is acting alone



$$Z_T = 4 + \left[(2+j5) \,//\, (2-j2)\right]$$

$$= 4 + \frac{(2+j5)(2-j2)}{4+j3} = 4 + \frac{4+j6-j^2 10}{4+j3}$$

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$$Z_T = 4 + \frac{4 + j6 + 10}{4 + j3}$$

$$= 4 + \frac{14 + j6}{4 + j3}$$

$$= 7 \angle -5.91^\circ$$

$$I_T' = \frac{10\angle 0}{7\angle -5.91} = 1.43 \angle 5.91^\circ \text{ A}$$

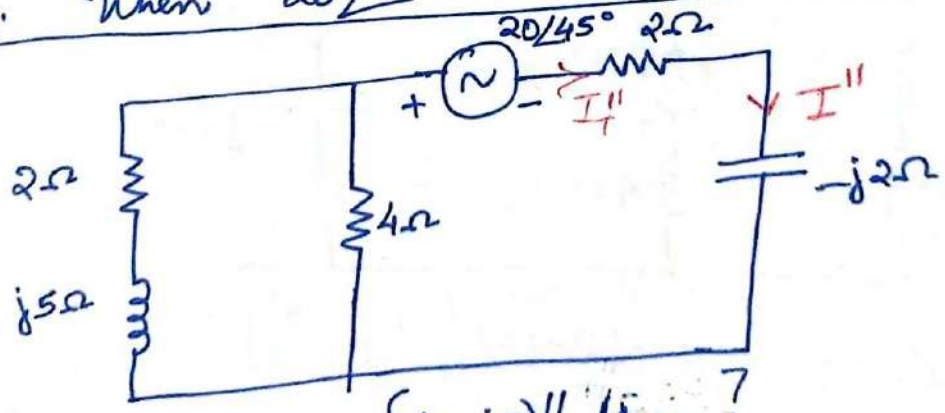
By Current division rule

$$I' = I_T' \times \frac{(2+j5)}{(2+j5)+(2-j2)}$$

$$= 1.43\angle 5.91 \times \frac{(2+j5)}{(2+j5)+(2-j2)}$$

$$= 1.54 \angle 37.42^\circ \ (\downarrow)$$

2. When $20\angle 45^\circ$ V source is acting alone



$$Z_T = (2-j2) + \left[(2+j5) \parallel 4\right]$$

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$$Z_T = (2-j2) + \frac{(2+j5)(4-j2)}{(6+j5)}$$

$$= 4.48\angle -8.84^\circ$$

$$I''_T = I'' = \frac{-20\angle 45}{4.48\angle -8.84}$$

$$= -4.46\angle 53.84 \ (\downarrow)$$

3. By Superposition Theorem

$$I = I' + I''$$

$$= 1.54\angle 37.42 - 4.46\angle 53.84$$

$$= 3.01\angle -117.86^\circ \ A$$

$$V_c = (-j2)I = (-j2)(3.01\angle -117.86^\circ)$$

$$= 6.02\angle 152.14^\circ \ V$$

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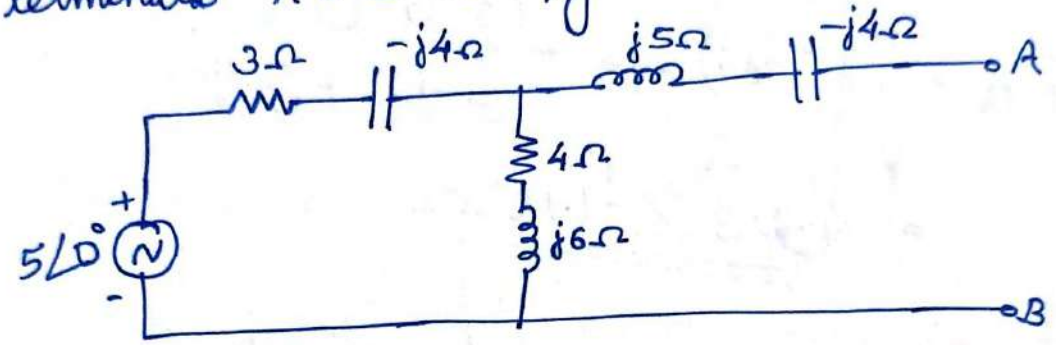
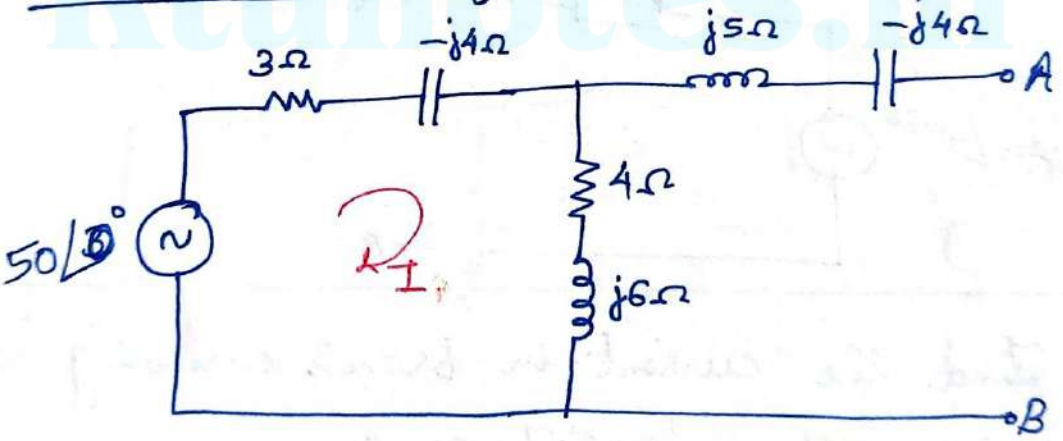
THEVENIN'S THEOREM

Any linear network can be replaced by a voltage source V_{Th} in series with an impedance Z_{Th}.

1. Obtain Thevenin equivalent network for the terminals A & B in Fig



1. Calculation of V_{Th}



$$I = \frac{50\angle 0}{3-j4+4+j6} = \frac{50\angle 0}{7+j2}$$

$$= 6.87\angle -15.95^\circ$$

$$V_{Th} = (4+j6)I = (4+j6)(6.87\angle -15.95^\circ)$$

$$= 49.5\angle 40.35^\circ \text{ V}$$

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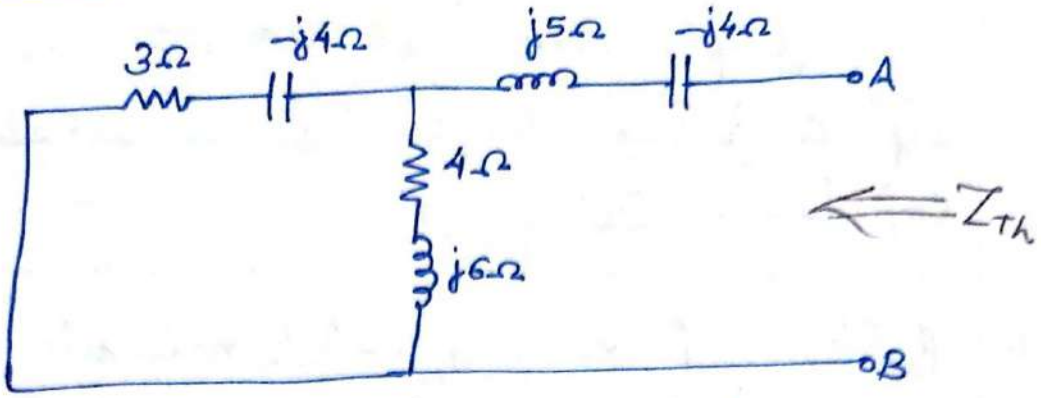
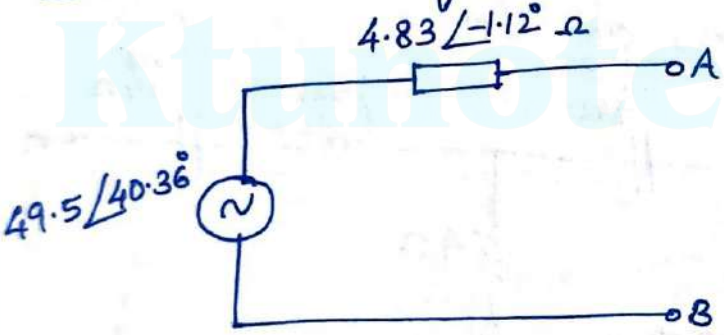
2. Calculation of Z_{Th}



$$Z_{Th} = (j5 - j4) + \frac{(3-j4)(4+j6)}{(3-j4)+(4+j6)}$$

$$= 4.83 \angle -1.12^\circ \ \Omega$$

3. Thevenin's equivalent network



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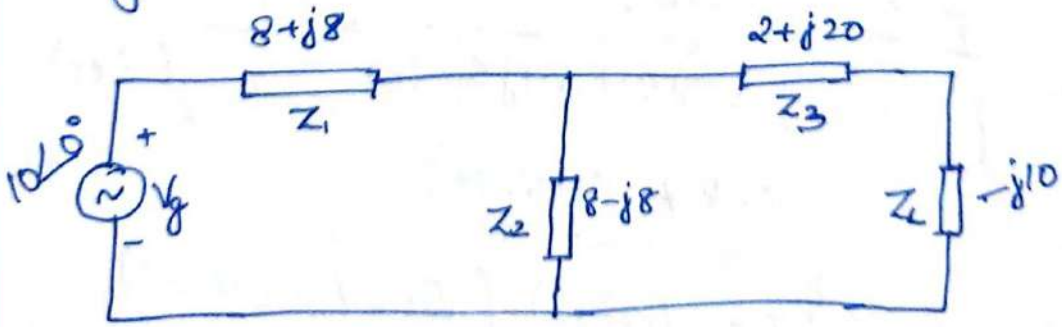
2. Find the current in branch containing Z_L using Thevenin's Theorem.



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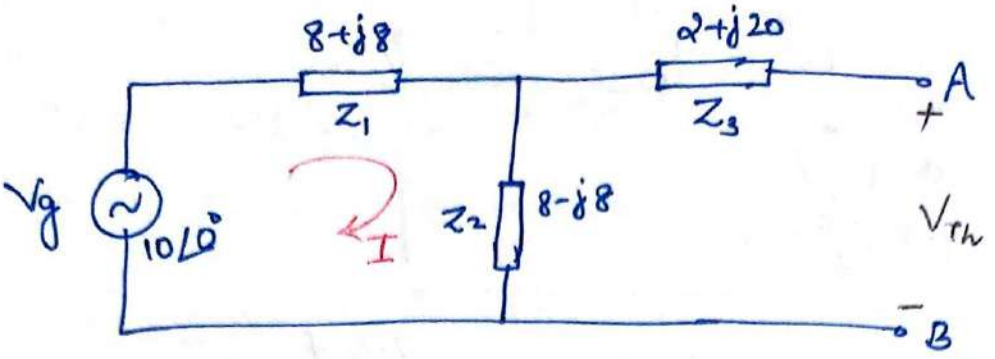
1. Calculation of V_{Th}



$$I = \frac{10}{(8+j8)+(8-j8)} = \frac{10}{16} A.$$

$$V_{Th} = I Z_2 = \frac{10}{16} \times (8-j8)$$

$$= 7.07 \angle -45^\circ$$

2. Calculation of Z_{Th}

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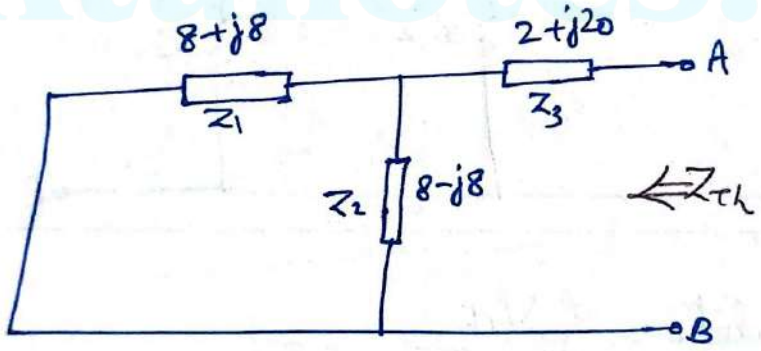

$$Z_{Th} = (2+j20) + \left((8+j8) // (8-j8)\right)$$

$$= (2+j20) + \frac{(8+j8)(8-j8)}{8+j8+8-j8}$$

$$= 22.36 \angle 63.43^\circ$$

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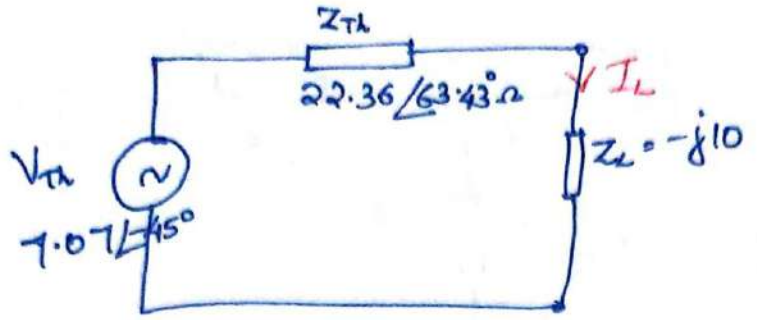
3. Thevenin's equivalent circuit

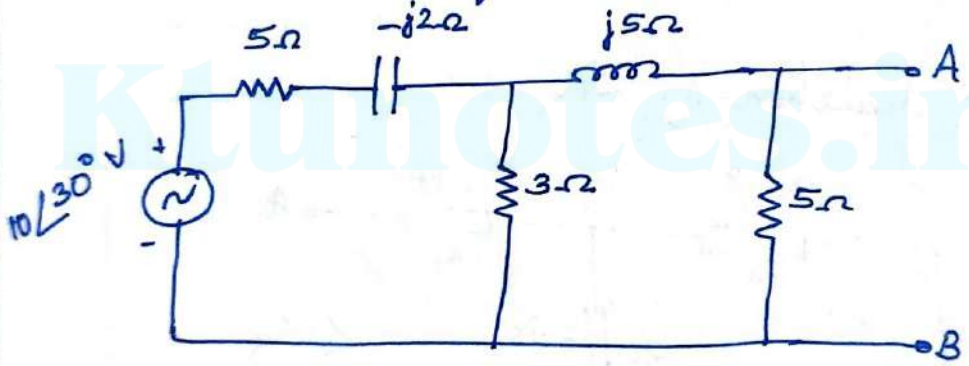


$$I_L = \frac{V_{Th}}{Z_{Th}+Z_L} = \frac{7.07\angle -45}{(22.36\angle 63.43^\circ - j10)}$$

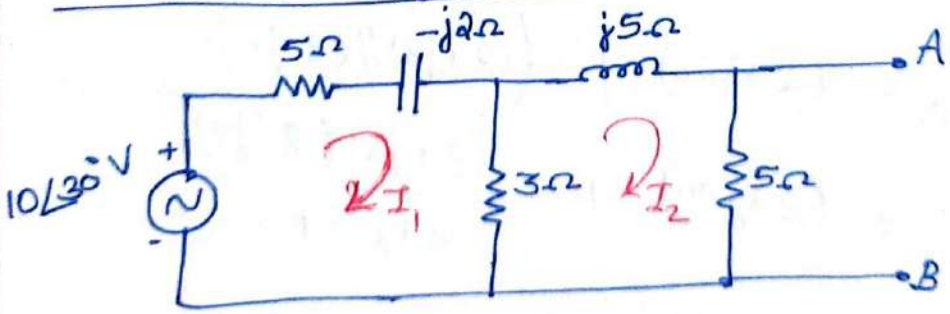
$$= \underline{\underline{0.5\angle -90^\circ}}\ A$$

---

3. Find Thevenin equivalent network



1. Calculation of V_{Th}



KVL for mesh 1

$$10\angle 30^\circ - (5 + -j2)I_1 - 3(I_1 - I_2) = 0$$

$$(8-j2)I_1 - 3I_2 = 10\angle 30^\circ \quad \text{——— (1)}$$

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KVL to Mesh 2

$$-3(I_2 - I_1) - j5 I_2 - 5 I_2 = 0$$

$$-3 I_1 + (8+j5) I_2 = 0 \quad \text{———} \quad (2)$$

Eqns (1) & (2) in Matrix Form

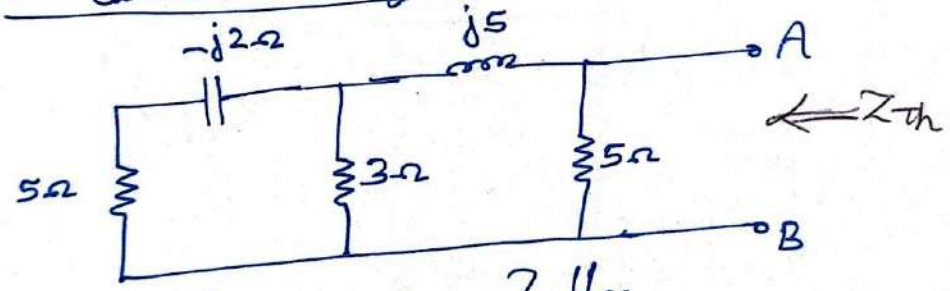
$$\begin{bmatrix} (8-j2) & -3 \\ -3 & 8+j5 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} 10\angle 30^\circ \\ 0 \end{bmatrix}$$

By Cramer's Rule

$$I_2 = \frac{\begin{vmatrix} (8-j2) & 10\angle 30 \\ -3 & 0 \end{vmatrix}}{\begin{vmatrix} (8-j2) & -3 \\ -3 & (8+j5) \end{vmatrix}} = \frac{3 \times 10\angle 30}{(8-j2)(8+j5)-9}$$

$$= \underline{\underline{0.433\angle 9.7^\circ \text{ V}}}$$

2. Calculation of Z_{Th}



$$Z_{Th} = \left\{ \left[(5-j2) \| 3 \right] + j5 \right\} \| 5$$

$$= \left(\frac{(5-j2) \times 3}{8-j^2} + j5 \right) \| 5 = (1.94 + j4.735) \| 5$$

$$= \frac{(1.94 + j4.735)5}{1.94 + j4.735 + 5} = \underline{\underline{3.04\angle 33.4^\circ \Omega}}$$

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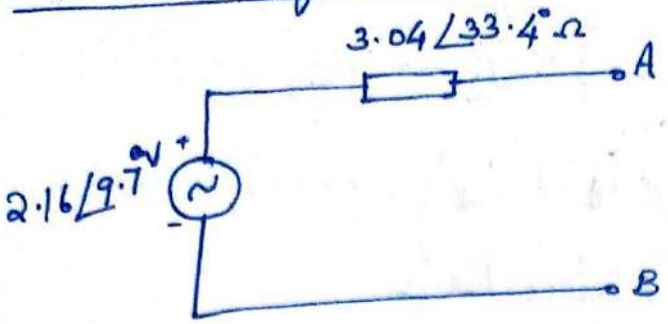
3. Thevenin's equivalent network



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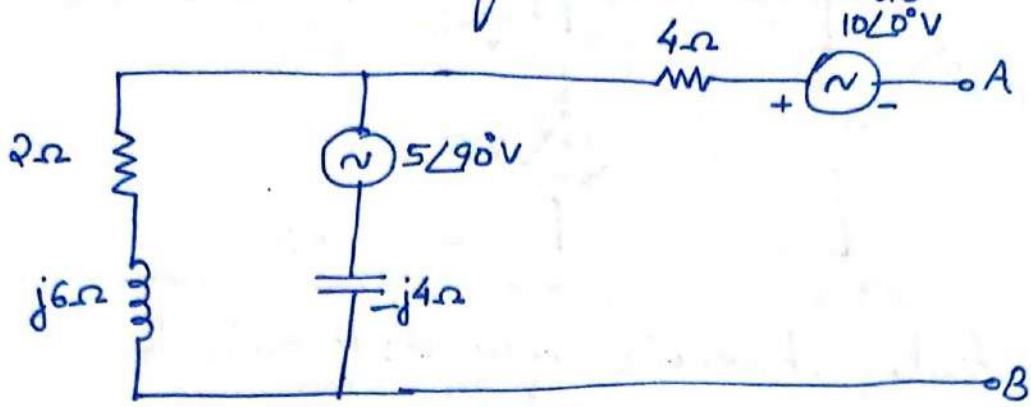
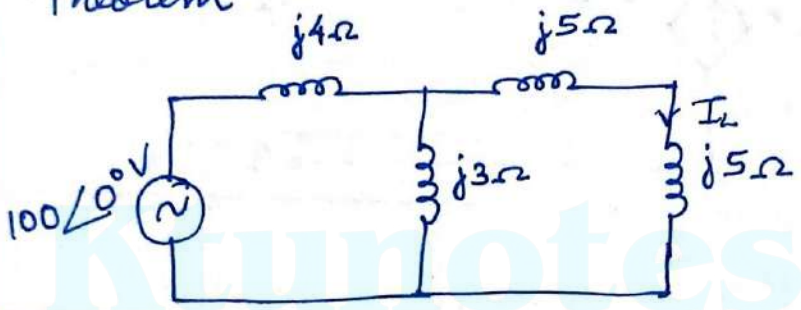
Tutorial Questions from Thevenins (AC)

1. Obtain Thevenin's equivalent network



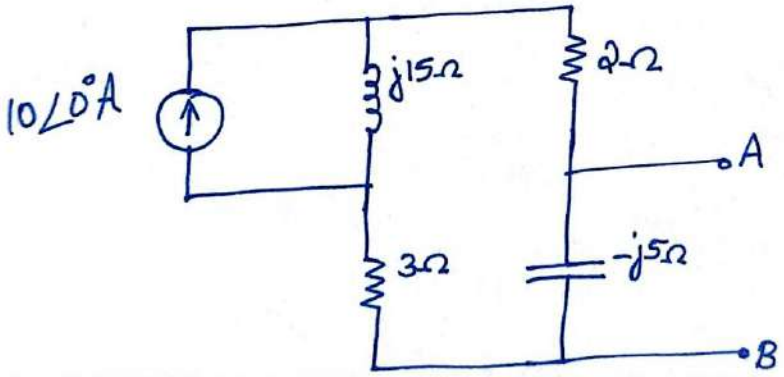
$V_{Th} = 5\angle -180^\circ$
$Z_{Th} = 11.3\angle -45^\circ$

2. Determine the load current by applying Thevenin's Theorem



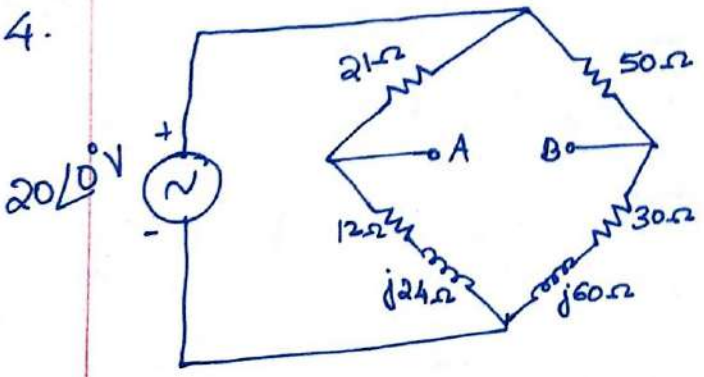
$V_{Th} = 42.86\angle 0$,
$Z_{Th} = 6.71\angle 90^\circ$
$I_L = 3.66\angle -90^\circ$

3. Obtain Thevenin's equivalent network.



$V_{Th} = 67.\cancel{3}\angle -63.43^\circ$
$Z_{Th} = 7.07\angle -81.86^\circ\Omega$

4.



Obtain Thevenin's equivalent network.

$V_{Th} = 0.33\angle -8.97^\circ$ V
$Z_{Th} = 47.35\angle 26.8^\circ\Omega$

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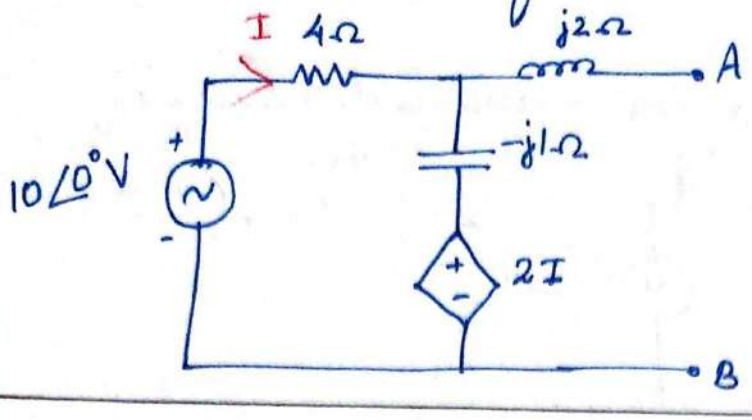
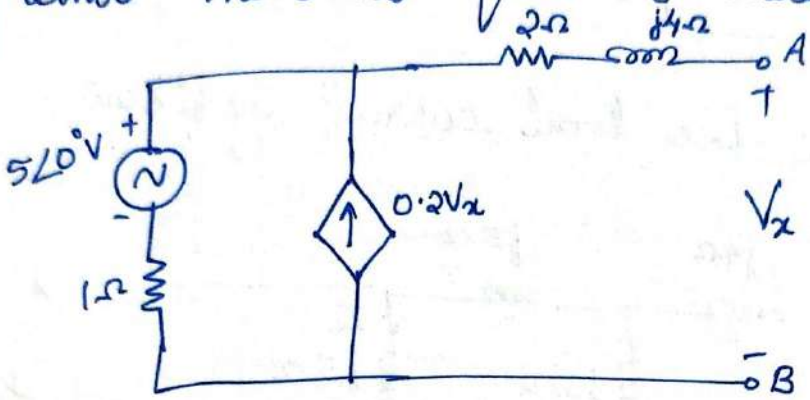
5. Obtain Thevenin's equivalent network across A & B



6. Find Thevenin's equivalent network



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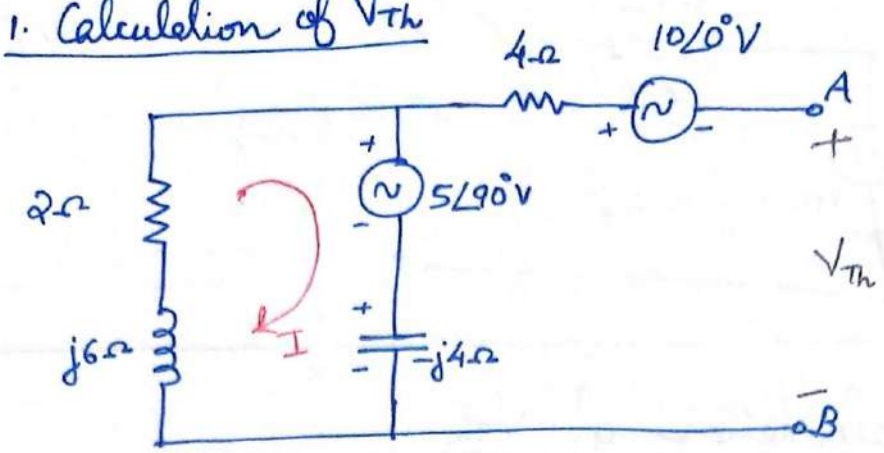
Answers to Tutorial Questions from Thevenin's Theorem (AC)

1. 1. Calculation of V_{Th}



$$-(2+j6-j4)I + 5\angle 90^\circ = 0$$

$$I = \frac{5\angle 90^\circ}{2+j2} = \underline{\underline{1.77\angle 45^\circ A}}$$

$$(-j4)I + 5\angle 90 + 0 - 10\angle 0^\circ - V_{Th} = 0$$

$$V_{Th} = -j4I + 5\angle 90 - 10\angle 0$$

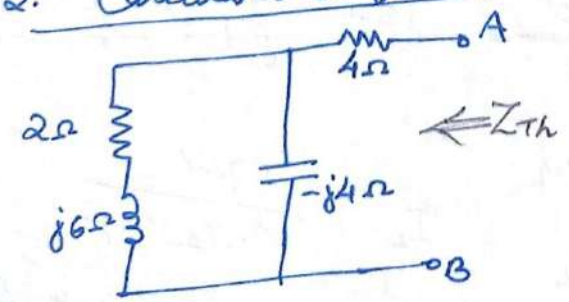
$$= (-j4) \times 1.77\angle 45^\circ + 5\angle 90^\circ - 10\angle 0^\circ$$

$$= 4\angle -90 \times 1.77\angle 45 + j5 - 10$$

$$= 7.08\angle -45 + j5 - 10 = 5 - j5 + j5 - 10$$

$$= \underline{\underline{5\angle -180^\circ}} \qquad \underline{\underline{-5}}$$

2. Calculation of Z_{Th}



$$Z_{Th} = 4 + \frac{(2+j6) \times -j4}{(2+j6) - j4}$$

$$= \underline{\underline{11.3\angle -45^\circ}}$$

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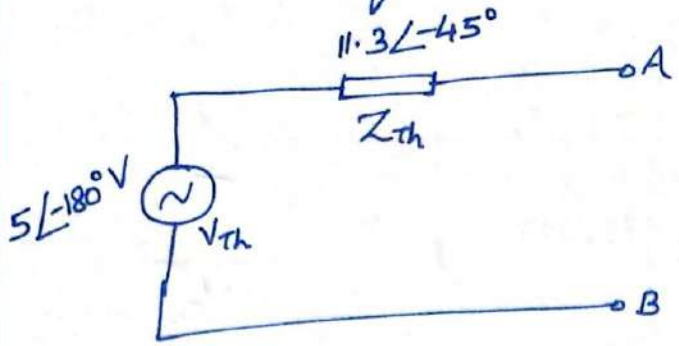
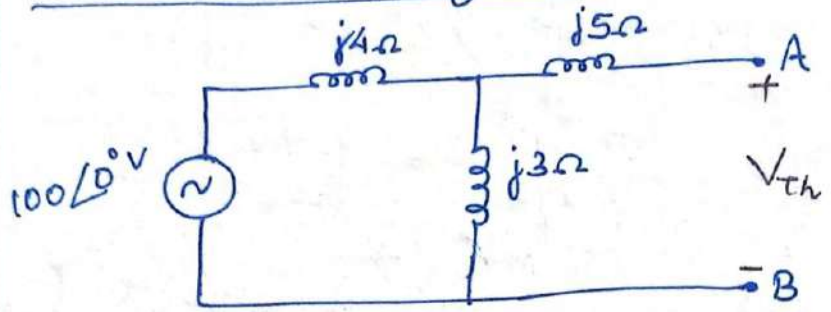
3. Thevenin's equivalent Circuit



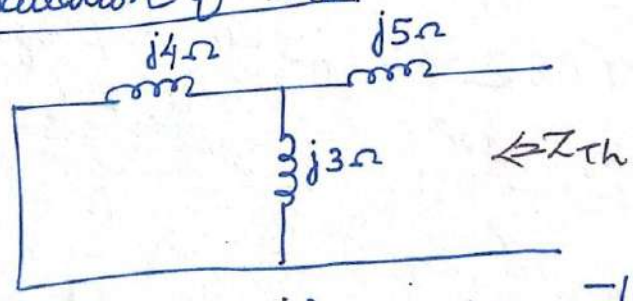
2. 1. Calculation of V_{Th}



$$V_{Th} = \frac{100\angle 0}{(j4+j3)} \times j3 = \frac{100\angle 0}{j7} \times j3$$

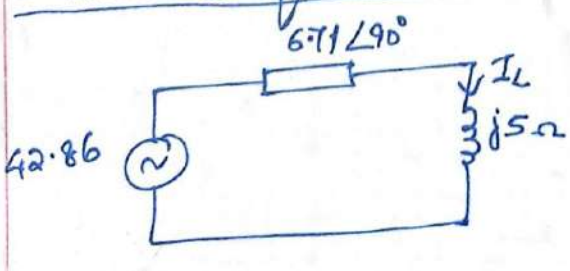
$$= \frac{100 \times 3}{7} = \underline{\underline{42.86\angle 0^\circ}}\ V$$

2. Calculation of Z_{Th}



$$Z_{Th} = j5 + \frac{j4 \times j3}{j4+j3} = j5 + \frac{-12}{j7} = \underline{\underline{6.71\angle 90^\circ}}$$

3. Thevenin's equivalent N/w



$$I_L = \frac{42.86}{6.71\angle 90 + j5}$$

$$= \underline{\underline{3.66\angle -90^\circ}}\ A$$

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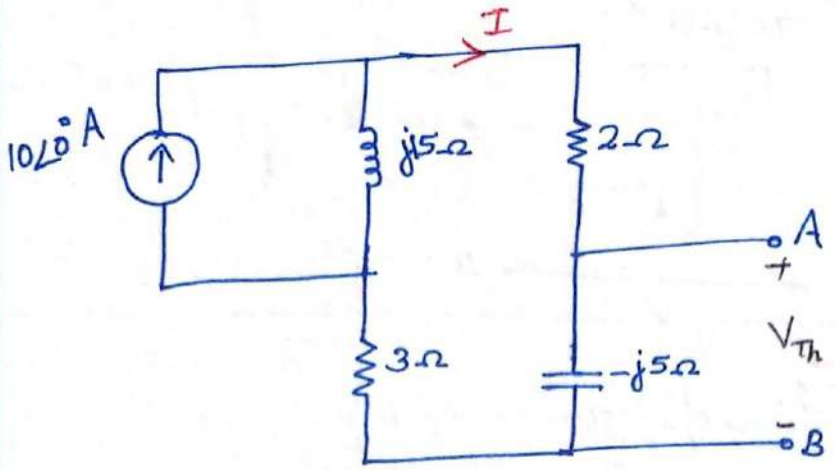
3.

1. Calculation of V_{Th}



$$I = \frac{10\angle 0 \times j15}{j15 + 2 - j5 + 3} = \frac{10\angle 0 \times j15}{(5+j10)}$$

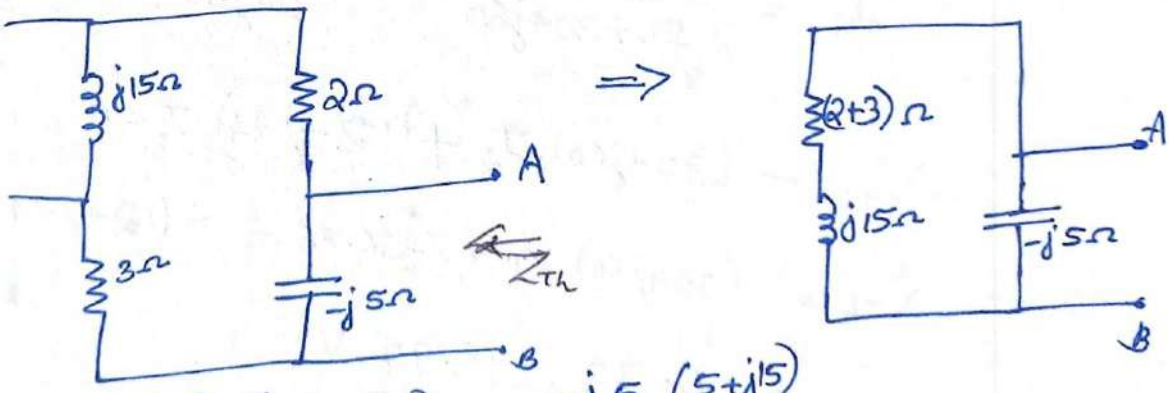
$$= \underline{\underline{13.42\angle 26.57^\circ \text{ A}}}$$

$$V_{Th} = (-j5)\, I$$

$$= (-j5)\,(13.42\angle 26.57^\circ)$$

$$= \underline{\underline{67.1\angle -63.43^\circ}}$$

2. Calculation of Z_{Th}



$$Z_{Th} = (-j5) \,//\, (5+j15) = \frac{-j5\,(5+j15)}{-j5+5+j15}$$

$$= \frac{-j5\,(5+j15)}{(5+j10)} = \underline{\underline{7.07\angle -81.86^\circ\,\Omega}}$$

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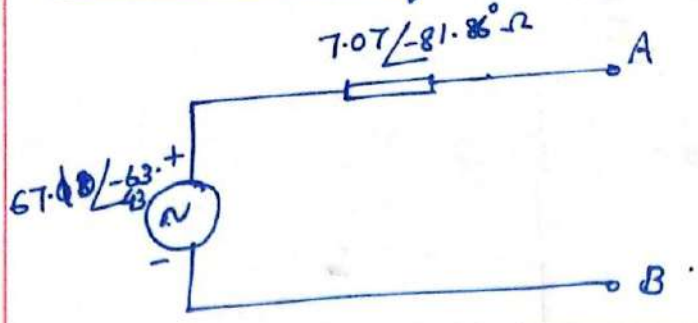
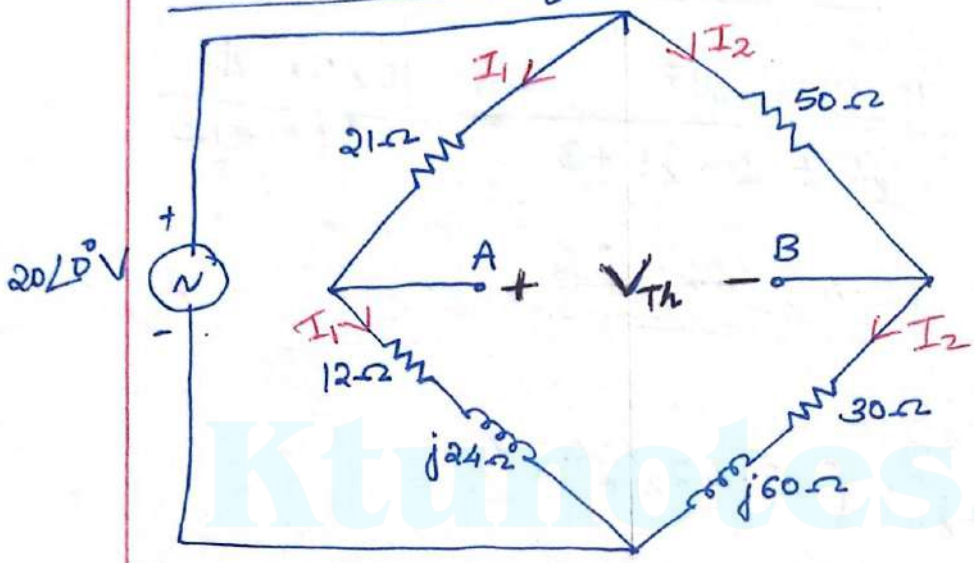
3. Thevenin's Equivalent Network



4. 1. Calculation of V_{Th}



$$I_1 = \frac{20\angle 0}{21+12+j24} = \frac{20\angle 0}{33+j24} = \underline{\underline{0.49\angle -36.02^\circ A}}$$

$$I_2 = \frac{20\angle 0}{50+30+j60} = \frac{20\angle 0}{80+j60} = \underline{\underline{0.2\angle -36.86^\circ A}}$$

$$-V_{Th} - (30+j60)\, I_2 + (12+j24)\, I_1 = 0$$

$$V_{Th} = (30+j60)\, 0.2\angle -36.86 - (12+j24)\, 0.49\angle -36.02$$

$$= \underline{\underline{0.33\angle -8.97^\circ V}}$$

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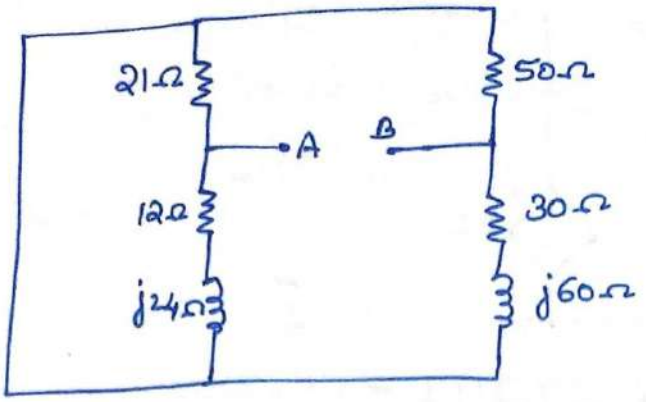
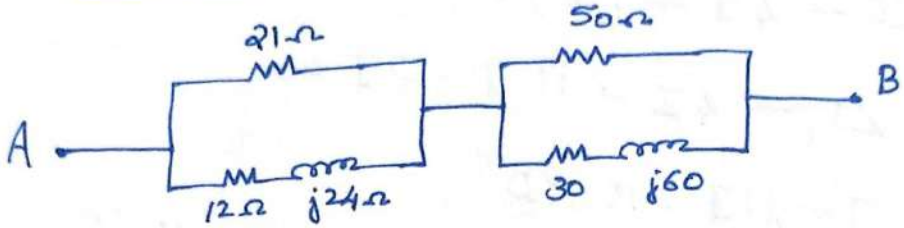
2. Calculation of Z_{Th}





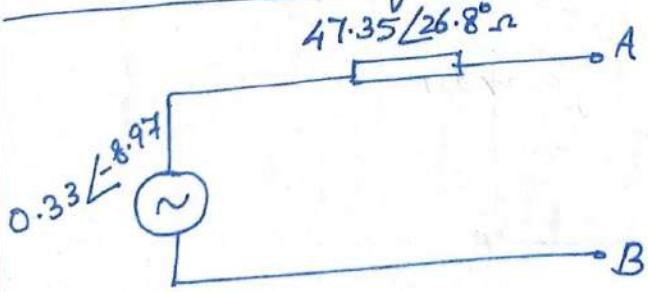
$$Z_{Th} = 21 \,||\, (12+j24) + 50 \,||\, (30+j60)$$

$$= \frac{21(12+j24)}{21+12+j24} + \frac{50(30+j60)}{50+30+j60}$$

$$= \frac{21(12+j24)}{33+j24} + \frac{50(30+j60)}{80+j60}$$

$$= \underline{\underline{47.35 \angle 26.8^\circ \, \Omega}}$$

3. Thevenin's equivalent network



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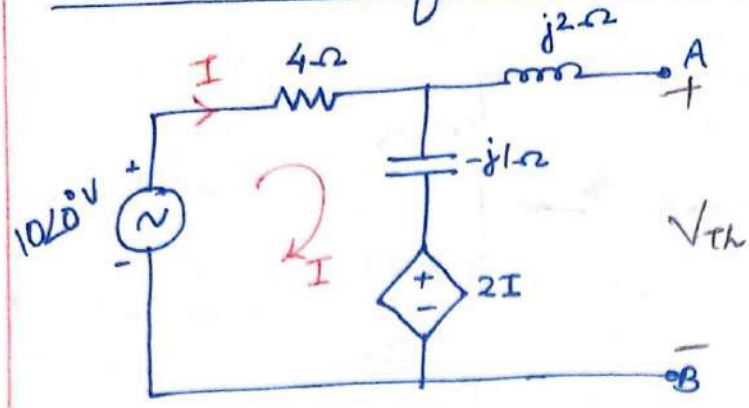
5. 1. Calculation of V_{Th}



$$10\angle 0 - 4I - (-j1)I - 2I = 0$$

$$10\angle 0 - 4I + j1\,I - 2I = 0$$

$$6I - j1I = 10\angle 0$$

$$(6-j1)I = 10\angle 0; \qquad I = \frac{10\angle 0}{(6-j1)} = \underline{\underline{1.64\angle 9.46^\circ A}}$$

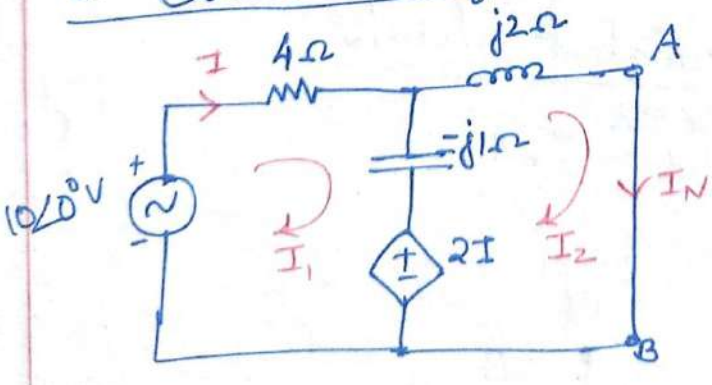
V_{Th} equation

$$10\angle 0 - 4I - 0 - V_{Th} = 0$$

$$V_{Th} = 10\angle 0 - 4I = 10\angle 0 - (1.64\angle 9.46)4$$

$$= \underline{\underline{3.69\angle -17}}\ V$$

2. Calculation of I_N



KVL to Mesh 1

$\cancel{10\angle 0 - 4I_1 + jI_1 - 2I_1 = 0}$

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KVL to Mesh 1

$$10\angle 0 - 4I_1 - -j1(I_1 - I_2) - 2I_1 = 0$$

$$10\angle 0 - 4I_1 + j1(I_1 - I_2) - 2I_1 = 0$$

$$(6 - j1)I_1 + j1 I_2 = 10\angle 0 \quad \text{———} \quad (1)$$

KVL to mesh 2

$$2I - -j1(I_2 - I_1) - j2 I_2 = 0$$

$$2I_1 + j(I_2 - I_1) - j2 I_2 = 0$$

$$(2 - j1)I_1 - j1 I_2 = 0 \quad \text{———} \quad (2)$$

$$\begin{bmatrix} 6-j1 & j1 \\ 2-j1 & -j1 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} 10\angle 0 \\ 0 \end{bmatrix}$$

$$I_2 = \frac{\begin{vmatrix} 6-j1 & 10\angle 0 \\ 2-j1 & 0 \end{vmatrix}}{\begin{vmatrix} 6-j1 & j1 \\ 2-j1 & -j1 \end{vmatrix}} = \frac{-10\angle 0\,(2-j1)}{-j(6-j1) - j1(2-j1)}$$

$$= \frac{-10\angle 0\,(2-j1)}{-6j - 1 - j2 - 1} = \frac{-10\angle 0\,(2-j1)}{(-2-j8)}$$

$$= \underline{\underline{2.71\angle -102.53^\circ A}}$$

$$I_N = I_2 = \underline{\underline{2.71\angle -102.53^\circ A}}.$$

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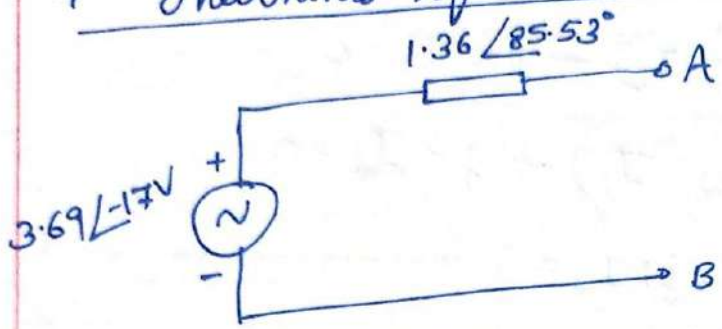
3. Calculation of Z_{Th}

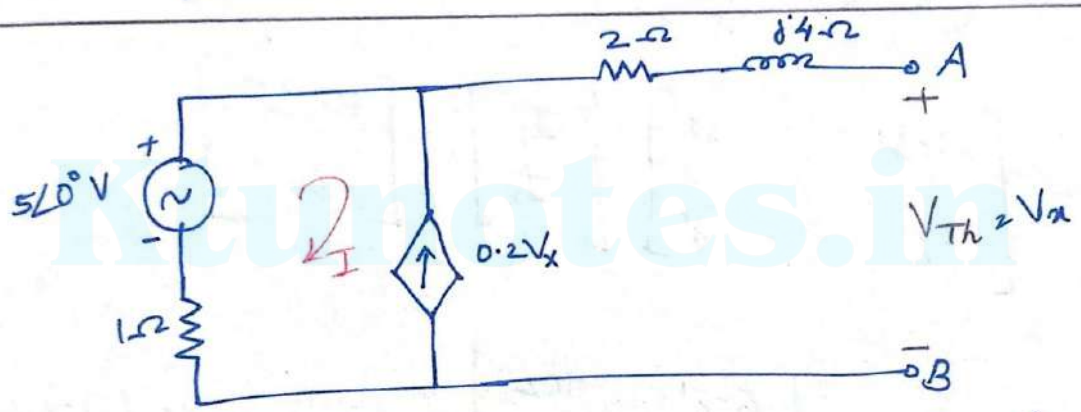
$$Z_{Th} = \frac{V_{Th}}{I_N} = \frac{3.69\angle -17}{2.71\angle -102.53^\circ}$$

$$= 1.36\angle 85.53^\circ\ \Omega$$

4. Thevenins Equivalent Circuit



6.



$V_{Th} = V_x$

$$I = -0.2\,V_x$$

V_{Th} eqn

$$-I + 5\angle 0 - 0 - V_{Th} = 0$$

$$V_{Th} = 5\angle 0 - I$$

$$V_x = 5 + 0.2V_x$$

$$V_x = \frac{5\angle 0}{0.8} = 6.25\angle 0^\circ$$

$$V_{Th} = V_x = 6.25\angle 0^\circ\ V$$

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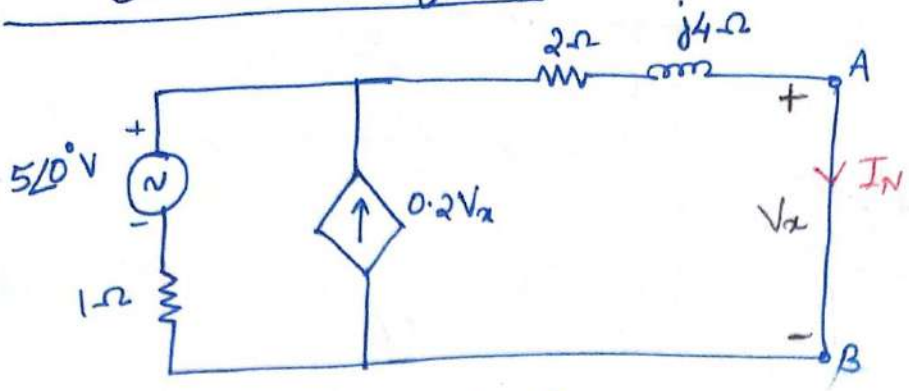
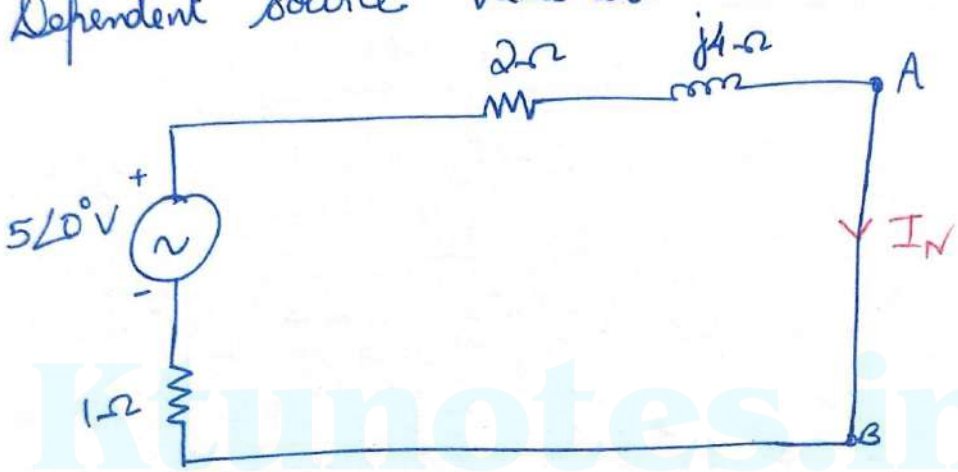
2. Calculation of I_N



As $V_x = 0$, $0.2V_x = 0$.

Dependent source vanishes



$$I_N = \frac{5\angle 0}{1+2+j4} = \frac{5\angle 0}{3+j4} = \underline{\underline{1\angle -53.13^\circ \text{ A}}}$$

3. Calculation of Z_{Th}

$$Z_{Th} = \frac{V_{Th}}{I_N} = \frac{6.25\angle 0^\circ}{1\angle -53.13^\circ}$$

$$= \underline{\underline{6.25\angle 53.13^\circ\ \Omega}}$$

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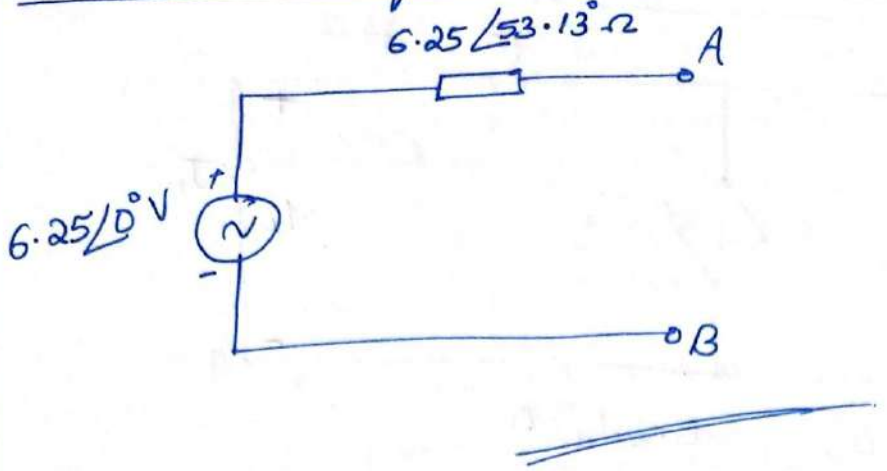
4. Thevenin's equ. network.



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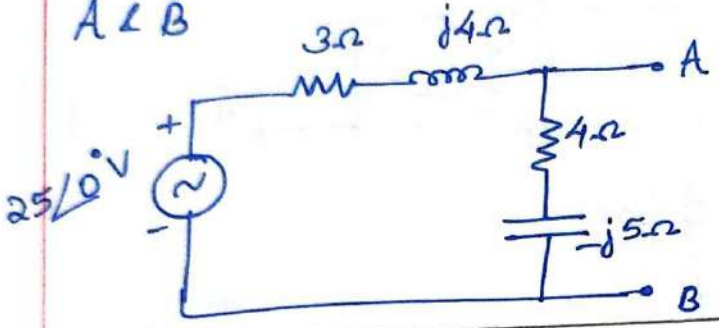
NORTON'S THEOREM (AC)

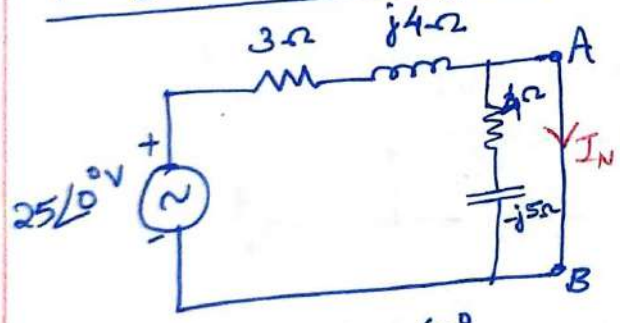
Any linear network can be replaced by a current source I_N parallel with an impedance Z_N, where I_N is the current flowing through the short circuited path placed across the terminals.

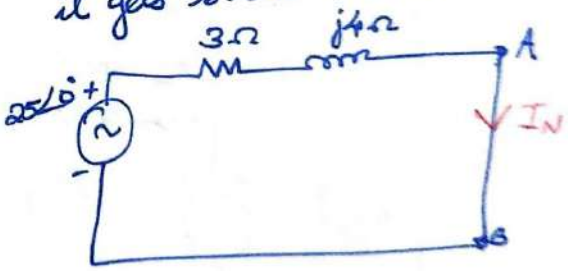
1. Obtain Nortons equivalent network between A & B



1. Calculation of I_N

when a short circuit is placed across $(4-j5)\Omega$, it gets short circuited.

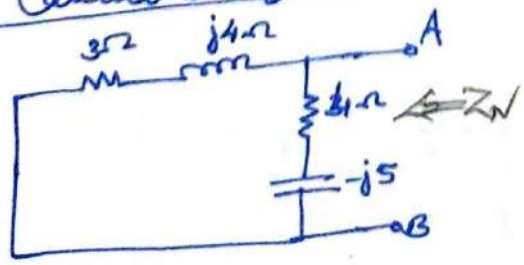




$$I_N = \frac{25\angle 0^\circ}{3+j4}$$

$$= 5\angle -53.13^\circ \text{ A}$$

2. Calculation of Z_N



$$Z_N = \frac{(3+j4)(4-j5)}{(3+j4)+(4-j5)}$$

$$= \frac{(3+j4)(4-j5)}{(7-j1)}$$

$$= 4.53\angle 9.92^\circ\ \Omega$$

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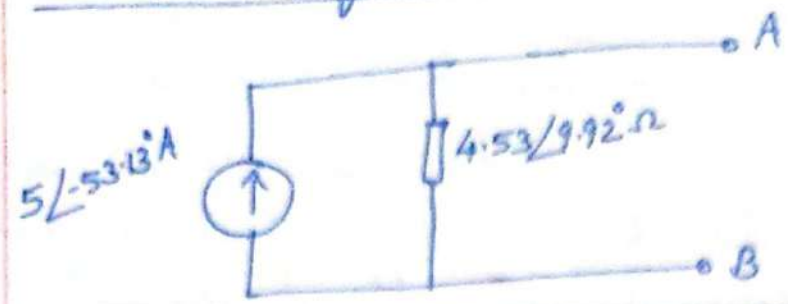
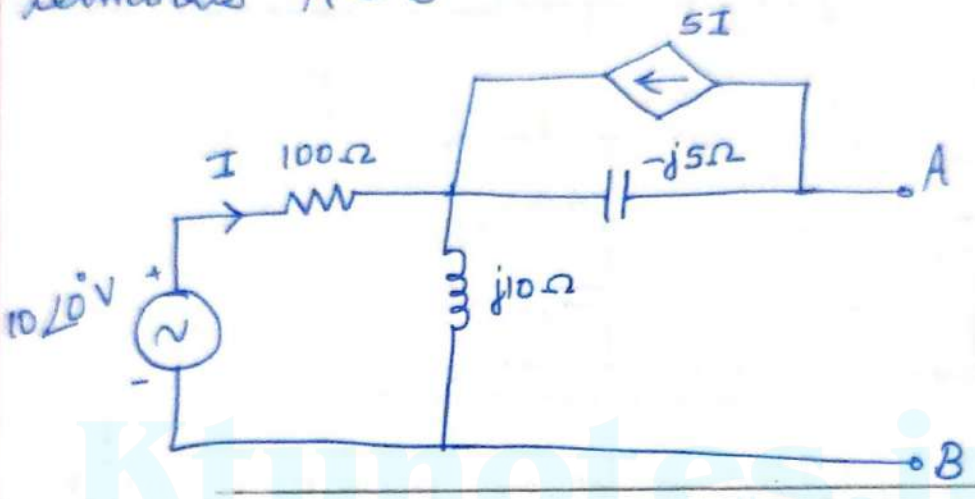
3. Norton's Equivalent Network



2. Obtain Norton's equivalent network across terminals A & B.



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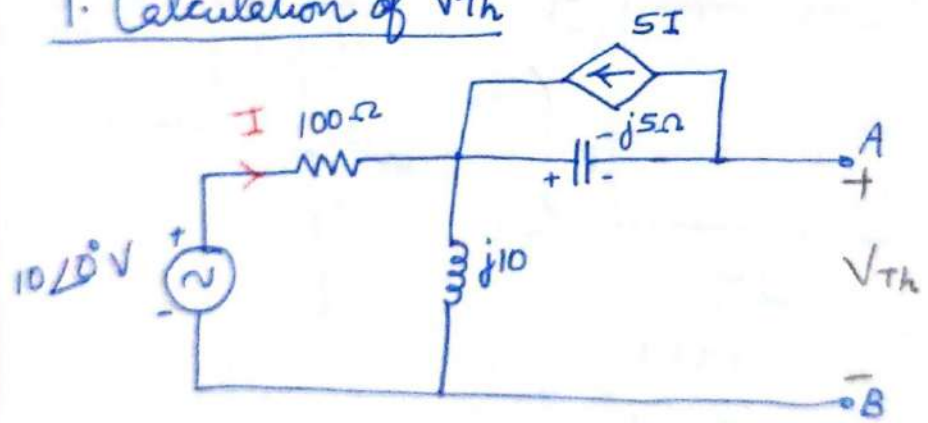
1. Calculation of V_{Th}



$$I = \frac{10\angle 0}{100 + j10} = 0.1\angle -5.71\ A$$

V_{Th} equation

$$10\angle 0 - 100I - (-j5)\,5I - V_{Th} = 0$$

$$10\angle 0 - 100I + j25I = V_{Th}$$

$$V_{Th} = 10\angle 0 - (100 - j25)\,0.1\angle -5.71 = 3.5\angle 85.1^\circ\ V$$

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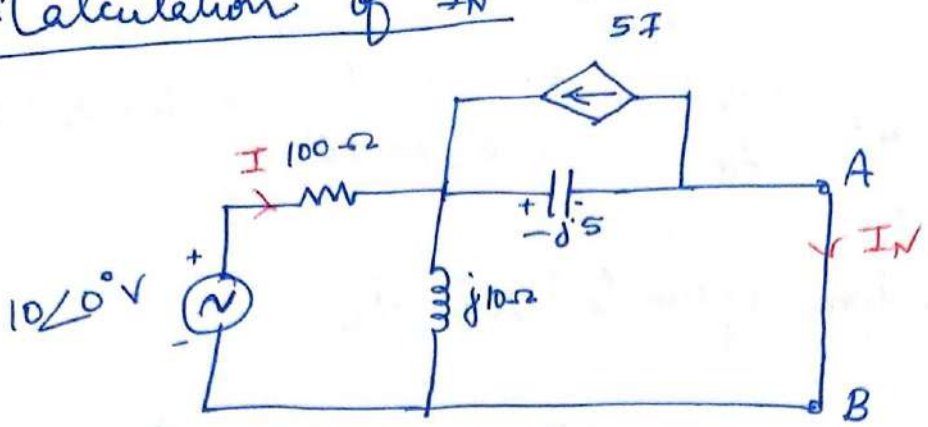
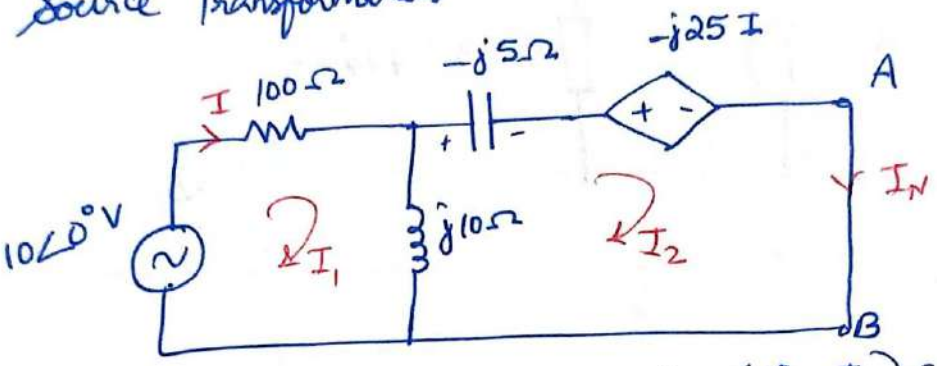
2. Calculation of I_N



Source Transformation



$I = I_1$

Mesh 1 → $10\angle 0 - 100 I_1 - j10(I_1 - I_2) = 0$

$$(100 + j10) I_1 - j10 I_2 = 10\angle 0^\circ \quad \text{——— (1)}$$

Mesh 2 → $-j10(I_2 - I_1) - -j5 I_2 - -j25 I = 0$

$$-j10 I_2 + j10 I_1 + j5 I_2 + j25 I_1 = 0$$

$$j35 I_1 - j5 I_2 = 0 \quad \text{——— (2)}$$

Matrix Form

$$\begin{bmatrix} 100+j10 & -j10 \\ j35 & -j5 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} 10\angle 0 \\ 0 \end{bmatrix}$$

$$I_N = I_2 = \frac{\begin{vmatrix} 100+j10 & 10\angle 0 \\ j35 & 0 \end{vmatrix}}{\begin{vmatrix} 100+j10 & -j10 \\ j35 & -j5 \end{vmatrix}} = \frac{-j350}{(100+j10)(-j5) + j^2 350}$$

$$= 0.6\angle 30.96^\circ \text{ A}$$

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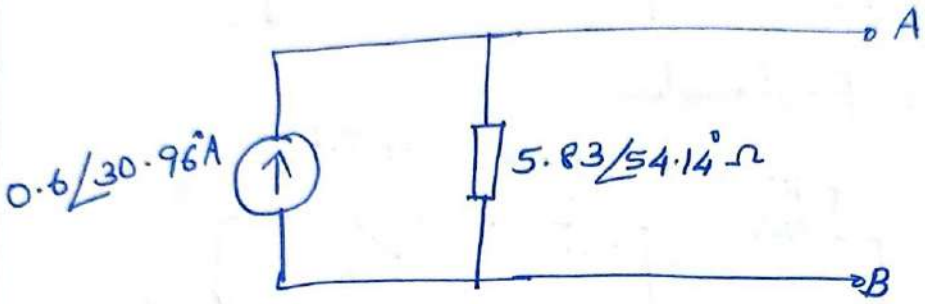
3. Calculation of Z_N

$$Z_N = \frac{V_{Th}}{I_N} = \frac{3.5\angle 85.1}{0.6\angle 30.96^\circ} = \underline{\underline{5.83\angle 54.14^\circ\ \Omega}}$$

4. Norton's equivalent Network



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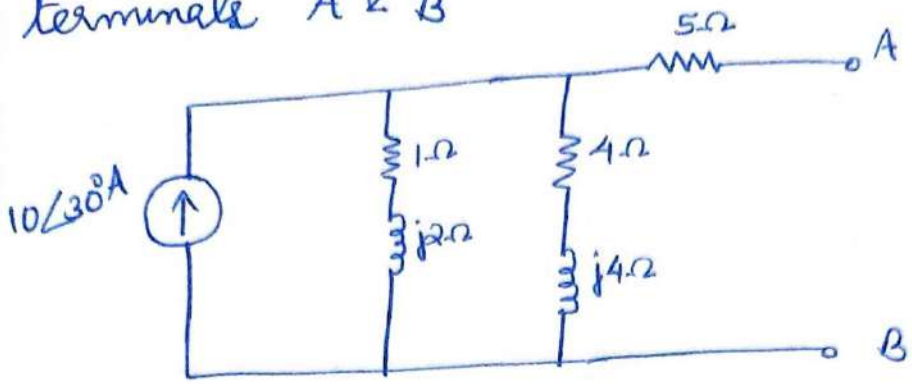
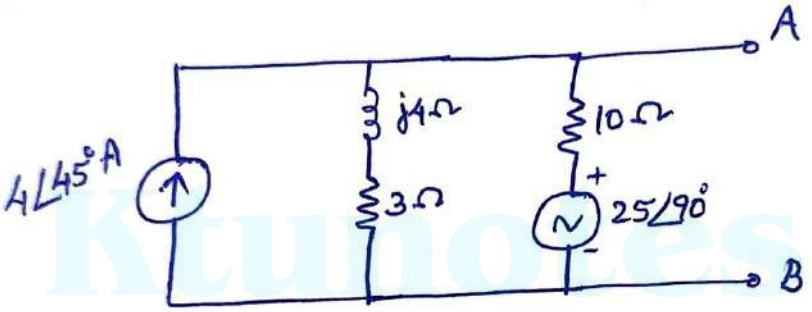
Tutorial Questions – Nortons (AC)

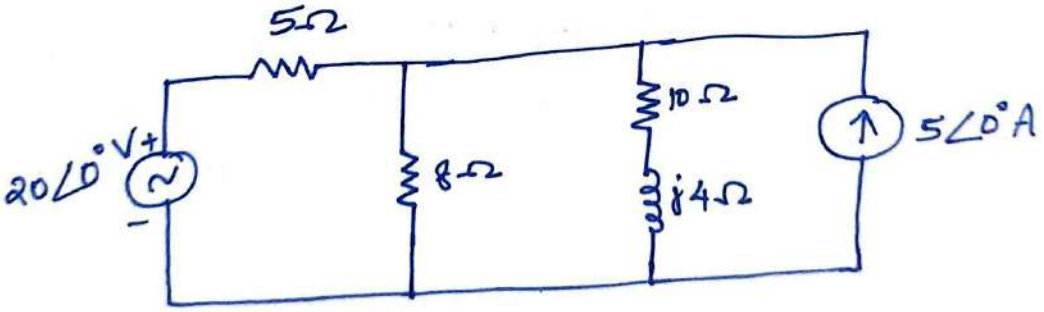
1. Obtain Nortons equivalent network at the terminals A & B



2. Find Nortons equivalent network



3. Find the current through 8Ω resistor



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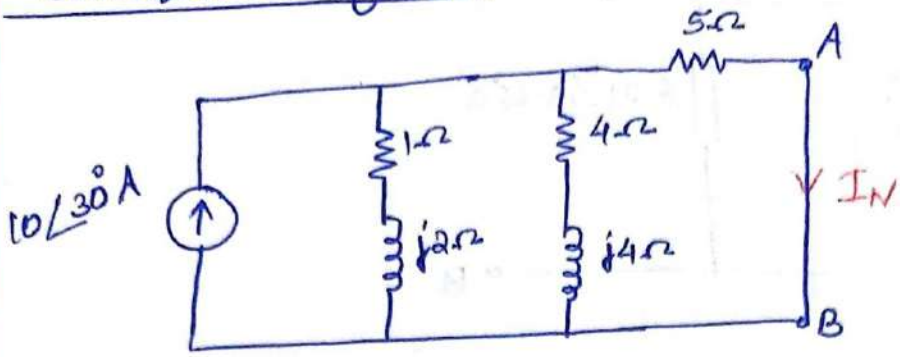
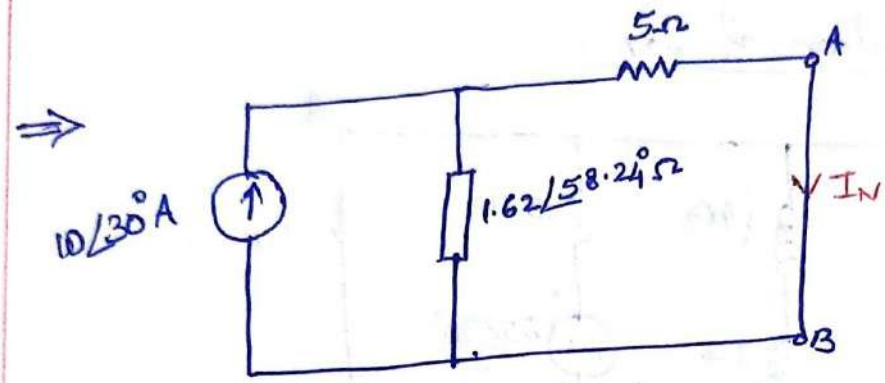
Answers → Tutorial - Nortons (AC)

1. 1. Calculation of I_N



$$\frac{(4+j4)(1+j2)}{5+j6}$$

⇒



$$I_N = 10\angle 30 \times \frac{1.62\angle 58.24}{1.62\angle 58.24 + 5}$$

$$= \underline{\underline{2.69\angle 75^\circ}}\ A$$

2. Calculation of Z_N

$$Z_N = 5 + \frac{(1+j2)(4+j4)}{(5+j6)} = \underline{\underline{6.01\angle 13.24^\circ}}$$

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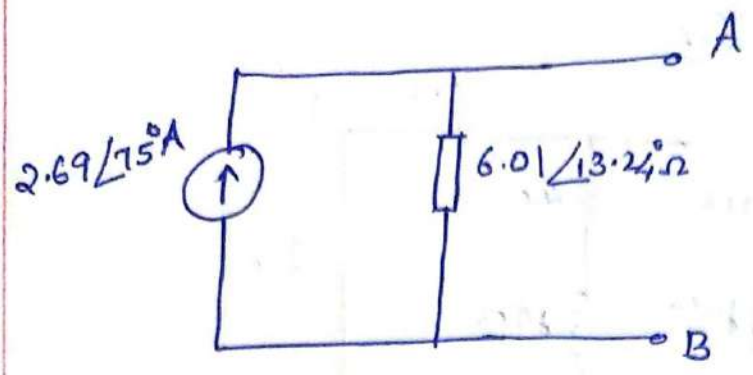
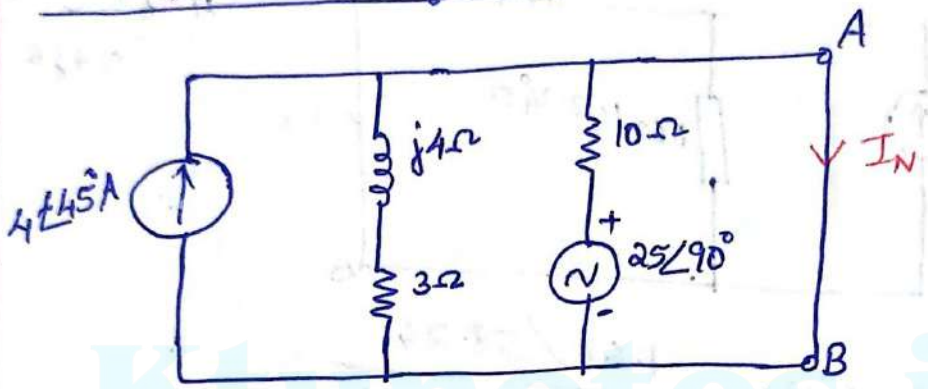
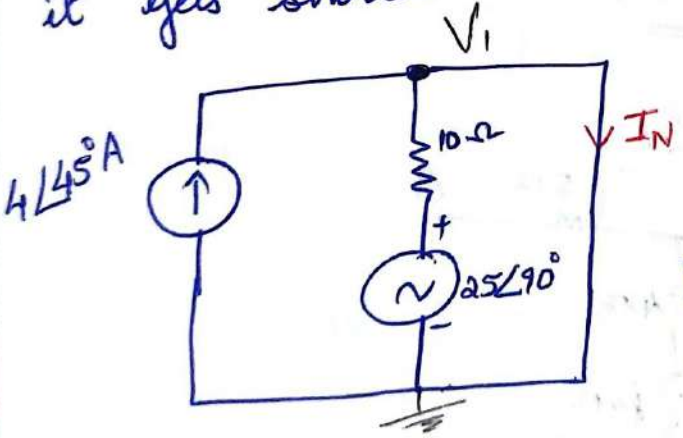
3. Nortons equivalent Circuit



2. 1. Calculation of I_N



when a short Circuit is placed across $(3+j4)\Omega$, it gets shorted.



KCL at Node 1

$$4\angle 45^\circ = \frac{V_1 - 25\angle 90^\circ}{10} + I_N$$

From fig. it is clear that $V_1 = 0$

So $$4\angle 45 = \frac{-25\angle 90^\circ}{10} + I_N$$

$$I_N = 4\angle 45 + \frac{25\angle 90^\circ}{10} = 4\angle 45 - 2.5\angle 90^\circ$$

$$= \underline{\underline{6.03\angle 62.04^\circ A}}$$

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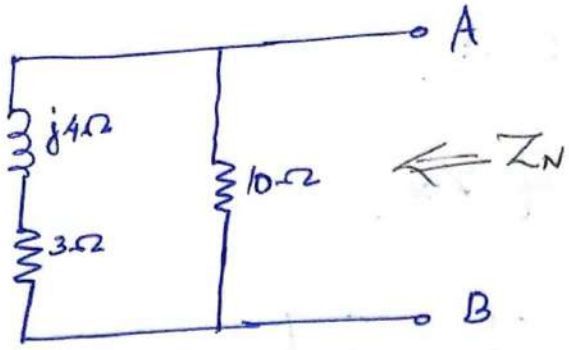
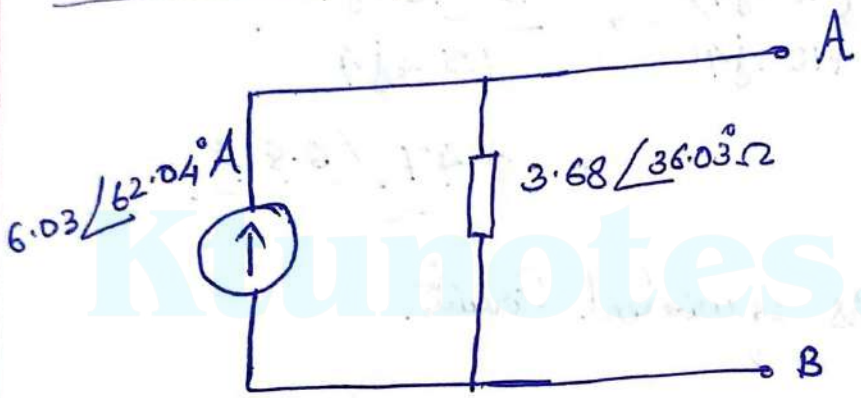
2. Calculation of Z_N



$$Z_N = \frac{10(3+j4)}{10+(3+j4)} = \underline{\underline{3.68\angle 36.03^\circ\,\Omega}}$$

3. Norton equ. Circuit



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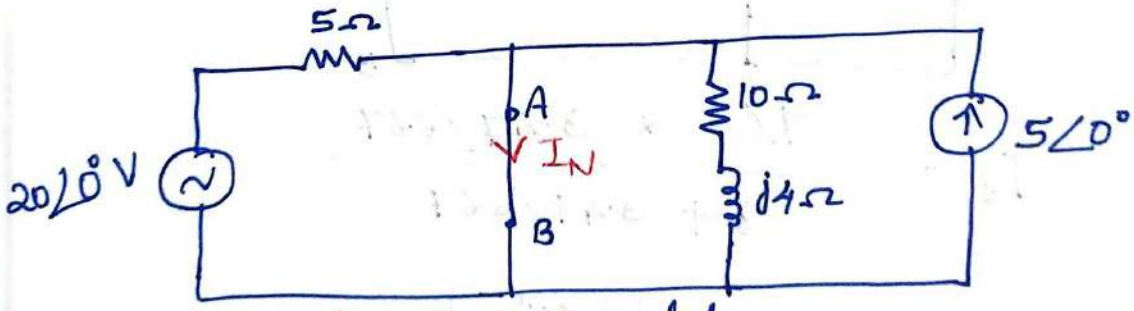
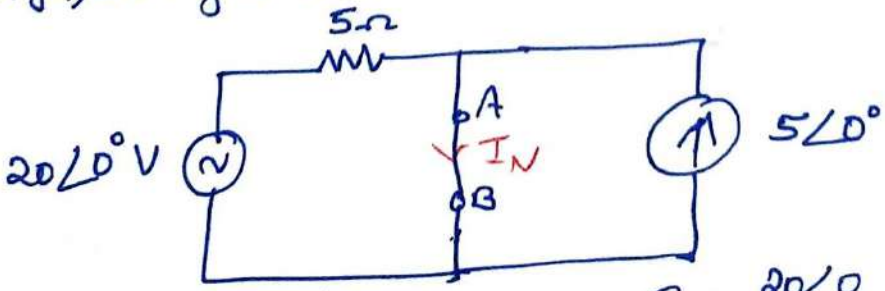
3. 1. Calculation of I_N



$(10+j4)\,\Omega$, gets short circuited



$$I_N = \frac{20\angle 0}{5} + 5\angle 0^\circ$$

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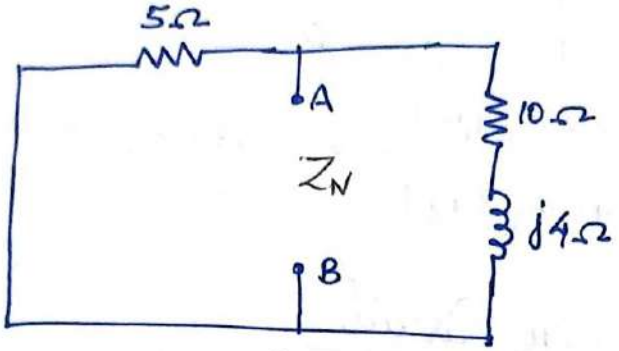
$$I_N = 4\angle 0^\circ + 5\angle 0^\circ$$

$$= \underline{\underline{9\angle 0^\circ \text{ A}}}$$

2. Calculation of Z_N



$$Z_N = \frac{5(10+j4)}{5+10+j4} = \frac{5(10+j4)}{15+j4}$$

$$= \underline{\underline{3.47\angle 6.87^\circ}}$$

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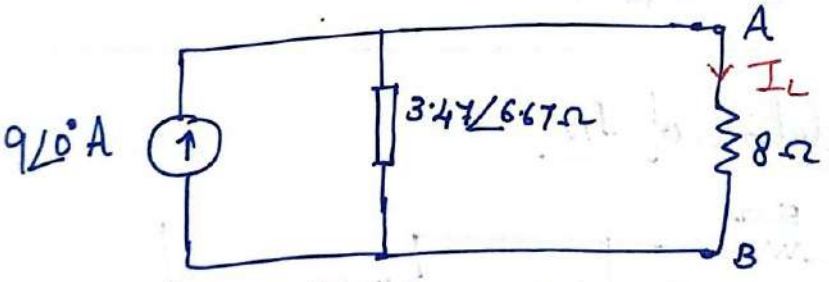
3. Norton's equivalent Circuit



$$I_L = \frac{9\angle 0 \times 3.47\angle 6.67}{8 + 3.47\angle 6.67}$$

$$= \underline{\underline{2.73\angle 4.65^\circ \text{ A}}}$$

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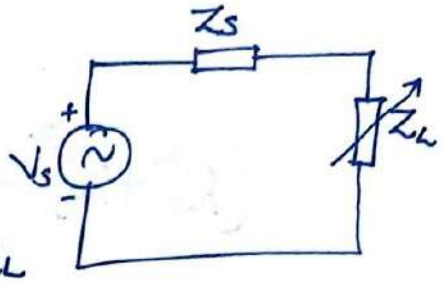
MAXIMUM POWER TRANSFER THEOREM (AC)

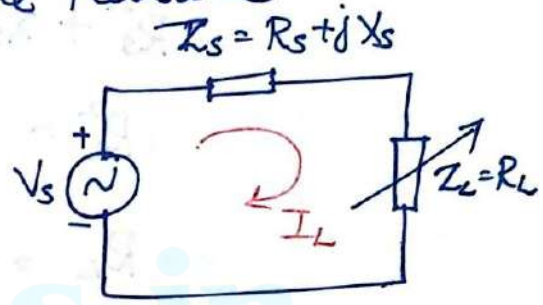
This theorem is used to determine the value of load impedance for which the source will transfer maximum power.

Consider a simple network



There are 3 possible cases for Z_L

Case 1 When Z_L is variable Resistance



$$I_L = \frac{V_S}{Z_S + Z_L}$$

$$= \frac{V_S}{R_S + jX_S + R_L}$$

$$|I_L| = \frac{|V_S|}{\sqrt{(R_S+R_L)^2 + X_S^2}}$$

Power delivered to the load is

$$P_L = |I_L|^2 R_L = \frac{|V_S|^2 R_L}{\sqrt{(R_S+R_L)^2 + X_S^2}}$$

For Power to be maximum,

$$\frac{dP}{dR_L} = 0$$

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$$\frac{dP}{dR_L} = 0$$

$$|V_s|^2 \cdot \frac{\{(R_s+R_L)^2 + X_s^2\} \times 1 - 2R_L(R_s+R_L)}{[(R_s+R_L)^2 + X_s^2]^2} = 0$$

$$(R_s+R_L)^2 + X_s^2 - 2R_L(R_s+R_L) = 0$$

$$R_s^2 + 2R_sR_L + R_L^2 + X_s^2 - 2R_sR_L - 2R_L^2 = 0$$

$$R_s^2 + X_s^2 - R_L^2 = 0$$

$$R_L^2 = R_s^2 + X_s^2$$

$$R_L = \sqrt{R_s^2 + X_s^2} = |Z_s|$$

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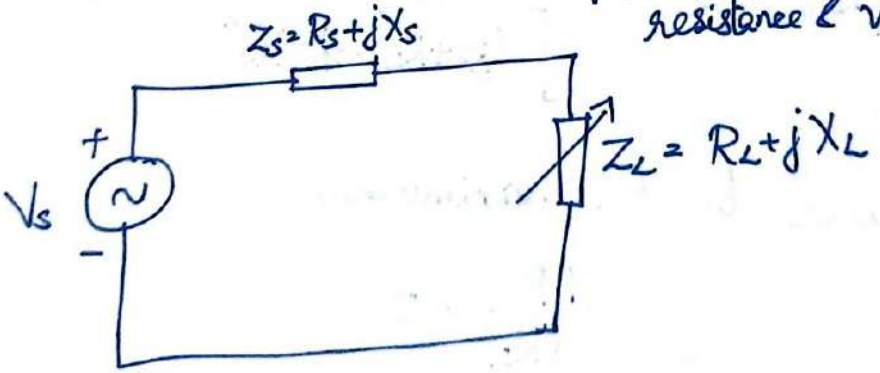
Hence, load resistance R_L should be equal to the magnitude of the source impedance for maximum power transfer.

Case 2:

When Z_L is a complex impedance with variable resistance & variable reactance



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$$I_L = \frac{V_s}{Z_s + Z_L}$$

$$|I_L| = \frac{|V_s|}{\sqrt{(R_s + R_L)^2 + (X_s + X_L)^2}}$$

Power delivered to the load is

$$P_L = |I_L|^2 R_L$$

$$= \frac{|V_s|^2 R_L}{(R_s + R_L)^2 + (X_s + X_L)^2}$$

For maximum P_L, denominator of the equation should be small, ie; $X_L = -X_s$

$$P_L = \frac{|V_s|^2 R_L}{(R_s + R_L)^2}$$

Diff. above equation w.r.to R_L & equating to zero,

$$\frac{dP_L}{dR_L} = |V_s|^2 \left\{ \frac{(R_s + R_L)^2 - 2R_L(R_s + R_L)}{(R_s + R_L)^4} \right\} = 0$$

$$(R_s + R_L)^2 - 2R_L(R_s + R_L) = 0$$

$$R_s^2 + R_L^2 + 2R_sR_L - 2R_sR_L - 2R_L^2 = 0$$

$$R_s^2 - R_L^2 = 0 \quad ; \quad \underline{\underline{R_L = R_s}}$$

$$\underline{\underline{R_L = R_s}} \quad \boxed{R_L = R_s \ \& \ X_L = -X_s}$$

So $\boxed{Z_L = Z_s^* = \underline{\underline{R_s - jX_s}}}$

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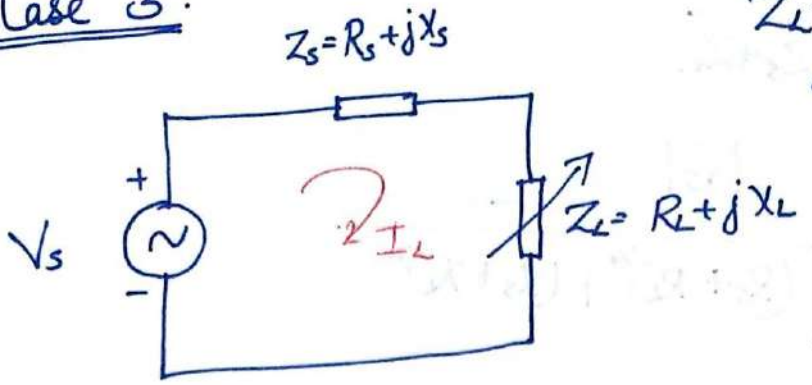
Case 3:

$Z_L \rightarrow$ Variable R_L & fixed X_L



$$I_L = \frac{V_s}{Z_s + Z_L}$$

$$|I_L| = \frac{|V_s|}{\sqrt{(R_s+R_L)^2 + (X_s+X_L)^2}}$$

Power delivered to load is

$$P_L = |I_L|^2 R_L$$

$$= \frac{|V_s|^2 R_L}{(R_s+R_L)^2 + (X_s+X_L)^2}$$

For maximum power $\frac{dP_L}{dR_L} = 0$

$$|V_s|^2 \frac{\left[(R_s+R_L)^2 + (X_s+X_L)^2\right] \times 1 - 2R_L(R_s+R_L)}{\left\{(R_s+R_L)^2 + (X_s+X_L)^2\right\}^2} = 0$$

$$(R_s+R_L)^2 + (X_s+X_L)^2 - 2R_L(R_s+R_L) = 0$$

$$R_s^2 + 2R_sR_L + R_L^2 + (X_s+X_L)^2 - 2R_sR_L - 2R_L^2 = 0$$

$$R_s^2 + (X_s+X_L)^2 - R_L^2 = 0$$

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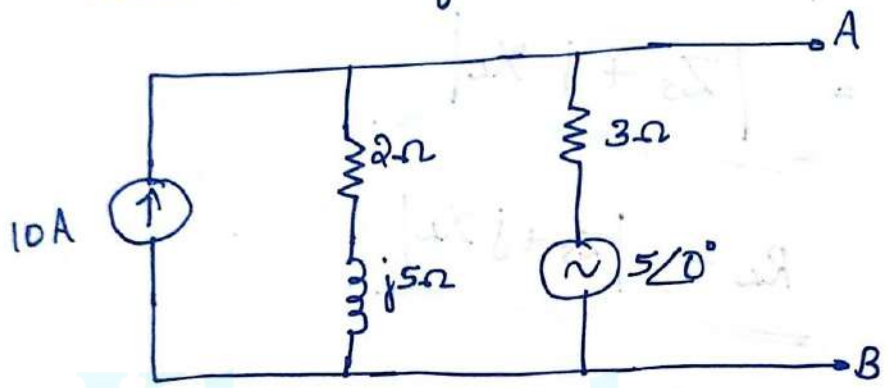
2. For the maximum power transfer, find the value of Z_L in the network for the following cases

i) Z_L is a variable resistance

ii) Z_L is complex impedance with variable resistance and variable reactance and

iii) Z_L is complex impedance with variable resistance & fixed reactance of $j5\Omega$.



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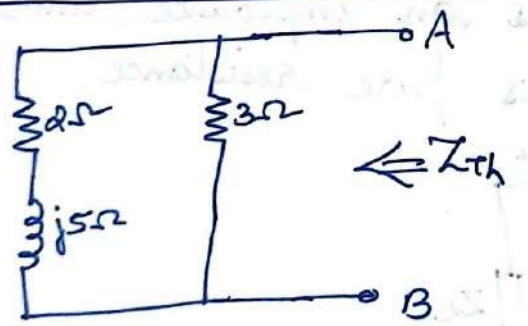
1. Calculation of Z_{Th}



$$Z_{Th} = \frac{3(2+j5)}{5+j5}$$

$$= (2.1 + j0.9)\,\Omega$$

i) Z_L is variable resistance; $Z_L = |Z_{Th}|$

$$= (2.1 + j0.9)\,\Omega = 2.28\angle 23.2^\circ$$

ii) R_L & X_L are variable; $Z_L = Z_{Th}^* = (2.1 - j0.9)\,\Omega$

iii) R_L variable, X_L fixed
$X_L = j5\Omega$

$$Z_L = (Z_{Th} + X_L)\,\Omega$$

$$= 2.1 + j0.9 + j5 = (2.1 + j5.9)\,\Omega$$

$$= 6.26\angle 70.40\,\Omega$$

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$$R_L^2 = R_s^2 + (X_s + X_L)^2$$

$$R_L = \sqrt{R_s^2 + (X_s + X_L)^2}$$

$$= \left| R_s + j(X_s + X_L) \right|$$

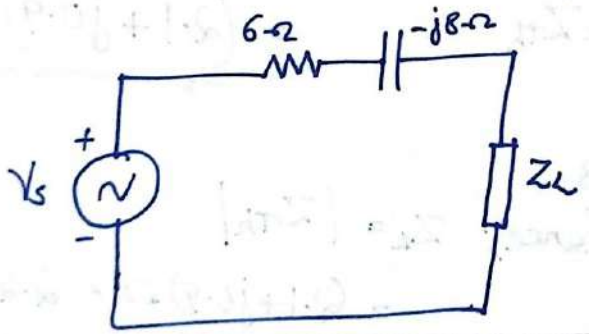
$$= \left| R_s + jX_s + jX_L \right|$$

$$= \underline{\underline{\left| Z_s + jX_L \right|}}.$$

Hence; $\underline{\underline{R_L = \left| Z_s + jX_L \right|}}$.

---

1. For maximum power transfer, find the value of Z_L if (i) Z_L is an impedance and (ii) Z_L is pure resistance



i) If Z_L is an impedance

$Z_L = Z_s^* = \underline{\underline{(6+j8)\Omega}}$

ii) If Z_L is a resistance

$Z_L = |Z_s| = |6+j8|$

$= \underline{\underline{10\Omega}}$

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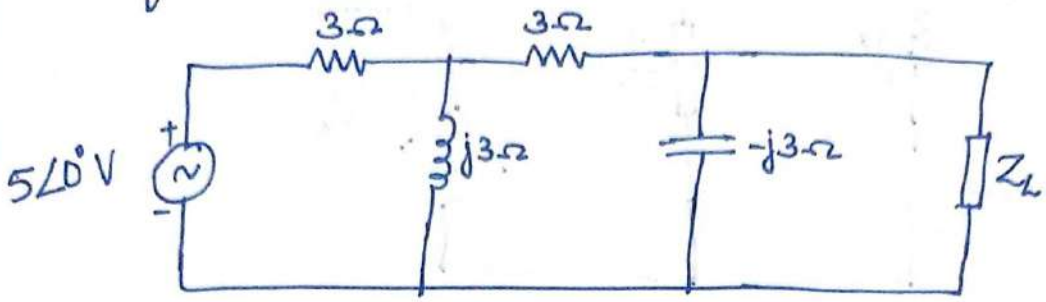
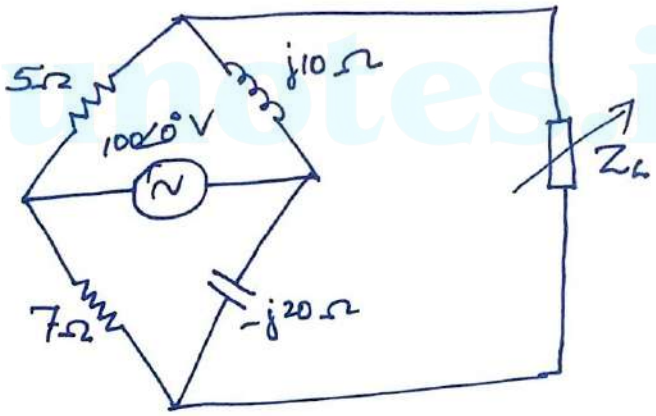
Tutorial Questions → Maximum Power Transfer Theorem (AC)

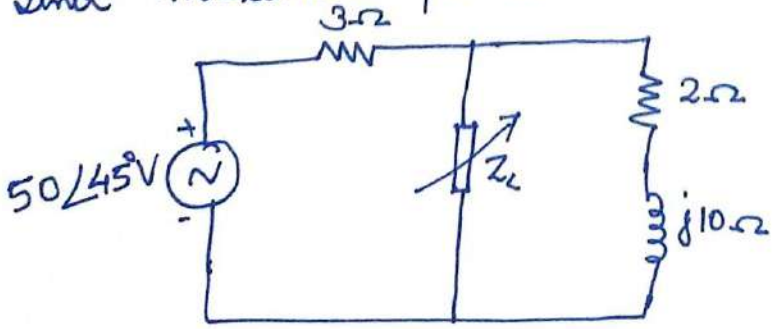
1. Find Z_L so that maximum power can be transferred to it. Find maximum power



2. Find the value of Z_L for maximum power transfer, and find maximum Power



3. Find the value of Load impedence Z_L so that max. power can be transferred to it. Find maximum power



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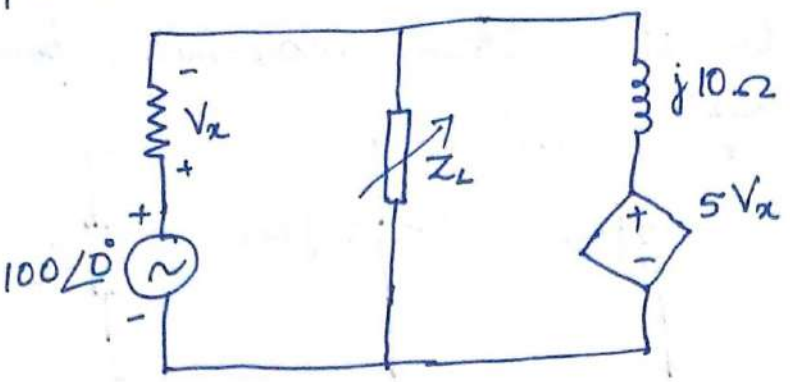
4. Find the value of Z_L that will transfer max. power from the source. Also find maximum power.



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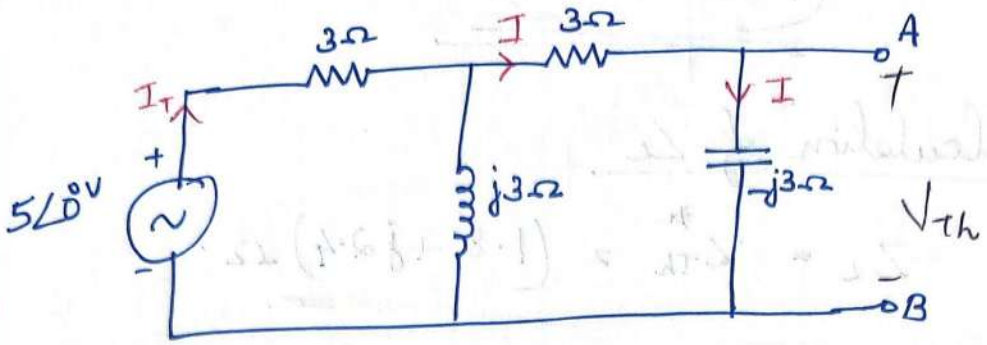
Answers (Tutorial) Max. Power Transfer (AC)

1. 1) Calculation of V_{Th}



$$Z_T = 3 + \frac{(3-j3)\,+j3}{3+j3-j3} = 3 + \frac{(3-j3)\times +j3}{3}$$

$$= \underline{\underline{6.71\angle +26.57^\circ}}$$

$$I_T = \frac{5\angle 0^\circ}{6.71\angle 26.57} = \underline{\underline{0.75\angle -26.57^\circ\ A}}$$

By Current division rule

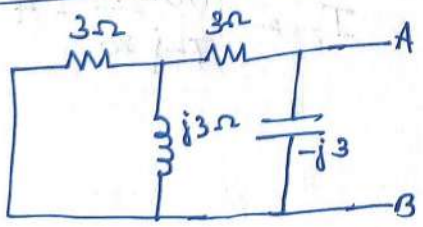
$$I = 0.75\angle -26.57^\circ \times \frac{j3}{j3+(3-j3)}$$

$$= \underline{\underline{0.75\angle 63.43^\circ A}}$$

$$V_{Th} = (-j3)\,I$$

$$= (-j3)\,(0.75\angle 63.43) = \underline{\underline{2.24\angle -26.57^\circ V}}$$

2) Calculation of Z_{Th}



$$Z_{Th} = \left[(3\|j3)+3\right]\|-j3$$

$$= \left(\frac{3\times j3}{3+j3}+3\right)\|-j3$$

$$= (4.7\angle 18.43)\|-j3$$

$$= \underline{\underline{3\angle -53^\circ\Omega}}$$

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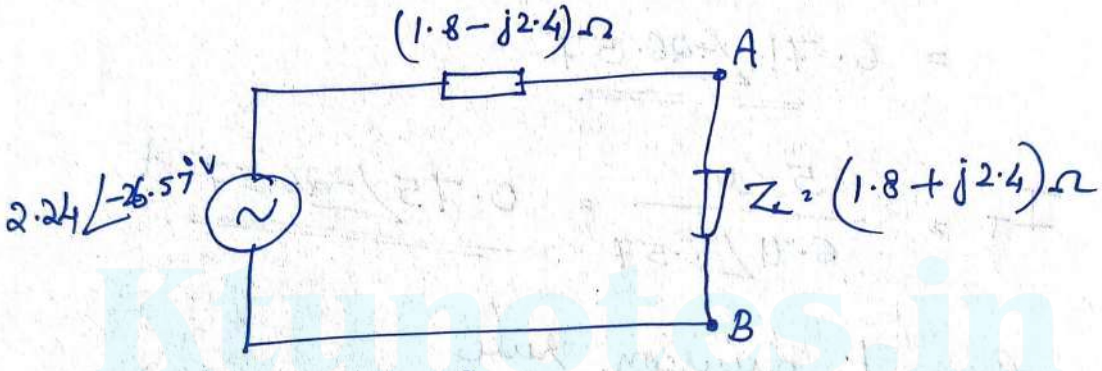
$Z_{Th} = 3\angle -53^\circ$

$= (1.8 - j2.4)\,\Omega$

3) Calculation of Z_L

$Z_L = Z_{Th}^* = (1.8 + j2.4)\,\Omega$.

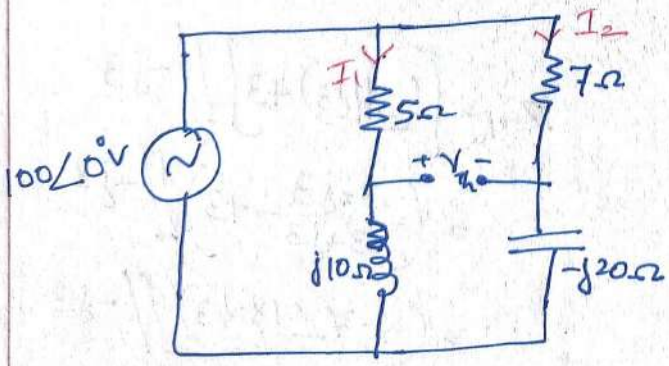
4) Calculation of P_{max}



$$P_{max} = \frac{|V_{Th}|^2}{4R_L} = \frac{(2.24)^2}{4\times 1.8}$$

$$= 0.7\,W$$

2. 1. Calculation of V_{Th}



$$I_1 = \frac{100\angle 0}{5+j10} = 8.94\angle -63.43^\circ\,A$$

$$I_2 = \frac{100\angle 0}{7-j20} = 4.72\angle 70.7^\circ\,A$$

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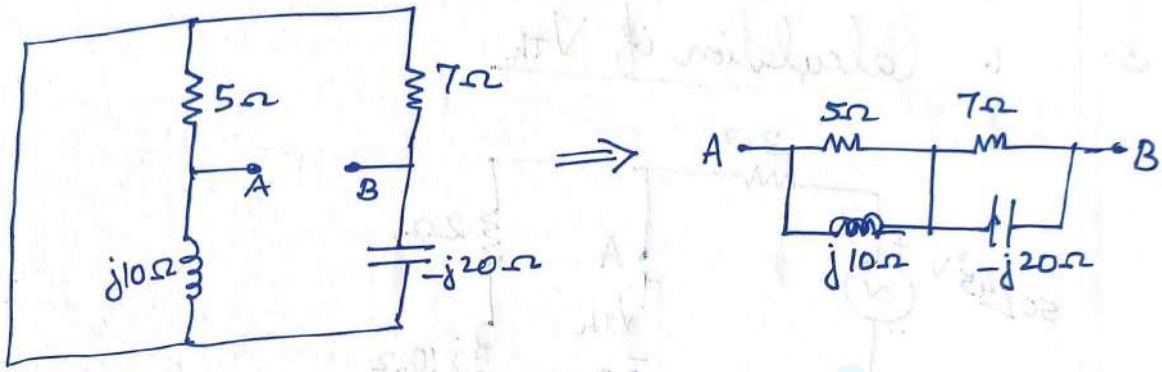
$$V_{Th} = V_A - V_B$$

$$= (8.94\angle -63.43^\circ)(j10) - (4.72\angle 70.7)(-j20)$$

$$= \underline{\underline{71.77\angle 97.3^\circ \text{ V}}}$$

2. Calculation of Z_{Th}



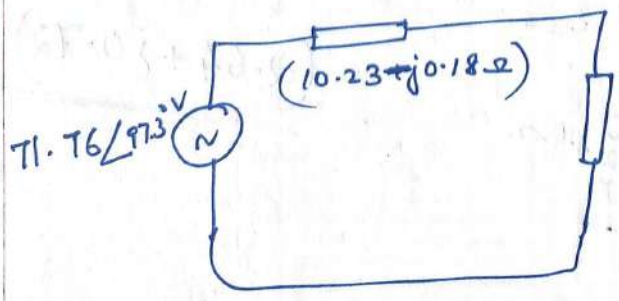
$$Z_{Th} = (5 // j10) + (7 // -j20)$$

$$= \frac{5 \times j10}{5 + j10} + \frac{7 \times -j20}{7 - j20}$$

$$= \underline{\underline{(10.23 - j0.18)\,\Omega}}$$

3. For Max. Power Transfer $Z_L = Z_{Th}^*$

$$Z_L = (10.23 + j0.18)\,\Omega$$



$Z_L = (10.23 + j0.18)\,\Omega$.

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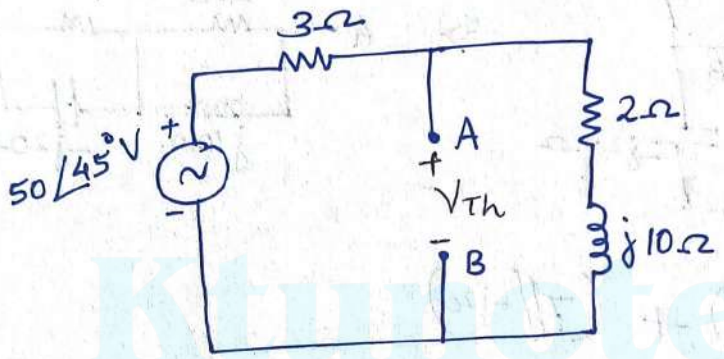
4. Calculation of P_{max}

$$P_{max} = \frac{|V_{Th}|^2}{4R_L} = \frac{|71.76|^2}{4 \times 10.23}$$

$$= \underline{\underline{125.84\ W}}$$

3. 1. Calculation of V_{Th}

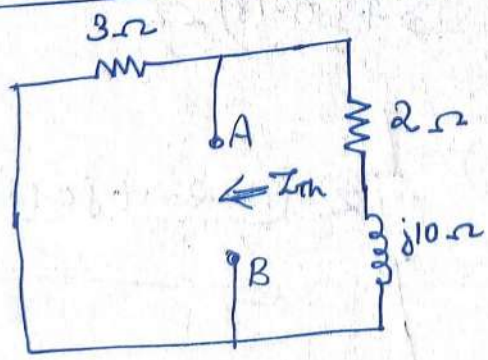


$$I = \frac{50\angle 45}{3+2+j10} = 4.47\angle -18.43$$

$$V_{Th} = (2+j10)\,I = (2+j10) \times 4.47\angle -18.43$$

$$= \underline{\underline{45.6\angle 60.26^\circ\ V}}$$

2. Calculation of Z_{Th}



$$Z_{Th} = \frac{3(2+j10)}{3+2+j10} = (2.64 + j0.72)\,\Omega$$

$$= \underline{\underline{(2.64 + j0.72)\,\Omega}}$$

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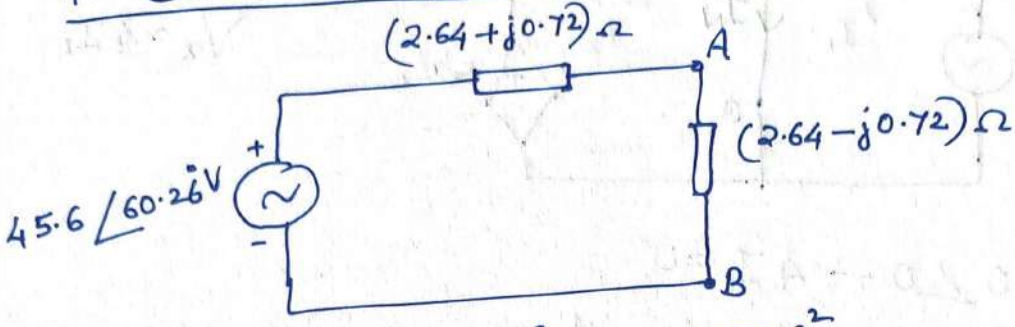
3. Calculation of Z_L

$$Z_L = |Z_S|^* = (2.64 - j\,0.72)\,\Omega$$

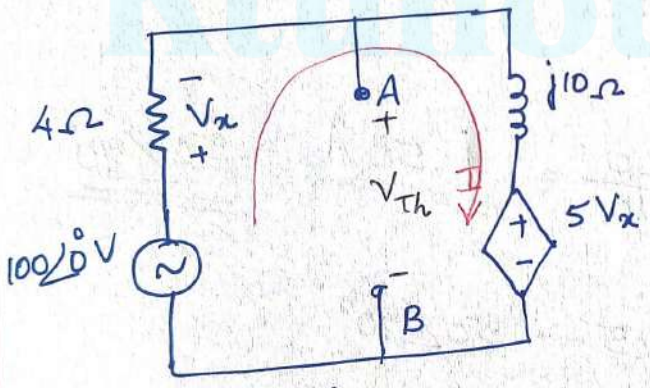
4. Calculation of P_{max}



$$P_{max} = \frac{|V_{Th}|^2}{4R_L} = \frac{(45.6)^2}{4 \times 2.64} = 196.91\,W$$

---

4. 1. Calculation of V_{Th}



$V_x = 4I$

KVL to mesh!

$$100\angle 0 - 4I - j10I - 5V_x = 0$$

$$100\angle 0 - 4I - j10I - 5(4I) = 0$$

$$I = \frac{100\angle 0}{24 + j10} = 3.85\angle -22.62^\circ\,A$$

V_{Th} eqn

$$100\angle 0 - 4I - V_{Th} = 0$$

$$V_{Th} = 100\angle 0 - 4(3.85\angle -22.62^\circ A)$$

$$= 86\angle 3.95^\circ\,V$$

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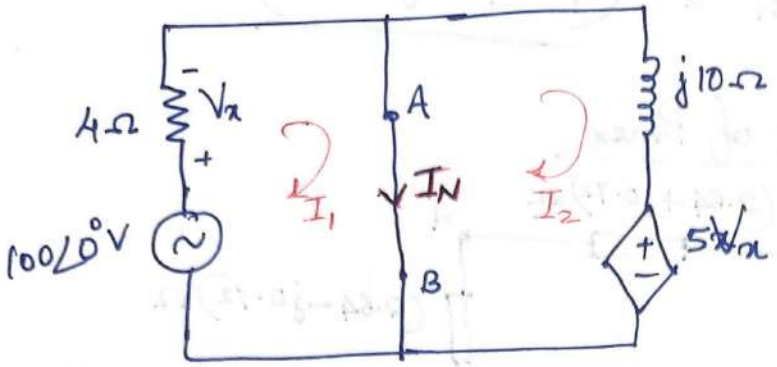
2. Calculation of I_N



$V_x = 4 I_1$

Mesh 1

$$100\angle 0 - 4 I_1 = 0$$

$$I_1 = \underline{\underline{25 A}}$$

Mesh 2

$$-j10 I_2 - 5 V_x = 0$$

$$-j10 I_2 - 5(4 I_1) = 0$$

$$-j10 I_2 = 20 I_1 = 20 \times 25$$

$$= 500$$

$$I_2 = -\frac{500}{j10} = \underline{\underline{j\,50 A}} = \underline{\underline{50\angle 90^\circ}}$$

$$I_N = I_1 - I_2 = 25 - 50\angle 90^\circ$$

$$= \underline{\underline{55.9\angle -63.43^\circ A}}$$

3. Calculation of Z_{Th}

$$Z_{Th} = \frac{V_{Th}}{I_N} = \frac{86\angle 3.95}{55.9\angle -63.43^\circ}$$

$$= 1.53\angle 67.38^\circ \Omega$$

$$= \underline{\underline{(0.59 + j1.42)\Omega}}$$

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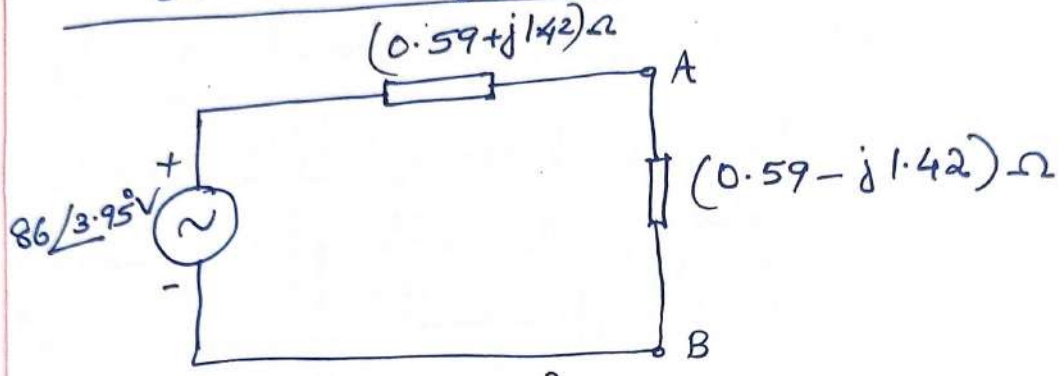
4. Calculation of Z_L

$$Z_L = Z_{Th}^* = (0.59 - j\,1.42)\,\Omega$$

5. Calculation of P_{max}



$$P_{max} = \frac{|V_{Th}|^2}{4 R_L}$$

$$= \frac{(86)^2}{4 \times 0.59} = 3133.9\,W$$

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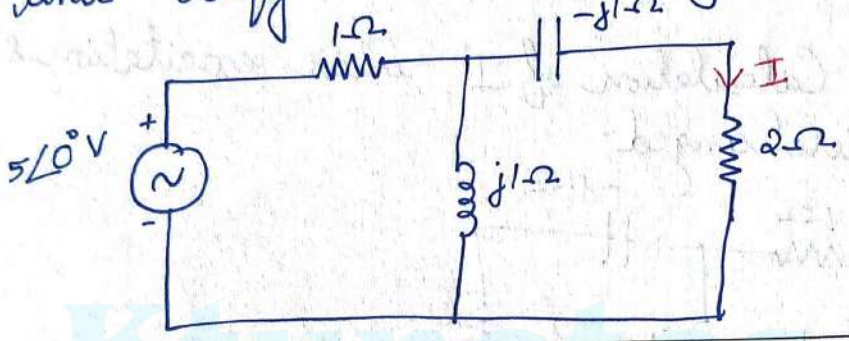
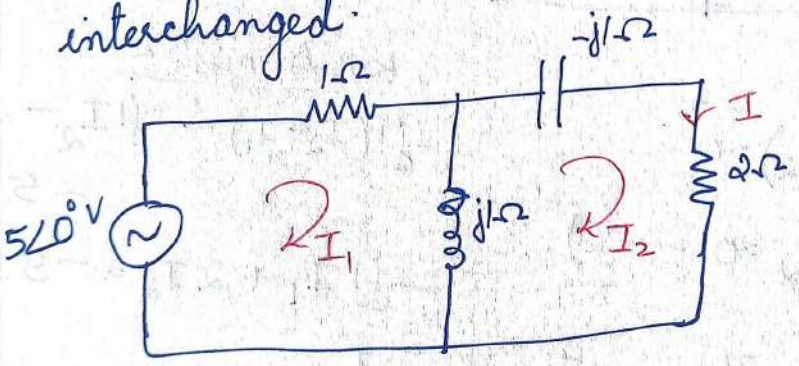
RECIPROCITY THEOREM (AC)

The Reciprocity Theorem states that "In a linear, bilateral, active, single-source network, the ratio of excitation to response remains same when the positions of exitation & Response are interchanged."

1. Find the current through the 2Ω resistor and verify the reciprocity Theorem.



Case 1: When excitation & response are not interchanged.



KVL to Mesh 1

$50 - 1I_1 - j1(I_1 - I_2) = 0$

$(1+j1)I_1 - j1 I_2 = 5\angle 0$ —— (1)

KVL to mesh 2

$-j1(I_2 - I_1) - j1 I_2 - 2I_2 = 0$

$-j1 I_1 + 2I_2 = 0$ —— (2)

Eqn (1) & (2) in matrix Form

$$\begin{bmatrix} 1+j1 & -j1 \\ -j1 & 2 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} 5\angle 0 \\ 0 \end{bmatrix}$$

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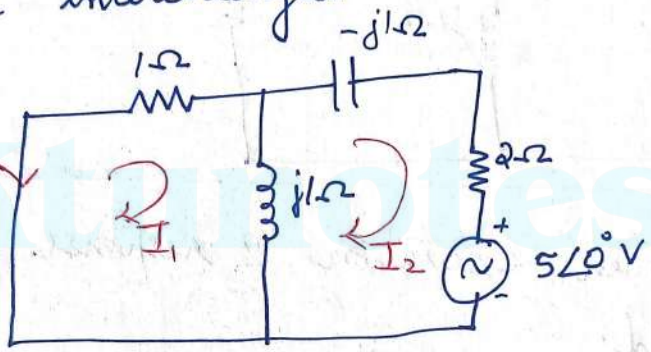
$$I_2 = \frac{\Delta_2}{\Delta}$$

$$I_2 = \frac{\begin{vmatrix} 1+j1 & 5\angle 0^\circ \\ -j1 & 0 \end{vmatrix}}{\begin{vmatrix} 1+j1 & -j1 \\ -j1 & 2 \end{vmatrix}} = \frac{j1 \times 5\angle 0}{2(1+j1) - j^2 1}$$

$$= \frac{j5}{2+j2+1} = \frac{j5}{3+j2} = \underline{\underline{1.39\angle 56.31^\circ A}}$$

<u>Case 2</u>: Calculation of I when excitation & response are interchanged.



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<u>KVL to mesh 1</u>

$$1 I_1 - j1(I_1 - I_2) = 0$$

$$(1+j1) I_1 - j1 I_2 = 0 \quad \text{— (1)}$$

<u>KVL to mesh 2</u>

$$-j1(I_2 - I_1) - -j1 I_2 - 2I_2 - 5\angle 0 = 0$$

$$-j1 I_1 + 2 I_2 = -5\angle 0^\circ \quad \text{— (2)}$$

$$\begin{bmatrix} (1+j1) & -j1 \\ -j1 & 2 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} 0 \\ -5\angle 0^\circ \end{bmatrix}$$

$$I_1 = \frac{\begin{vmatrix} 0 & -j1 \\ -5\angle 0 & 2 \end{vmatrix}}{\begin{vmatrix} (1+j1) & -j1 \\ -j1 & 2 \end{vmatrix}} = \frac{-j1 \times 5\angle 0}{2(1+j1) - j^2 1} = \frac{-j5}{2+j2+1}$$

$$= \underline{\underline{-1.38\angle 56.31^\circ A}}$$

$I = -I_1 = \underline{\underline{1.38\angle 56.31^\circ A}}$

Since I remains same, Reciprocity Theorem is verified.

$I = -I_1 = \underline{\underline{1.38\angle 56.31}}$

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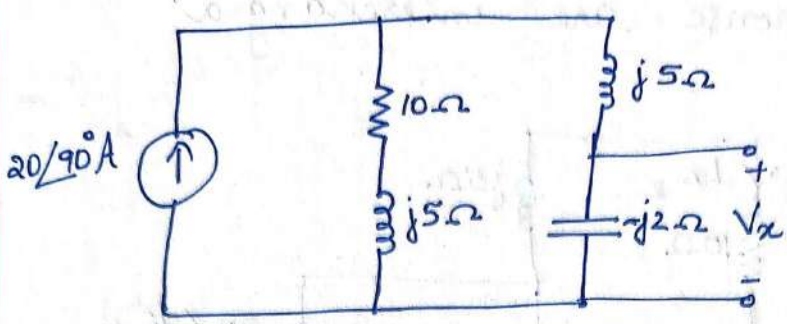
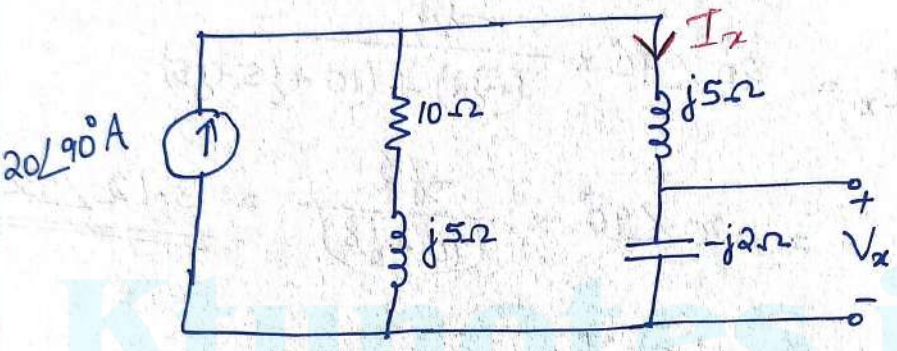
2. Find V_x and verify reciprocity theorem



Case 1: Calculation of V_x when excitation & response are interchanged.



By Current division rule,

$$I_x = \frac{20\angle 90^\circ \times (10+j5)}{(10+j5)+(j5-j2)}$$

$$= \frac{20\angle 90 \times (10+j5)}{(10+j8)} = \underline{\underline{17.46\angle 77.91^\circ A}}$$

$$V_x = (-j2)\, I_x = (-j2) \times 17.46\angle 77.91^\circ A$$

$$= \underline{\underline{34.92\angle 12.09 V}}$$

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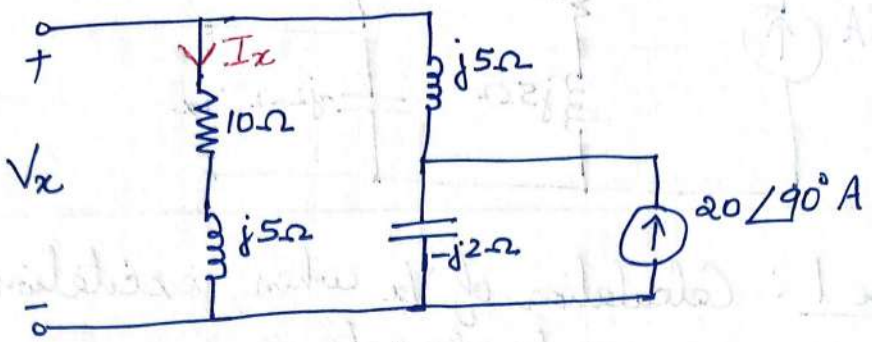
Case 2: Calculation of V_x when excitation and response are interchanged.



By Current division rule

$$I_x = 20\angle 90^\circ \times \frac{-j2}{(-j2)+(10+j5+j5)}$$

$$= 20\angle 90^\circ \times \frac{-j2}{(10+j8)} = \underline{\underline{3.12\angle -38.66^\circ}}$$

$$V_x = (10+j5)\, I_x$$

$$= (10+j5) \times 3.12\angle -38.66^\circ$$

$$= \underline{\underline{34.88\angle -12.09^\circ \text{ V}}}$$

Since V_x is same in both the cases, the Reciprocity Theorem is verified.

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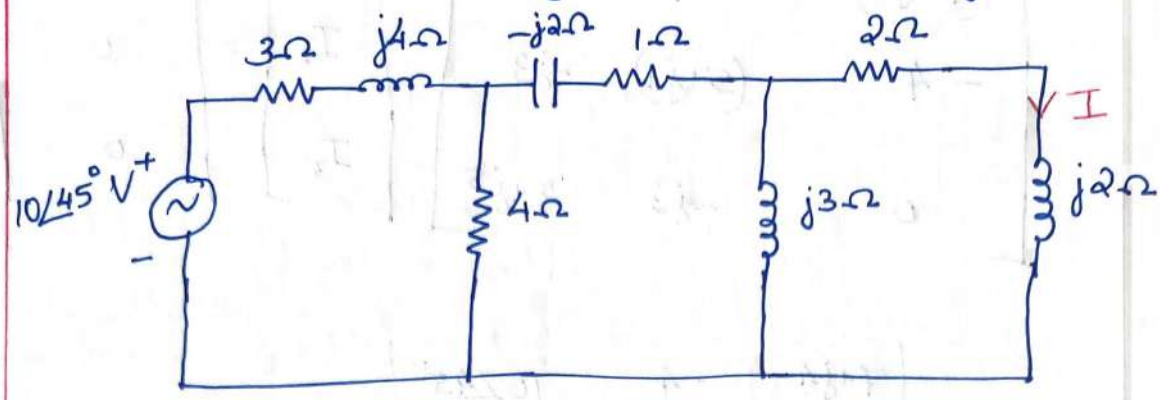
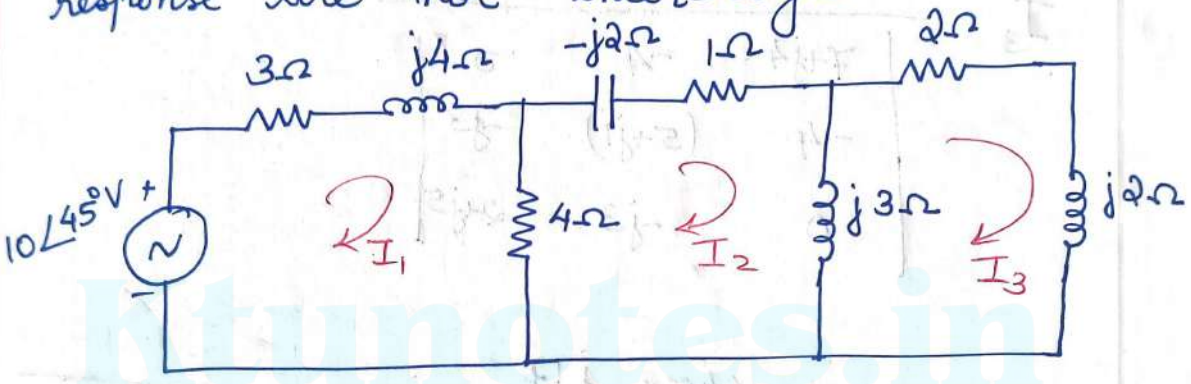
3. Find I and verify the Reciprocity theorem



Case 1: Calculation of I when excitation & response are not interchanged.



KVL to mesh 1

$$10\angle 45 - (3+j4)\,I_1 - 4(I_1 - I_2) = 0$$

$$(7+j4)\,I_1 - 4\,I_2 = 10\angle 45^\circ \quad \text{——— (1)}$$

KVL to mesh 2

$$-4(I_2 - I_1) - (1-j2)\,I_2 - j3(I_2 - I_1) = 0$$

$$-4I_1 + (5+j1)\,I_2 - j3\,I_3 = 0 \quad \text{——— (2)}$$

KVL to mesh 3

$$-j3(I_3 - I_2) - 2I_2 - j2\,I_3 = 0$$

$$-j3\,I_2 + (2+j5)\,I_3 = 0 \quad \text{——— (3)}$$

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$$\begin{bmatrix} (7+j4) & -4 & 0 \\ -4 & (5+j1) & -j3 \\ 0 & -j3 & 2+j5 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} 10\angle 45^\circ \\ 0 \\ 0 \end{bmatrix}$$

$$I_3 = \frac{\begin{vmatrix} 7+j4 & -4 & 10\angle 45^\circ \\ -4 & 5+j1 & 0 \\ 0 & -j3 & 0 \end{vmatrix}}{\begin{vmatrix} 7+j4 & -4 & 0 \\ -4 & (5+j1) & -j3 \\ 0 & -j3 & 2+j5 \end{vmatrix}}$$

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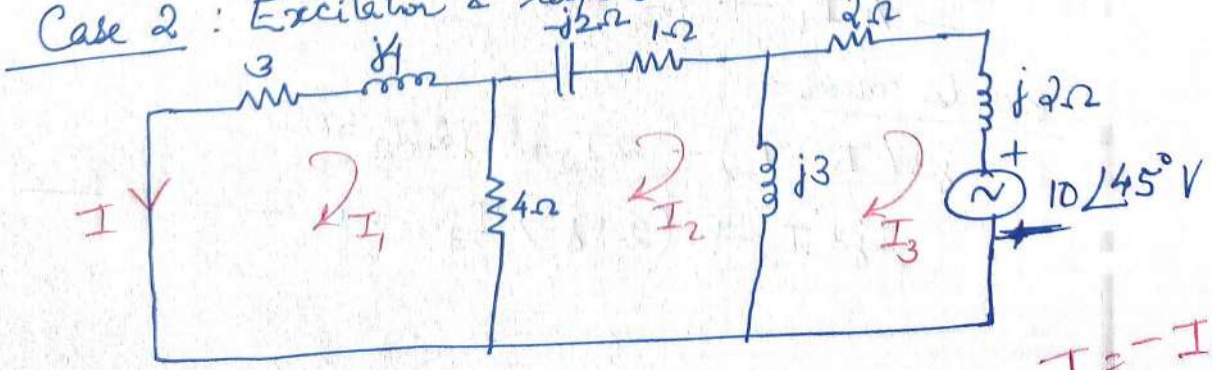
$$= \frac{10\angle 45 \times j12}{(7+j4)\left[(5+j1)(2+j5) - j^2 9\right] + 4\left[-4(2+j5)\right]}$$

$$= \frac{10\angle 45 \times j12}{(7+j4)(14+j27) - (+32+j80)}$$

$$= \underline{\underline{0.704\angle 30.72^\circ \text{ A}}}$$

Case 2 : Excitation & response are interchanged



$I = -I_1$

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KVL to mesh 1

$$(3+j4)I_1 - 4(I_1 - I_2) = 0$$

$$(7+j4)I_1 - 4I_2 = 0 \quad \text{——— (1)}$$

KVL to mesh 2

$$-4(I_2 - I_1) - (1-j2)I_2 - j3(I_2 - I_3) = 0$$

$$-4I_1 + (5+j1)I_2 - j3I_3 = 0 \quad \text{——— (2)}$$

KVL to mesh 3

$$-j3(I_3 - I_2) - (2+j2)I_2 - 10\angle 45^\circ = 0$$

$$-j3I_2 + (2+j5)I_3 = -10\angle 45^\circ \quad \text{——— (3)}$$

$$\begin{bmatrix} 7+j4 & -4 & 0 \\ -4 & 5+j1 & -j3 \\ 0 & -j3 & 2+j5 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ -10\angle 45^\circ \end{bmatrix}$$

$$I_1 = \frac{\begin{vmatrix} 0 & -4 & 0 \\ 0 & 5+j1 & -j3 \\ -10\angle 45 & -j3 & 2+j5 \end{vmatrix}}{\begin{vmatrix} 7+j4 & -4 & 0 \\ -4 & 5+j1 & -j3 \\ 0 & -j3 & 2+j5 \end{vmatrix}} = \cancel{\frac{10\angle 45\{(5+j1)(2+j5)-j^2 9\}}{(7+j4)(14+j27)-(32+j80)}}$$

$$= \frac{-10\angle 45 \times \{j12\}}{(7+j4)(14+j27)-(32+j80)} = \frac{-120\angle 135}{170.26\angle +104.28}$$

$$= -0.705\angle -30.72^\circ \text{ A}$$

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$I = -I_1$

$= 0.705 \angle 30.72^\circ \text{ A}$

$= \underline{\underline{0.705 \angle 30.72^\circ \text{ A}}}$

Since, the Current I is same in both the cases, the Reciprocity Theorem is verified.

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